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A — B
reactant

X — A
reactant

A — A
product

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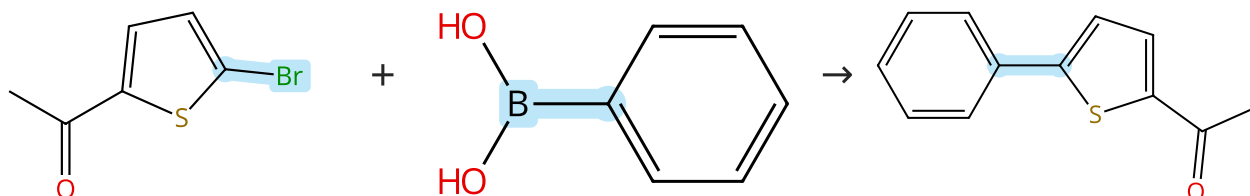
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Reactions (500)

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Scheme 1 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (85)

Suppliers (132)

Suppliers (58)

31-614-CAS-38412797

Steps: 1 Yield: 97%

Diimidazolium Salt HBDIM: An Easily Available, Low-Cost, CageCarbene Precursor with Broad Applications in Transition Metal-Catalyzed Reactions

By: Jiang, Weikang; et al

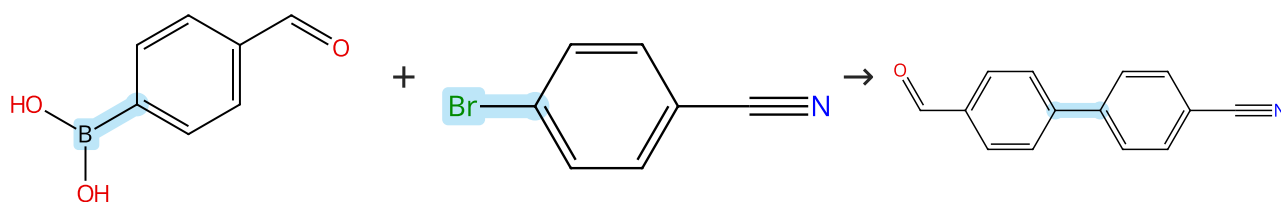
Chemistry - A European Journal (2023), 29(40), e202300991.

1.1 **Reagents:** Potassium carbonate
Catalysts: 3002982-91-1
Solvents: Ethanol, Toluene, Water; 5 h, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (106)

Suppliers (93)

Suppliers (45)

31-614-CAS-36331490

Steps: 1 Yield: 97%

Automated, Capsule-Based Suzuki-Miyaura Cross Couplings

By: Coin, Guillaume; et al

Organic Letters (2024), 26(14), 2708-2712.

1.1 **Reagents:** Cesium carbonate
Catalysts: Silica, Phosphine, diphenyl-, palladium(2+) salt (2:1)
Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 3.5 h, 60 °C; 60 °C → rt

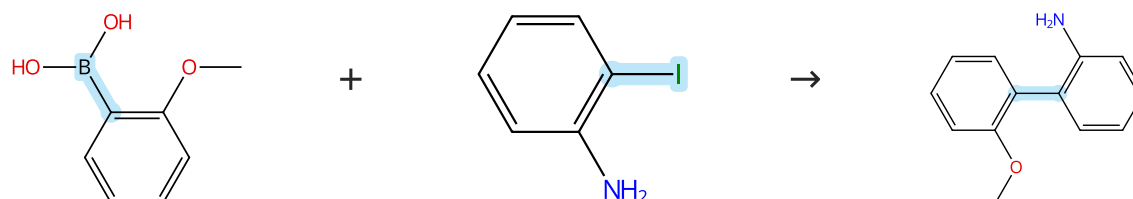
1.2 **Reagents:** Silica

1.3 **Reagents:** Methanaminium, *N,N,N*-trimethyl-, carbonate (2:1) (reagent supported); 2 h, rt

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (110)

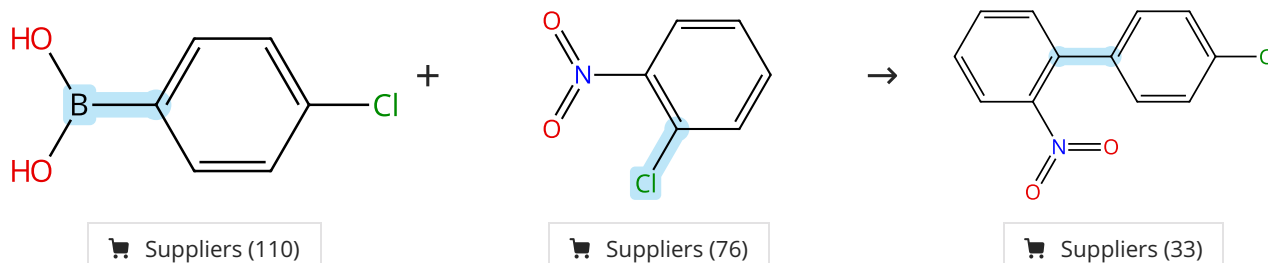
Suppliers (98)

Suppliers (41)

31-179-CAS-6496227	Steps: 1 Yield: 97%	Radical Fluoroalkylation of Isocyanides with Fluorinated Sulfones by Visible-Light Photoredox Catalysis
1.1 Reagents: Potassium carbonate Solvents: 1,2-Dimethoxyethane, Water; 30 min, rt		By: Rong, Jian; et al
1.2 Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane; rt; 12 h, 80 °C		Angewandte Chemie, International Edition (2016), 55(8), 2743-2747.
Experimental Protocols		

Scheme 4 (2 Reactions)

Steps: 1 Yield: 97%

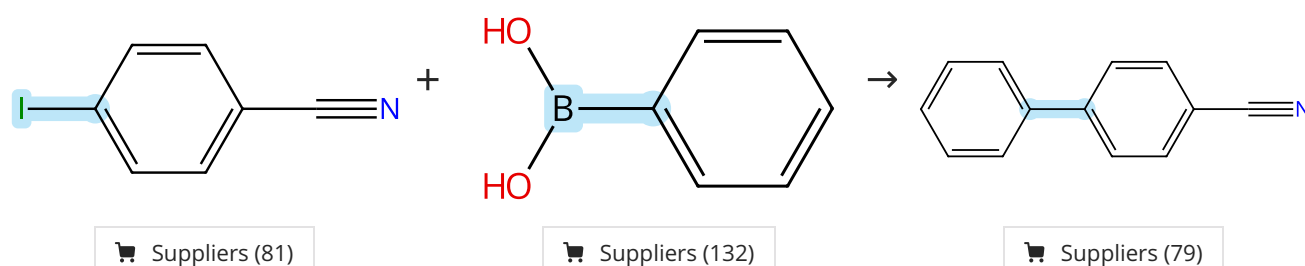


31-614-CAS-24285563	Steps: 1 Yield: 97%	Pd-Catalysed Suzuki-Miyaura cross-coupling of aryl chlorides at low catalyst loadings in water for the synthesis of industrially important fungicides
1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, <i>tert</i> -Butyldicyclohexylphosphine Solvents: Cyclohexane, Water; 3 h, 100 °C		By: Orecchia, Patrizio; et al
Experimental Protocols		Green Chemistry (2021), 23(20), 8169-8180.

31-179-CAS-15629119	Steps: 1 Yield: 97%	Tri(1-adamantyl)phosphine: Expanding the Boundary of Electron-Releasing Character Available to Organophosphorus Compounds
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(tricyclo[3.3.1.1 ^{3,7}]dec-1-yl)phosphine, Bis[2-[(acetyl- κ O)amino]phenyl- κ C]bis[μ -(4-methylbenzenesulfonato- κ O: κ O)]dipalladium Solvents: Tetrahydrofuran; 10 min, 100 °C		By: Chen, Liye; et al
Experimental Protocols		Journal of the American Chemical Society (2016), 138(20), 6392-6395.

Scheme 5 (1 Reaction)

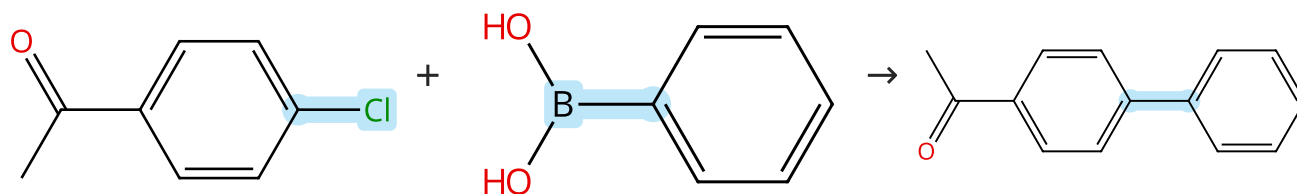
Steps: 1 Yield: 97%



31-614-CAS-45067485	Steps: 1 Yield: 97%	Continuous photocatalytic C-C coupling achieved using palladium anchored to a covalent organic framework
1.1 Reagents: Potassium carbonate Catalysts: 1,2-Acenaphthylenedione, polymer with 4,4',4''-(1,3,5-triazine-2,4,6-triyl)tris[benzenamine] (complex with palladium dichloride) Solvents: Ethanol; 6 h, rt		By: Ma, Chao; et al
Experimental Protocols		Chemical Communications (Cambridge, United Kingdom) (2025), 61(39), 7085-7088.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (68)

Suppliers (132)

Suppliers (100)

31-179-CAS-16111845

Steps: 1 Yield: 97%

Liquid-Assisted Grinding Accelerating: Suzuki-Miyaura Reaction of Aryl Chlorides under High-Speed Ball-Milling Conditions

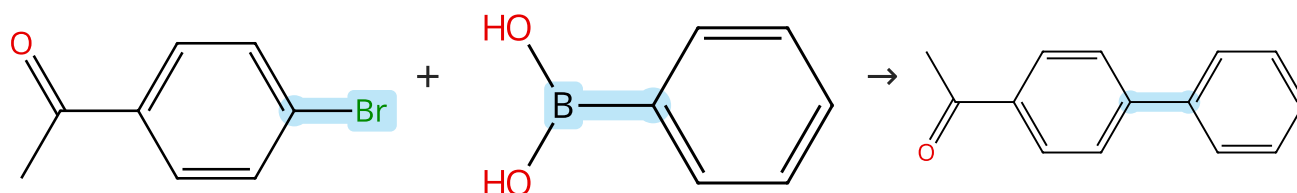
By: Jiang, Zhi-Jiang; et al

Journal of Organic Chemistry (2016), 81(20), 10049-10055.

1.1 **Reagents:** Methanol, Potassium carbonate
Catalysts: Palladium diacetate, Phosphine, tricyclohexyl-, tetrafluoroborate(1-) (1:1); 99 min

Scheme 7 (7 Reactions)

Steps: 1 Yield: 97%



Suppliers (84)

Suppliers (132)

Suppliers (100)

31-179-CAS-18618060

Steps: 1 Yield: 97%

One-Pot Tandem Photoredox and Cross-Coupling Catalysis with a Single Palladium Carbodicarbene Complex

By: Hsu, Yu-Cheng; et al

Angewandte Chemie, International Edition (2018), 57(17), 4622-4626.

1.1 **Reagents:** Potassium carbonate
Catalysts: 2213470-68-7
Solvents: Dimethylformamide; 20 h, 120 °C

31-179-CAS-17995103

Steps: 1 Yield: 97%

Synthesis of sterically hindered N-benzyladamantyl substituted benzimidazol-2-ylidene palladium complexes and investigation of their catalytic activity in aqueous medium

By: Dehimat, Zineb Imene; et al

Tetrahedron (2017), 73(40), 5940-5945.

1.1 **Reagents:** Potassium carbonate
Catalysts: 2130871-13-3
Solvents: Isopropanol, Water; 10 min, rt

1.2 **Reagents:** Water

Experimental Protocols

31-179-CAS-17995095

Steps: 1 Yield: 97%

Synthesis of sterically hindered N-benzyladamantyl substituted benzimidazol-2-ylidene palladium complexes and investigation of their catalytic activity in aqueous medium

By: Dehimat, Zineb Imene; et al

Tetrahedron (2017), 73(40), 5940-5945.

1.1 **Reagents:** Potassium carbonate
Catalysts: 2130871-09-7
Solvents: Isopropanol, Water; 10 min, rt

1.2 **Reagents:** Water

Experimental Protocols

31-179-CAS-17995099

Steps: 1 Yield: 97%

Synthesis of sterically hindered N-benzyladamantyl substituted benzimidazol-2-ylidene palladium complexes and investigation of their catalytic activity in aqueous medium

By: Dehimat, Zineb Imene; et al

Tetrahedron (2017), 73(40), 5940-5945.

1.1 **Reagents:** Potassium carbonate
Catalysts: 2130871-11-1
Solvents: Isopropanol, Water; 10 min, rt

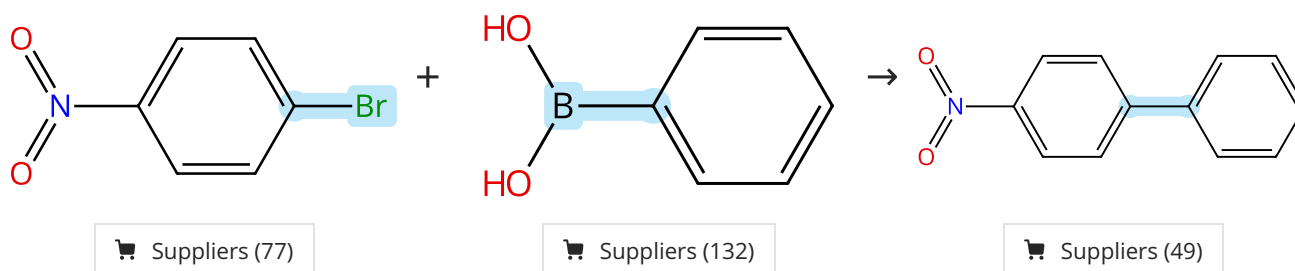
1.2 **Reagents:** Water

Experimental Protocols

31-179-CAS-18398973 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate, Oxygen Catalysts: 2157460-65-4 Solvents: <i>N</i>-Methyl-2-pyrrolidone, Water; 23 h, rt Experimental Protocols	Synthesis of Tetranuclear Palladium(II) Complexes and Their Catalytic Activity for Cross-Coupling Reactions By: Puls, Florian; et al Chemistry - A European Journal (2017), 23(69), 17576-17583.
31-179-CAS-7146872 Steps: 1 Yield: 97% 1.1 Catalysts: Palladium diacetate; 15 min, rt Experimental Protocols	A novel green protocol for ligand free Suzuki- Miyaura cross-coupling reactions in WEB at room temperature By: Boruah, Preeti Rekha; et al Green Chemistry (2015), 17(3), 1442-1445.
31-179-CAS-3748696 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate Solvents: 1,4-Dioxane, Water; 10 min, 40 °C 1.2 Solvents: 1,4-Dioxane, Water; 15 h, 40 °C Experimental Protocols	Suzuki cross-coupling in aqueous media By: Hoffmann, Ina; et al Green Chemistry (2015), 17(7), 3844-3857.

Scheme 8 (6 Reactions)

Steps: 1 Yield: 97%

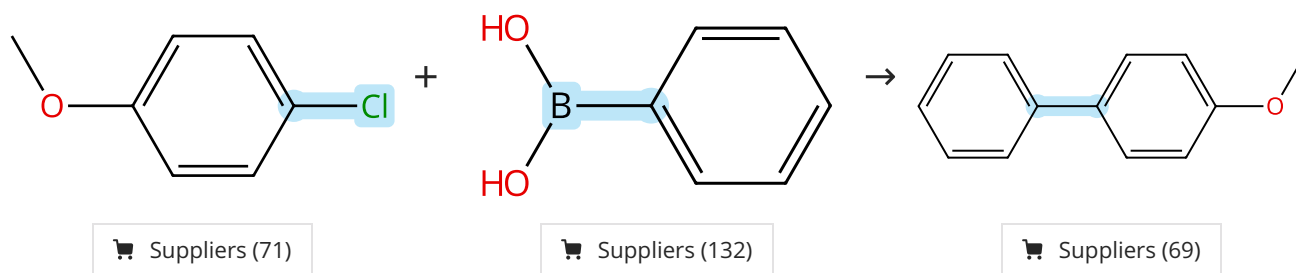


31-614-CAS-39816176 Steps: 1 Yield: 97% 1.1 Reagents: Propanoic acid, 2,2-dimethyl-, potassium salt (1:1) Catalysts: Palladium Solvents: γ-Valerolactone, Water; 90 - 120 °C Experimental Protocols	Microwave assisted batch and continuous flow Suzuki-Miyaura reactions in GVL using a Pd/PiNe biowaste-derived heterogeneous catalyst By: Valentini, Federica; et al Green Chemistry (2024), 26(8), 4871-4879.
31-614-CAS-24552986 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: 4,8,12-Trioxa-2,10-diazatetradecane-5,17-diamine (palladium acetate complex) Solvents: <i>p</i>-Xylene; 2.5 h, 150 °C	Postsynthetic Metalation of a Robust Hydrogen-Bonded Organic Framework for Heterogeneous Catalysis By: Han, Bin; et al Journal of the American Chemical Society (2019), 141(22), 8737-8740.
31-179-CAS-20509069 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Palladium, 2375720-23-1 (complex with palladium) Solvents: Water; 2 h, rt Experimental Protocols	An N-heterocyclic carbene-functionalised covalent organic framework with atomically dispersed palladium for coupling reactions under mild conditions By: Yang, Junjuan; et al Green Chemistry (2019), 21(19), 5267-5273.

31-179-CAS-18398972 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate, Oxygen Catalysts: 2157460-64-3 Solvents: <i>N</i>-Methyl-2-pyrrolidone, Water; 4 h, rt Experimental Protocols	Synthesis of Tetranuclear Palladium(II) Complexes and Their Catalytic Activity for Cross-Coupling Reactions By: Puls, Florian; et al Chemistry - A European Journal (2017), 23(69), 17576-17583.
31-179-CAS-17211235 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Ruthenium (sulfur modified, Au supported) Solvents: 1,2-Dimethoxyethane; 6 h, 80 °C; 80 °C → rt 1.2 Solvents: Water; 10 h, 100 °C Experimental Protocols	Ligand-free Suzuki-Miyaura coupling using ruthenium(0) nanoparticles and a continuously irradiating microwave system By: Akiyama, Toshiki; et al Green Chemistry (2017), 19(14), 3357-3369.
31-179-CAS-16424877 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Palladium chloride, 1<i>H</i>-imidazolium, 3-dodecyl-1,2-dimethyl-, <i>P,P</i>-bis(2,4,4-trimethylpentyl)phosphinate (1:1) Solvents: Heptane, Water; 1 h, 80 °C Experimental Protocols	Ionic Liquid-Based Microemulsions in Catalysis By: Hejazifar, Mahtab; et al Journal of Organic Chemistry (2016), 81(24), 12332-12339.

Scheme 9 (2 Reactions)

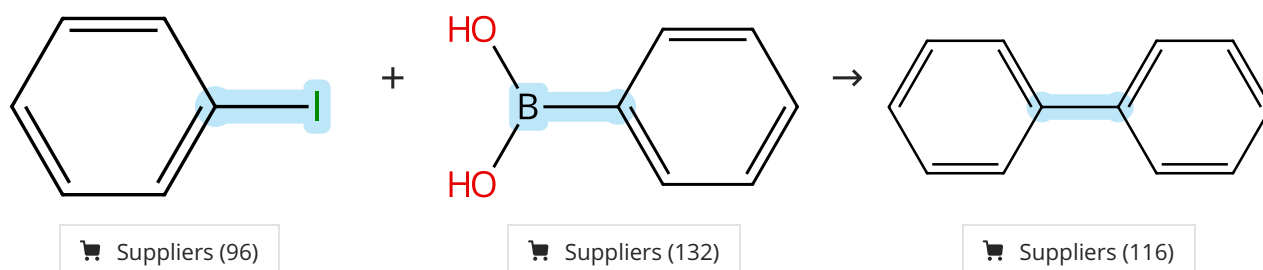
Steps: 1 Yield: 97%



31-614-CAS-40877960 Steps: 1 Yield: 97% 1.1 Reagents: Cesium carbonate Catalysts: 3049230-94-3 Solvents: 1,4-Dioxane; 8 h, 90 °C Experimental Protocols	N-Indole-Substituted Imidazolyldiene Pd-PEPPSI Precatalysts: Enhanced Performance with a 3,5-Diisopropyl-4-indolyl Moiety By: Peng, Jiahao; et al Journal of Organic Chemistry (2024), 89(13), 9322-9335.
31-179-CAS-12244068 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Palladium (complexes with tris(<i>p</i>-vinylphenyl) phosphine homopolymer), Phosphine, tris(4-ethenylphenyl)-, homopolymer Solvents: Trifluorotoluene; 10 h, 100 °C Experimental Protocols	A polymer-bound monodentate-P-ligated palladium complex as a recyclable catalyst for the Suzuki-Miyaura coupling reaction of aryl chlorides By: Zhou, Yun-Bing; et al Advanced Synthesis & Catalysis (2015), 357(11), 2503-2508.

Scheme 10 (1 Reaction)

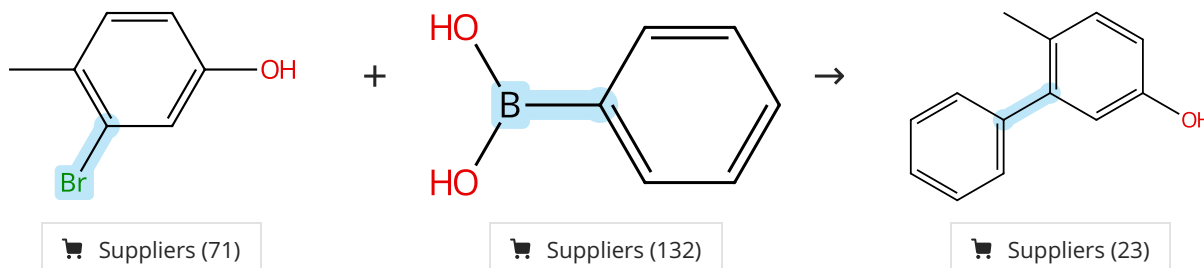
Steps: 1 Yield: 97%



31-179-CAS-20329738	Steps: 1 Yield: 97%	Stainless steel mesh-GO/Pd NPs: catalytic applications of Suzuki-Miyaura and Stille coupling reactions in eco-friendly media By: Feizi Mohazzab, Bahareh; et al Green Chemistry (2019), 21(12), 3319-3327.
1.1 Reagents: Potassium carbonate Catalysts: Palladium Solvents: Ethanol, Water; 90 min, 90 °C		
Experimental Protocols		

Scheme 11 (1 Reaction)

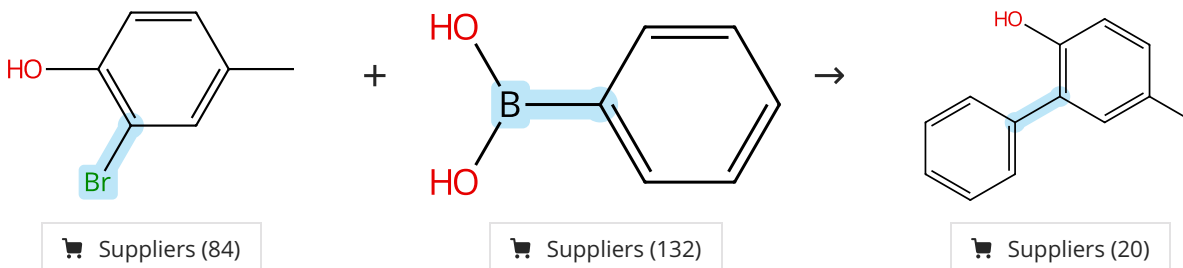
Steps: 1 Yield: 97%



31-614-CAS-24826666	Steps: 1 Yield: 97%	Atom Economical Photocatalytic Oxidation of Phenols and Site-Selective Epoxidation Toward Epoxyquinols By: Peault, Louis; et al Journal of Organic Chemistry (2021), 86(24), 18192-18203.
1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 16 h, reflux		
Experimental Protocols		

Scheme 12 (1 Reaction)

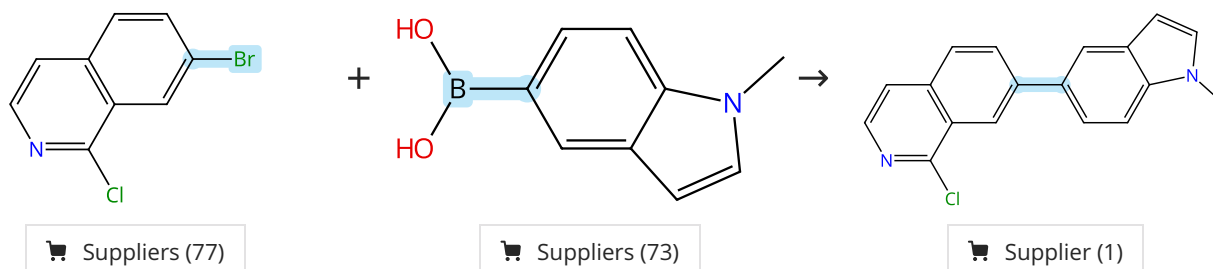
Steps: 1 Yield: 97%



31-614-CAS-31767656	Steps: 1 Yield: 97%	A recyclable self-supported nanoporous PdCu heterogeneous catalyst for aqueous Suzuki-Miyaura cross-coupling By: Du, Jiale; et al Chemical Communications (Cambridge, United Kingdom) (2021), 57(88), 11641-11644.
1.1 Reagents: Sodium carbonate Catalysts: Palladium alloy, base, Pd 83,Cu 17 Solvents: Ethanol, Water; 6 h, 80 °C		
Experimental Protocols		

Scheme 13 (1 Reaction)

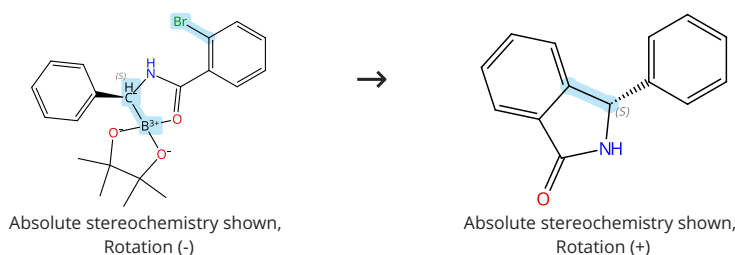
Steps: 1 Yield: 97%



31-179-CAS-20985252	Steps: 1 Yield: 97%	N,C-Disubstituted Biaryl palladacycles as Precatalysts for ppm Pd-Catalyzed Cross Couplings in Water under Mild Conditions
1.1 Reagents: Tripotassium phosphate, Poly(oxy-1,2-ethanediyl), α-[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyl tridecyl)-2<i>H</i>-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy- Catalysts: [Dicyclohexyl[2,6-dimethoxy-3-(2-methoxy-1-naphthalenyl)phenyl]phosphine-κP][4-(1-methylethyl)-2'-[(1-methylethyl)amino-κM][1,1'-biphenyl]-2-yl-κC](1,1,1-trifluoro methanesulfonato-κO)palladium Solvents: Tetrahydrofuran, Water; 26 h, 55 °C		By: Thakore, Ruchita R.; et al ACS Catalysis (2019), 9(12), 11647-11657.
Experimental Protocols		

Scheme 14 (1 Reaction)

Steps: 1 Yield: 97%

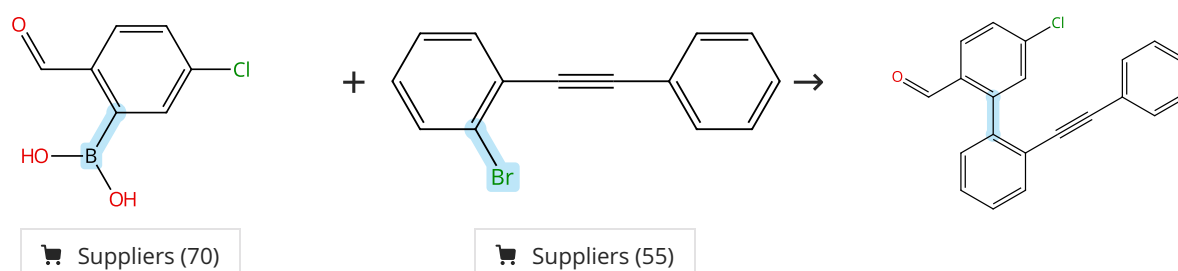


Suppliers (3)

31-179-CAS-22896955	Steps: 1 Yield: 97%	Enantioconvergent Cu-Catalyzed Intramolecular C-C Coupling at Boron-Bound C(sp³) Atoms of α-Aminoalkylboronates Using a C₁-Symmetrical 2,2'-Bipyridyl Ligand Attached to a Helically Chiral Macromolecular Scaffold
1.1 Catalysts: Cupric chloride, 5-[2,2'-Bipyridin]-6-yl-8-methyl-2,3-diphenylquinoxaline Solvents: Ethanol, Toluene; 1 h, 50 °C 1.2 Reagents: Cesium carbonate Solvents: Chloroform, Toluene, Water; 24 h, 10 °C		By: Yoshinaga, Yukako; et al Journal of the American Chemical Society (2020), 142(43), 18317-18323.
Experimental Protocols		

Scheme 15 (1 Reaction)

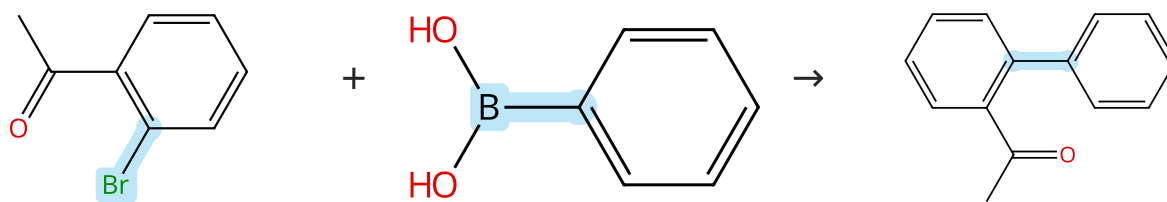
Steps: 1 Yield: 97%



31-179-CAS-22788088	Steps: 1 Yield: 97%	Selective Synthesis of Phenanthrenes and Dihydrophe nanthrenes via Gold-Catalyzed Cycloisomerization of Biphenyl Embedded Trienynes
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene; 24 h, 90 °C		By: Milian, Ana; et al Organic Letters (2020), 22(21), 8464-8469.

Scheme 16 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (89)

Suppliers (132)

Suppliers (62)

31-614-CAS-35310270

Steps: 1 Yield: 97%

o-Nitrobenzyl Oxime Ethers Enable Photoinduced Cyclization Reaction to Provide Phenanthridines under Aqueous Conditions

By: Okamura, Hidenori; et al

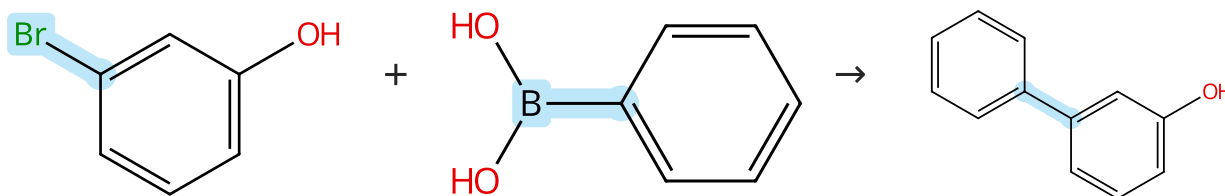
Organic Letters (2023), 25(3), 466-470.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene; 19 h, reflux

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (74)

Suppliers (132)

Suppliers (73)

31-179-CAS-17010185

Steps: 1 Yield: 97%

Design, Synthesis, Structure-Activity Relationship Studies, and Three-Dimensional Quantitative Structure-Activity Relationship (3D-QSAR) Modeling of a Series of O-Biphenyl Carbamates as Dual Modulators of Dopamine D3 Receptor and Fatty Acid Amide Hydrolase

By: De Simone, Alessio; et al

Journal of Medicinal Chemistry (2017), 60(6), 2287-2304.

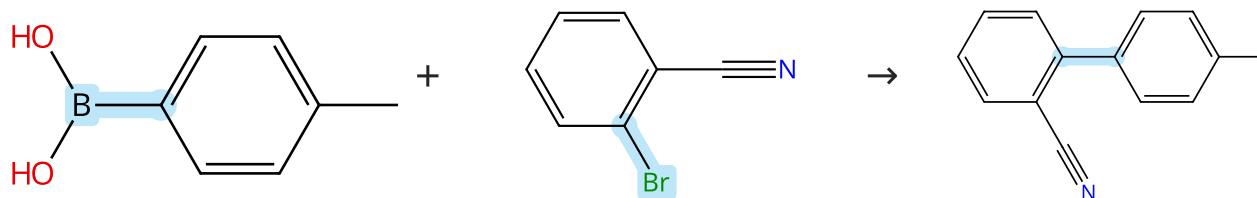
1.1 **Reagents:** Potassium carbonate
Catalysts: Palladium diacetate
Solvents: 2-Methoxyethanol, Water; > 1 min, rt; 5 min, rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; acidified, rt

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (114)

Suppliers (79)

Suppliers (75)

31-179-CAS-772584

Steps: 1 Yield: 97%

In Situ Trapped and Immobilized Palladium Nanoparticles as Active and Clean Catalysts for Suzuki-Miyaura Reaction

By: Li, Yan; et al

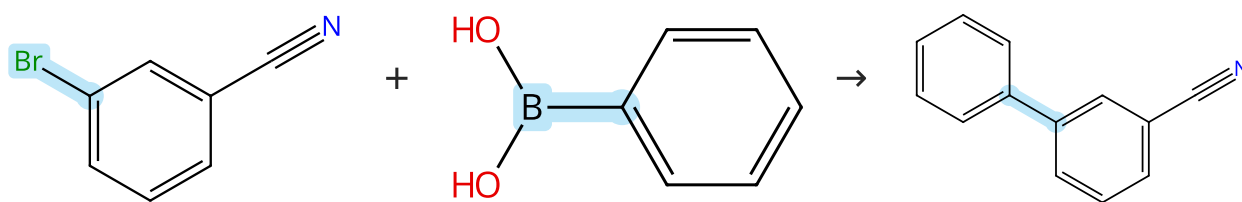
Advanced Synthesis & Catalysis (2015), 357(13), 2827-2832.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium
Solvents: Dimethylformamide; 12 h, 120 °C

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (80)

Suppliers (132)

Suppliers (53)

31-179-CAS-16686608

Steps: 1 Yield: 97%

Suzuki-Miyaura Micellar Cross-Coupling in Water, at Room Temperature, and under Aerobic Atmosphere

By: Mattiello, Sara; et al

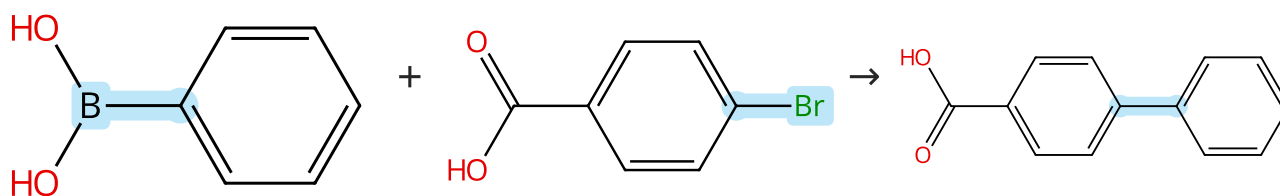
Organic Letters (2017), 19(3), 654-657.

1.1 Reagents: Triethylamine
 Catalysts: (SP-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino-κ^P]ferrocene]dichloropalladium
 Solvents: Water; 1 h, rt

Experimental Protocols

Scheme 20 (3 Reactions)

Steps: 1 Yield: 97%



Suppliers (132)

Suppliers (97)

Suppliers (102)

31-614-CAS-37908810

Steps: 1 Yield: 97%

Reduced Tiara-like Palladium Complex for Suzuki Cross-Coupling Reactions

By: Daka, Mario; et al

Chemistry - A European Journal (2023), 29(61), e202301740.

1.1 Reagents: Potassium carbonate
 Catalysts: Alumina, Palladium
 Solvents: Ethanol, Water; 2 min, rt

1.2 30 min, 50 °C

Experimental Protocols

31-614-CAS-36968182

Steps: 1 Yield: 97%

Continuous-Flow Suzuki-Miyaura Coupling in Water and Organic Solvents Promoted by Blends of Stabilized Convuluted Polymeric Palladium Catalysts and Polymeric Auxiliary Materials

By: Zhang, Zhenzhong; et al

Chemistry - A European Journal (2023), 29(34), e202300494.

1.1 Reagents: Tripotassium phosphate
 Catalysts: Benzene, diethenyl-, polymer with 1-(1,1-dimethylethyl)-4-ethenylbenzene, Pyridine, 4-ethenyl-, polymer with 1-(1,1-dimethylethyl)-4-ethenylbenzene
 Solvents: Ethanol, Tetrahydrofuran, Water; 5 h, 70 °C

1.2 Solvents: Water

Experimental Protocols

31-179-CAS-959175

Steps: 1 Yield: 97%

Activation of Aryl Chlorides in Water under Phase-Transfer Agent-Free and Ligand-Free Suzuki Coupling by Heterogeneous Palladium Supported on Hybrid Mesoporous Carbon

By: Duan, Linlin; et al

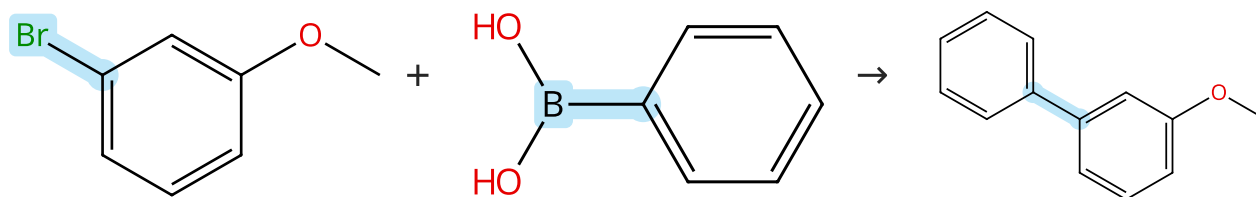
ACS Catalysis (2015), 5(2), 575-586.

1.1 Reagents: Potassium carbonate, Oxygen
 Catalysts: Palladium
 Solvents: Water; 8 h, 80 °C

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (80)

Suppliers (132)

Suppliers (51)

31-179-CAS-23512425

Steps: 1 Yield: 97%

Selective C-O Bond Reduction and Borylation of Aryl Ethers Catalyzed by a Rhodium-Aluminum Heterobimetallic Complex

By: Seki, Rin; et al

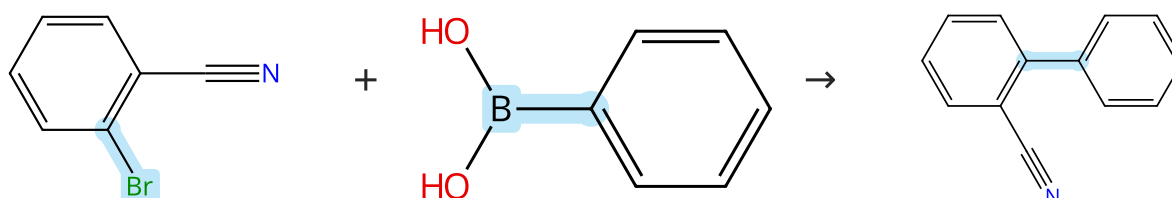
Journal of the American Chemical Society (2021), 143(17), 6388-6394.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; 12 h, 80 °C

Experimental Protocols

Scheme 22 (3 Reactions)

Steps: 1 Yield: 97%



Suppliers (79)

Suppliers (132)

Suppliers (63)

31-179-CAS-21608051

Steps: 1 Yield: 97%

Pd@COF-QA: A phase transfer composite catalyst for aqueous Suzuki-Miyaura coupling reaction

By: Wang, Jian-Cheng; et al

Green Chemistry (2020), 22(4), 1150-1155.

1.1 **Reagents:** Triethylamine
Catalysts: Palladium (supported on paraffin-chain quaternary ammonium salt decorated covalen...), 2412753-38-7
Solvents: Water; 6 h, 50 °C

31-179-CAS-20952874

Steps: 1 Yield: 97%

A palladium-carbon-connected organometallic framework and its catalytic application

By: Dong, Ying; et al

Chemical Communications (Cambridge, United Kingdom) (2019), 55(96), 14414-14417.

1.1 **Reagents:** Potassium carbonate
Catalysts: Stereoisomer of bis[μ-[1,3-bis[2-[4-(isocyano-κC)-3,5-bis(1-methylethyl)phenyl]ethynyl]-5-[2-[4-isocyano-3,5-bis(1-methylethyl)phenyl]ethynyl]benzene]]hexaiodotripalladium homopolymer
Solvents: Methanol; 8 h, 70 °C

31-179-CAS-16958554

Steps: 1 Yield: 97%

Di- and Tetranuclear Palladium(II) Complexes Containing C,N-Bidentate Furoylhydrazone for Suzuki-Miyaura Reactions

By: Qian, Hengyu; et al

European Journal of Organic Chemistry (2017), 2017(10), 1337-1342.

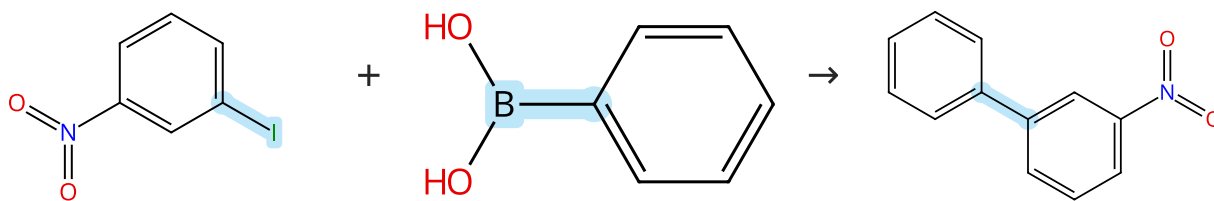
1.1 **Reagents:** Tripotassium phosphate
Catalysts: 2089585-08-8
Solvents: Toluene; 2 h, 60 °C

1.2 **Reagents:** Water

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (52)

Suppliers (132)

Suppliers (66)

31-614-CAS-42942535

Steps: 1 Yield: 97%

Heterogenization of palladium trimer and nanoparticles through polymerization boosted catalytic efficiencies in recyclable coupling and reduction reactions

By: Wang, Miaomiao; et al

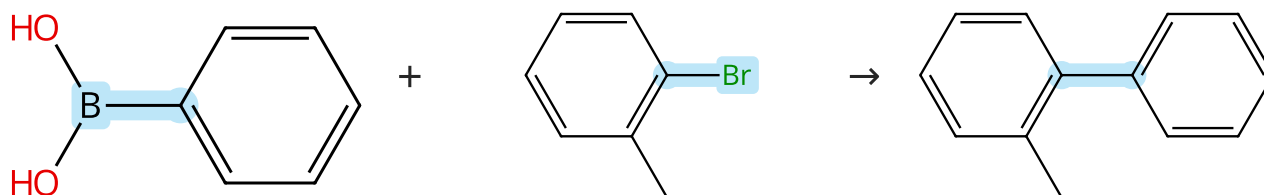
Chemistry - A European Journal (2024), 30(72), e202403447.

1.1 Reagents: Potassium carbonate
Catalysts: 3066141-61-2
Solvents: Acetonitrile, Water; 80 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

Suppliers (60)

Suppliers (63)

31-179-CAS-10488473

Steps: 1 Yield: 97%

In Situ Trapped and Immobilized Palladium Nanoparticles as Active and Clean Catalysts for Suzuki-Miyaura Reaction

By: Li, Yan; et al

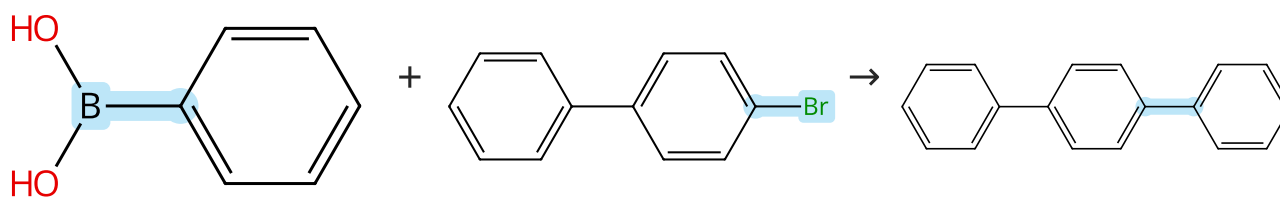
Advanced Synthesis & Catalysis (2015), 357(13), 2827-2832.

1.1 Reagents: Tripotassium phosphate
Catalysts: Palladium
Solvents: Dimethylformamide; 6 h, 120 °C

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

Suppliers (89)

Suppliers (111)

31-614-CAS-37908788

Steps: 1 Yield: 97%

Reduced Tiara-like Palladium Complex for Suzuki Cross-Coupling Reactions

By: Daka, Mario; et al

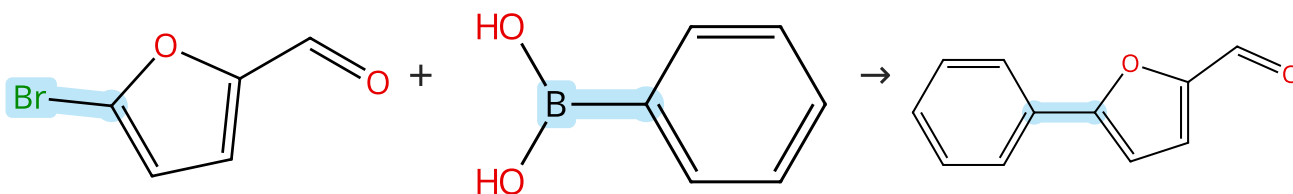
Chemistry - A European Journal (2023), 29(61), e202301740.

1.1 Reagents: Potassium carbonate
Catalysts: Alumina, Palladium
Solvents: Ethanol, Water; 2 min, rt
1.2 30 min, 50 °C

Experimental Protocols

Scheme 26 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (90)

Suppliers (132)

Suppliers (71)

31-179-CAS-18736658

Steps: 1 Yield: 97%

Access to Densely Functionalized Chalcone Derivatives with a 2-Pyridone Subunit via Pd/Cu-Catalyzed Oxidative Furan-Yne Cyclization of N-(2-Furanylmethyl) Alkynamides under Air

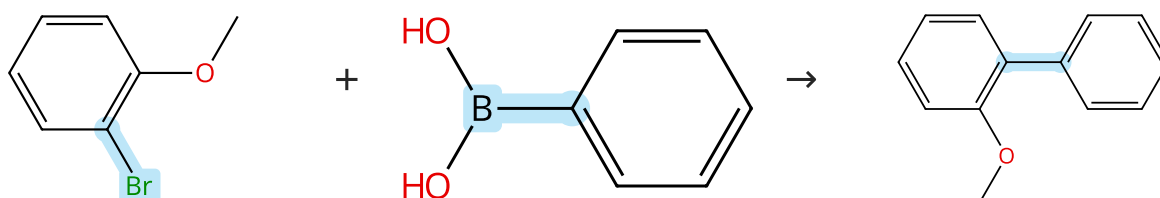
By: Yang, Yongjie; et al

Organic Letters (2018), 20(8), 2273-2277.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium

Scheme 27 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (84)

Suppliers (132)

Suppliers (66)

31-179-CAS-6612084

Steps: 1 Yield: 97%

In Situ Trapped and Immobilized Palladium Nanoparticles as Active and Clean Catalysts for Suzuki-Miyaura Reaction

By: Li, Yan; et al

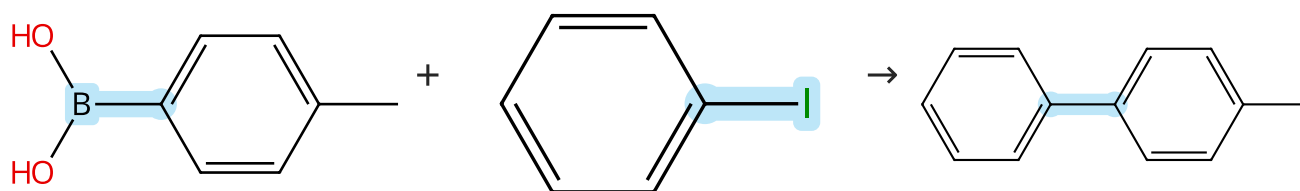
Advanced Synthesis & Catalysis (2015), 357(13), 2827-2832.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium
Solvents: Dimethylformamide; 6 h, 120 °C

Experimental Protocols

Scheme 28 (2 Reactions)

Steps: 1 Yield: 97%



Suppliers (114)

Suppliers (96)

Suppliers (83)

31-614-CAS-34076995

Steps: 1 Yield: 97%

A palladium polyaniline complex: a simple and efficient catalyst for batch and flow Suzuki-Miyaura cross-couplings

By: Zhang, Dongliang; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(77), 10845-10848.

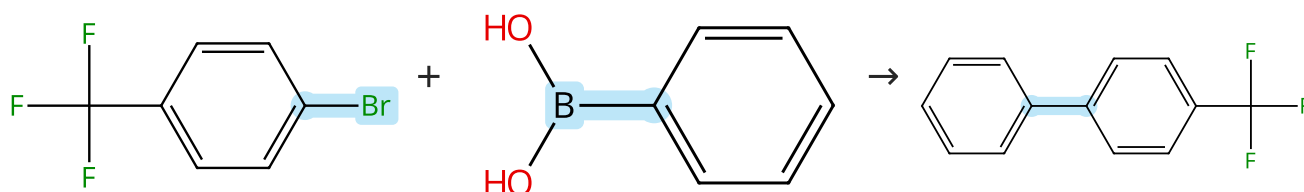
1.1 **Reagents:** Potassium carbonate
Catalysts: Palladium
Solvents: Ethanol; 12 h, 100 °C

Experimental Protocols

31-614-CAS-31041654	Steps: 1 Yield: 97%	Intermetallic ZrPd₃-Embedded Nanoporous ZrC as an Efficient and Stable Catalyst of the Suzuki Cross-Coupling Reaction
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, Zirconium oxychloride Solvents: Ethanol; 8 h, 25 °C		By: Lu, Yangfan; et al
Experimental Protocols		ACS Catalysis (2020), 10(24), 14366-14374.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (79)

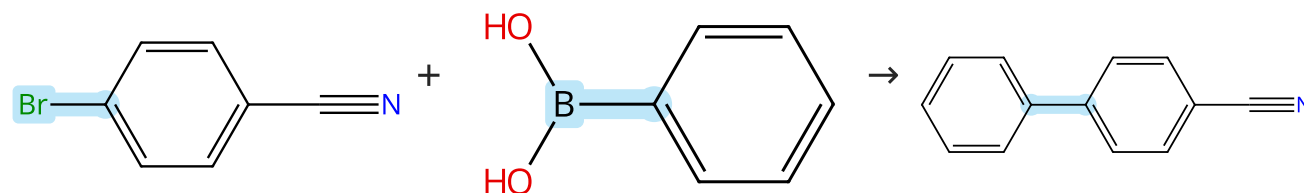
Suppliers (132)

Suppliers (61)

31-614-CAS-36968179	Steps: 1 Yield: 97%	Continuous-Flow Suzuki-Miyaura Coupling in Water and Organic Solvents Promoted by Blends of Stabilized Convuluted Polymeric Palladium Catalysts and Polymeric Auxiliary Materials
1.1 Reagents: Tripotassium phosphate Catalysts: Benzene, diethenyl-, polymer with 1-(1,1-dimethylethyl)-4-ethenylbenzene, Pyridine, 4-ethenyl-, polymer with 1-(1,1-dimethylethyl)-4-ethenylbenzene Solvents: Ethanol, Tetrahydrofuran, Water; 5 h, 70 °C		By: Zhang, Zhenzhong; et al
1.2 Solvents: Water		Chemistry - A European Journal (2023), 29(34), e202300494.
Experimental Protocols		

Scheme 30 (3 Reactions)

Steps: 1 Yield: 97%



Suppliers (93)

Suppliers (132)

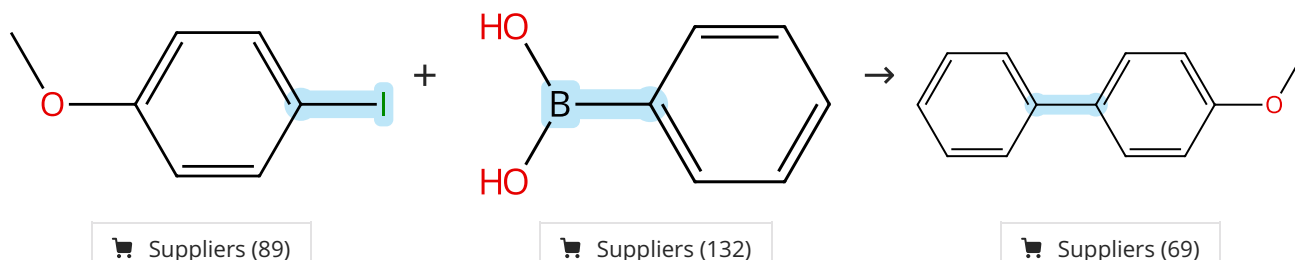
Suppliers (79)

31-614-CAS-24552971	Steps: 1 Yield: 97%	Postsynthetic Metalation of a Robust Hydrogen-Bonded Organic Framework for Heterogeneous Catalysis
1.1 Reagents: Potassium carbonate Catalysts: 4,8,12-Trioxa-2,10-diazatetradecane-5,17-diamine (palladium acetate complex) Solvents: <i>p</i> -Xylene; 1.5 h, 150 °C		By: Han, Bin; et al
		Journal of the American Chemical Society (2019), 141(22), 8737-8740.
31-179-CAS-18618061	Steps: 1 Yield: 97%	One-Pot Tandem Photoredox and Cross-Coupling Catalysis with a Single Palladium Carbodicarbene Complex
1.1 Reagents: Potassium carbonate Catalysts: 2213470-68-7 Solvents: Dimethylformamide; 20 h, 120 °C		By: Hsu, Yu-Cheng; et al
		Angewandte Chemie, International Edition (2018), 57(17), 4622-4626.

31-179-CAS-16683791	Steps: 1 Yield: 97%	Suzuki-Miyaura Micellar Cross-Coupling in Water, at Room Temperature, and under Aerobic Atmosphere
1.1 Reagents: Triethylamine Catalysts: (<i>SP</i> -4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino- κ P]ferrocene]dichloropalladium Solvents: Water; 1 h, rt		By: Mattiello, Sara; et al Organic Letters (2017), 19(3), 654-657.
Experimental Protocols		

Scheme 31 (2 Reactions)

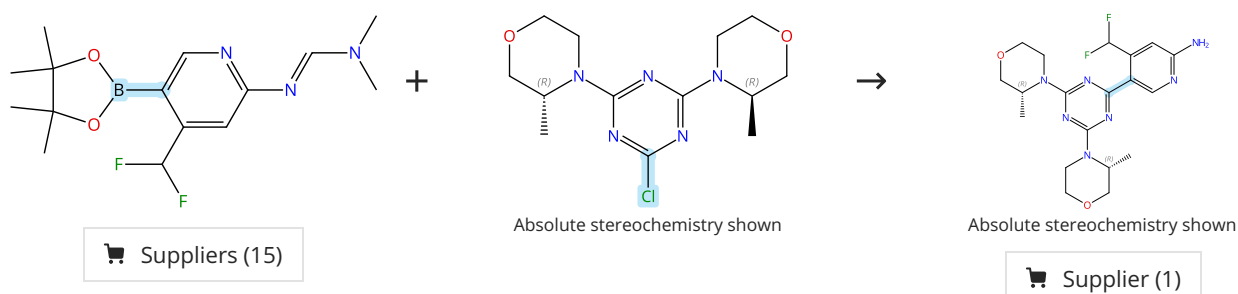
Steps: 1 Yield: 97%



31-179-CAS-20411115	Steps: 1 Yield: 97%	Preserving Porosity of Mesoporous Metal-Organic Frameworks through the Introduction of Polymer Guests
1.1 Reagents: Potassium carbonate Catalysts: Palladium nitrate, 2365148-60-1 Solvents: Dimethylformamide, Water; 1 h, 50 °C		By: Peng, Li; et al Journal of the American Chemical Society (2019), 141(31), 12397-12405.
31-614-CAS-24552972	Steps: 1 Yield: 97%	Postsynthetic Metalation of a Robust Hydrogen-Bonded Organic Framework for Heterogeneous Catalysis
1.1 Reagents: Potassium carbonate Catalysts: 4,8,12-Trioxa-2,10-diazatetradecane-5,17-diamine (palladium acetate complex) Solvents: <i>p</i> -Xylene; 2 h, 150 °C		By: Han, Bin; et al Journal of the American Chemical Society (2019), 141(22), 8737-8740.

Scheme 32 (2 Reactions)

Steps: 1 Yield: 97%

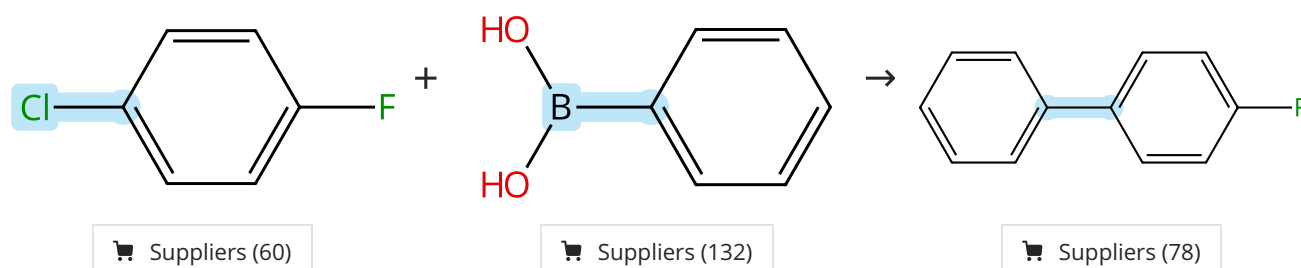


31-614-CAS-24633152	Steps: 1 Yield: 97%	4-(Difluoromethyl)-5-(4-((3R,5S)-3,5-dimethylmorpholino)-6-((R)-3-methylmorpholino)-1,3,5-triazin-2-yl)pyridin-2-amine (P QR626), a Potent, Orally Available, and Brain-Penetrant mTOR Inhibitor for the Treatment of Neurological Disorders
1.1 Reagents: Tripotassium phosphate Catalysts: (<i>SP</i> -4-4)-[2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C]chloro [dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine]palladium Solvents: 1,4-Dioxane, Water; 2 - 16 h, 95 °C		By: Borsari, Chiara; et al Journal of Medicinal Chemistry (2020), 63(22), 13595-13617.
1.2 Reagents: Hydrochloric acid Solvents: Water; 3 - 16 h, 60 °C		
1.3 Reagents: Sodium hydroxide Solvents: Water; pH 9 - 10		
Experimental Protocols		

31-179-CAS-19761473 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: (<i>SP</i>-4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: 1,4-Dioxane, Water; 2 - 5 h, 95 °C; 95 °C → rt 1.2 Reagents: Hydrochloric acid Solvents: Water; 3 - 15 h, 60 °C 1.3 Reagents: Sodium hydroxide Solvents: Water; pH 10 Experimental Protocols	Discovery and Preclinical Characterization of 5-[4,6-Bis({3-oxa-8-azabicyclo[3.2.1]octan-8-yl})-1,3,5-triazin-2-yl]-4-(difluoromethyl)pyridin-2-amine (PQR620), a Highly Potent and Selective mTORC1/2 Inhibitor for Cancer and Neurological Disorders By: Rageot, Denise; et al Journal of Medicinal Chemistry (2018), 61(22), 10084-10105.
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Scheme 33 (1 Reaction)

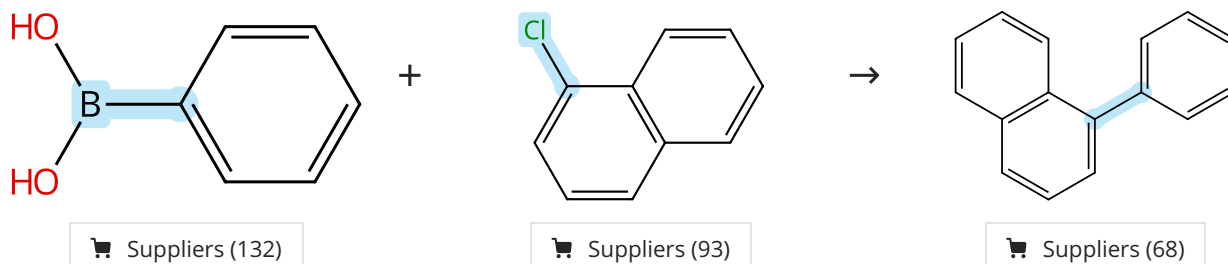
Steps: 1 Yield: 97%



31-614-CAS-34388114 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, 2849469-49-2 Solvents: 1,4-Dioxane, Water; 24 h, 60 °C Experimental Protocols	Ylide-Functionalized Diisopropyl Phosphine (prYPhos): A Ligand for Selective Suzuki-Miyaura Couplings of Aryl Chlorides By: Wei, Xiao-Jing; et al Advanced Synthesis & Catalysis (2022), 364(19), 3336-3341.
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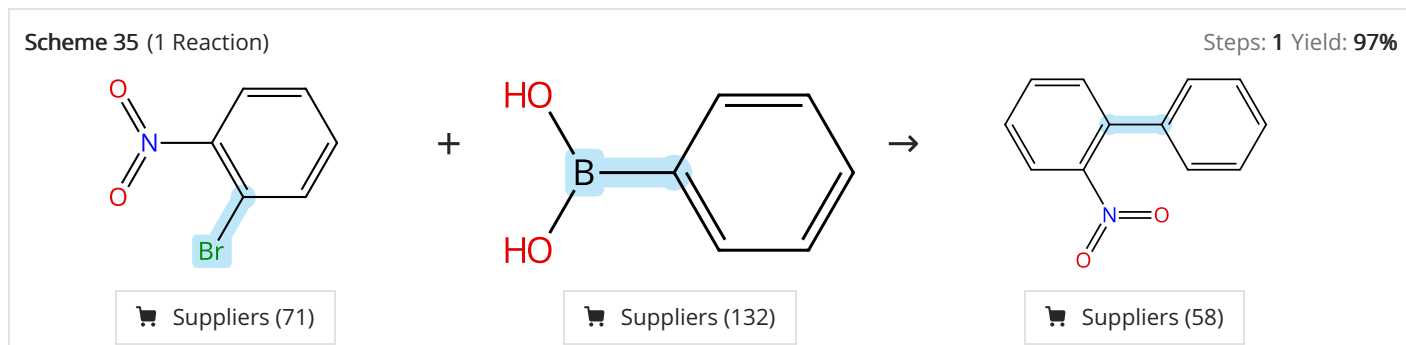
Scheme 34 (2 Reactions)

Steps: 1 Yield: 97%

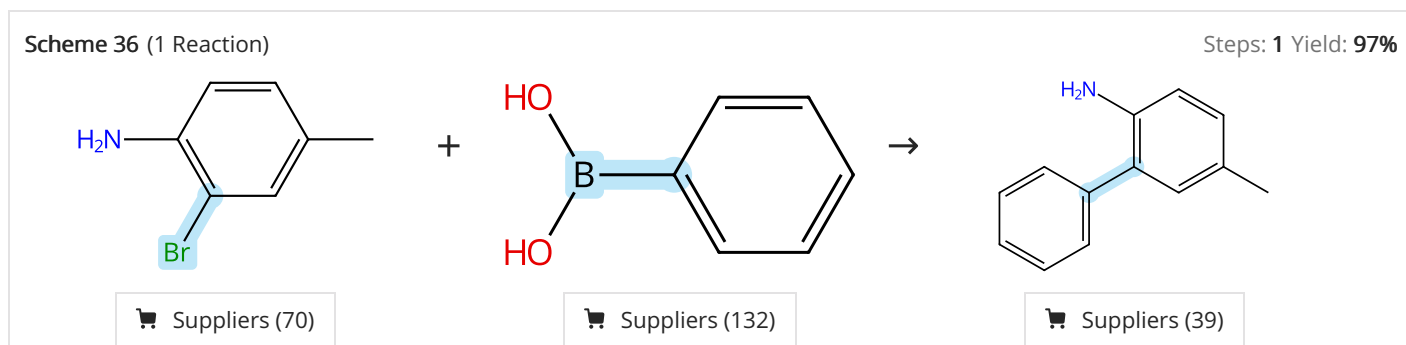


31-179-CAS-16474863 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Triphenylphosphine, Chloro(η⁵-2,4-cyclopentadien-1-yl)[<i>rel</i>-(3<i>aR</i>,9<i>aS</i>)-1,3,3<i>a</i>,4,9,9<i>a</i>-hexahydro-1,3-bis(1-methylethyl)-4,9[1',2']-benzeno-2<i>H</i>-naphth[2,3-<i>d'</i>]imidazol-2-ylidene]nickel Solvents: <i>tert</i>-Butanol, Toluene; 25 °C; 24 h, 90 °C 1.2 Solvents: Water; 0 °C Experimental Protocols	An N-Heterocyclic Carbene-Nickel Half-Sandwich Complex as a Precatalyst for Suzuki-Miyaura Coupling of Aryl/Heteroaryl Halides with Aryl/Heteroarylboronic Acids By: Ando, Shin; et al Journal of Organic Chemistry (2017), 82(2), 1266-1272.
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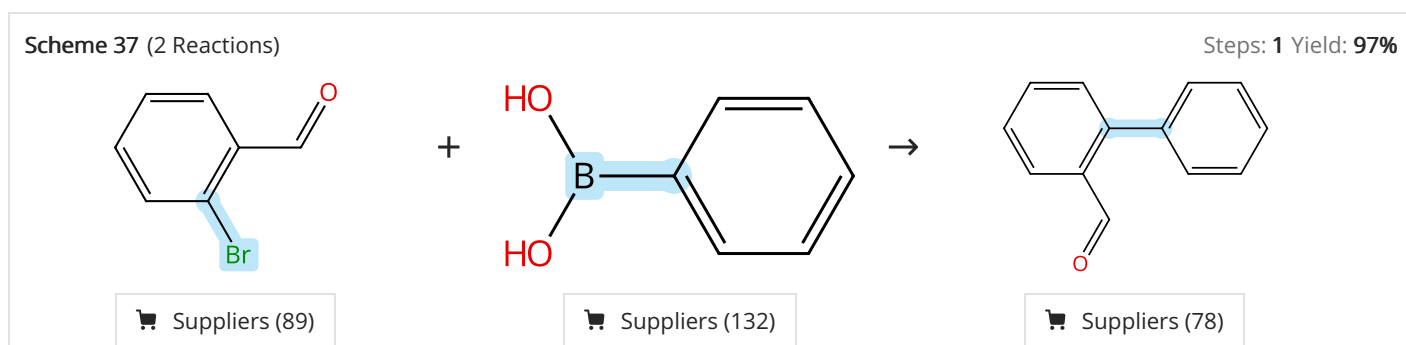
31-179-CAS-17791093 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: 2133818-09-2 Solvents: Isopropanol, Water; 30 min, 90 °C 1.2 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Cyclopalladation in the Periphery of a NHC Ligand as the Crucial Step in the Synthesis of Highly Active Suzuki-Miyaura Cross-Coupling Catalysts By: Fizia, Agnes; et al Chemistry - A European Journal (2017), 23(58), 14563-14575.
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31-614-CAS-24489508 Steps: 1 Yield: 97% 1.1 Reagents: Sodium hydroxide Catalysts: Graphene (post-treated with Pd(OAc)₂) Solvents: Water; 60 min, 100 °C Experimental Protocols	Palladium nanoparticles supported on graphene acid: a stable and eco-friendly bifunctional C-C homo- and cross-coupling catalyst By: Blanco, Matias; et al Green Chemistry (2019), 21(19), 5238-5247.
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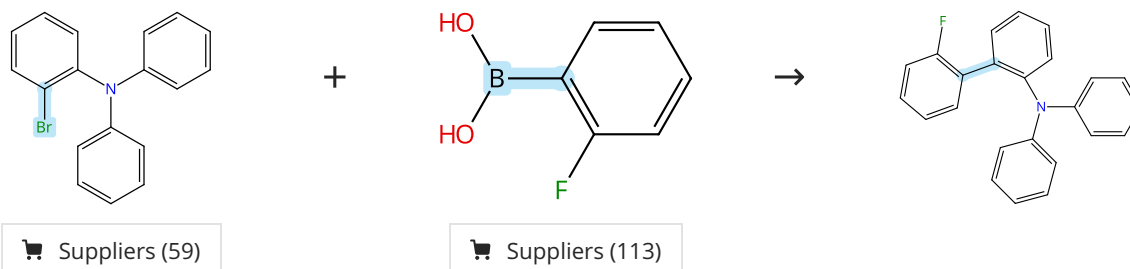
31-179-CAS-19395334 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Solvents: 1,2-Dimethoxyethane; 30 min, rt 1.2 Catalysts: Dichlorobis(triphenylphosphine)palladium; 12 h, 80 °C; 80 °C → rt	Visible Light-Induced Radical Cyclization of Tertiary Bromides with Isonitriles To Construct Trifluoromethylated Quaternary Carbon Center By: Wang, Wengui; et al Journal of Organic Chemistry (2018), 83(23), 14588-14599.
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31-614-CAS-35417995 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate Solvents: Dimethylformamide, Water; 12 h, rt 1.2 Reagents: Water; rt Experimental Protocols	Construction of Alkylidene Fluorene Scaffolds Using Pd-Catalyzed Direct Arene/Alkene Coupling Strategy By: Matsuyama, Hidenori; et al Organic Letters (2023), 25(5), 800-804.
31-614-CAS-24484881 Steps: 1 Yield: 97% 1.1 Reagents: Cesium carbonate Solvents: 1,4-Dioxane, Water; 5 min, rt 1.2 Catalysts: Tetrakis(triphenylphosphine)palladium; 6 h, reflux Experimental Protocols	Synthesis of Bicyclic N-Heterocycles via Photoredox Cycloaddition of Imino-Alkynes and Imino-Alkenes By: Oh, Hyeonji; et al ACS Catalysis (2021), 11(21), 13670-13679.

Scheme 38 (1 Reaction)

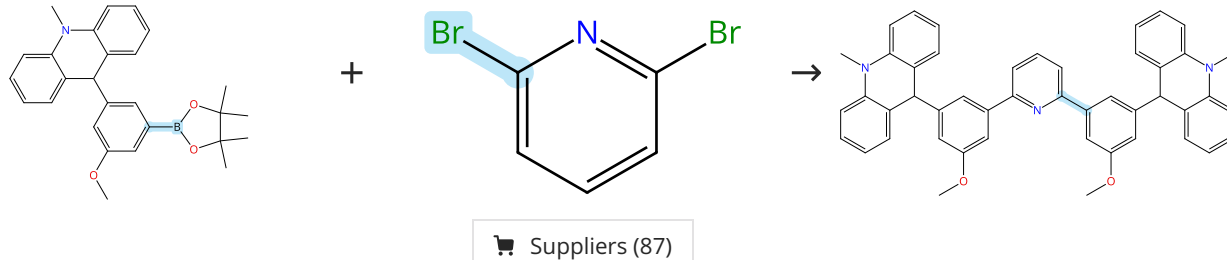
Steps: 1 Yield: 97%



31-179-CAS-22695168 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate Solvents: 1,2-Dimethoxyethane, Water; 2 h, 80 °C 1.2 Reagents: Water Experimental Protocols	Peraryl-X-onium ions of nitrogen and oxygen By: Lu, Mengsi; et al Organic Chemistry Frontiers (2019), 6(15), 2640-2646.
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Scheme 39 (1 Reaction)

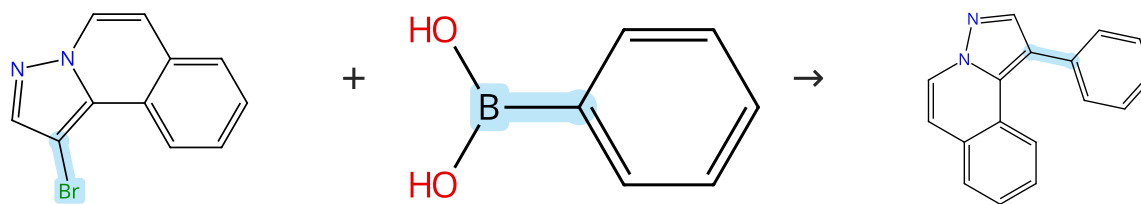
Steps: 1 Yield: 97%



31-614-CAS-41680245 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide; 16 h, 80 °C Experimental Protocols	Self-Assembled Bis-Acrininium Tweezer Equilibria Controlled by Multi-Responsive Properties By: Hu, Johnny; et al Chemistry - A European Journal (2024), 30(44), e202401866.
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Scheme 40 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (2)

Suppliers (132)

31-614-CAS-45760387

Steps: 1 Yield: 97%

[3+2] Cycloaddition of N-Amino(iso)quinolinium Salts with Vinylsulfonium Salt: Synthesis of Pyrazolo[1,5-a]quinoline and Pyrazolo[5,1-a]isoquinoline Derivatives

By: Wu, Hui; et al

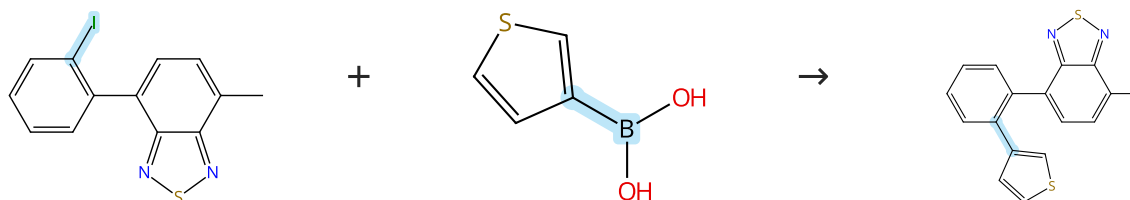
Journal of Organic Chemistry (2025), 90(21), 6969-6979.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 12 h, 105 °C

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (101)

31-179-CAS-21542233

Steps: 1 Yield: 97%

Palladium-Catalyzed Direct Mono- or Poly- Halogenation of Benzothiadiazole Derivatives

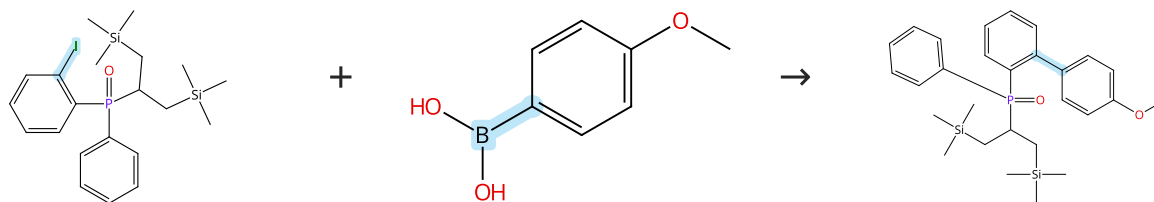
By: He, Hui; et al

Journal of Organic Chemistry (2020), 85(5), 3788-3798.

1.1 **Reagents:** Sodium carbonate
Solvents: Dimethylformamide; 12 h, 90 °C

Scheme 42 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (112)

31-179-CAS-19588366

Steps: 1 Yield: 97%

New Phosphine Ligand Architectures Lead to Efficient Gold Catalysts for Cycloisomerization Reactions at Very Low Loading

By: Mahamulkar, Shraddha G.; et al

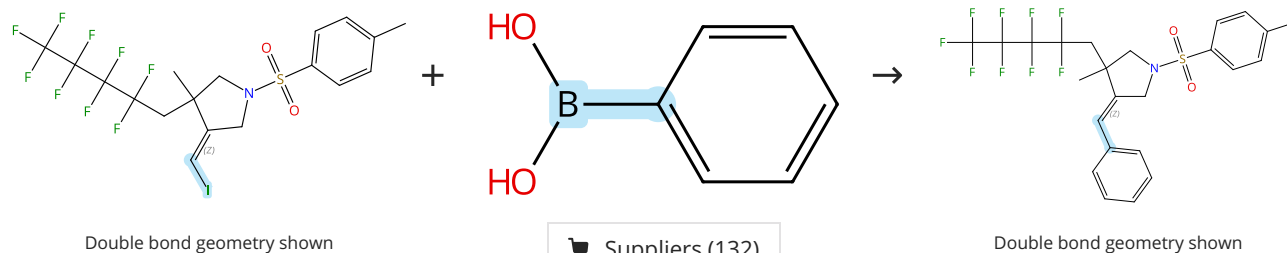
Advanced Synthesis & Catalysis (2018), 360(21), 4215-4224.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 12 - 16 h, 85 °C

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-179-CAS-18397835

Steps: 1 Yield: 97%

Copper/Iron-Cocatalyzed Cascade Perfluoroalkylation/Cyclization of 1,6-Enynes with Iodoperfluoroalkanes

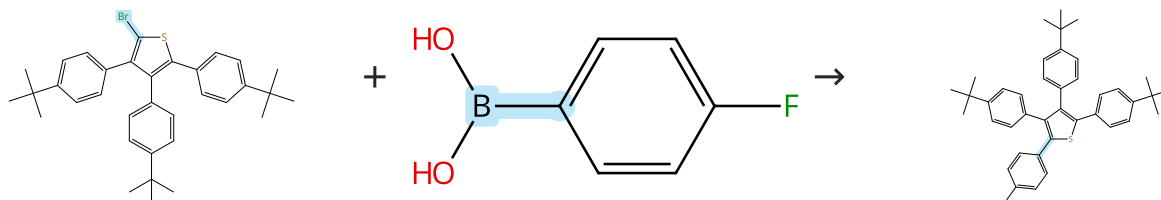
By: Xia, Xiao-Feng; et al

Advanced Synthesis & Catalysis (2018), 360(3), 562-567.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 90 °C

Scheme 44 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (104)

31-614-CAS-38944225

Steps: 1 Yield: 97%

Crafting diverse tetraphenylthiophene designs through comprehensive 'classical' Suzuki-Miyaura synthesis and electrochemical exploration

By: Maddala, Sudhakar; et al

Tetrahedron (2024), 151, 133807.

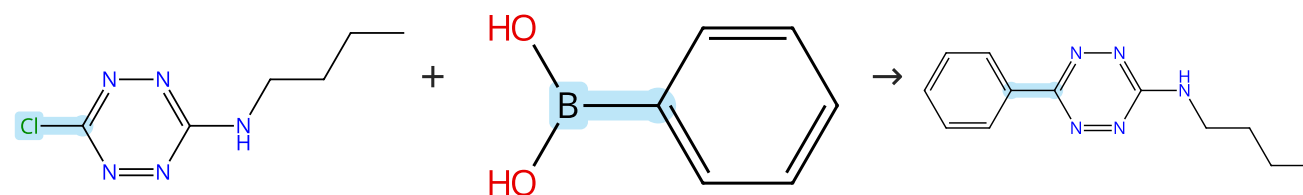
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Methanol, Toluene; 16 h, 80 °C; 80 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (3)

Suppliers (132)

Suppliers (5)

31-179-CAS-17864050

Steps: 1 Yield: 97%

Preparation of Unsymmetrical 1,2,4,5-Tetrazines via a Mild Suzuki Cross-Coupling Reaction

By: Bender, Aaron M.; et al

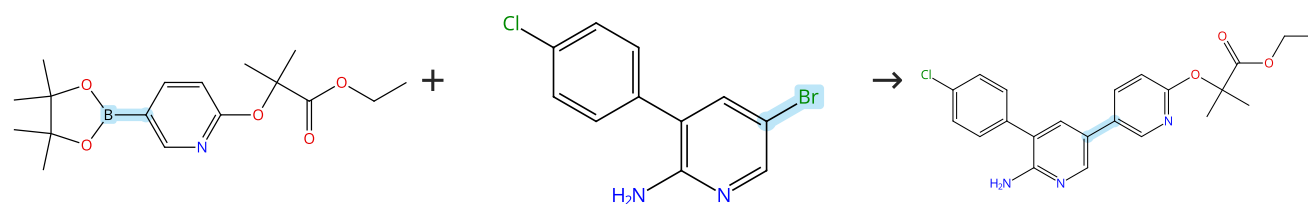
Organic Letters (2017), 19(20), 5693-5696.

1.1 **Reagents:** Cesium carbonate
Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC](dicyclohexyl [3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine-κP)(methanesulfonato-κO)palladium
Solvents: 1,4-Dioxane, Water; 2 h, 70 °C

Experimental Protocols

Scheme 46 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (16)

Suppliers (20)

Suppliers (5)

31-179-CAS-18915674

Steps: 1 Yield: 97%

2-Aminopyridine-Based Mitogen-Activated Protein Kinase Kinase Kinase 4 (MAP4K4) Inhibitors: Assessment of Mechanism-Based Safety

By: Dow, Robert L.; et al

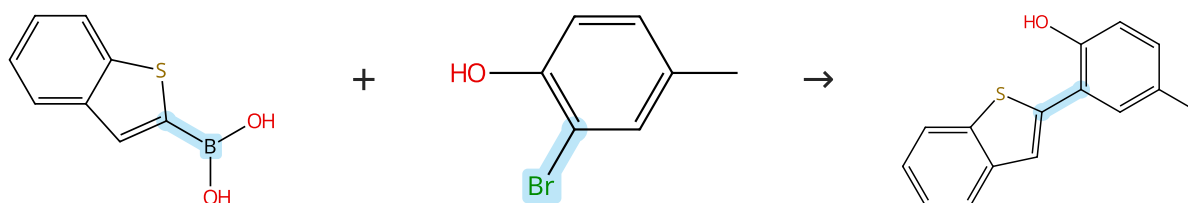
Journal of Medicinal Chemistry (2018), 61(7), 3114-3125.

1.1 **Reagents:** Sodium carbonate
Catalysts: (*SP*-4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino- κ P]-*N,N*-dimethylbenzenamine]dichloropalladium
Solvents: Ethanol, Water; 1 h, 60 °C

Experimental Protocols

Scheme 47 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (90)

Suppliers (84)

31-179-CAS-23678750

Steps: 1 Yield: 97%

Cu-Catalyzed Dehydrogenative C-O Cyclization for the Synthesis of Furan-Fused Thienoacenes

By: Mitsudo, Koichi; et al

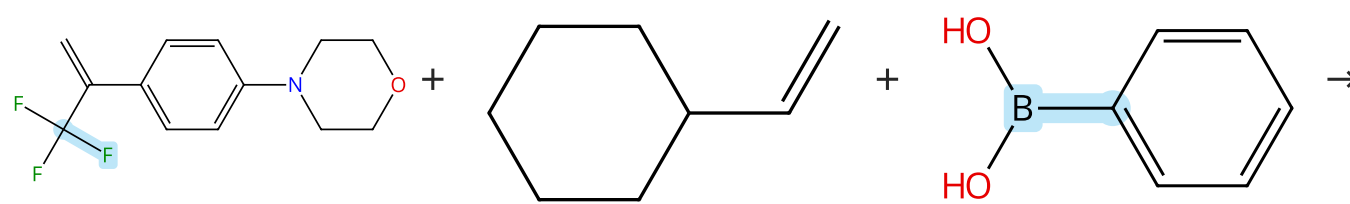
Organic Letters (2021), 23(11), 4322-4326.

1.1 **Reagents:** Potassium carbonate
Catalysts: Bis(tri-*tert*-butylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 24 h, 80 °C; 80 °C → rt
 1.2 **Reagents:** Hydrochloric acid
Solvents: Water; acidified, rt

Experimental Protocols

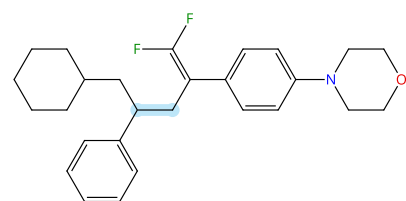
Scheme 48 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (49)

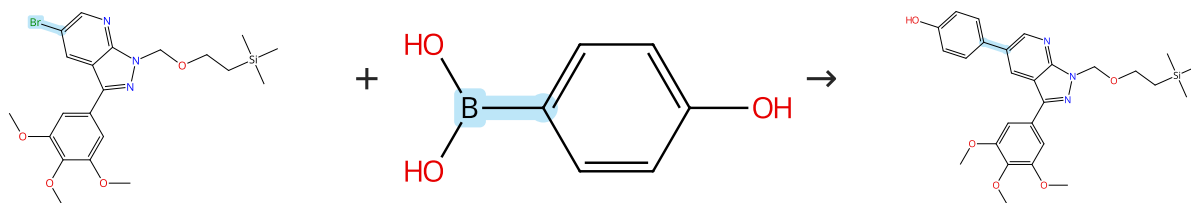
Suppliers (132)



31-614-CAS-37918681 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Bis(acetylacetonato)nickel, 6,6'-Dimethyl-2,2'-bipyridine Solvents: Tetrahydrofuran; 36 h, 80 °C 1.2 Reagents: Water Experimental Protocols	Alkene 1,1-difunctionalizations via organometallic-radical relay By: Wu, Dong; et al Nature Catalysis (2023), 6(11), 1030-1041.
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Scheme 49 (1 Reaction)

Steps: 1 Yield: 97%

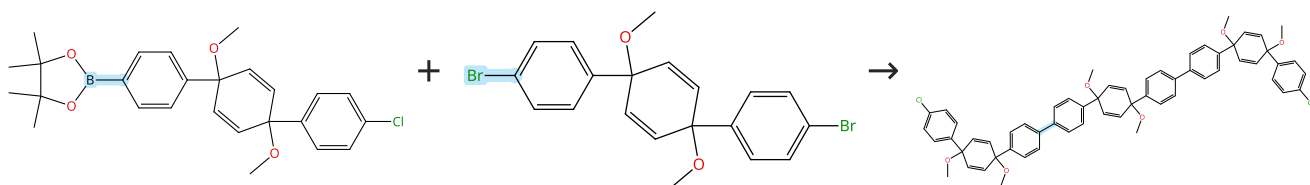


Suppliers (103)

31-614-CAS-37350998 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 100 °C Experimental Protocols	Design, synthesis and biological evaluation of 3, 5-disubstituted-1H-pyrazolo[3,4-b]pyridines as multiacting inhibitors against microtubule and kinases By: Ning, Chengqing; et al European Journal of Medicinal Chemistry (2023), 259, 115687.
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Scheme 50 (1 Reaction)

Steps: 1 Yield: 97%

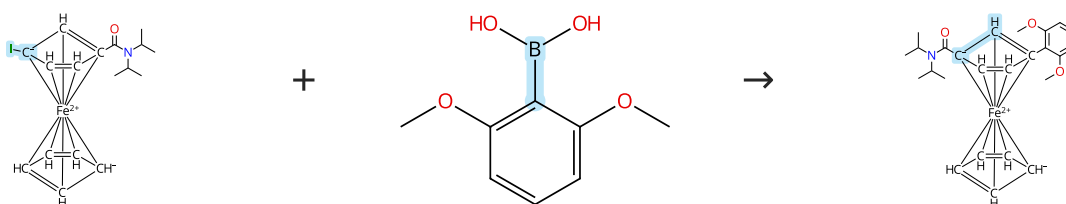


Suppliers (3)

31-614-CAS-37947708 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 16 h, reflux Experimental Protocols	Efficient manipulation of Forster resonance energy transfer through host-guest interaction enables tunable white-light emission and devices in heterotopic bisnanohoops By: Fan, Yanqing; et al Chemical Science (2023), 14(40), 11121-11130.
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Scheme 51 (1 Reaction)

Steps: 1 Yield: 97%

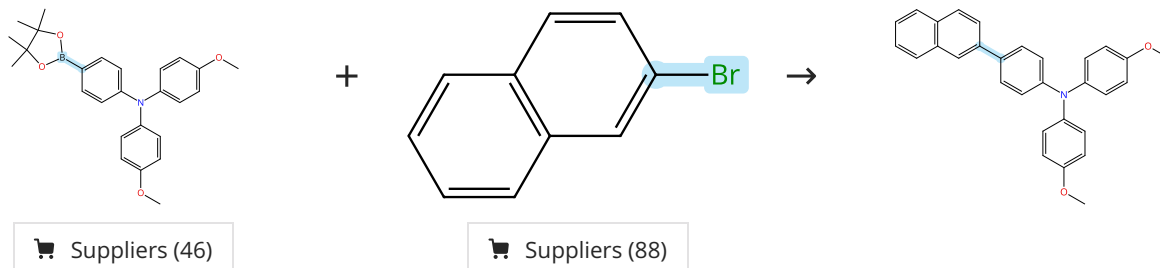


Suppliers (91)

31-179-CAS-21802113 Steps: 1 Yield: 97% 1.1 Reagents: Cesium fluoride Catalysts: Bis(dibenzylideneacetone)palladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: Toluene; overnight, 110 °C	Functionalization of 3-Iodo-N,N-Diisopropylferrocene-Carboxamide, a Pivotal Substrate to Open the Chemical Space to 1, 3-Disubstituted Ferrocenes By: Erb, William; et al Advanced Synthesis & Catalysis (2020), 362(4), 832-850.
Experimental Protocols	

Scheme 52 (1 Reaction)

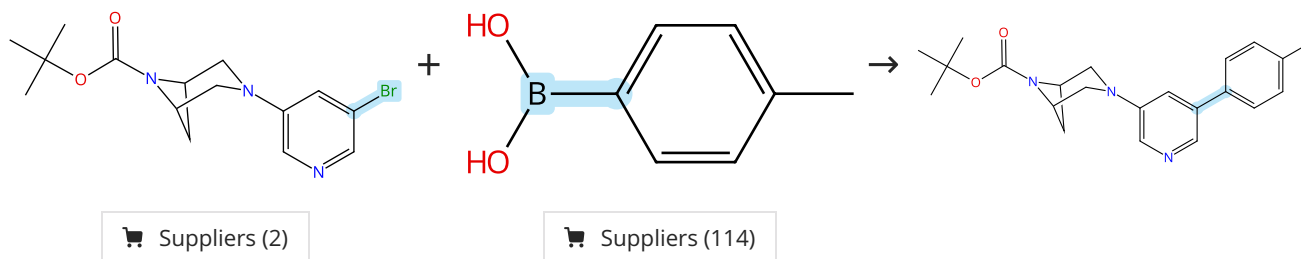
Steps: 1 Yield: 97%



31-179-CAS-4812918 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; overnight, reflux	Charge Transfer Pathways in Three Isomers of Naphthalene-Bridged Organic Mixed Valence Compounds By: Schmidt, Hauke C.; et al Journal of Organic Chemistry (2016), 81(2), 595-602.
Experimental Protocols	

Scheme 53 (1 Reaction)

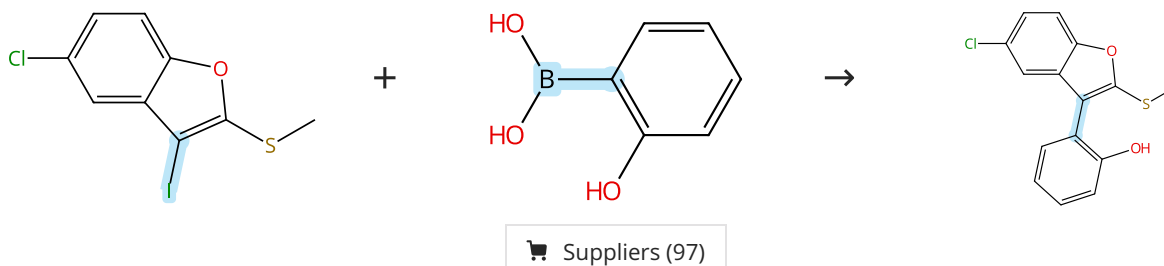
Steps: 1 Yield: 97%



31-179-CAS-1962345 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, Tri- <i>o</i> -tolylphosphine Solvents: 1,2-Dimethoxyethane, Water; 15 min, rt; 2 - 5 h, reflux	Design of novel 3,6-diazabicyclo[3.1.1]heptane derivatives with potent and selective affinities for $\alpha_4\beta_2$ neuronal nicotinic acetylcholine receptors By: Deligia, Francesco; et al European Journal of Medicinal Chemistry (2015), 103, 429-437.
Experimental Protocols	

Scheme 54 (1 Reaction)

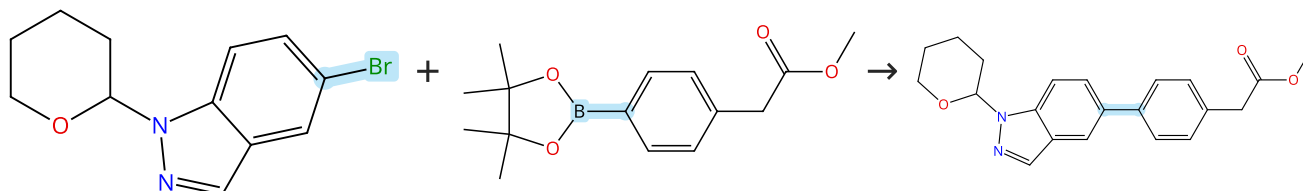
Steps: 1 Yield: 97%



31-179-CAS-20024435	Steps: 1 Yield: 97%	Synthesis of Alkynyl Sulfides by Copper-Catalyzed Thiolation of Terminal Alkynes Using Thiosulfonates
1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i> -4-1)-Bis[4-[bis(1,1-dimethylethyl)phosphino- κP]- <i>N,N</i> -dimethylbenzenamine]dichloropalladium Solvents: 1,4-Dioxane, Water; 12 h, 90 °C	By: Kanemoto, Kazuya; et al Organic Letters (2019), 21(9), 3172-3177.	
Experimental Protocols		

Scheme 55 (1 Reaction)

Steps: 1 Yield: 97%



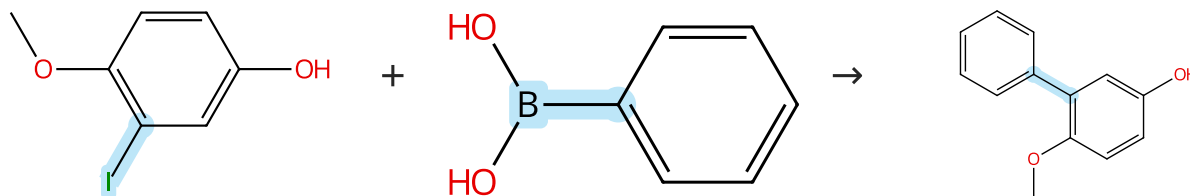
Suppliers (56)

Suppliers (84)

31-614-CAS-42307418	Steps: 1 Yield: 97%	Tuning RXR Modulators for PGC1α Recruitment
1.1 Reagents: Tripotassium phosphate Catalysts: (<i>SP</i> -4-4)-[2'-(Amino- κM)[1,1'-biphenyl]-2-yl- κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: 1,4-Dioxane, Water; 1.5 - 24 h, reflux	By: Nawa, Felix; et al Journal of Medicinal Chemistry (2024), 67(18), 16338-16354.	
Experimental Protocols		

Scheme 56 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (41)

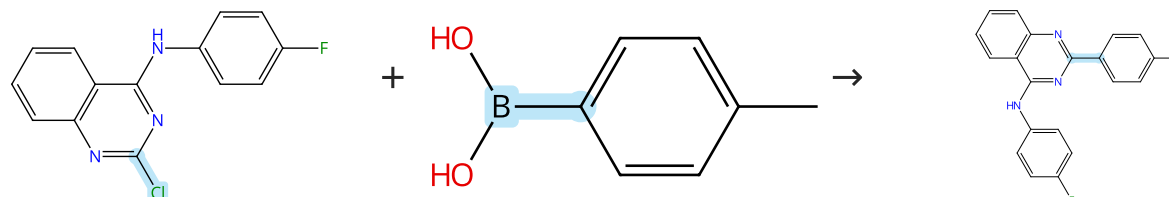
Suppliers (132)

Suppliers (16)

31-179-CAS-17391137	Steps: 1 Yield: 97%	Dearomatization Approach to 2-Trifluoromethylated Benzofuran and Dihydrobenzofuran Products
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate Solvents: Dimethylformamide, Water; rt; rt $\rightarrow$ 45 °C; overnight, 45 °C	By: Smith, David T.; et al Organic Letters (2017), 19(13), 3508-3511.	
1.2 Reagents: Water		
Experimental Protocols		

Scheme 57 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (2)

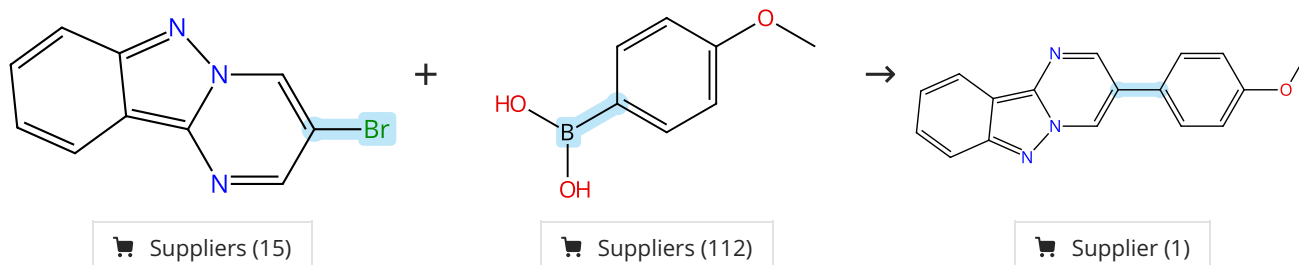
Suppliers (114)

Suppliers (2)

31-179-CAS-22667267	Steps: 1 Yield: 97%	N-heterocyclic carbene-Pd(II)-1-methylimidazole complex-catalyzed Suzuki-Miyaura coupling of 2-chloro-4-aminoquinazolines with arylboronic acids
1.1 Reagents: Potassium hydroxide Catalysts: (<i>SP</i> -4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2 <i>H</i> -imidazol-2-ylidene]dichloro(1-methyl-1 <i>H</i> -imidazole- κN^3)palladium Solvents: Ethanol; 24 h, rt		By: Bao, Zhen; et al Tetrahedron (2020), 76(44), 131548.
Experimental Protocols		

Scheme 58 (1 Reaction)

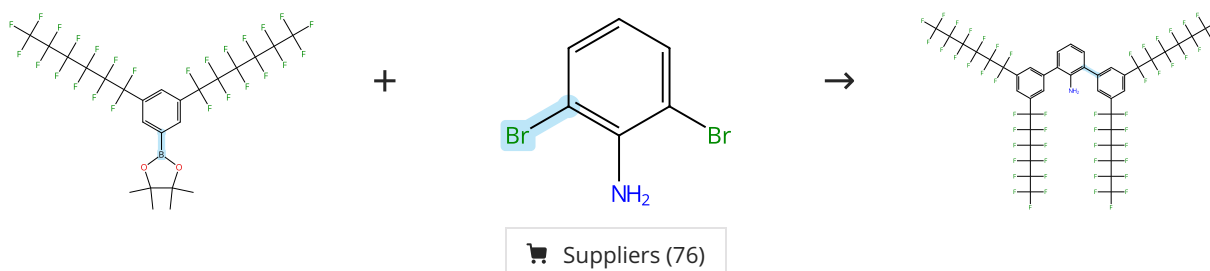
Steps: 1 Yield: 97%



31-179-CAS-22869142	Steps: 1 Yield: 97%	Pyrimido[1,2-b]indazole derivatives: Selective inhibitors of human monoamine oxidase B with neuroprotective activity
1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 3 h, rt → 110 °C		By: Jismy, Badr; et al European Journal of Medicinal Chemistry (2021), 209, 112911.
Experimental Protocols		

Scheme 59 (1 Reaction)

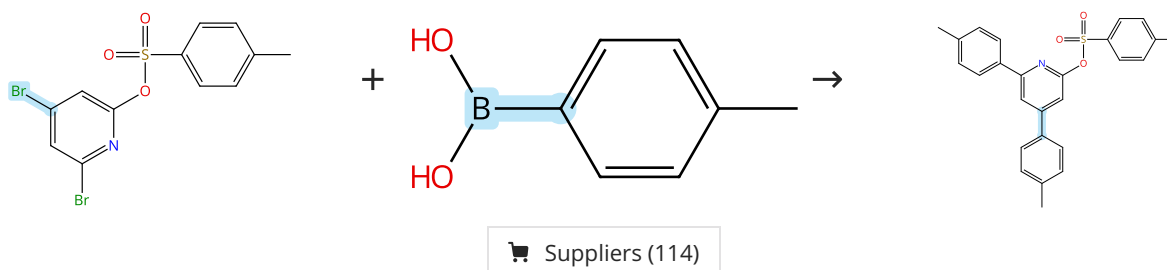
Steps: 1 Yield: 97%



31-179-CAS-21677585	Steps: 1 Yield: 97%	Uniform shape monodisperse single chain nanocrystals by living aqueous catalytic polymerization
1.1 Reagents: Cesium fluoride Catalysts: Triphenylphosphine, Bis(dibenzylideneacetone)palladium Solvents: 1,4-Dioxane; 2 h, rt → 130 °C; overnight, 100 °C		By: Schnitte, Manuel; et al Nature Communications (2019), 10(1), 1-6.

Scheme 60 (1 Reaction)

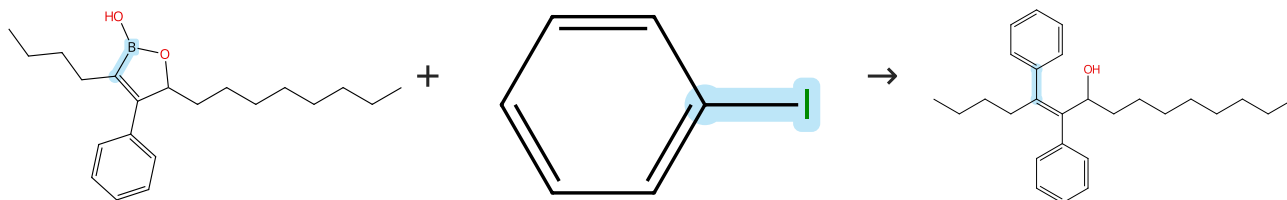
Steps: 1 Yield: 97%



31-179-CAS-16617512	Steps: 1 Yield: 97%	Synthesis of Trisubstituted Pyridines via Chemoselective Suzuki-Miyaura Coupling of 3,5- and 4,6-Dibromo-2-tosyloxypyrindines By: Park, Cho-Hee; et al Advanced Synthesis & Catalysis (2017), 359(1), 107-119.
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, Di-1-adamantylbenzylphosphine Solvents: Toluene; 8 h, 50 °C		
Experimental Protocols		

Scheme 61 (1 Reaction)

Steps: 1 Yield: 97%

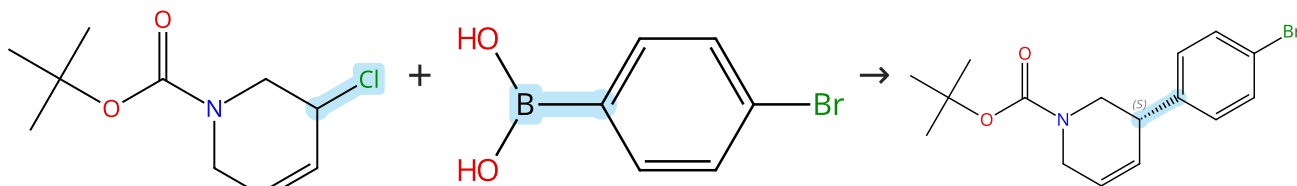


Suppliers (96)

31-004-CAS-19035272	Steps: 1 Yield: 97%	Copper-Catalyzed Borocarbonylative Coupling of Internal Alkynes with Unactivated Alkyl Halides: Modular Synthesis of Tetrasubstituted β-Borylenones By: Cheng, Li-Jie; et al Angewandte Chemie, International Edition (2018), 57(32), 10328-10332.
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 2 h, 80 °C		
Experimental Protocols		

Scheme 62 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (20)

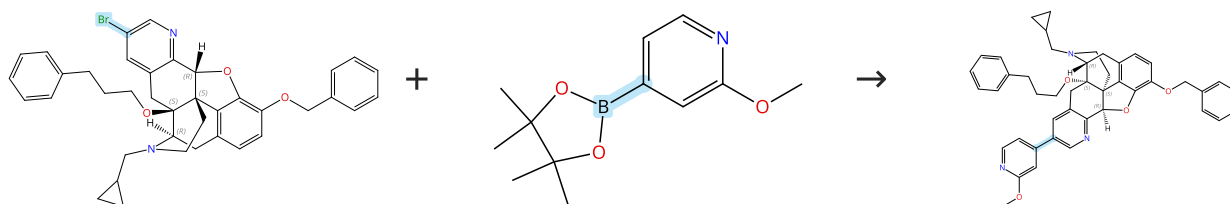
Suppliers (100)

 Absolute stereochemistry shown,
Rotation (-)

31-179-CAS-17758729	Steps: 1 Yield: 97%	Asymmetric Suzuki-Miyaura coupling of heterocycles via Rhodium-catalysed allylic arylation of racemates By: Schafer, Philipp; et al Nature Communications (2017), 8, 15762pp..
1.1 Reagents: Cesium carbonate Catalysts: Bis[(1,2,5,6- η)-1,5-cyclooctadiene]di- μ -hydroxydi rhodium, [(1 <i>S</i>)-5,5'-Dichloro-6,6'-dimethoxy[1,1'-biphenyl]-2,2'-diyl]bis[diphenylphosphine] Solvents: Tetrahydrofuran; 30 min, 60 °C 1.2 Solvents: Tetrahydrofuran; 16 h, 80 °C		
Experimental Protocols		

Scheme 63 (1 Reaction)

Steps: 1 Yield: 97%



Absolute stereochemistry shown

Supplier (1)

Suppliers (81)

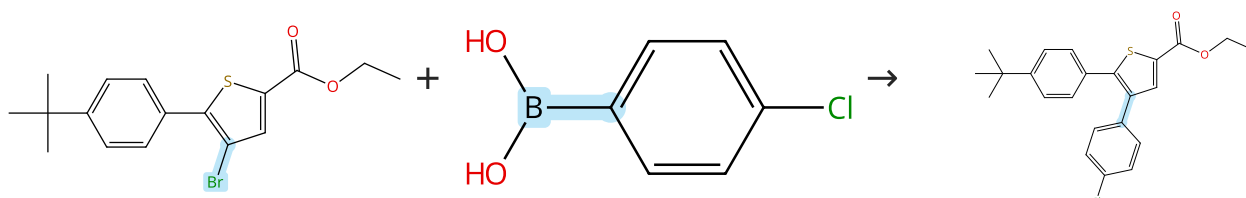
Absolute stereochemistry shown

Supplier (1)

31-179-CAS-22767370	Synthesis and Structure-Activity Relationships of 5'-Aryl-14-alkoxy-pyridomorphinans: Identification of a μ Opioid Receptor Agonist/δ Opioid Receptor Antagonist Ligand with Systemic Antinociceptive Activity and Diminished Opioid Side Effects
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 1 h, 100 °C	By: Vekariya, Rakesh H.; et al Journal of Medicinal Chemistry (2020), 63(14), 7663-7694.

Scheme 64 (1 Reaction)

Steps: 1 Yield: 97%



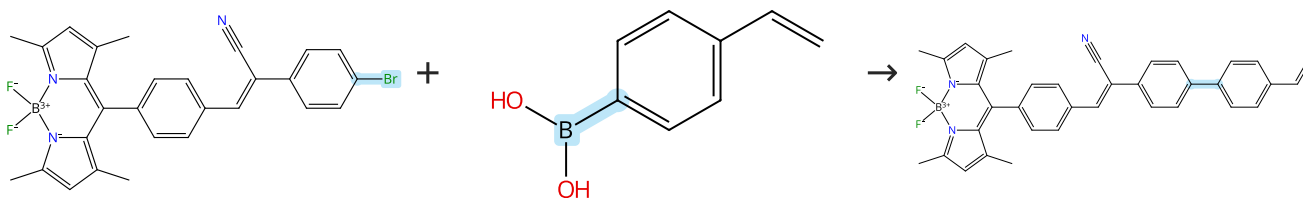
Suppliers (110)

31-614-CAS-24059229	Programmed Synthesis of Tetra-Aryl Thiophenes with Stepwise, Ester-Controlled Regioselectivity
1.1 Reagents: Potassium carbonate Catalysts: Palladium chloride Solvents: Ethanol, Water; 30 min, 60 °C	By: Messina, Cynthia; et al Organic Letters (2021), 23(19), 7348-7352.

Experimental Protocols

Scheme 65 (1 Reaction)

Steps: 1 Yield: 97%

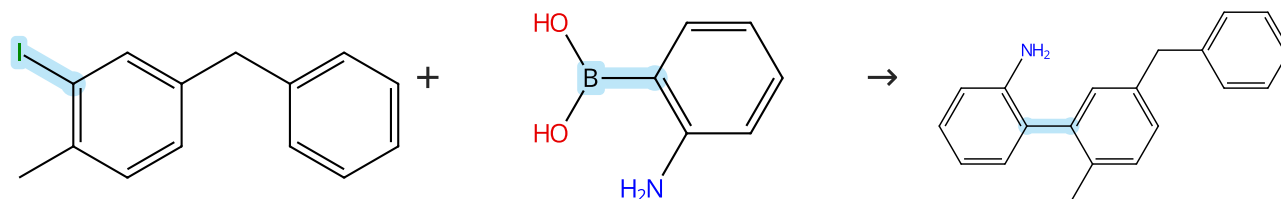


Suppliers (97)

31-179-CAS-20182426	A versatile scaffold for facile synthesis of fluore scent cyano-substituted stilbenes
1.1 Reagents: Potassium carbonate Catalysts: Palladium, Water Solvents: Toluene; 2 h, 100 °C	By: Hayashi, Shotaro; et al Tetrahedron (2019), 75(8), 1079-1084.

Scheme 66 (1 Reaction)

Steps: 1 Yield: 97%

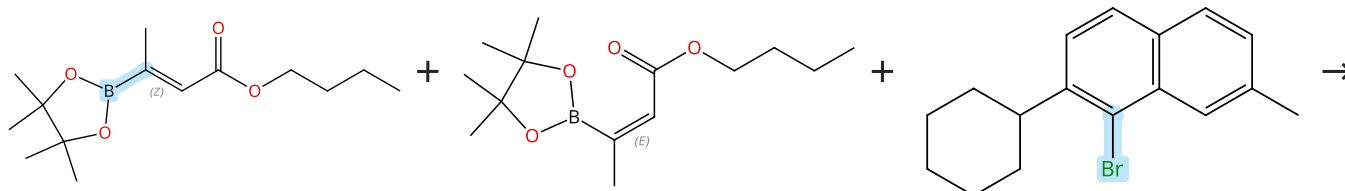


Suppliers (80)

31-179-CAS-23422612	Steps: 1 Yield: 97%	Modular metal-free catalytic radical annulation of cyclic diaryliodoniums to access π-extended arenes
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 12 h, reflux		By: Zhu, Daqian; et al
Experimental Protocols		Green Chemistry (2021), 23(5), 1972-1977.

Scheme 67 (1 Reaction)

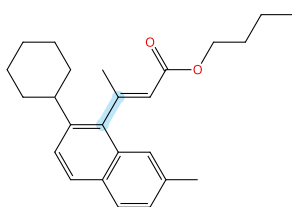
Steps: 1 Yield: 97%



Double bond geometry shown

Double bond geometry shown

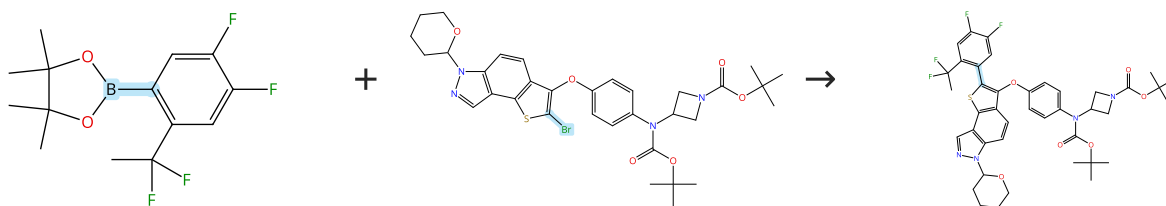
Suppliers (3)



31-614-CAS-34486186	Steps: 1 Yield: 97%	Asymmetric Construction of Aryl-Alkene Axis by Palladium-Catalyzed Suzuki-Miyaura Coupling Reaction
1.1 Reagents: Tripotassium phosphate Catalysts: Bis(dibenzylideneacetone)palladium, 2851421-56-0 Solvents: Diethyl ether, Ethyl acetate; 48 h, 35 °C		By: Qiu, Sheng-Qi; et al
Experimental Protocols		Angewandte Chemie, International Edition (2022), 61(45), e202211211.

Scheme 68 (1 Reaction)

Steps: 1 Yield: 97%

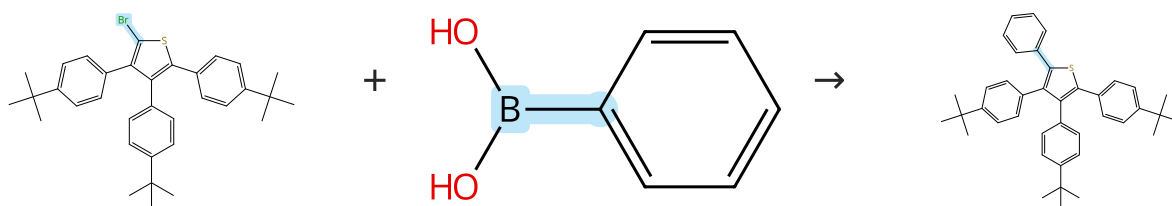


Suppliers (2)

31-614-CAS-39980615	Steps: 1 Yield: 97%	Discovery of Thieno[2,3-e]indazole Derivatives as Novel Oral Selective Estrogen Receptor Degradable with Highly Improved Antitumor Effect and Favorable Druggability
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos Solvents: 1,4-Dioxane, Water; 25 °C → 110 °C; 60 min, 110 °C		By: Lu, Zhengyu; et al
Experimental Protocols		Journal of Medicinal Chemistry (2022), 65(7), 5724-5750.

Scheme 69 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-614-CAS-38944223

Steps: 1 Yield: 97%

Crafting diverse tetraphenylthiophene designs through comprehensive 'classical' Suzuki-Miyaura synthesis and electrochemical exploration

By: Maddala, Sudhakar; et al

Tetrahedron (2024), 151, 133807.

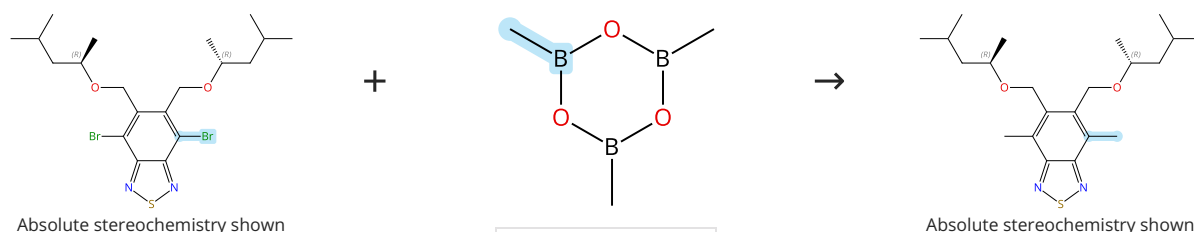
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Methanol, Toluene; 16 h, 80 °C; 80 °C → rt

1.2 **Reagents:** Water; rt

Experimental Protocols

Scheme 70 (1 Reaction)

Steps: 1 Yield: 97%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (61)

31-179-CAS-9548803

Steps: 1 Yield: 97%

Exerting Control over the Helical Chirality in the Main Chain of Sergeants-and-Soldiers-Type Poly(quinoxaline-2,3-diyl)s by Changing from Random to Block Copolymerization Protocols

By: Nagata, Yuuya; et al

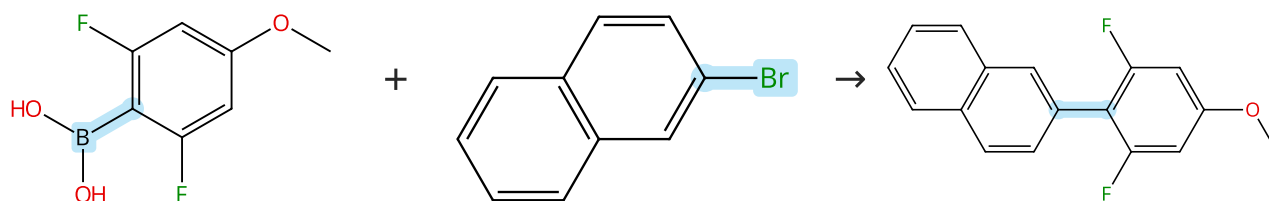
Journal of the American Chemical Society (2015), 137(12), 4070-4073.

1.1 **Reagents:** Potassium fluoride
Catalysts: Tri-*tert*-butylphosphine, Bis(dibenzylideneacetone) palladium
Solvents: Tetrahydrofuran, Water; 1 h, reflux

Experimental Protocols

Scheme 71 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (74)

Suppliers (88)

31-614-CAS-24895947

Steps: 1 Yield: 97%

Palladium-Catalyzed Solid-State Polyfluoroarylation of Aryl Halides Using Mechanochemistry

By: Takahashi, Rikuro; et al

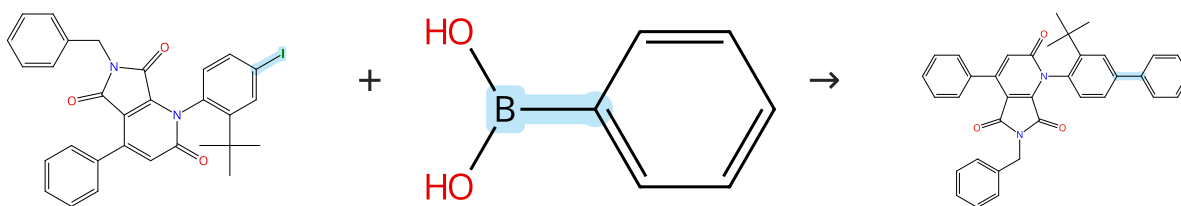
ACS Catalysis (2021), 11(24), 14803-14810.

1.1 **Reagents:** Triethylamine
Catalysts: Palladium diacetate, Tris(tricyclo[3.3.1.1^{3,7}]dec-1-yl) phosphine; 30 min, 80 °C

Experimental Protocols

Scheme 72 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-614-CAS-32273122

Steps: 1 Yield: 97%

N-Heterocyclic Carbene-Catalyzed Atroposelective Synthesis of Pyrrolo[3,4-b]pyridines with Configurationally Stable C-N Axial Chirality

By: Chu, Yunpeng; et al

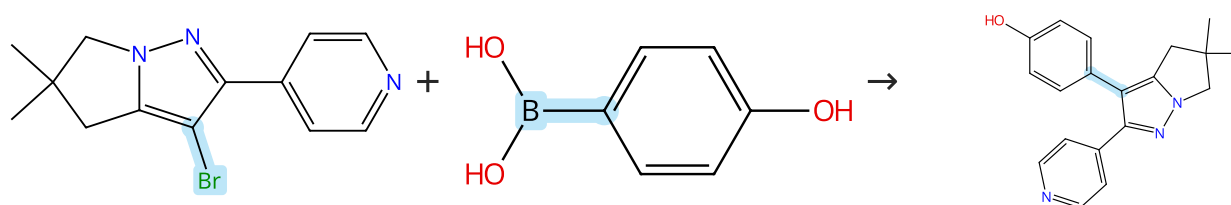
Organic Letters (2022), 24(21), 3884-3889.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: Toluene; 2 h, 80 °C

Experimental Protocols

Scheme 73 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (103)

31-179-CAS-16375359

Steps: 1 Yield: 97%

Application of Pd-Catalyzed Cross-Coupling Reactions in the Synthesis of 5,5-Dimethyl-5,6-dihydro-4H-pyrrolo[1,2-b]pyrazoles that Inhibit ALK5 Kinase

By: Tenora, Lukas; et al

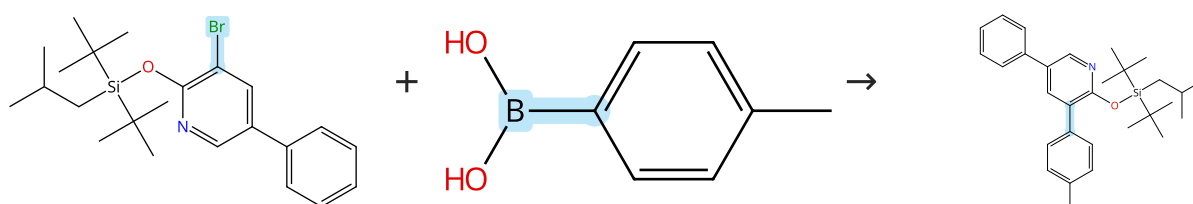
Journal of Organic Chemistry (2016), 81(23), 11841-11856.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide, Water; 1 - 2 h, 110 °C

Experimental Protocols

Scheme 74 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (114)

31-614-CAS-31695646

Steps: 1 Yield: 97%

Protecting Group-Controlled Regioselective Synthesis for Unsymmetrical 3,5-Disubstituted Pyridones

By: Kwon, Yong-Ju; et al

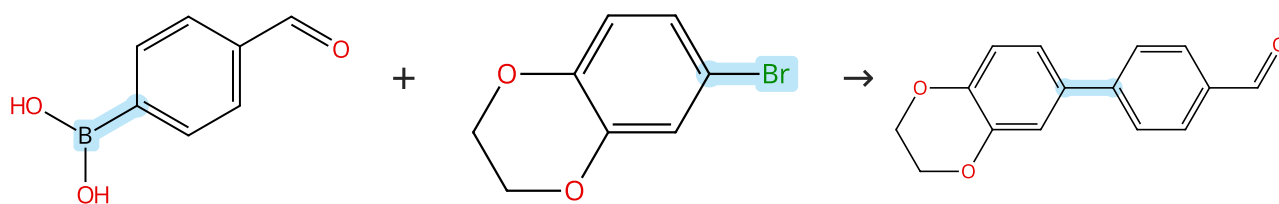
Advanced Synthesis & Catalysis (2022), 364(8), 1440-1449.

1.1 **Reagents:** Potassium carbonate
Catalysts: Palladium diacetate, Di-1-adamantylbenzylphosphine
Solvents: Toluene, Water; 2 h, 50 °C

Experimental Protocols

Scheme 75 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (106)

Suppliers (89)

Suppliers (14)

31-179-CAS-18669117

Steps: 1 Yield: 97%

A thiocyanopalladation/carbocyclization transformation identified through enzymatic screening: stereocontrolled tandem C-SCN and C-C bond formation

By: Malik, G.; et al

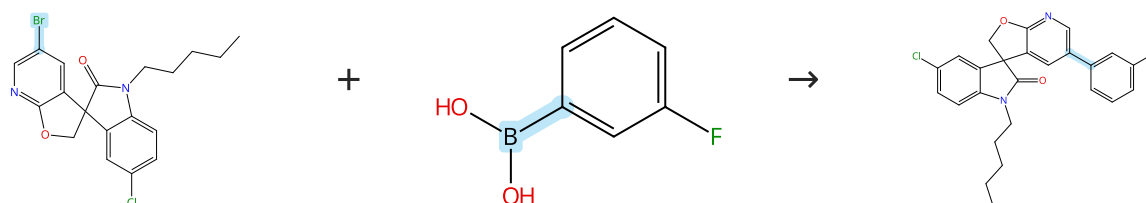
Chemical Science (2017), 8(12), 8050-8060.

1.1 **Reagents:** Potassium carbonate
Catalysts: Palladium diacetate, Tri-*o*-tolylphosphine
Solvents: Dimethylformamide, Water; rt → 80 °C

Experimental Protocols

Scheme 76 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (100)

31-179-CAS-3880269

Steps: 1 Yield: 97%

Suzuki-Miyaura Reactions of Halospirooxindole Derivatives

By: El Bouakher, Abderrahman; et al

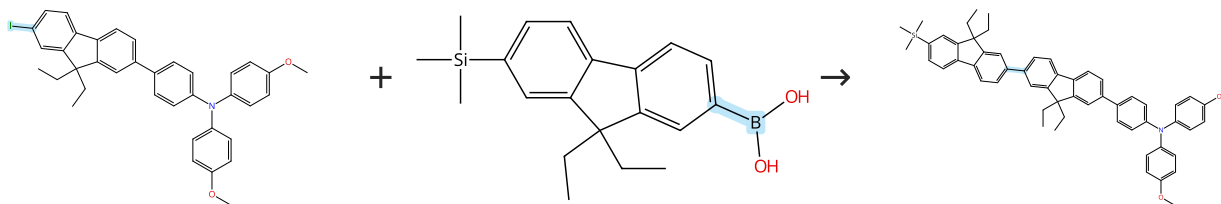
European Journal of Organic Chemistry (2015), 2015(16), 3450-3461.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene; 10 min, 150 °C

Experimental Protocols

Scheme 77 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-18849349

Steps: 1 Yield: 97%

Electron Transfer around a Molecular Corner

By: Schmidt, Hauke C.; et al

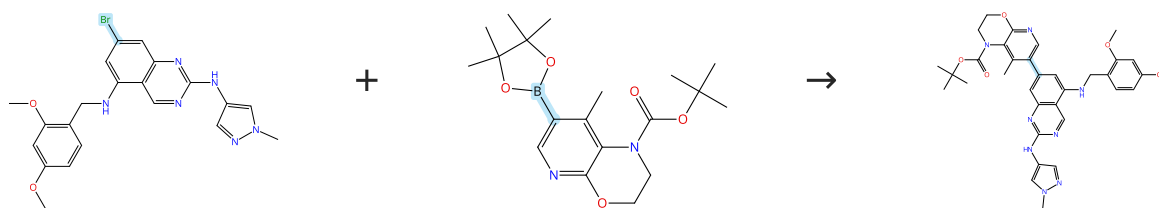
Angewandte Chemie, International Edition (2018), 57(22), 6696-6700.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; overnight, 85 °C

Experimental Protocols

Scheme 78 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (38)

31-614-CAS-35316122

Steps: 1 Yield: 97%

Development of a series of quinazoline-2,5-diamine derivatives as potent hematopoietic progenitor kinase 1 (HPK1) inhibitors

By: Shi, Huanyu; et al

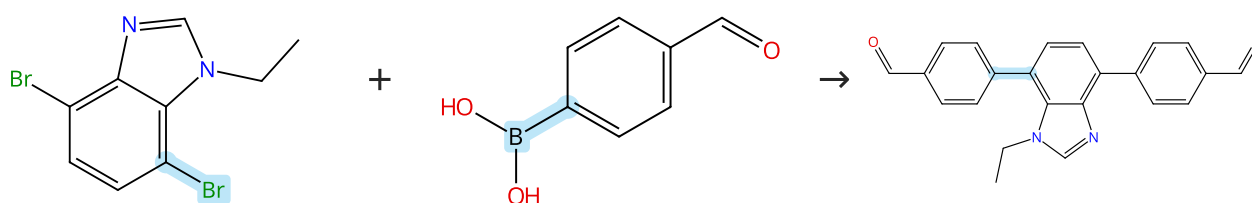
European Journal of Medicinal Chemistry (2023), 248, 115064.

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; rt → 80 °C; overnight, 80 °C

Experimental Protocols

Scheme 79 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (5)

Suppliers (106)

Suppliers (29)

31-614-CAS-25035586

Steps: 1 Yield: 97%

Synthesis and Catalytic Properties of Metal-N-Heterocyclic-Carbene-Decorated Covalent Organic Framework

By: Li, Yue; et al

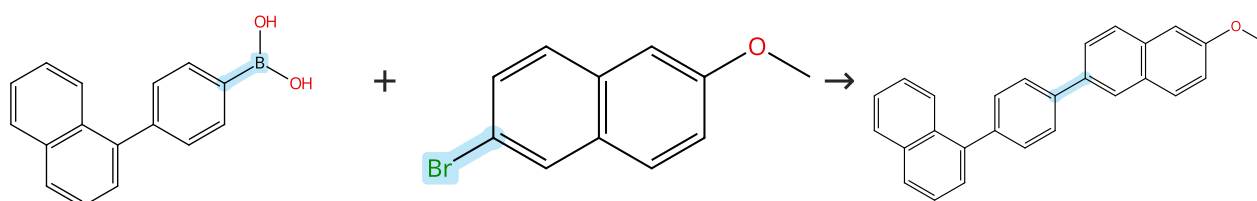
Organic Letters (2020), 22(18), 7363-7368.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 24 h, 90 °C

Experimental Protocols

Scheme 80 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (76)

Suppliers (107)

31-614-CAS-35769824

Steps: 1 Yield: 97%

Mechanochemistry-Directed Ligand Design: Development of a High-Performance Phosphine Ligand for Palladium-Catalyzed Mechanochemical Organoboron Cross-Coupling

By: Seo, Tamae; et al

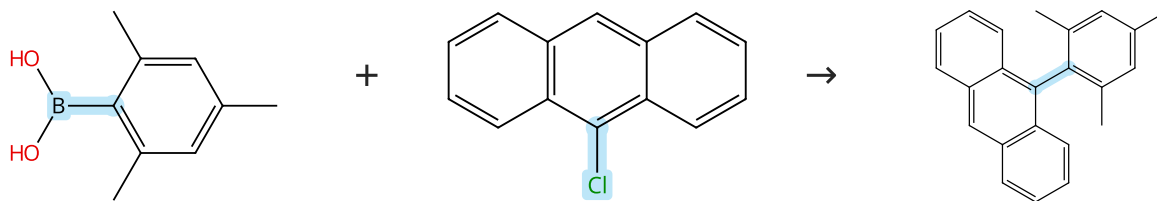
Journal of the American Chemical Society (2023), 145(12), 6823-6837.

1.1 **Reagents:** Water, Cesium fluoride
Catalysts: 1,5-Cyclooctadiene, Palladium diacetate, Poly(oxy-1, 2-ethanediyl), α-[2-(diphenylphosphino)[1,1'-biphenyl]-4-yl]-ω-methoxy-; 30 min, 45 °C

Experimental Protocols

Scheme 81 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (91)

Suppliers (64)

Suppliers (3)

31-179-CAS-17819413

Steps: 1 Yield: 97%

Bulky Yet Flexible Pd-PEPPSI-IPent^{An} for the Synthesis of Sterically Hindered Biaryls in Air

By: Lu, Dong-Dong; et al

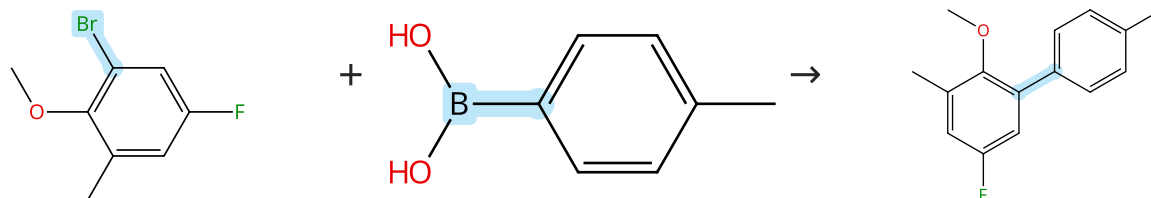
Journal of Organic Chemistry (2017), 82(20), 10898-10911.

1.1 **Reagents:** Potassium *tert*-butoxide
Catalysts: (SP-4-1)-[7,9-Bis[2,6-bis(1-ethylpropyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]dichloro(3-chloropyridine-κ*M*)palladium
Solvents: *tert*-Butanol; 4 h, 80 °C

Experimental Protocols

Scheme 82 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (19)

Suppliers (114)

31-614-CAS-36803602

Steps: 1 Yield: 97%

Radicals as Exceptional Electron-Withdrawing Groups: Nucleophilic Aromatic Substitution of Halophenols Via Homolysis-Enabled Electronic Activation

By: Shin, Nick Y.; et al

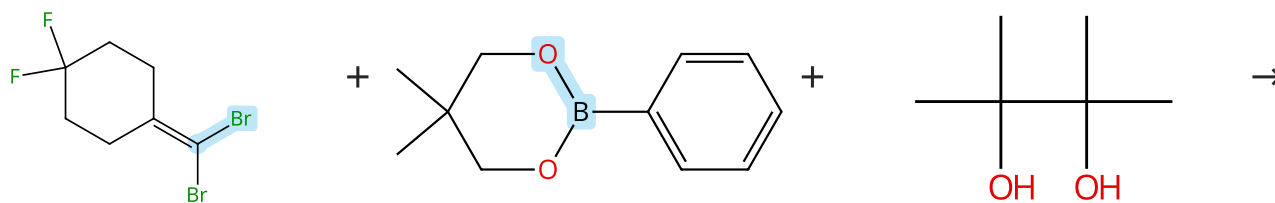
Journal of the American Chemical Society (2022), 144(47), 21783-21790.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: 1,4-Dioxane, Water; 20 h, 100 °C

Experimental Protocols

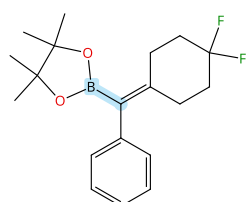
Scheme 83 (1 Reaction)

Steps: 1 Yield: 97%

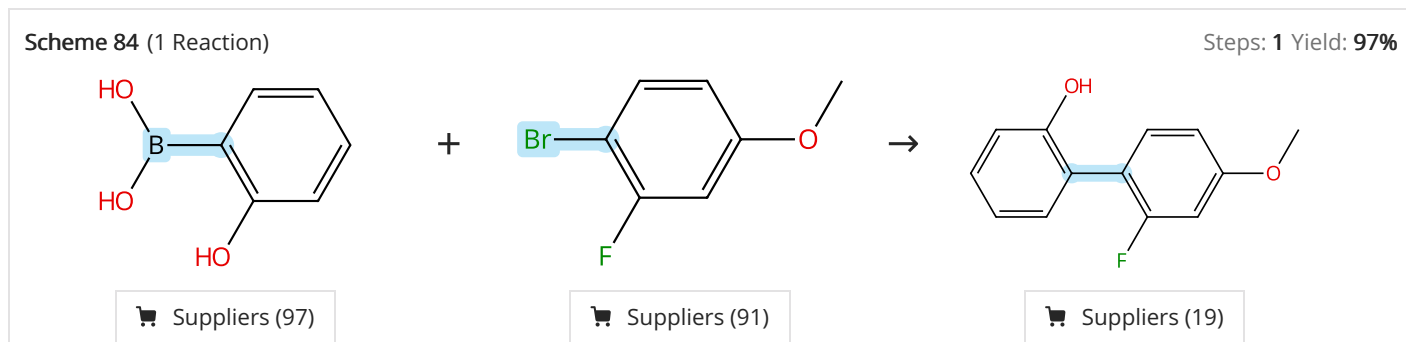


Suppliers (66)

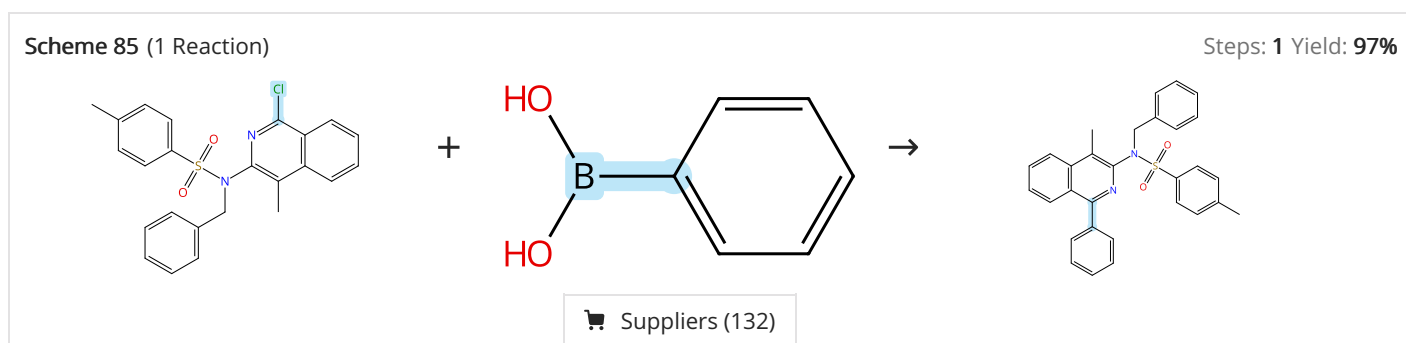
Suppliers (85)



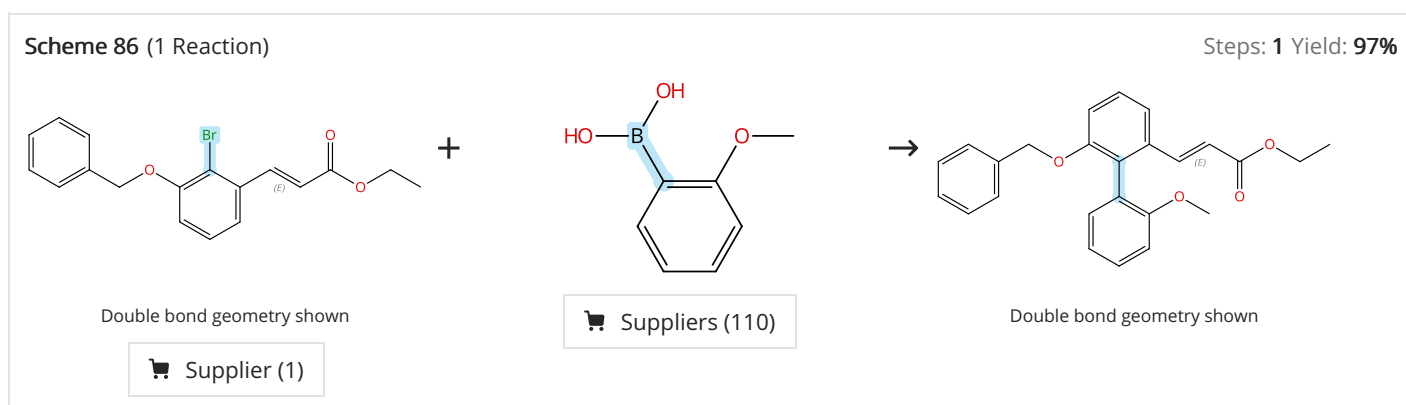
31-614-CAS-42386609 Steps: 1 Yield: 97% 1.1 Reagents: Butyllithium, 1,3-Dimethyl-3,4,5,6-tetrahydro-2(1<i>H</i>)-pyrimidinone, Lithium bromide Solvents: Diethyl ether, Hexane; -110 °C; 30 min, -110 °C; -110 °C → rt; 30 min, rt 1.2 Reagents: Triethylamine Solvents: Acetone; 12 h, rt → 65 °C Experimental Protocols	Stereospecific alkenylidene homologation of organoboronates by S_NV reaction By: Chen, Miao; et al Nature (London, United Kingdom) (2024), 631(8020), 328-334.
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31-614-CAS-35318634 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: Tetrahydrofuran, Water; 16 h, 70 °C Experimental Protocols	Dearomatization of Biaryls through Polarity Mismatched Radical Spirocyclization By: Azpilcueta-Nicolas, Carlos R.; et al Angewandte Chemie, International Edition (2023), 62(6), e202215422.
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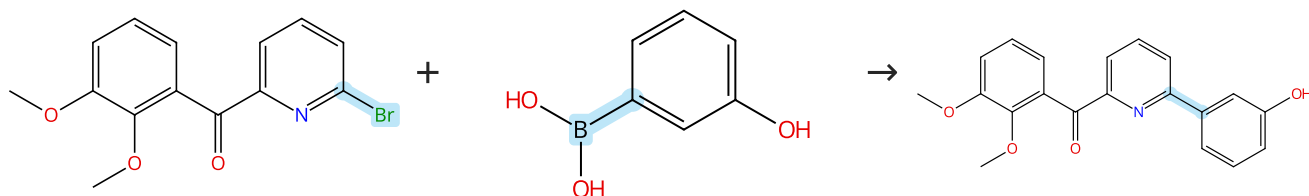
31-614-CAS-39340153 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Triphenylphosphine, Tetrakis(triphenylphosphine) palladium Solvents: Ethanol, Toluene; 9 h, 70 °C; 70 °C → rt 1.2 Reagents: Water; rt Experimental Protocols	Annulations of Ynamides with 1,2-Benzisothiazoles to Construct 1,4-Benzothiazepines and 3-Aminoisoquinolines By: Wang, Xiao-Na; et al Organic Letters (2024), 26(7), 1522-1527.
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31-614-CAS-35318619 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: Tetrahydrofuran, Water; 16 h, 70 °C Experimental Protocols	Dearomatization of Biaryls through Polarity Mismatched Radical Spirocyclization By: Azpilcueta-Nicolas, Carlos R.; et al Angewandte Chemie, International Edition (2023), 62(6), e202215422.
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Scheme 87 (1 Reaction)

Steps: 1 Yield: 97%

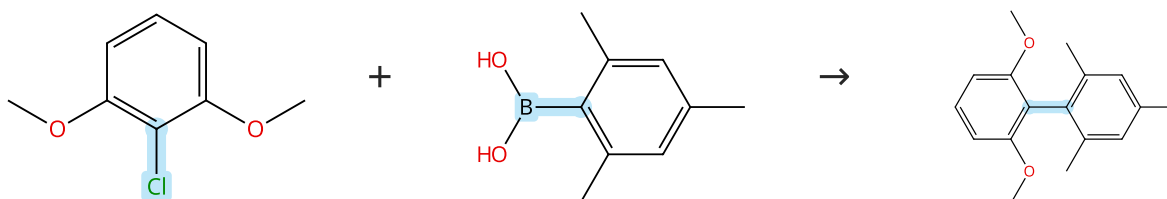


Suppliers (101)

31-179-CAS-16628541 Steps: 1 Yield: 97% 1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; overnight, 80 °C; 80 °C → rt Experimental Protocols	First Structure-Activity Relationship of 17β-Hydroxysteroid Dehydrogenase Type 14 Nonsteroidal Inhibitors and Crystal Structures in Complex with the Enzyme By: Braun, Florian; et al Journal of Medicinal Chemistry (2016), 59(23), 10719-10737.
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Scheme 88 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (55)

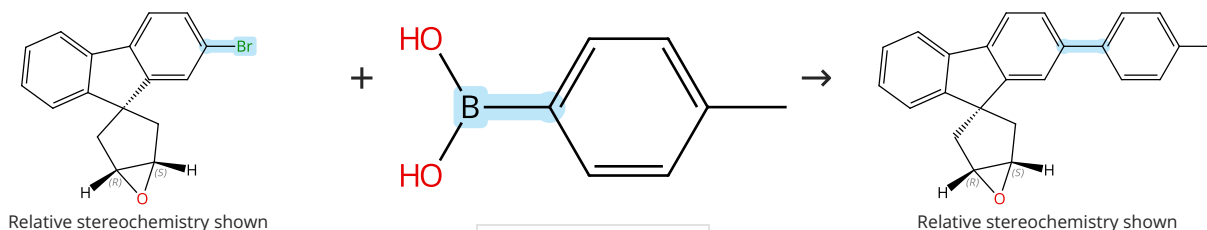
Suppliers (91)

Suppliers (2)

31-614-CAS-24822620 Steps: 1 Yield: 97% 1.1 Reagents: Tetradecane, Potassium hydroxide Catalysts: 1,3-Bis[2,6-bis(diphenylmethyl)-4-methoxyphenyl]-1,3-dihydro-2H-imidazol-2-ylidene, Palladium, di-μ-bromobis(η³-1-naphthalenylmethyl)di- Solvents: Tetrahydrofuran; 12 h, rt Experimental Protocols	Halogen-Bridged Methylphenyl Palladium Dimers as Versatile Catalyst Precursors in Coupling Reactions By: Sivendran, Nardana; et al Angewandte Chemie, International Edition (2021), 60(47), 25151-25160.
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Scheme 89 (1 Reaction)

Steps: 1 Yield: 97%



Relative stereochemistry shown

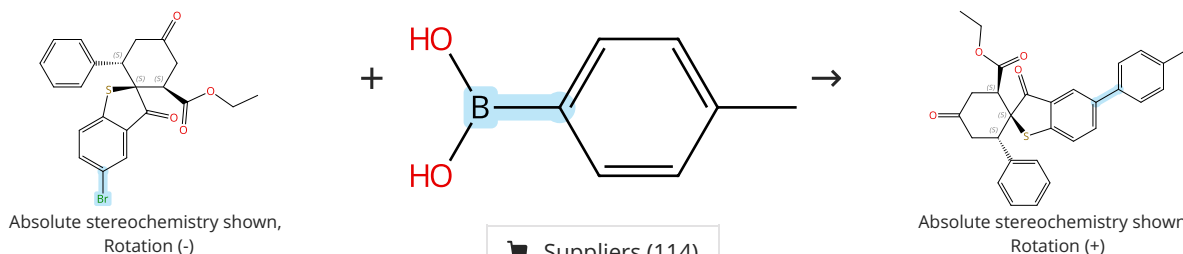
Relative stereochemistry shown

Suppliers (114)

31-614-CAS-37282665 Steps: 1 Yield: 97% 1.1 Reagents: Cesium carbonate Catalysts: Palladium, tris[μ-[(1,2-η:4,5-η)-(1<i>E</i>,4<i>E</i>)-1,5-diphenyl-1,4-pentadien-3-one]]di-, compd. with benzene (1:1), 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: Toluene; 3 h, 100 °C Experimental Protocols	Pseudo-Diastereodivergent Synthesis of Chiral Fluorenes Bearing Bis-1,3-Nonadjacent Stereogenic Centers via Organocatalytic Desymmetrization of meso-Epoxides By: Jiang, Feng; et al Organic Letters (2023), 25(32), 6006-6011.
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Scheme 90 (1 Reaction)

Steps: 1 Yield: 97%

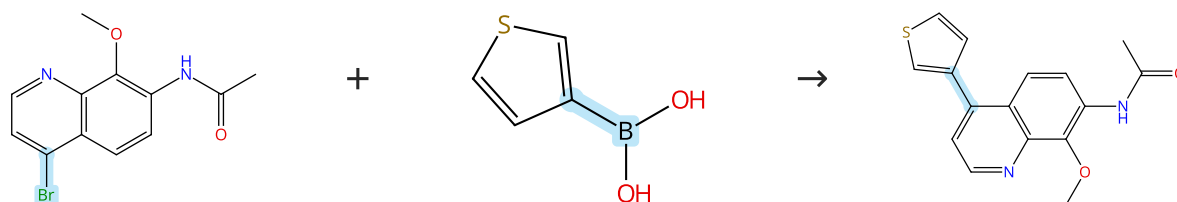


Suppliers (114)

31-179-CAS-22201947 Steps: 1 Yield: 97% 1.1 Reagents: Potassium acetate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane; overnight, 110 °C Experimental Protocols	Access to Spirocyclic Benzothiophenones with Multiple Stereocenters via an Organocatalytic Cascade Reaction By: Formanek, Bedrich; et al Journal of Organic Chemistry (2020), 85(13), 8510-8521.
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Scheme 91 (1 Reaction)

Steps: 1 Yield: 97%



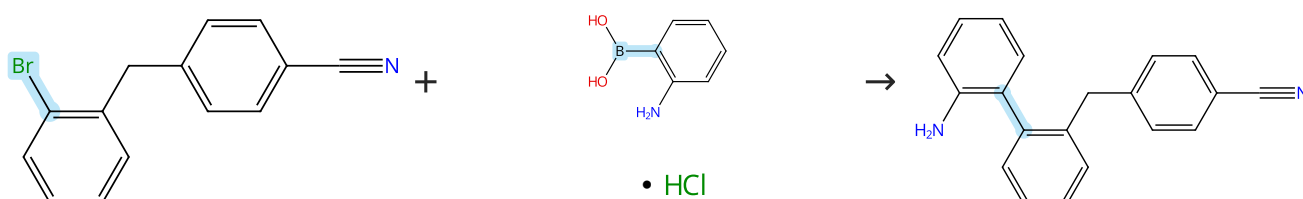
Suppliers (2)

Suppliers (101)

31-179-CAS-23109341 Steps: 1 Yield: 97% 1.1 Reagents: Cesium carbonate Catalysts: 1,2-Bis(diphenylphosphino)ethane, Palladium diacetate Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Drug Design Targeting T-Cell Factor-Driven Epithelial-Mesenchymal Transition as a Therapeutic Strategy for Colorectal Cancer By: Abraham, Adedoyin D.; et al Journal of Medicinal Chemistry (2019), 62(22), 10182-10203.
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Scheme 92 (1 Reaction)

Steps: 1 Yield: 97%



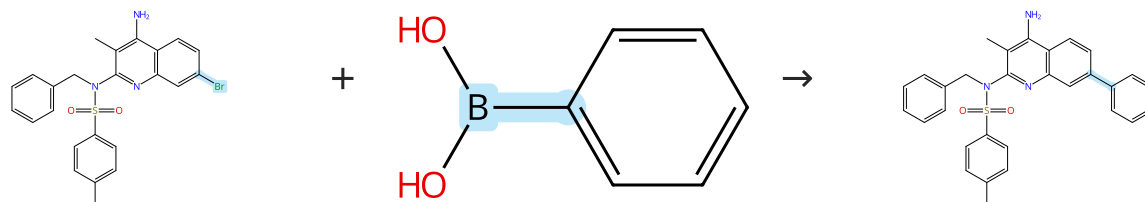
Supplier (1)

Suppliers (64)

31-179-CAS-17867790 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Solvents: Ethanol, Toluene; 0.5 h, rt 1.2 Catalysts: Tetrakis(triphenylphosphine)palladium; 8 h, reflux 1.3 Solvents: Ethyl acetate, Water; rt	Coupling N-H Deprotonation, C-H Activation, and Oxidation: Metal-Free C(sp³)-H Aminations with Unprotected Anilines By: Evoniuk, Christopher J.; et al Journal of the American Chemical Society (2017), 139(45), 16210-16221.
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Scheme 93 (1 Reaction)

Steps: 1 Yield: 97%

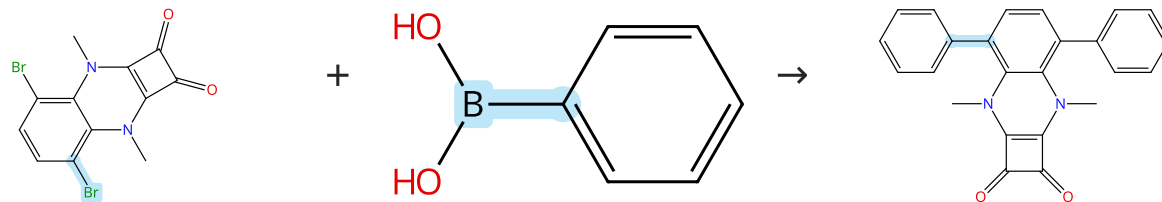


Suppliers (132)

31-614-CAS-44471208 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Triphenylphosphine, Tetrakis(triphenylphosphine) palladium Solvents: Ethanol, Toluene; 3 h, 80 °C 1.2 Solvents: Water Experimental Protocols	TMSOTf/TfOH-Promoted [4+2] Annulation of Ynamides with 2-Aminoarylnitriles To Construct 2,4-Diaminoquinolines By: Wang, Xiao-Na; et al Journal of Organic Chemistry (2025), 90(11), 3930-3935.
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Scheme 94 (1 Reaction)

Steps: 1 Yield: 97%

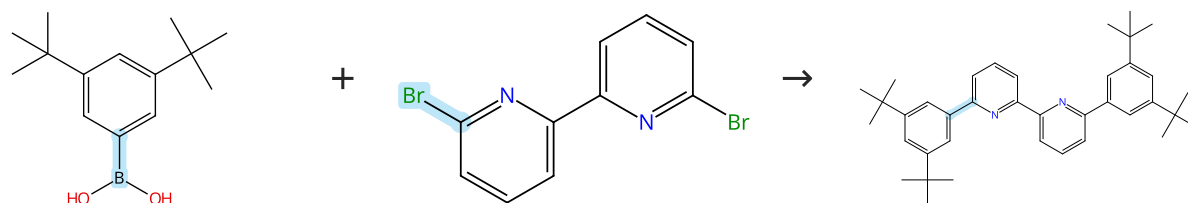


Suppliers (132)

31-614-CAS-38305989 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran; 20 h, 70 °C Experimental Protocols	From Squaric Acid Amides (SQAs) to Quinoxaline-Based SQAs-Evolution of a Redox-Active Cathode Material for Organic Polymer Batteries By: Baumert, Marcel E.; et al Journal of the American Chemical Society (2023), 145(42), 23334-23345.
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Scheme 95 (1 Reaction)

Steps: 1 Yield: 97%



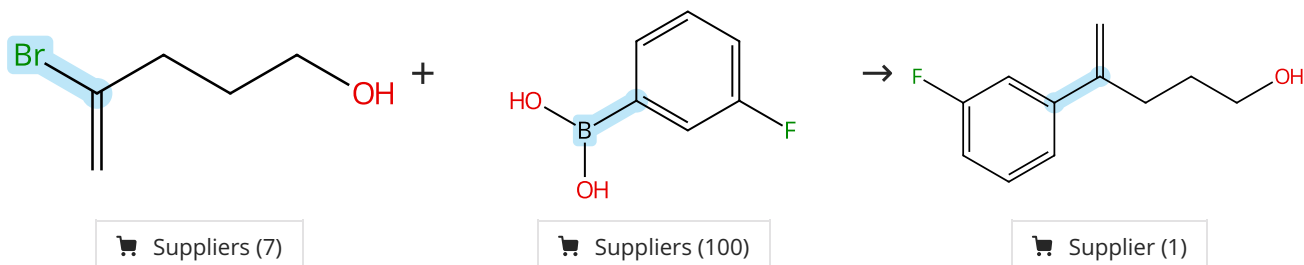
Suppliers (69)

Suppliers (68)

31-614-CAS-46572855	Steps: 1 Yield: 97%	Iron-Catalyzed Transfer Hydroalumination of Alkynes
1.1 Reagents: Barium hydroxide Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Toluene, Water; 30 °C → 120 °C		By: Li, Wen-Tao; et al Journal of the American Chemical Society (2025), 147(18), 15545-15553.
Experimental Protocols		

Scheme 96 (1 Reaction)

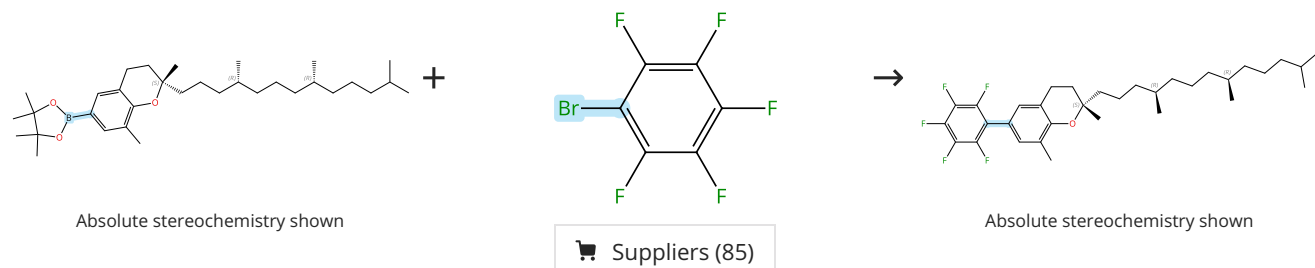
Steps: 1 Yield: 97%



31-614-CAS-44358496	Steps: 1 Yield: 97%	Emergence of a distinct mechanism of C-N bond formation in photoenzymes
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; overnight, 75 °C		By: Raps, Felix C.; et al Nature (London, United Kingdom) (2025), 637(8045), 362-368.
Experimental Protocols		

Scheme 97 (1 Reaction)

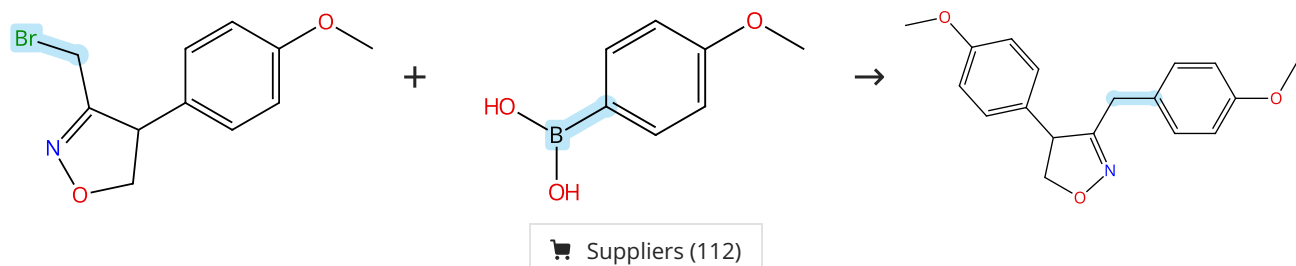
Steps: 1 Yield: 97%



31-614-CAS-43159131	Steps: 1 Yield: 97%	Pd-Catalyzed Site-Selective Defluorinative Etherification Between Unactivated Perfluoroarenes and Hydrobenzoxazoles
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene; 12 h, 90 °C		By: Wang, Yaping; et al Chemistry - A European Journal (2025), 31(8), e202403914.
Experimental Protocols		

Scheme 98 (1 Reaction)

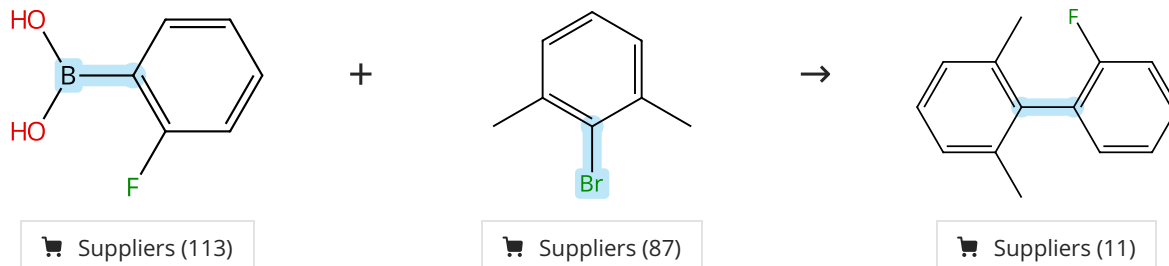
Steps: 1 Yield: 97%



31-179-CAS-23747121 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate, Water Catalysts: Palladium diacetate, 2-(Di-<i>tert</i>-butylphosphino) biphenyl Solvents: Tetrahydrofuran; 24 h, 60 °C Experimental Protocols	Sequential Formal [4+1]-Cycloaddition, C-H Functionalization and Suzuki-Miyaura Cross-Coupling for the Synthesis of Trisubstituted Isoxazolines By: Ushakov, Pavel Yu.; et al European Journal of Organic Chemistry (2021), 2021(18), 2680-2693.
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Scheme 99 (1 Reaction)

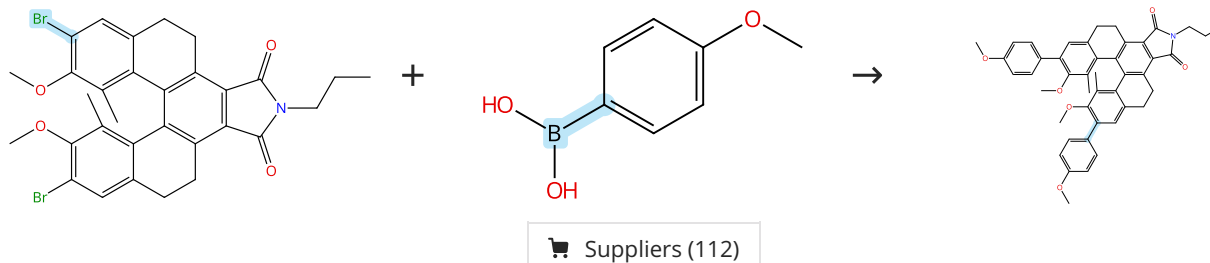
Steps: 1 Yield: 97%



31-614-CAS-40329239 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrabutylammonium bromide, Palladium diacetate, 3022418-51-2 Solvents: Dichloromethane, Water; 2 h, 100 °C Experimental Protocols	Design and Discovery of Water-Soluble Benzooxaphosphole-Based Ligands for Hindered Suzuki-Miyaura Coupling Reactions with Low Catalyst Load By: Shada, Arun D. R.; et al Organic Letters (2024), 26(14), 2751-2757.
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Scheme 100 (1 Reaction)

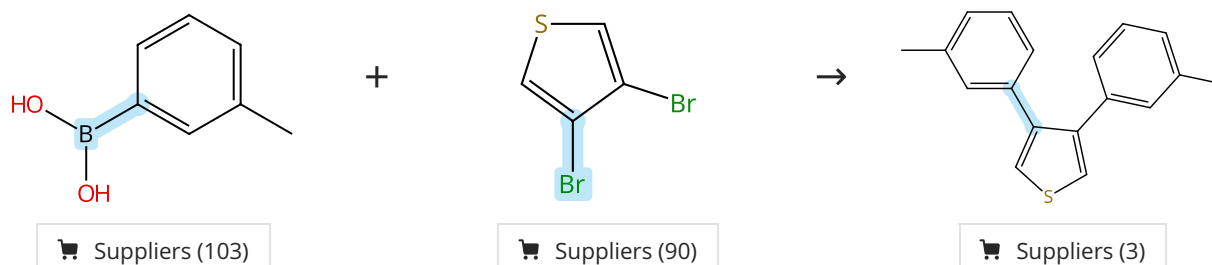
Steps: 1 Yield: 97%



31-179-CAS-16471797 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 12 h, 75 °C Experimental Protocols	Helical aromatic imide based enantiomers with full-color circularly polarized luminescence By: Li, Meng; et al Chemical Communications (Cambridge, United Kingdom) (2016), 52(64), 9921-9924.
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Scheme 101 (1 Reaction)

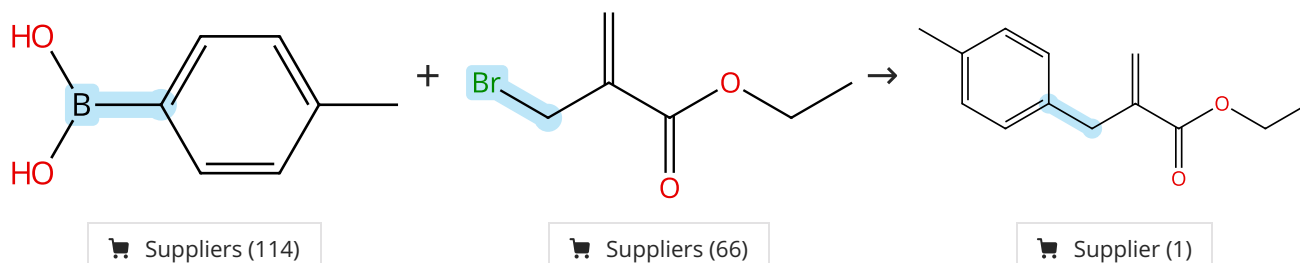
Steps: 1 Yield: 97%



31-614-CAS-38944339	Steps: 1 Yield: 97%	Crafting diverse tetraphenylthiophene designs through comprehensive 'classical' Suzuki-Miyaura synthesis and electrochemical exploration
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene; 24 h, 80 °C; 80 °C → rt		By: Maddala, Sudhakar; et al
1.2 Reagents: Water; rt		Tetrahedron (2024), 151, 133807.
Experimental Protocols		

Scheme 102 (1 Reaction)

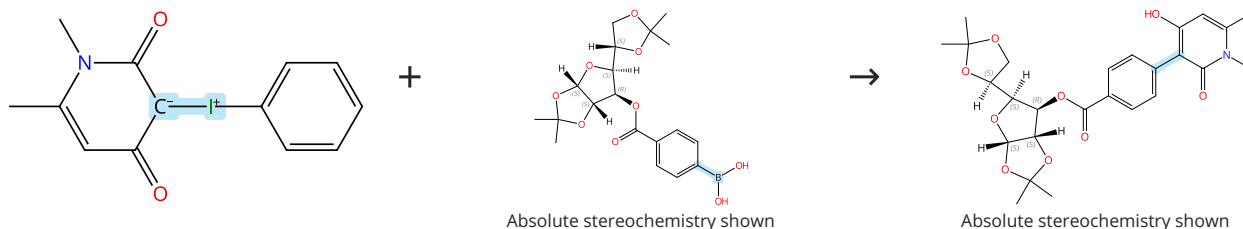
Steps: 1 Yield: 97%



31-614-CAS-37145888	Steps: 1 Yield: 97%	Versatile Biocatalytic C(sp³)-H Oxyfunctionalization for the Site- Selective and Stereodivergent Synthesis of α- and β- Hydroxy Acids
1.1 Reagents: Potassium hydroxide Catalysts: Palladium trifluoroacetate Solvents: Water; rt; 5 min, rt		By: Mao, Yingle; et al
1.2 3 h, 90 °C		Angewandte Chemie, International Edition (2023), 62(33), e202305250.
Experimental Protocols		

Scheme 103 (1 Reaction)

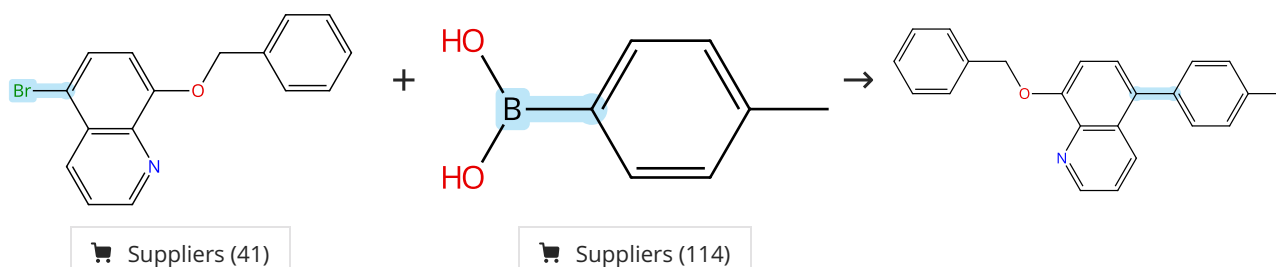
Steps: 1 Yield: 97%



31-614-CAS-39973639	Steps: 1 Yield: 97%	Catalytic and Base-free Suzuki-type α-Arylation of Cyclic 1,3- Dicarboxyls via a Cyclic Iodonium Ylide Strategy
1.1 Catalysts: Bis[dichloro[η^5 -(pentamethylcyclopentadienyl)] rhodium] Solvents: Water; 30 min, 50 °C		By: Liang, Mingxuan; et al
Experimental Protocols		Angewandte Chemie, International Edition (2024), 63(17), e202400741.

Scheme 104 (1 Reaction)

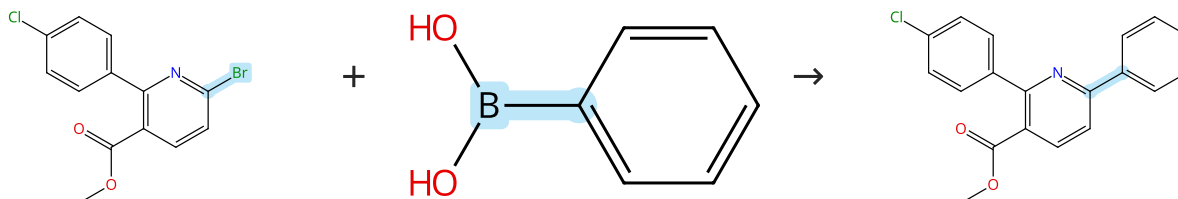
Steps: 1 Yield: 97%



31-179-CAS-19610865	Steps: 1 Yield: 97%	Optimization of 8-Hydroxyquinolines as Inhibitors of Catechol O-Methyltransferase
1.1 Reagents: Sodium bicarbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; overnight, 90 °C		By: Buchler, Ingrid; et al
Experimental Protocols		Journal of Medicinal Chemistry (2018), 61(21), 9647-9665.

Scheme 105 (1 Reaction)

Steps: 1 Yield: 97%



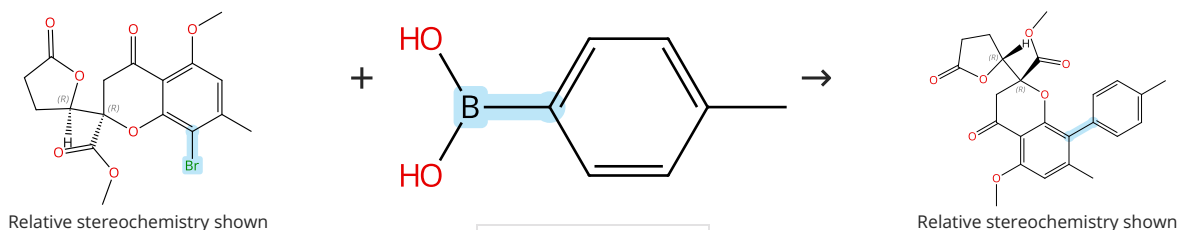
Supplier (1)

Suppliers (132)

31-179-CAS-17161545	Steps: 1 Yield: 97%	Fe(II)/Au(I) Relay Catalyzed Propargylisoxazole to Pyridine Isomerization: Access to 6-Halonicotinates
1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 2 h, reflux		By: Galenko, Alexey V.; et al
Experimental Protocols		Journal of Organic Chemistry (2017), 82(10), 5367-5379.

Scheme 106 (1 Reaction)

Steps: 1 Yield: 97%



Relative stereochemistry shown

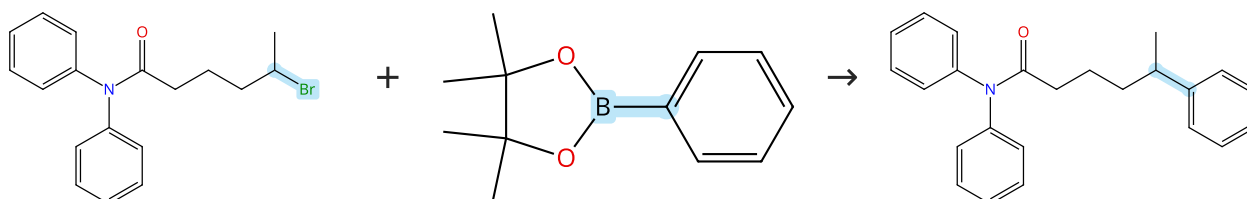
Relative stereochemistry shown

Suppliers (114)

31-179-CAS-17759674	Steps: 1 Yield: 97%	Suzuki-Miyaura Cross-Coupling Reactions of Tetrahydro xanthenes and 4-Chromanone Lactones to Heteromeric Biaryls
1.1 Reagents: Tripotassium phosphate Catalysts: (<i>SP</i> -4-3)-[2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C][dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato- κ O)palladium Solvents: 1,4-Dioxane, Water; 6.5 h, 60 °C		By: Geiger, Larissa; et al
Experimental Protocols		Advanced Synthesis & Catalysis (2017), 359(19), 3421-3427.

Scheme 107 (1 Reaction)

Steps: 1 Yield: 97%

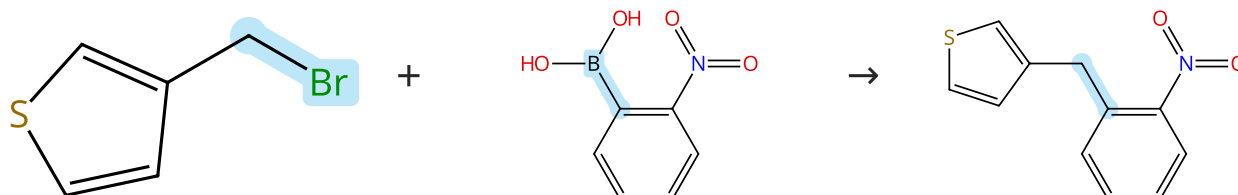


Suppliers (80)

31-614-CAS-40989485	Steps: 1 Yield: 97%	A Rationally Designed Iron(II) Catalyst for C(sp³)-C(sp²) and C(sp³)-C(sp³) Suzuki-Miyaura Cross-Coupling
1.1 Reagents: Ethanamine, <i>N</i> -methyl-, lithium salt Catalysts: 3050668-44-2 Solvents: <i>tert</i> -Amyl methyl ether; 24 h, 25 °C		By: Chen, Donghuang; et al
1.2 Reagents: Ammonium chloride Solvents: Water; 25 °C		Angewandte Chemie, International Edition (2024), 63(32), e202408419.
Experimental Protocols		

Scheme 108 (1 Reaction)

Steps: 1 Yield: 97%



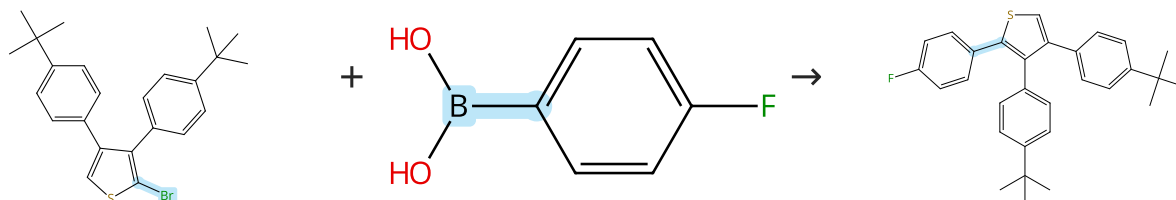
Suppliers (47)

Suppliers (93)

31-614-CAS-34078025	Steps: 1 Yield: 97%	A switchable redox annulation of 2-nitroarylethanols affording N-heterocycles: photoexcited nitro as a multifunctional handle
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 2 - 4 h, 60 - 90 °C		By: Wang, Bin; et al
Experimental Protocols		Chemical Science (2022), 13(37), 11074-11082.

Scheme 109 (1 Reaction)

Steps: 1 Yield: 97%

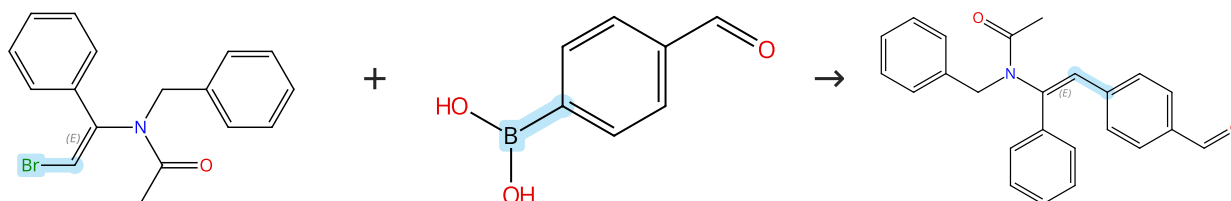


Suppliers (104)

31-614-CAS-38944364	Steps: 1 Yield: 97%	Crafting diverse tetraphenylthiophene designs through comprehensive 'classical' Suzuki-Miyaura synthesis and electrochemical exploration
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene; 8 h, 80 °C; 80 °C → rt		By: Maddala, Sudhakar; et al
1.2 Reagents: Water; rt		Tetrahedron (2024), 151, 133807.
Experimental Protocols		

Scheme 110 (1 Reaction)

Steps: 1 Yield: 97%



Double bond geometry shown

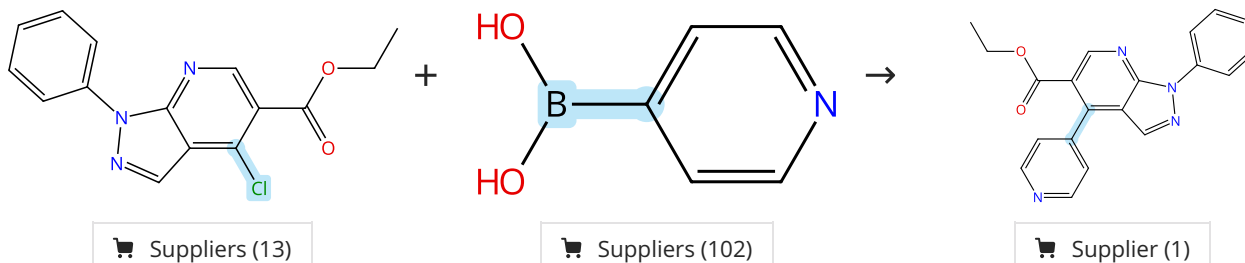
Suppliers (106)

Double bond geometry shown

31-614-CAS-39395721	Steps: 1 Yield: 97%	Electrochemical bromination of enamides with sodium bromide
1.1 Reagents: Cesium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Tetrahydrofuran, Water; 16 h, reflux		By: Luan, Shinan; et al Green Chemistry (2024), 26(6), 3429-3434.
Experimental Protocols		

Scheme 111 (1 Reaction)

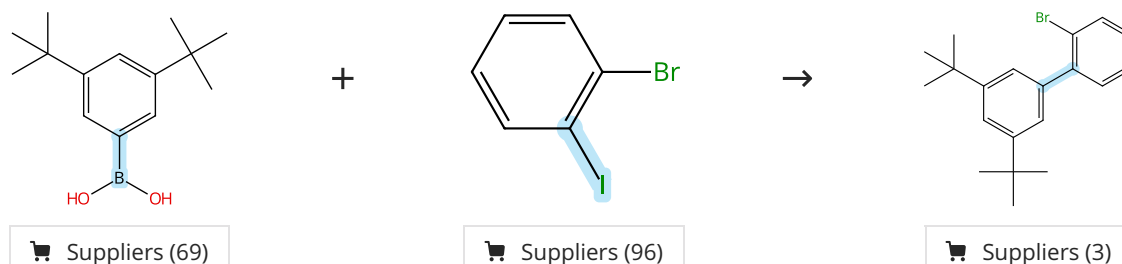
Steps: 1 Yield: 97%



31-179-CAS-16785817	Steps: 1 Yield: 97%	Harnessing Cascade Suzuki-Cyclization Reactions of Pyrazolo [3,4-b]pyridine for the Synthesis of Tetracyclic Fused Heteroaromatics
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: <i>tert</i> -Butanol, Toluene, Water; 5 - 18 h, 75 °C		By: Lavrard, Hubert; et al European Journal of Organic Chemistry (2017), 2017(3), 600-608.

Scheme 112 (1 Reaction)

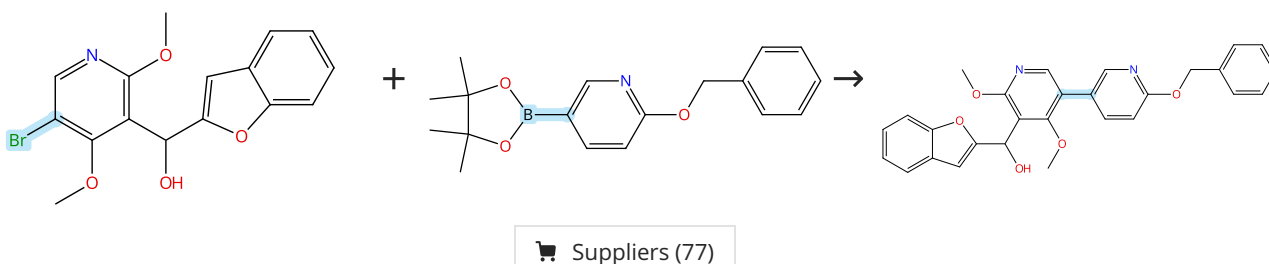
Steps: 1 Yield: 97%



31-179-CAS-22515626	Steps: 1 Yield: 97%	Design of Bowl-Shaped N-Hydroxyimide Derivatives as New Organoradical Catalysts for Site-Selective C(sp³)-H Bond Functionalization Reactions
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 16 h, 80 °C		By: Kato, Terumasa; et al Angewandte Chemie, International Edition (2020), 59(34), 14261-14264.
Experimental Protocols		

Scheme 113 (1 Reaction)

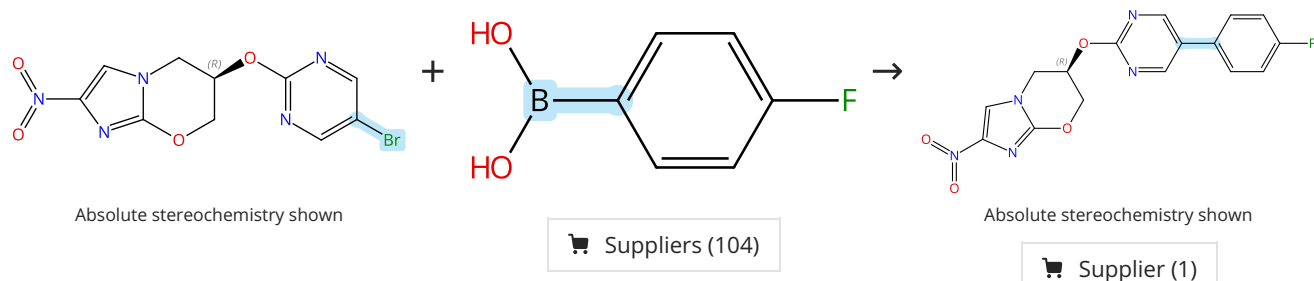
Steps: 1 Yield: 97%



31-179-CAS-19607213	Steps: 1 Yield: 97%	Discovery of 2,4-dimethoxypyridines as novel autophagy inhibitors
1.1 Reagents: Sodium carbonate Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tetrakis (triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 16 h, 110 °C		By: Robke, Lucas; et al Tetrahedron (2018), 74(35), 4531-4537.
Experimental Protocols		

Scheme 114 (1 Reaction)

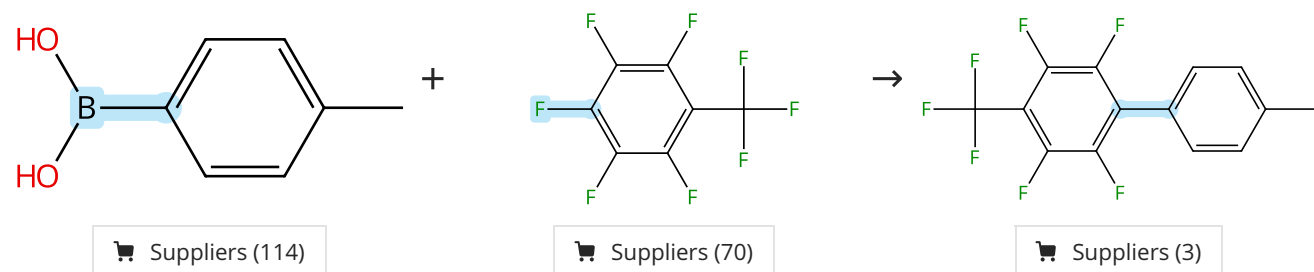
Steps: 1 Yield: 97%



31-179-CAS-18854671	Steps: 1 Yield: 97%	Development of (6R)-2-Nitro-6-[4-(trifluoromethoxy)phenoxy]-6,7-dihydro-5H-imidazo[2,1-b][1,3]oxazine (DNDI-8219): A New Lead for Visceral Leishmaniasis
1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Ethanol, Dimethylformamide, Toluene, Water; 80 min, 87 °C		By: Thompson, Andrew M.; et al Journal of Medicinal Chemistry (2018), 61(6), 2329-2352.
Experimental Protocols		

Scheme 115 (1 Reaction)

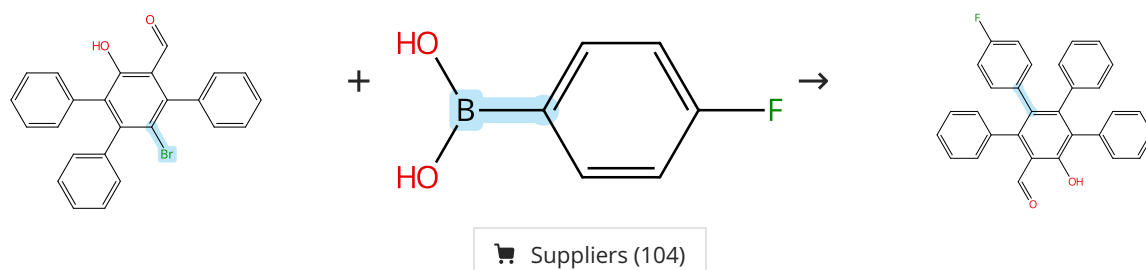
Steps: 1 Yield: 97%



31-179-CAS-15854765	Steps: 1 Yield: 97%	NHC Nickel-Catalyzed Suzuki-Miyaura Cross-Coupling Reactions of Aryl Boronate Esters with Perfluorobenzenes
1.1 Reagents: Cesium fluoride Catalysts: Stereoisomer of [μ -[(1,2:5,6- η)-1,5-cyclooctadiene]] tetrakis[1,3-dihydro-1,3-bis(1-methylethyl)-2H-imidazol-2-ylidene]dinickel Solvents: Toluene; 18 h, 100 °C		By: Zhou, Jing; et al Journal of Organic Chemistry (2016), 81(13), 5789-5794.
Experimental Protocols		

Scheme 116 (1 Reaction)

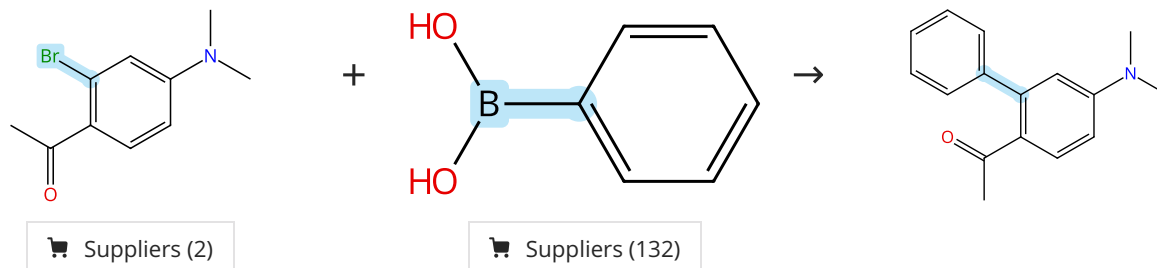
Steps: 1 Yield: 97%



31-179-CAS-22028041	Steps: 1 Yield: 97%	Short and modular synthesis of tetraaryl salicylaldehydes
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide; 16 h, 90 °C		By: Tejedor, David; et al Chemical Communications (Cambridge, United Kingdom) (2020), 56(28), 4019-4022.
Experimental Protocols		

Scheme 117 (1 Reaction)

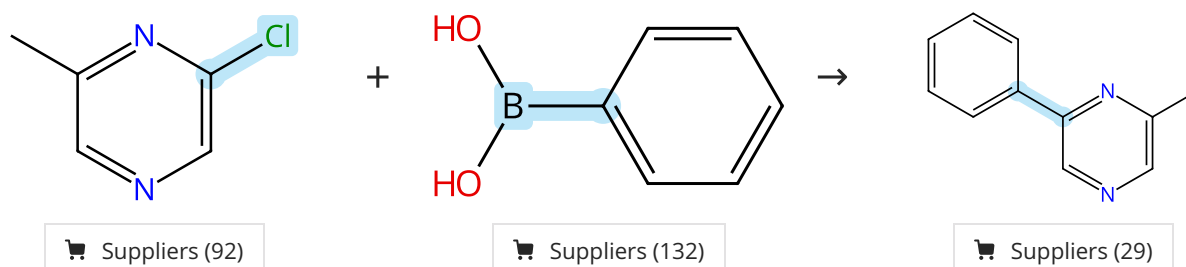
Steps: 1 Yield: 97%



31-614-CAS-35310273	Steps: 1 Yield: 97%	o-Nitrobenzyl Oxime Ethers Enable Photoinduced Cyclization Reaction to Provide Phenanthridines under Aqueous Conditions
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene; 14 h, reflux		By: Okamura, Hidenori; et al Organic Letters (2023), 25(3), 466-470.
Experimental Protocols		

Scheme 118 (1 Reaction)

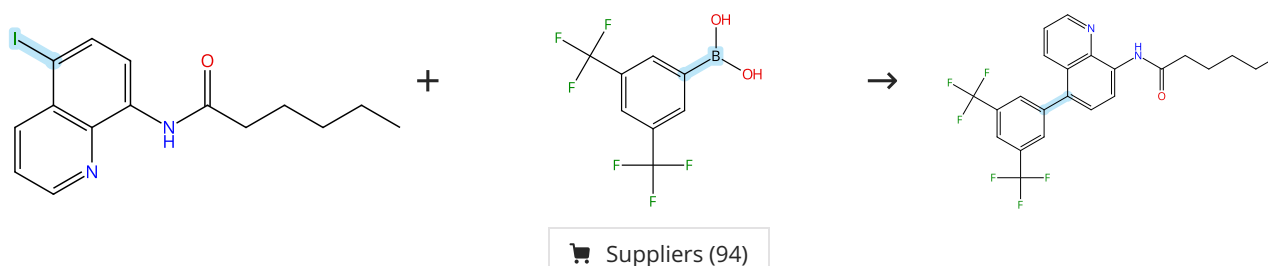
Steps: 1 Yield: 97%



31-179-CAS-15740784	Steps: 1 Yield: 97%	Synthesis of Chiral Piperazines via Hydrogenation of Pyrazines Activated by Alkyl Halides
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; 10 min, rt → 95 °C; 95 °C → rt		By: Huang, Wen-Xue; et al Organic Letters (2016), 18(13), 3082-3085.
1.2 Reagents: Potassium fluoride; rt → 95 °C		
Experimental Protocols		

Scheme 119 (1 Reaction)

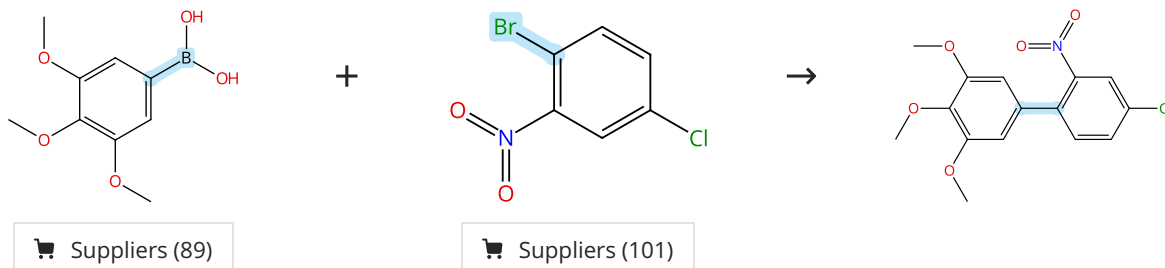
Steps: 1 Yield: 97%



31-614-CAS-38928329	Steps: 1 Yield: 97%	Palladium-Catalyzed Enantioselective Directed C(sp³)-H Functionalization Using C₅-Substituted 8-Aminoquinoline Auxiliaries By: Huang, Yi; et al Organic Letters (2024), 26(1), 94-99.
1.1 Reagents: Sodium carbonate Catalysts: Triphenylphosphine, Palladium diacetate Solvents: 1,4-Dioxane, Water; 12 h, 90 °C		

Scheme 120 (1 Reaction)

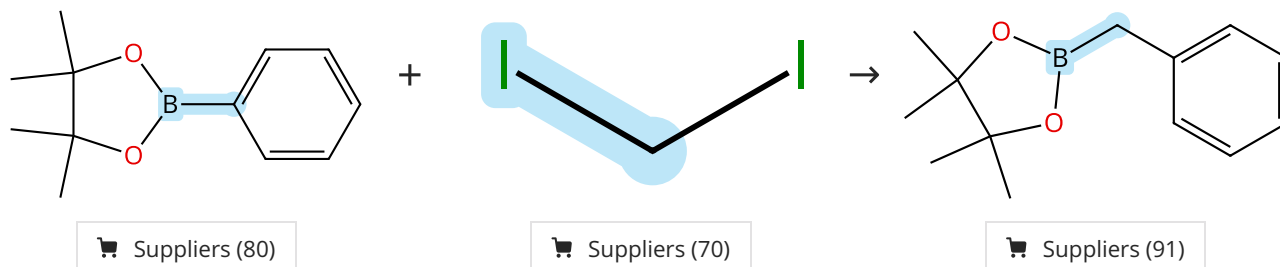
Steps: 1 Yield: 97%



31-614-CAS-24081412	Steps: 1 Yield: 97%	Mo-Catalyzed One-Pot Synthesis of N-Polyheterocycles from Nitroarenes and Glycols with Recycling of the Waste Reduction Byproduct. Substituent-Tuned Photophysical Properties By: Hernandez-Ruiz, Raquel; et al Chemistry - A European Journal (2021), 27(54), 13613-13623.
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; overnight, 80 °C		
Experimental Protocols		

Scheme 121 (1 Reaction)

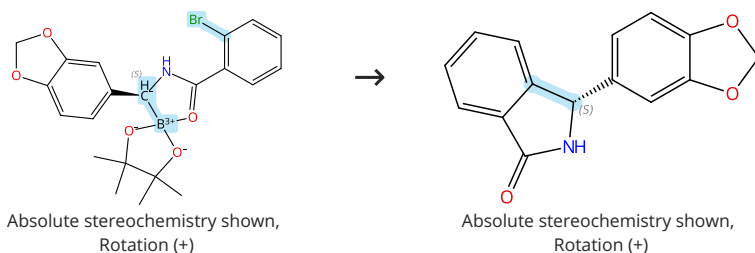
Steps: 1 Yield: 97%



31-614-CAS-37228946	Steps: 1 Yield: 97%	External Flash Generation of Carbenoids Enables Monodeuteration of Dihalomethanes By: Okamoto, Kazuhiro; et al Chemistry - A European Journal (2023), 29(47), e202301738.
1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; 37 s, -60 °C; 30 min, rt		
Experimental Protocols		

Scheme 122 (1 Reaction)

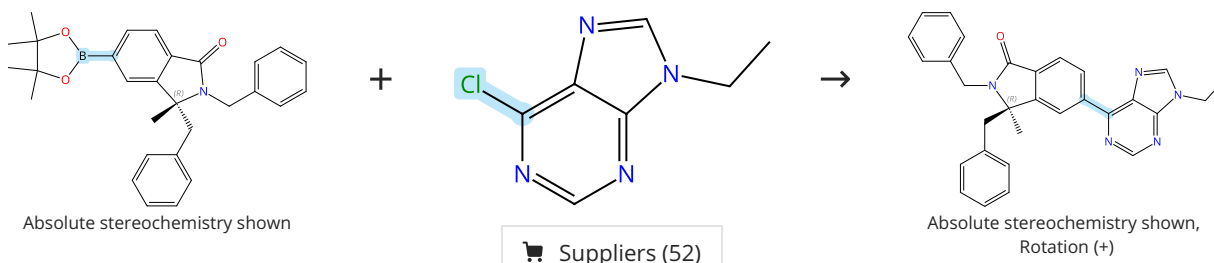
Steps: 1 Yield: 97%



31-179-CAS-21791501 Steps: 1 Yield: 97% 1.1 Catalysts: Cupric chloride, 6-Phenyl-2,2'-bipyridine Solvents: Ethanol, Toluene; 1 h, 50 °C 1.2 Reagents: Cesium carbonate, Water Solvents: Chloroform, Toluene; 24 h, 10 °C	Stereoinvertive C-C Bond Formation at the Boron- Bound Stereogenic Centers through Copper-Bipyridine-Catalyzed Intramolecular Coupling of α-Aminobenzylboronic Esters By: Yoshinaga, Yukako; et al Angewandte Chemie, International Edition (2020), 59(18), 7251-7255.
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Scheme 123 (1 Reaction)

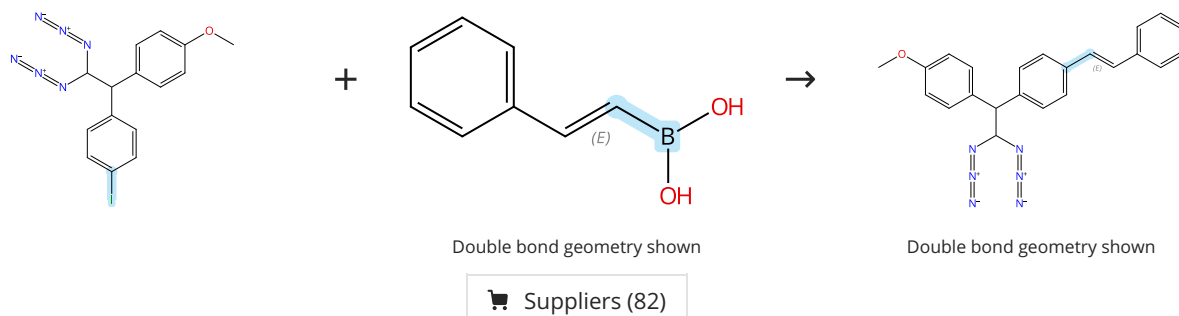
Steps: 1 Yield: 97%



31-179-CAS-23913297 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 16 h, 90 °C Experimental Protocols	Asymmetric synthesis of 3-benzyl and allyl isoindolinones by Pd-catalyzed dicarbofunctionalization of 1,1-disubstituted enamides By: Huang, Wenji; et al Organic Chemistry Frontiers (2021), 8(15), 4106-4111.
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Scheme 124 (1 Reaction)

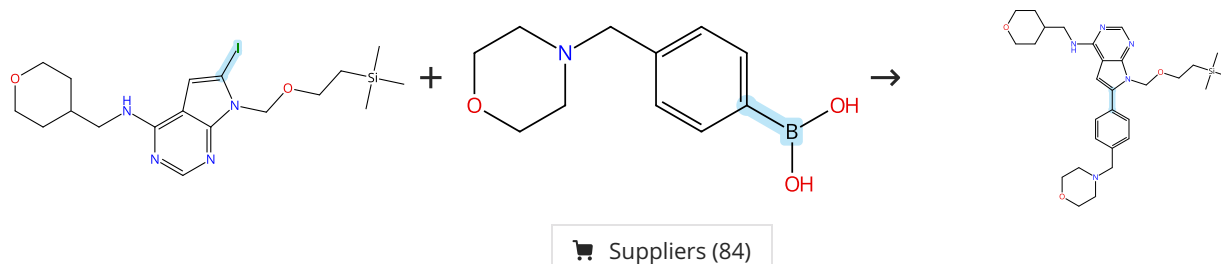
Steps: 1 Yield: 97%



31-614-CAS-40326865 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Phenylurea, Palladium diacetate Solvents: Methanol, Water; 4 h, rt Experimental Protocols	Catalytic 1,1-diazidation of alkenes By: Qiu, Wangzhen; et al Nature Communications (2024), 15(1), 3632.
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Scheme 125 (1 Reaction)

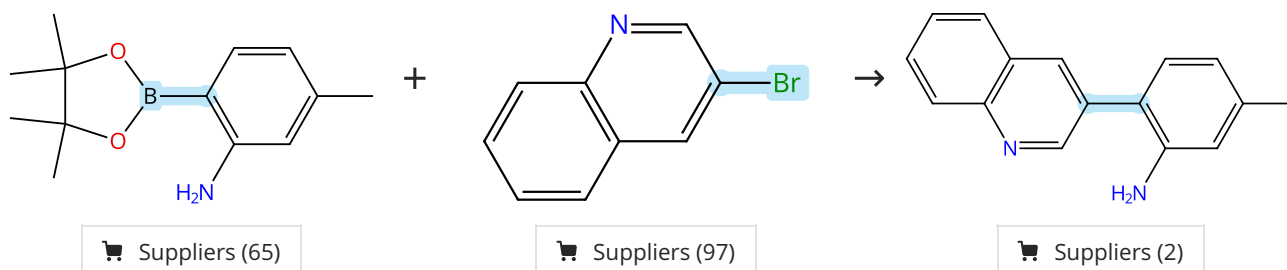
Steps: 1 Yield: 97%



31-614-CAS-39020351	Steps: 1 Yield: 97%	Pyrrolopyrimidine based CSF1R inhibitors: Attempted departure from Flatland By: Bjoernstad, Frithjof; et al European Journal of Medicinal Chemistry (2024), 265, 116053.
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 11 min, 80 °C; 5 min, cooled		
Experimental Protocols		

Scheme 126 (1 Reaction)

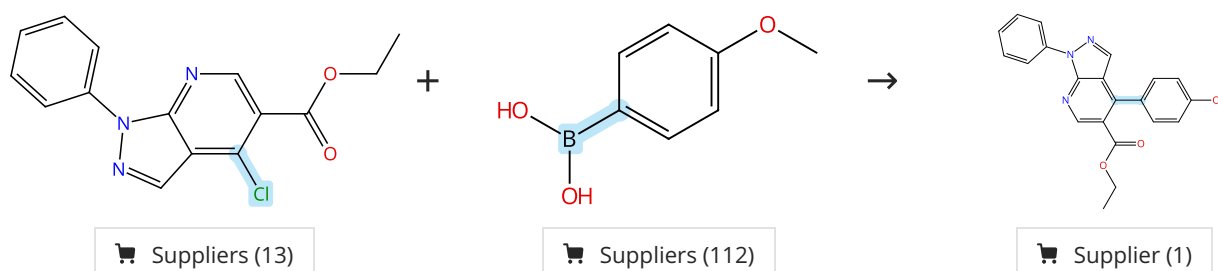
Steps: 1 Yield: 97%



31-614-CAS-35965805	Steps: 1 Yield: 97%	Regiodivergent Synthesis of 11 H-Indolo[3,2-c]quinolines and Neocryptolepine from a Common Starting Material By: Haaheim, Katja S.; et al European Journal of Organic Chemistry (2023), 26(13), e202300137.
1.1 Reagents: Cesium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 24 h, 80 °C		
Experimental Protocols		

Scheme 127 (1 Reaction)

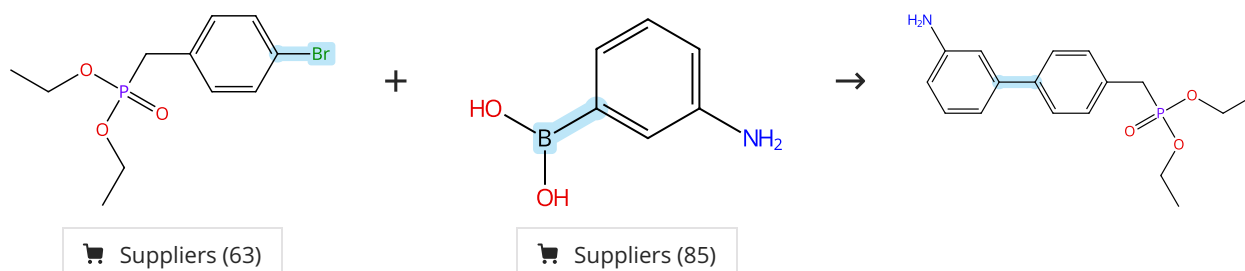
Steps: 1 Yield: 97%



31-179-CAS-16785808	Steps: 1 Yield: 97%	Harnessing Cascade Suzuki-Cyclization Reactions of Pyrazolo [3,4-b]pyridine for the Synthesis of Tetracyclic Fused Heteroaromatics By: Lavrard, Hubert; et al European Journal of Organic Chemistry (2017), 2017(3), 600-608.
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: <i>tert</i> -Butanol, Toluene, Water; 5 - 18 h, 75 °C		

Scheme 128 (1 Reaction)

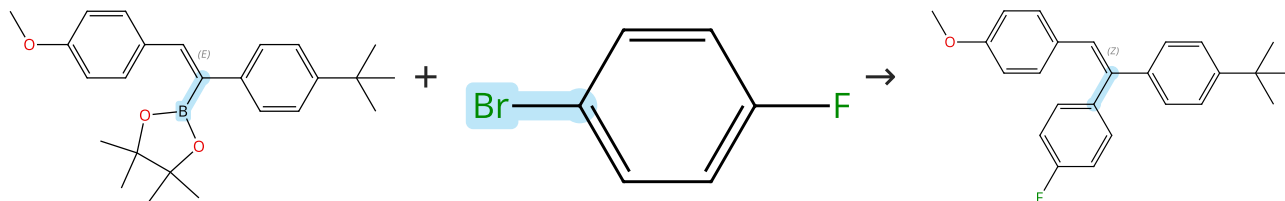
Steps: 1 Yield: 97%



31-179-CAS-4128357	Steps: 1 Yield: 97%	Modular Synthesis of Elongated Phosphonate Bipyridines
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; 1 d, reflux; 24 h, reflux; reflux → rt		By: Braumueller, Markus; et al European Journal of Organic Chemistry (2015), 2015(27), 5987-5994.
Experimental Protocols		

Scheme 129 (1 Reaction)

Steps: 1 Yield: 97%



Double bond geometry shown

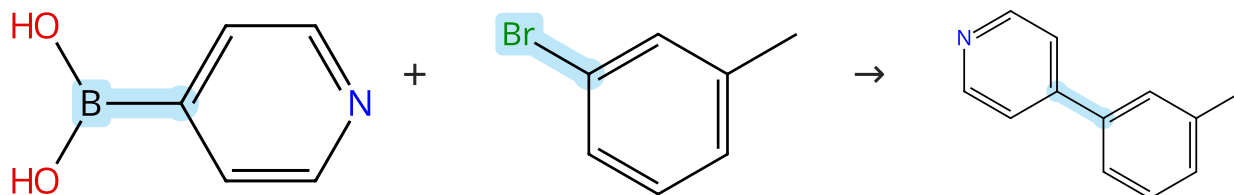
Suppliers (103)

Double bond geometry shown

31-177-CAS-18370805	Steps: 1 Yield: 97%	Stereoselective Synthesis of Trisubstituted Alkenes via Cobalt-Catalyzed Double Dehydrogenative Borylations of 1-Alkenes
1.1 Reagents: Tripotassium phosphate Catalysts: Bis(tri- <i>tert</i> -butylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; rt → 80 °C; 12 h, 80 °C		By: Wen, Huanan; et al ACS Catalysis (2017), 7(10), 6419-6425.
Experimental Protocols		

Scheme 130 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

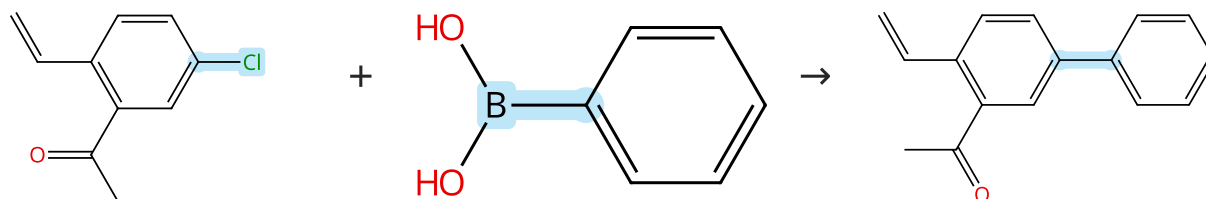
Suppliers (58)

Suppliers (15)

31-614-CAS-32528426	Steps: 1 Yield: 97%	Facile Synthesis of 2-Methylnicotinonitrile through Degenerate Ring Transformation of Pyridinium Salts
1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i> -4-2)-Dichloro[1,1'-(1,3-propanediyl)bis[1,1-diphenylphosphine-κ ^P]]palladium Solvents: 1,4-Dioxane, Water; overnight, reflux; reflux → rt		By: Duan, Xiaoxia; et al Journal of Organic Chemistry (2022), 87(12), 7975-7988.
1.2 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 131 (1 Reaction)

Steps: 1 Yield: 97%



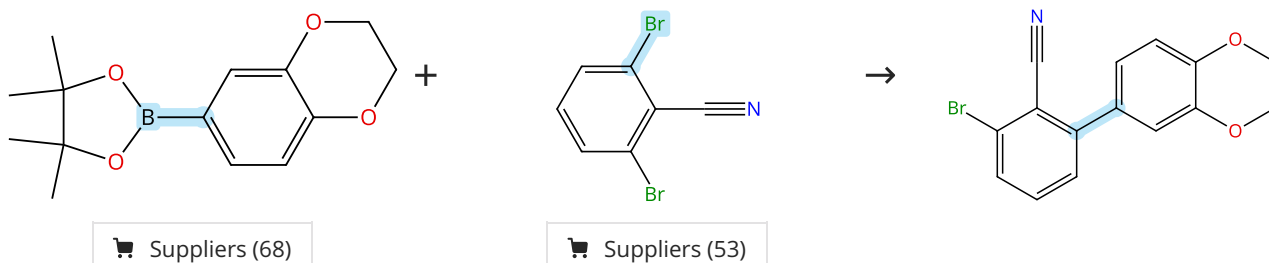
Suppliers (4)

Suppliers (132)

31-614-CAS-31627499	Steps: 1 Yield: 97%	Intramolecular One-Carbon Homologation of Unstrained Ketones via C-C Activation-Enabled 1,1-Insertion of Alkenes By: Huang, Jiangkun; et al Organic Letters (2022), 24(12), 2436-2440.
1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: 1,4-Dioxane; 10 h, 100 °C		
Experimental Protocols		

Scheme 132 (1 Reaction)

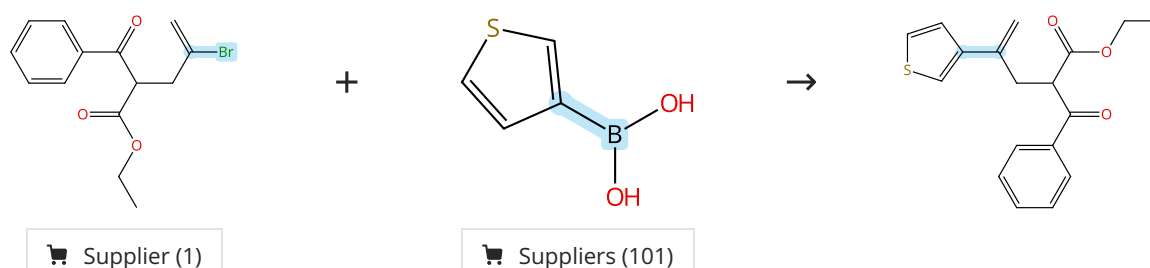
Steps: 1 Yield: 97%



31-614-CAS-44945170	Steps: 1 Yield: 97%	Design, Synthesis, Evaluation, and SAR of 5-Phenylisoindoline Derivatives, a Potent Class of Small-Molecule Inhibitors Targeting the Programmed Cell Death-1/Programmed Cell Death-Ligand 1 (PD-1/PD-L1) Interaction By: Lu, Tian; et al Journal of Medicinal Chemistry (2025), 68(7), 7291-7312.
1.1 Reagents: Potassium acetate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane; 20 h, 100 °C		
Experimental Protocols		

Scheme 133 (1 Reaction)

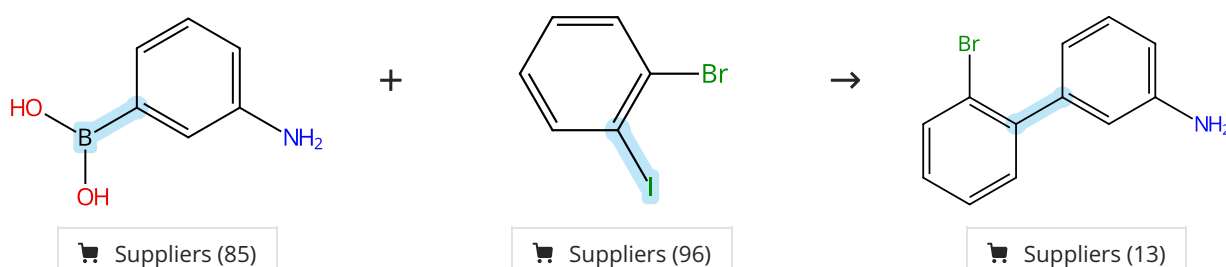
Steps: 1 Yield: 97%



31-614-CAS-43128612	Steps: 1 Yield: 97%	Catalytic Intermolecular Asymmetric [2π + 2σ] Cycloadditions of Bicyclo[1.1.0]butanes: Practical Synthesis of Enantioenriched Highly Substituted Bicyclo[2.1.1]hexanes By: Li, Ying-Jie; et al Journal of the American Chemical Society (2024), 146(50), 34427-34441.
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; 10 h, 70 °C		
Experimental Protocols		

Scheme 134 (1 Reaction)

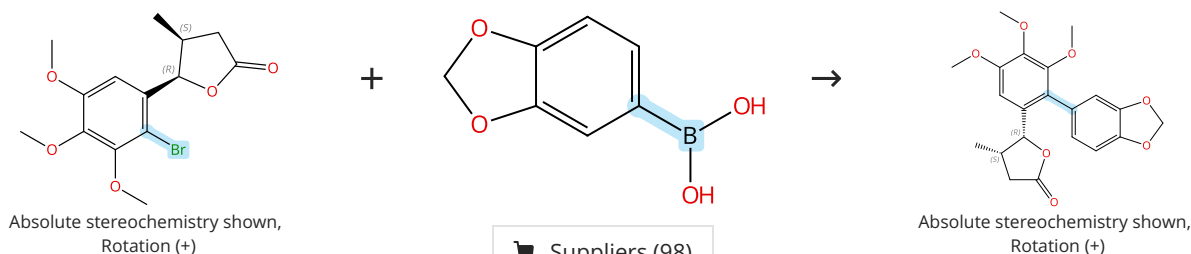
Steps: 1 Yield: 97%



31-179-CAS-19013563	Steps: 1 Yield: 97%	A Phenyl-Ethynyl-Macrocyclic: A Model Compound for "Gelae nder" Oligomers Comprising Reactive Conjugated Banisters By: Bannwart, Linda Maria; et al European Journal of Organic Chemistry (2018), 2018(26), 3391-3402.
1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Ethanol, Toluene; 18 h, 80 °C		

Scheme 135 (1 Reaction)

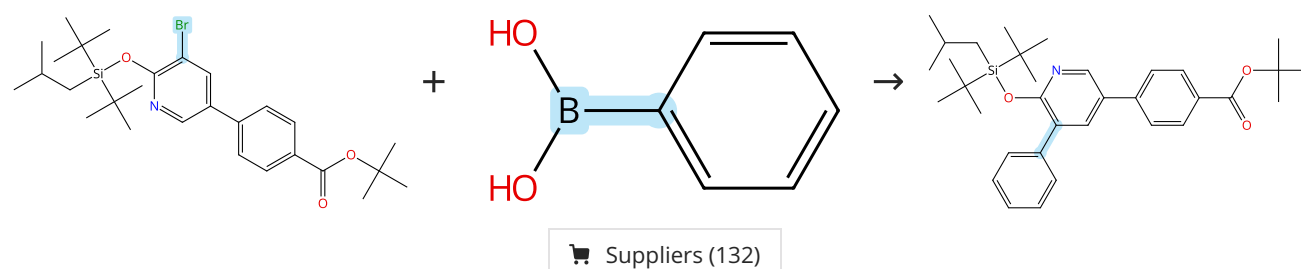
Steps: 1 Yield: 97%



31-179-CAS-20728651	Steps: 1 Yield: 97%	Synthesis of γ-Lactones via the Kowalski Homologation Reaction: Protecting-Group-Free Divergent Total Syntheses of Eupomatilones-2,5,6, and 3-epi-Eupomatilone-6 By: Choi, Hosam; et al Organic Letters (2019), 21(19), 7857-7862.
1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane; 1 h, 100 °C Experimental Protocols		

Scheme 136 (1 Reaction)

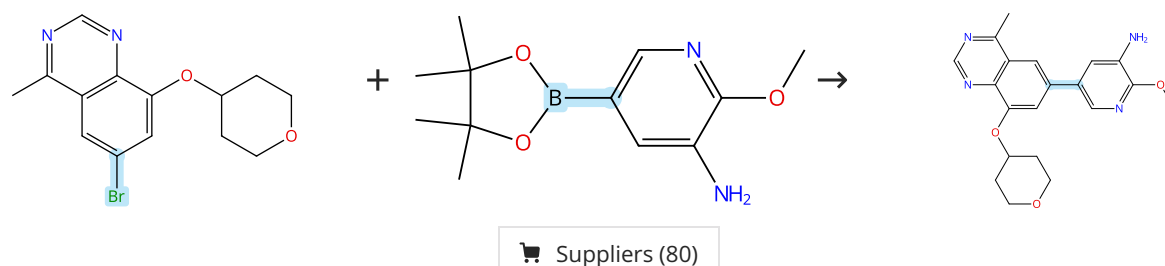
Steps: 1 Yield: 97%



31-614-CAS-31695648	Steps: 1 Yield: 97%	Protecting Group-Controlled Regioselective Synthesis for Unsymmetrical 3,5-Disubstituted Pyridones By: Kwon, Yong-Ju; et al Advanced Synthesis & Catalysis (2022), 364(8), 1440-1449.
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, Di-1-adamantylbenzylphosphine Solvents: Toluene, Water; 40 min, 50 °C Experimental Protocols		

Scheme 137 (1 Reaction)

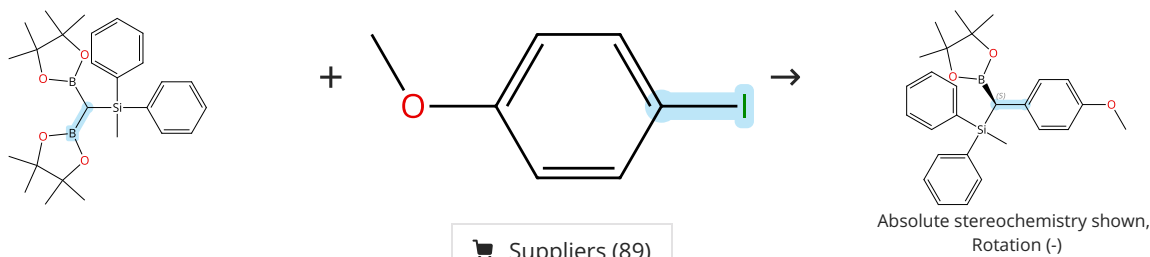
Steps: 1 Yield: 97%



31-614-CAS-39388345	Steps: 1 Yield: 97%	Design, Synthesis, and Bioevaluation of Novel Reversibly Photoswitchable PI3K Inhibitors Based on Phenylazo pyridine Derivatives Toward Light-Controlled Cancer Treatment By: Zhang, Yan; et al Journal of Medicinal Chemistry (2024), 67(5), 3504-3519.
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 6 h, 100 °C; 100 °C → rt		
1.2 Reagents: Water		
1.3 Reagents: Sodium carbonate Solvents: Water; pH 8		
Experimental Protocols		

Scheme 138 (1 Reaction)

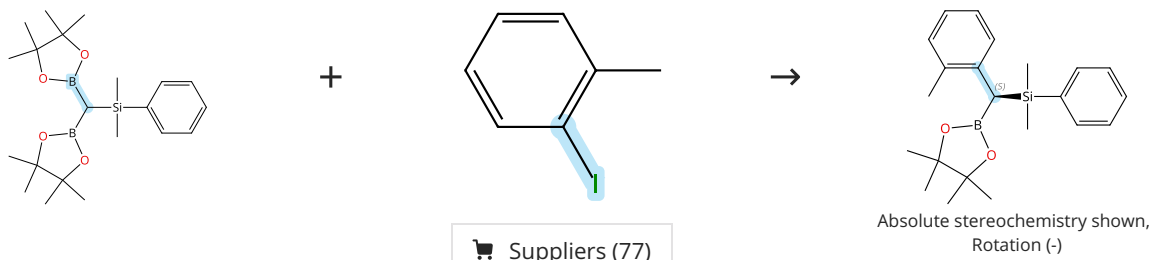
Steps: 1 Yield: 97%



31-179-CAS-19670043	Steps: 1 Yield: 97%	Access to Enantioenriched Benzylic 1,1-Silylboronate Esters by Palladium-Catalyzed Enantiotopic-Group Selective Suzuki-Miyaura Coupling of (Diborylmethyl)silanes with Aryl Iodides By: Kim, Junghoon; et al ACS Catalysis (2019), 9(1), 230-235.
1.1 Catalysts: Methanesulfonic acid, 1,1,1-trifluoro-, palladium(2+) salt (2:1), (2 <i>R</i>)-1-[(1 <i>R</i>)-1-[Bis[3,5-bis(trifluoromethyl)phenyl]phosphino]ethyl]-2-(dicyclohexylphosphino)ferrocene Solvents: 1,4-Dioxane; 30 min, rt		
1.2 Reagents: Sodium methoxide Catalysts: Sodium iodide; 3 h, rt		
Experimental Protocols		

Scheme 139 (1 Reaction)

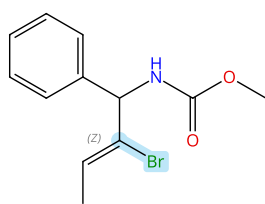
Steps: 1 Yield: 97%



31-179-CAS-19670055	Steps: 1 Yield: 97%	Access to Enantioenriched Benzylic 1,1-Silylboronate Esters by Palladium-Catalyzed Enantiotopic-Group Selective Suzuki-Miyaura Coupling of (Diborylmethyl)silanes with Aryl Iodides By: Kim, Junghoon; et al ACS Catalysis (2019), 9(1), 230-235.
1.1 Catalysts: Methanesulfonic acid, 1,1,1-trifluoro-, palladium(2+) salt (2:1), (2 <i>R</i>)-1-[(1 <i>R</i>)-1-[Bis[3,5-bis(trifluoromethyl)phenyl]phosphino]ethyl]-2-(dicyclohexylphosphino)ferrocene Solvents: 1,4-Dioxane; 30 min, rt		
1.2 Reagents: Sodium methoxide Catalysts: Sodium iodide; 3 h, rt		
Experimental Protocols		

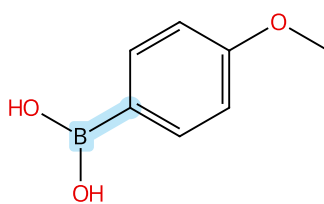
Scheme 140 (1 Reaction)

Steps: 1 Yield: 97%

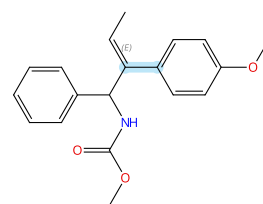


Double bond geometry shown

+



→



Double bond geometry shown

Suppliers (112)

31-614-CAS-36258056

Steps: 1 Yield: 97%

Synthesis of Chiral α -Amino Ketones via Transition Metal Catalyzed or Photoredox Cross-Coupling and Olefin Photo-Cleavage Reaction Sequence

By: Kobus-Bartoszewicz, Dominika; et al

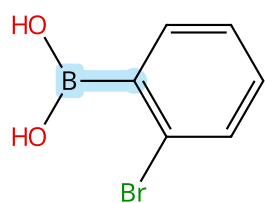
Advanced Synthesis & Catalysis (2023), 365(8), 1224-1237.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 60 min, 100 °C

Experimental Protocols

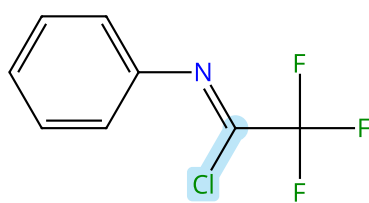
Scheme 141 (1 Reaction)

Steps: 1 Yield: 97%

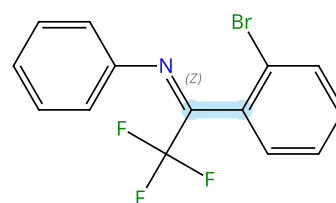


Suppliers (101)

+



→



Double bond geometry shown

Suppliers (56)

31-179-CAS-19885043

Steps: 1 Yield: 97%

Palladium-catalyzed tandem cyclization of fluorinated imidoyl chlorides with 2-bromophenylboronic acid: synthesis of 6-fluoroalkyl-phenanthridines

By: Bao, Yinwei; et al

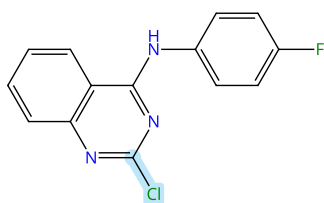
Tetrahedron (2019), 75(10), 1450-1456.

1.1 **Reagents:** Cesium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Toluene; 6 h, 60 °C

Experimental Protocols

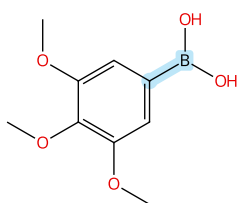
Scheme 142 (1 Reaction)

Steps: 1 Yield: 97%

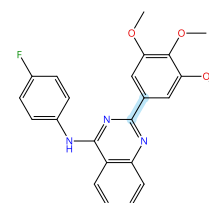


Suppliers (2)

+



→



Suppliers (3)

31-179-CAS-22667257

Steps: 1 Yield: 97%

N-heterocyclic carbene-Pd(II)-1-methylimidazole complex-catalyzed Suzuki-Miyaura coupling of 2-chloro-4-aminoquinazolines with arylboronic acids

By: Bao, Zhen; et al

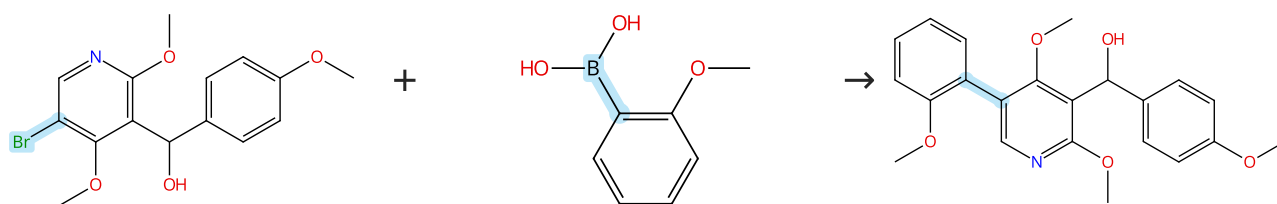
Tetrahedron (2020), 76(44), 131548.

1.1 **Reagents:** Potassium hydroxide
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(1-methyl-1*H*-imidazole- κ^N)palladium
Solvents: Ethanol; 24 h, rt

Experimental Protocols

Scheme 143 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (110)

31-179-CAS-19607233

Steps: 1 Yield: 97%

Discovery of 2,4-dimethoxypyridines as novel autophagy inhibitors

1.1 **Reagents:** Sodium carbonate
Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tetrakis (triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 16 h, 110 °C

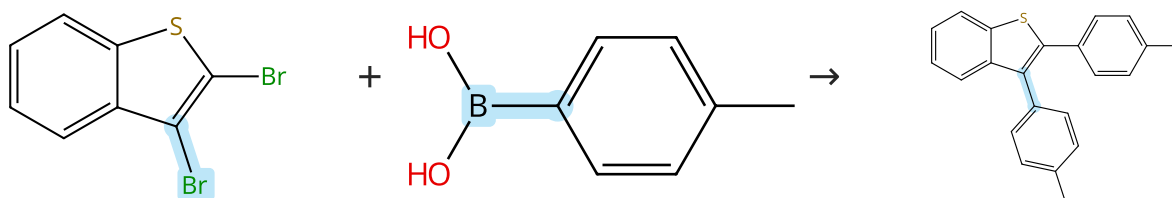
By: Robke, Lucas; et al

Tetrahedron (2018), 74(35), 4531-4537.

Experimental Protocols

Scheme 144 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (60)

Suppliers (114)

Suppliers (2)

31-179-CAS-5351819

Steps: 1 Yield: 97%

Transition-Metal-Free Synthesis of Carbazoles and Indoles by an S_NAr -Based "Aromatic Metamorphosis" of Thiaarenes

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 48 h, 80 °C

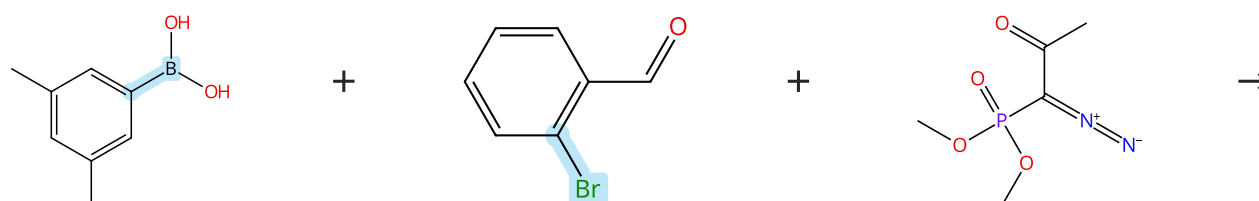
By: Bhanuchandra, M.; et al

Angewandte Chemie, International Edition (2015), 54(35), 10234-10238.

Experimental Protocols

Scheme 145 (1 Reaction)

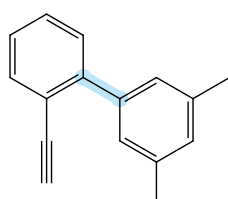
Steps: 1 Yield: 97%



Suppliers (95)

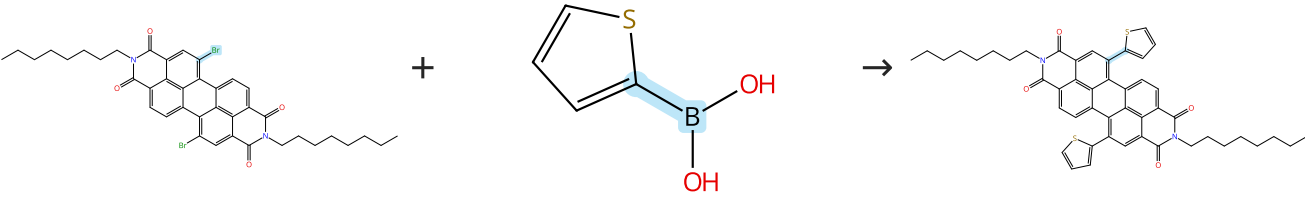
Suppliers (89)

Suppliers (88)

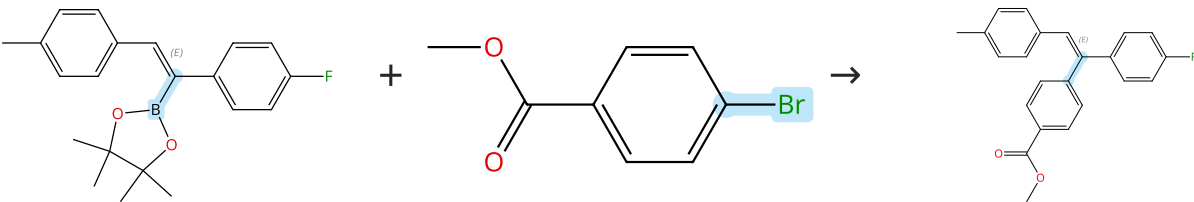


Suppliers (14)

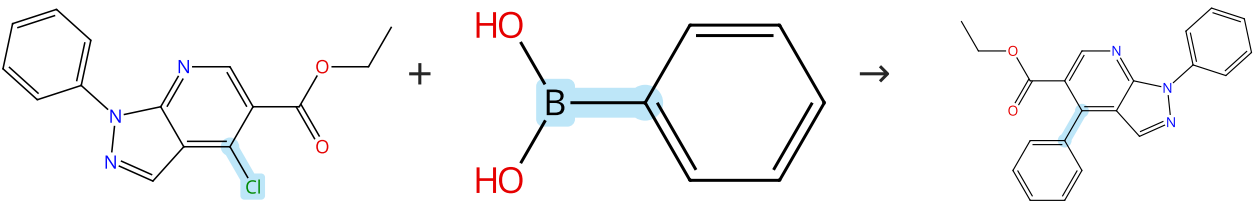
31-614-CAS-39680240 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; rt → 105 °C; overnight, 105 °C 1.2 Reagents: Potassium carbonate Solvents: Methanol; overnight, rt Experimental Protocols	One-Pot Synthesis of Polycyclic Sultams Enabled by Gold-Catalyzed Hydroamination of o-Alkynylbenzenesulfonamides and Subsequent Iodine(III)-Mediated Counteranion-Assisted Intramolecular Annulation By: Zhang, Lijun; et al Organic Letters (2024), 26(11), 2192-2196.
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Scheme 146 (1 Reaction) Steps: 1 Yield: 97%   Suppliers (38)  Suppliers (102)	
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31-179-CAS-20431877 Steps: 1 Yield: 97% 1.1 Catalysts: (<i>SP</i>-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino-κP]ferrocene]dichloropalladium, 2366258-45-7 Solvents: Water; 5 min, 25 °C 1.2 Reagents: Triethylamine; 12 h, 25 °C	Efficient synthesis of organic semiconductors by Suzuki-Miyaura coupling in an aromatic micellar medium By: Sanzone, Alessandro; et al Green Chemistry (2019), 21(16), 4400-4405.
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Scheme 147 (1 Reaction) Steps: 1 Yield: 97%  Double bond geometry shown  Suppliers (96) Double bond geometry shown	
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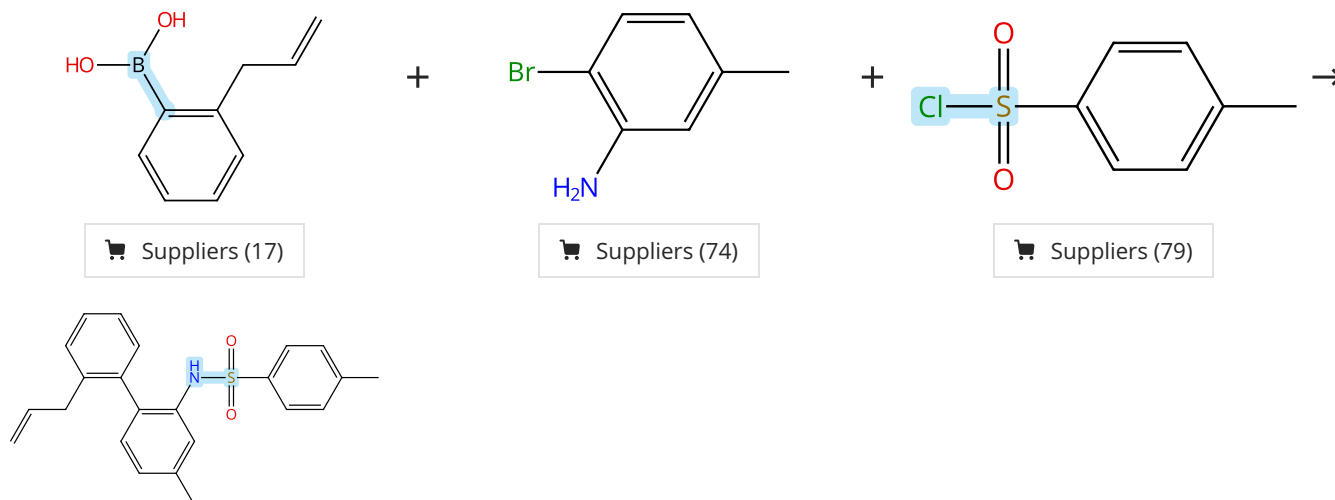
31-177-CAS-18370811 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Bis(tri-<i>tert</i>-butylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; rt → 80 °C; 12 h, 80 °C Experimental Protocols	Stereoselective Synthesis of Trisubstituted Alkenes via Cobalt-Catalyzed Double Dehydrogenative Borylations of 1-Alkenes By: Wen, Huanan; et al ACS Catalysis (2017), 7(10), 6419-6425.
--	---

Scheme 148 (1 Reaction) Steps: 1 Yield: 97%   Suppliers (13)  Suppliers (132)  Supplier (1)	
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31-179-CAS-16784155	Steps: 1 Yield: 97%	Harnessing Cascade Suzuki-Cyclization Reactions of Pyrazolo [3,4-b]pyridine for the Synthesis of Tetracyclic Fused Heteroaromatics
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: <i>tert</i> -Butanol, Toluene, Water; 5 - 18 h, 75 °C		By: Lavrard, Hubert; et al European Journal of Organic Chemistry (2017), 2017(3), 600-608.

Scheme 149 (1 Reaction)

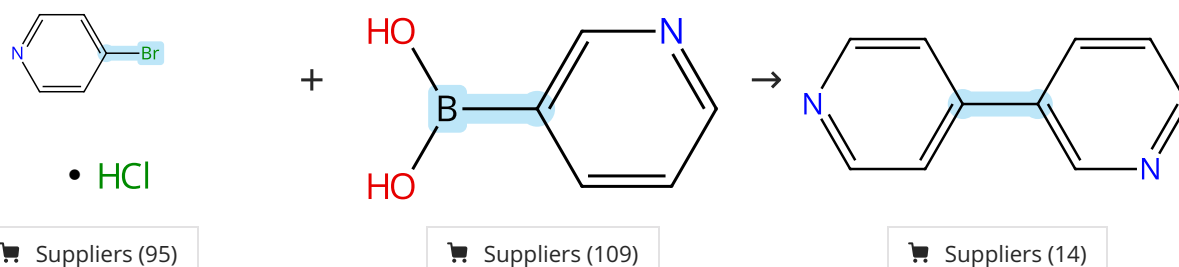
Steps: 1 Yield: 97%



31-614-CAS-44912828	Steps: 1 Yield: 97%	Enhanced Pyridine-Oxazoline Ligand-Enabled Pd(II)-Catalyzed Aminoacetoxylation of Alkenes for the Asymmetric Synthesis of Biaryl-Bridged 7-Membered N-Heterocycles and Atropisomers
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 24 h, 70 °C		By: Guo, Beibei; et al Journal of the American Chemical Society (2025), 147(15), 12614-12626.
1.2 Reagents: Water		
1.3 Reagents: Pyridine Solvents: Dichloromethane; overnight, rt		
1.4 Reagents: Hydrochloric acid Solvents: Water; pH 3 - 4, rt		
Experimental Protocols		

Scheme 150 (1 Reaction)

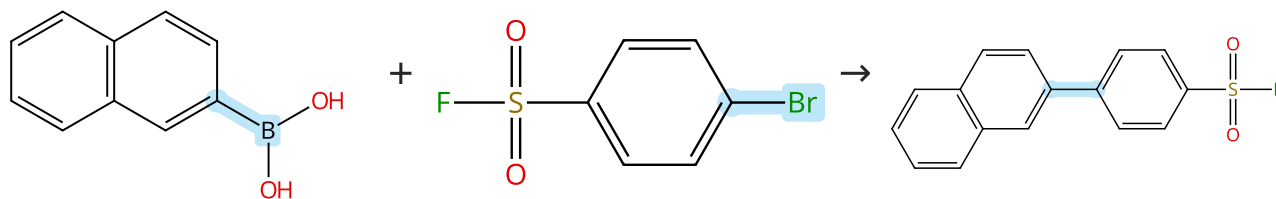
Steps: 1 Yield: 97%



31-179-CAS-14787544	Steps: 1 Yield: 97%	Fluorinated Boronic Acid-Appended Bipyridinium Salts for Diol Recognition and Discrimination via ¹⁹F NMR Barcodes
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: 1,4-Dioxane, Water; 15 min, rt; 18 h, 100 °C		By: Axthelm, Joerg; et al Journal of the American Chemical Society (2015), 137(49), 15402-15405.
Experimental Protocols		

Scheme 151 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (103)

Suppliers (51)

31-179-CAS-1117886

Steps: 1 Yield: 97%

On-Water Synthesis of Biaryl Sulfonyl Fluorides

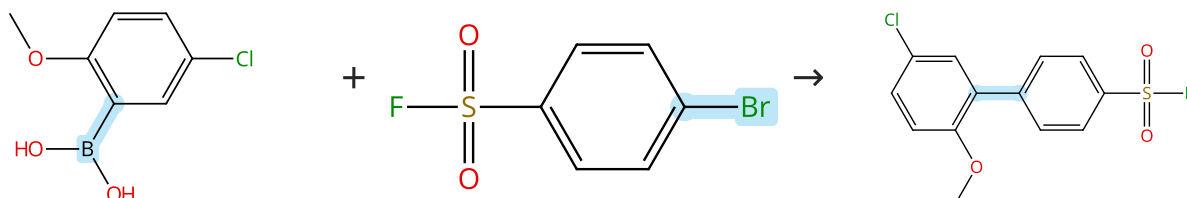
1.1 Reagents: Triethylamine
Catalysts: Palladium diacetate
Solvents: Water; 30 - 120 min, rt

By: Chinthakindi, Praveen K.; et al
Journal of Organic Chemistry (2016), 81(6), 2618-2623.

Experimental Protocols

Scheme 152 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (89)

Suppliers (51)

31-179-CAS-14185998

Steps: 1 Yield: 97%

On-Water Synthesis of Biaryl Sulfonyl Fluorides

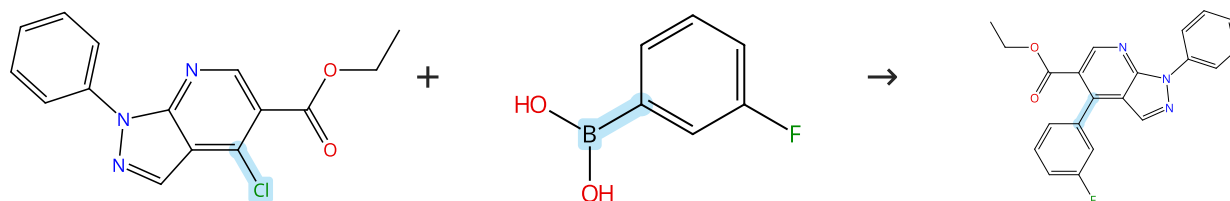
1.1 Reagents: Triethylamine
Catalysts: Palladium diacetate
Solvents: Water; 30 - 120 min, rt

By: Chinthakindi, Praveen K.; et al
Journal of Organic Chemistry (2016), 81(6), 2618-2623.

Experimental Protocols

Scheme 153 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (13)

Suppliers (100)

Supplier (1)

31-179-CAS-16785823

Steps: 1 Yield: 97%

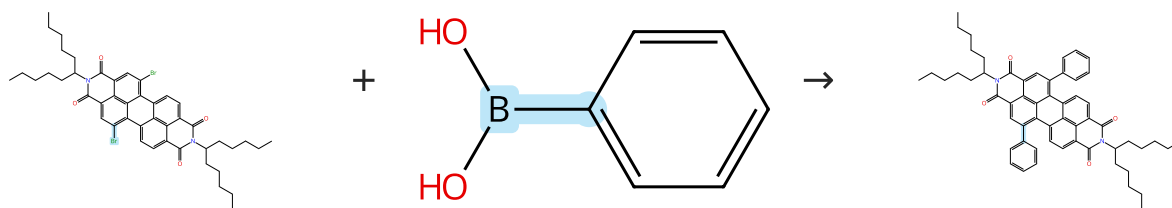
Harnessing Cascade Suzuki-Cyclization Reactions of Pyrazolo [3,4-b]pyridine for the Synthesis of Tetracyclic Fused Heteroaromatics

1.1 Reagents: Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: *tert*-Butanol, Toluene, Water; 5 - 18 h, 75 °C

By: Lavrard, Hubert; et al
European Journal of Organic Chemistry (2017), 2017(3), 600-608.

Scheme 154 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-179-CAS-20712823

Steps: 1 Yield: 97%

The importance of intramolecular conductivity in three dimensional molecular solids

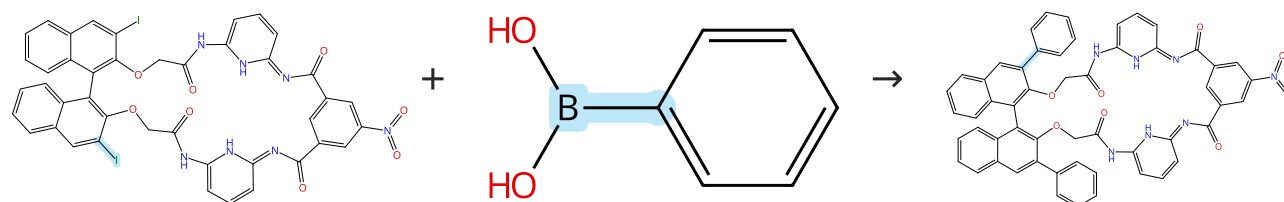
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; overnight, 100 °C

By: Ball, Melissa L.; et al

Chemical Science (2019), 10(40), 9339-9344.

Scheme 155 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-179-CAS-20723770

Steps: 1 Yield: 97%

Cross-Coupling Approach to an Array of Macrocyclic Receptors Functioning as Chiral Solvating Agents

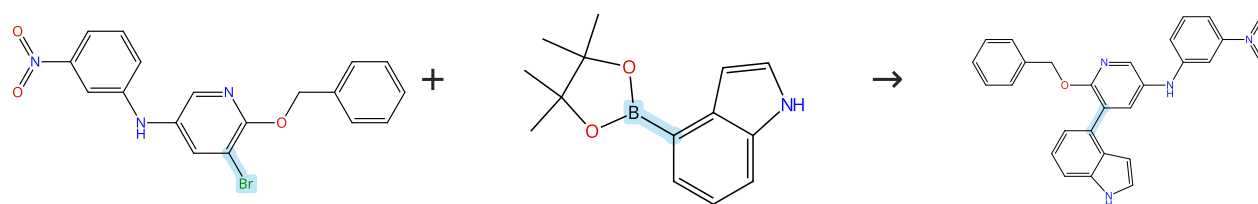
1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 6 h, 65 °C

By: Ema, Tadashi; et al

Journal of Organic Chemistry (2018), 83(18), 10762-10769.

Scheme 156 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (67)

31-614-CAS-31654127

Steps: 1 Yield: 97%

Improved potency of pyridin-2(1H)one derivatives for the treatment of mechanical allodynia

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 15 h, reflux

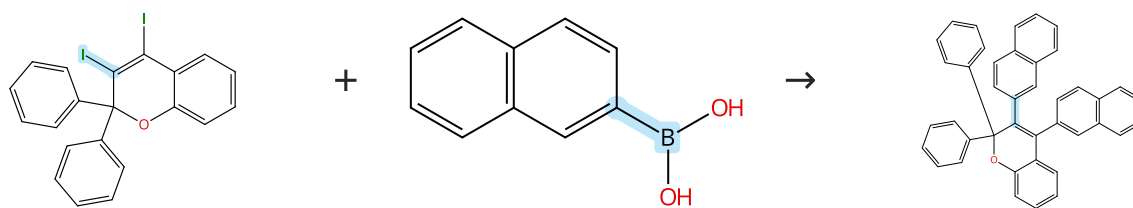
By: Visseque, Alexia; et al

European Journal of Medicinal Chemistry (2021), 225, 113748.

Experimental Protocols

Scheme 157 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (103)

31-179-CAS-13191721

Steps: 1 Yield: 97%

Convenient and Highly Efficient Routes to 2H-Chromene and 4-Chromanone Derivatives: Iodine-Promoted and p-Toluenesulfonic Acid Catalyzed Cascade Cyclizations of Propynols

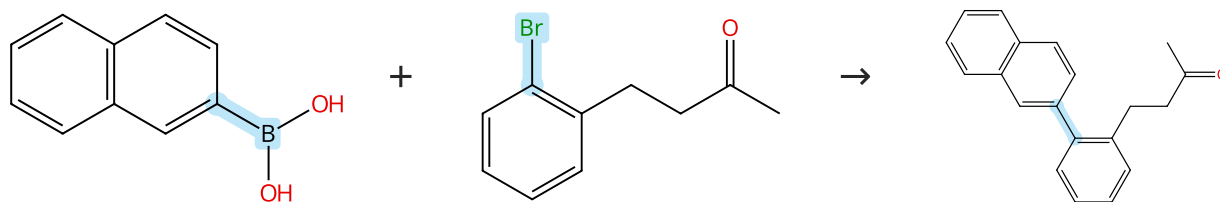
By: Qiu, Yi-Feng; et al

Chemistry - A European Journal (2015), 21(8), 3480-3487.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Acetonitrile; 7 h, 80 °C

Scheme 158 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (103)

Suppliers (20)

31-614-CAS-24349136

Steps: 1 Yield: 97%

Deacylation-aided C-H alkylative annulation through C-C cleavage of unstrained ketones

By: Zhou, Xukai; et al

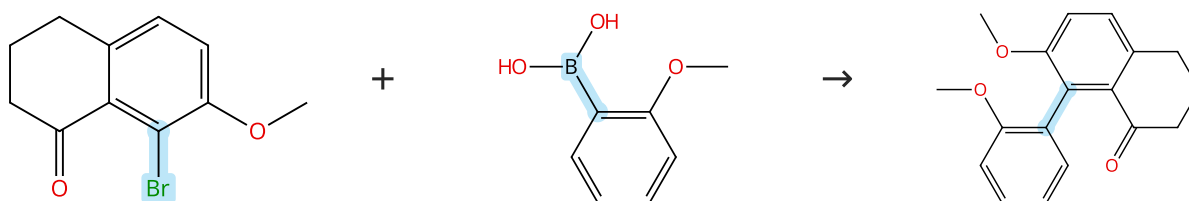
Nature Catalysis (2021), 4(8), 703-710.

1.1 **Reagents:** Cesium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: Tetrahydrofuran, Water; 90 °C

Experimental Protocols

Scheme 159 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (35)

Suppliers (110)

31-179-CAS-19487838

Steps: 1 Yield: 97%

Central-to-Helical-to-Axial-to-Central Transfer of Chirality with a Photoresponsive Catalyst

By: Pizzolato, Stefano F.; et al

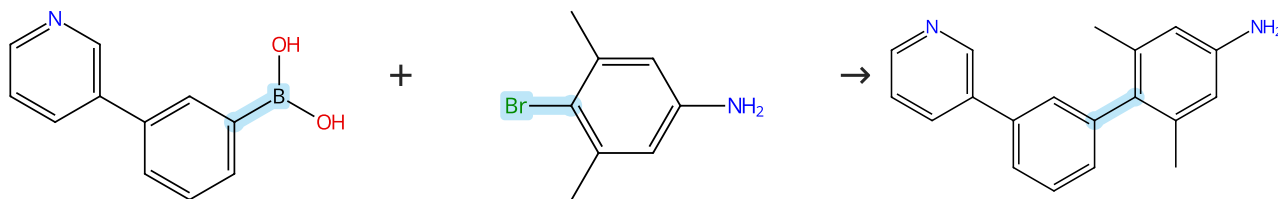
Journal of the American Chemical Society (2018), 140(49), 17278-17289.

1.1 **Reagents:** Tripotassium phosphate
Solvents: Toluene; 2 min, rt
 1.2 **Catalysts:** Tris(dibenzylideneacetone)dipalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl; 24 h, 100 °C

Experimental Protocols

Scheme 160 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (61)

Suppliers (65)

31-614-CAS-35055304

Steps: 1 Yield: 97%

Pseudo-heterolepticity in Low-Symmetry Metal-Organic Cages

1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 18 h, 100 °C

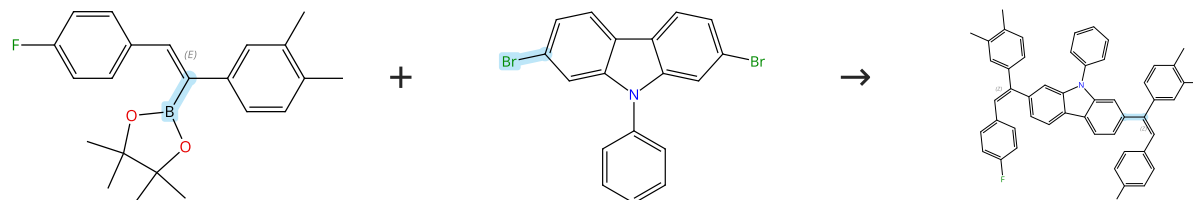
By: Lewis, James E. M.

Angewandte Chemie, International Edition (2022), 61(44),
 e202212392.

Experimental Protocols

Scheme 161 (1 Reaction)

Steps: 1 Yield: 97%



Double bond geometry shown

Suppliers (59)

Double bond geometry shown

31-177-CAS-18370816

Steps: 1 Yield: 97%

Stereoselective Synthesis of Trisubstituted Alkenes via Cobalt-Catalyzed Double Dehydrogenative Borylations of 1-Alkenes

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Bis(tri-*tert*-butylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; rt → 80 °C; 12 h, 80 °C

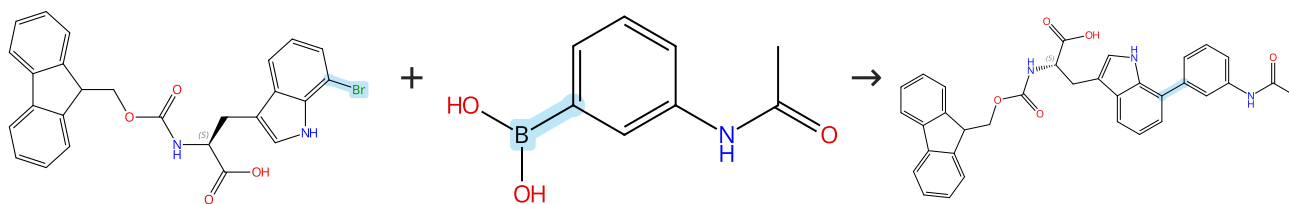
By: Wen, Huanan; et al

ACS Catalysis (2017), 7(10), 6419-6425.

Experimental Protocols

Scheme 162 (1 Reaction)

Steps: 1 Yield: 97%



Absolute stereochemistry shown

Suppliers (31)

Suppliers (86)

Absolute stereochemistry shown

31-179-CAS-23072198

Steps: 1 Yield: 97%

Suzuki-Miyaura cross-coupling of bromotryptophan derivatives at ambient temperature

1.1 **Reagents:** Tripotassium phosphate
Solvents: Water; rt → 40 °C

By: Dachwitz, Steffen; et al

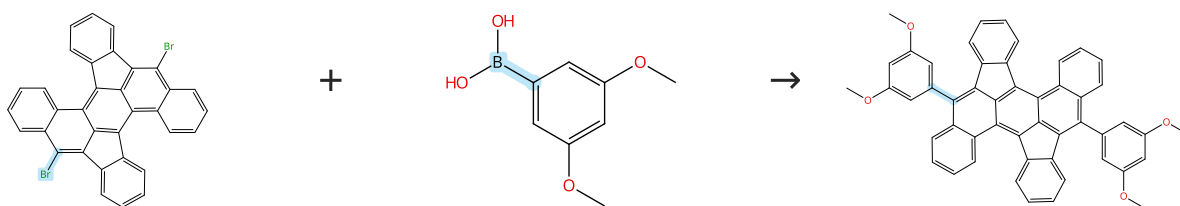
1.2 **Catalysts:** Palladium
Solvents: Isopropanol; 2 h, 40 °C

Chemistry - A European Journal (2020), 26(69), 16357-16364.

Experimental Protocols

Scheme 163 (1 Reaction)

Steps: 1 Yield: 97%



Supplier (1)

Suppliers (100)

31-179-CAS-2527237

Steps: 1 Yield: 97%

Synthesis, Molecular Packing, and Thin Film Transistors of Dibenzo[a,m]rubicenes

By: Gu, Xiao; et al

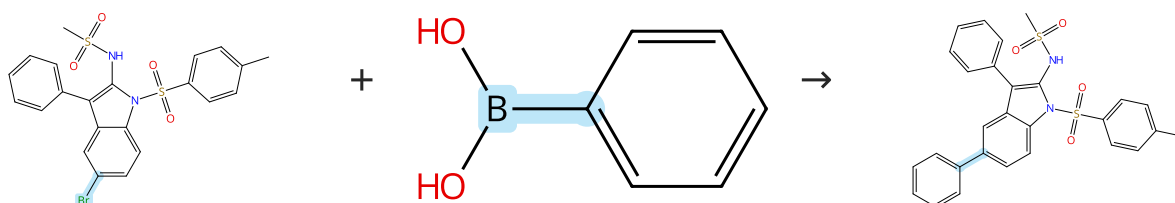
Journal of the American Chemical Society (2015), 137(51), 16203-16208.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 5 min, rt; 24 h, reflux

Experimental Protocols

Scheme 164 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-614-CAS-38946656

Steps: 1 Yield: 97%

TMSOTf-Catalyzed Reactions of N-Arylnamides with Sulfilimines To Construct 2-Aminoindoles and α -Arylated Amidines

By: Wang, Xiao-Na; et al

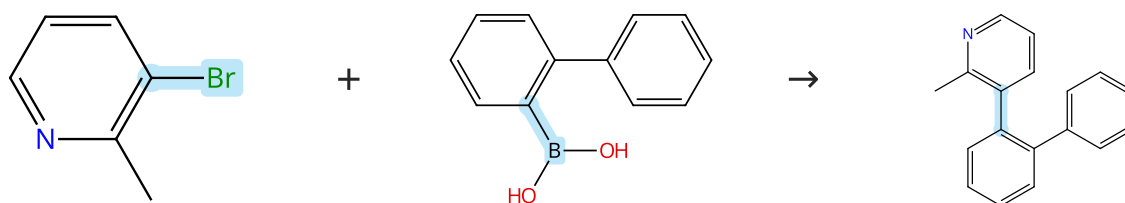
Journal of Organic Chemistry (2024), 89(2), 1193-1198.

1.1 **Reagents:** Cesium carbonate
Catalysts: Triphenylphosphine, Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 8 h, rt $\rightarrow$ 100 °C

Experimental Protocols

Scheme 165 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (104)

Suppliers (91)

31-614-CAS-40329246

Steps: 1 Yield: 97%

Design and Discovery of Water-Soluble Benzooxaphosphole-Based Ligands for Hindered Suzuki-Miyaura Coupling Reactions with Low Catalyst Load

By: Shada, Arun D. R.; et al

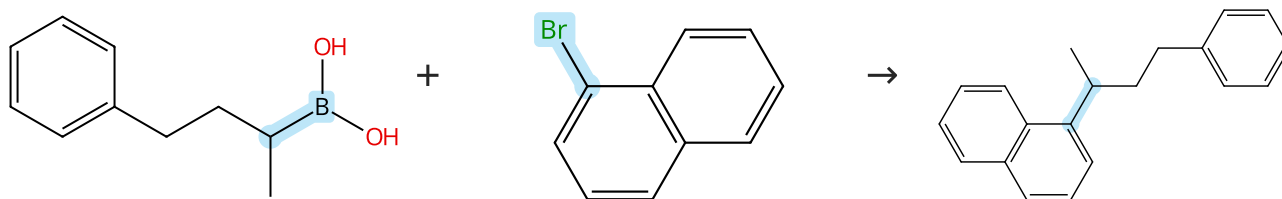
Organic Letters (2024), 26(14), 2751-2757.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetraglyme, Palladium diacetate, 3022418-51-2
Solvents: Dichloromethane, Water; 16 h, 105 °C

Experimental Protocols

Scheme 166 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (88)

31-179-CAS-2530749

Steps: 1 Yield: 97%

Efficient Synthesis of Sterically Hindered Arenes Bearing Acyclic Secondary Alkyl Groups by Suzuki-Miyaura Cross-Couplings

By: Li, Chengxi; et al

Angewandte Chemie, International Edition (2015), 54(12), 3792-3796.

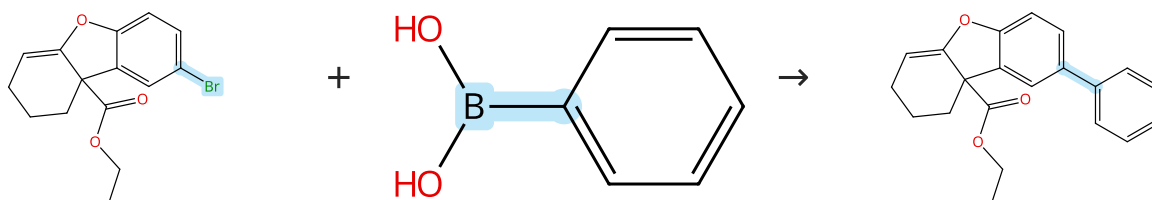
1.1 Reagents: Tripotassium phosphate

Catalysts: Di-μ-chlorobis[(1,2,3-η)-1-phenyl-2-propen-1-yl]dipalladium, *rel*-(2*R*,3*R*)-2-[Bis(1,1-dimethylethyl)phosphinyl]-3-(1,1-dimethylethyl)-2,3-dihydro-4-methoxy-1,3-benzoxaphosphole

Solvents: Toluene; 12 h, 100 °C

Scheme 167 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-614-CAS-34895323

Steps: 1 Yield: 97%

Synthesis of Annulated Benzofuran Derivatives from α-(Iodophenyl)-β-oxoesters

By: Fliegel, Lukas; et al

Organic Letters (2022), 24(46), 8526-8530.

1.1 Reagents: Potassium carbonate

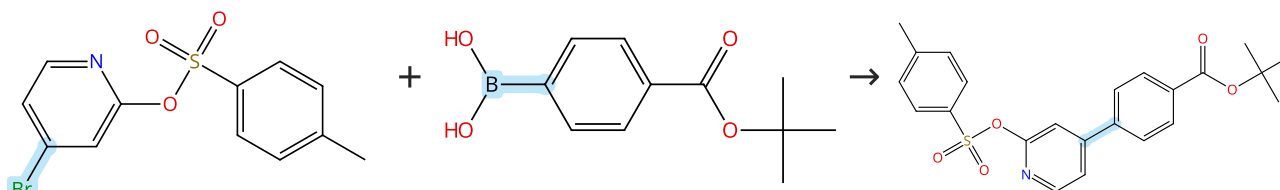
Catalysts: Tris(dibenzylideneacetone)dipalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl

Solvents: Toluene, Water; 17 h, 140 °C

Experimental Protocols

Scheme 168 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (4)

Suppliers (83)

31-179-CAS-22045417

Steps: 1 Yield: 97%

Highly Selective Room-Temperature Suzuki-Miyaura Coupling of Bromo-2-sulfonyloxypyridines for Unsymmetrical Diarylpyridines

By: Jeon, Young-Kyo; et al

Journal of Organic Chemistry (2020), 85(11), 7399-7412.

1.1 Reagents: Potassium carbonate

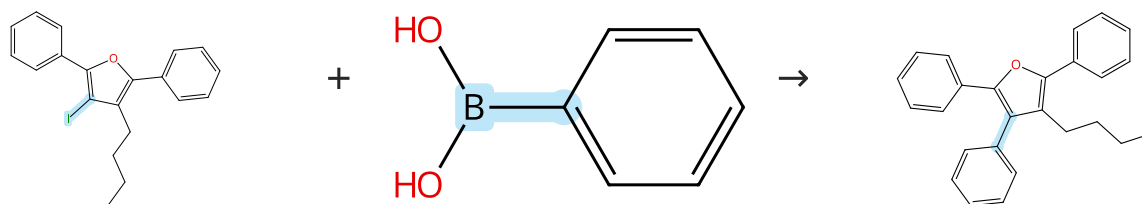
Catalysts: Palladium diacetate, Di-1-adamantylbenzylphosphine

Solvents: Toluene, Water; 3.5 h, 25 °C

Experimental Protocols

Scheme 169 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

Supplier (1)

31-179-CAS-21630091

Steps: 1 Yield: 97%

Synthesis of Tetrasubstituted Furans through One-Pot Formal [3+2] Cycloaddition Utilizing [1,2]-Phospha-Brook Rearrangement

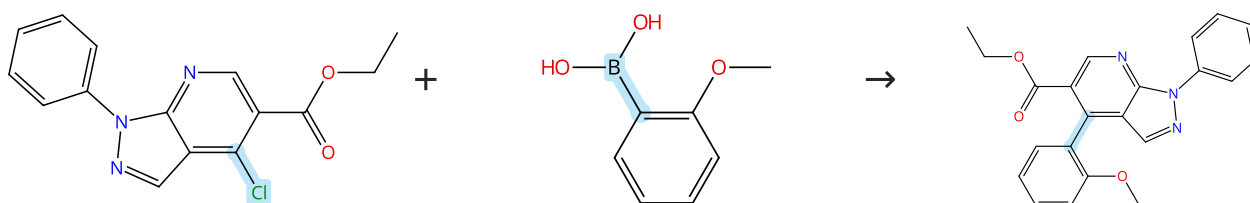
By: Kondoh, Azusa; et al

Organic Letters (2020), 22(5), 2105-2110.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 24 h, 100 °C; 100 °C → rt
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Scheme 170 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (13)

Suppliers (110)

Supplier (1)

31-179-CAS-16785837

Steps: 1 Yield: 97%

Harnessing Cascade Suzuki-Cyclization Reactions of Pyrazolo [3,4-b]pyridine for the Synthesis of Tetracyclic Fused Heteroaromatics

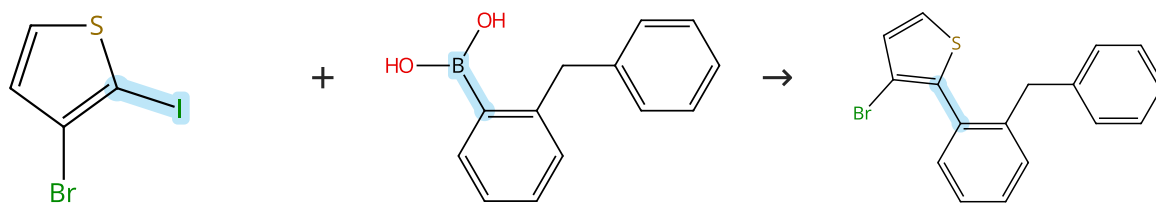
By: Lavrard, Hubert; et al

European Journal of Organic Chemistry (2017), 2017(3), 600-608.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: *tert*-Butanol, Toluene, Water; 5 - 18 h, 75 °C

Scheme 171 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (57)

Suppliers (54)

31-614-CAS-38458468

Steps: 1 Yield: 97%

Synthesis of Heptagon-Containing Polyarenes by Catalytic C-H Activation

By: Yamada, Keigo E.; et al

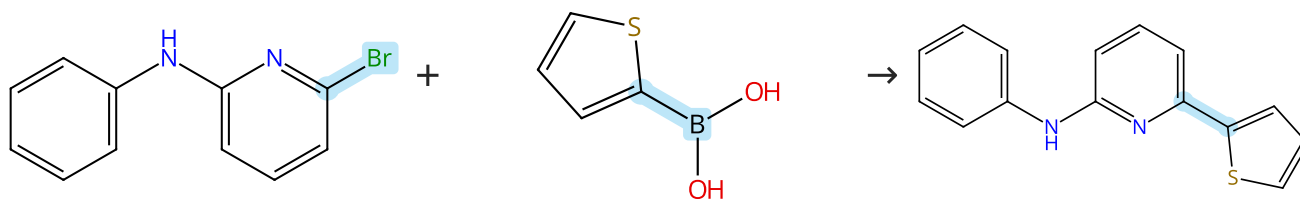
Angewandte Chemie, International Edition (2023), 62(51), e202311770.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 36 h, 80 °C

Experimental Protocols

Scheme 172 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (33)

Suppliers (102)

31-614-CAS-35993067

Steps: 1 Yield: 97%

Nonsymmetric [Pt(CNN'*Ĉ')] Complexes: Aggregation-Induced Emission in the Solid State and in Nanoparticles Tuned by Ligand Structure**

By: Solomatina, Anastasia I.; et al

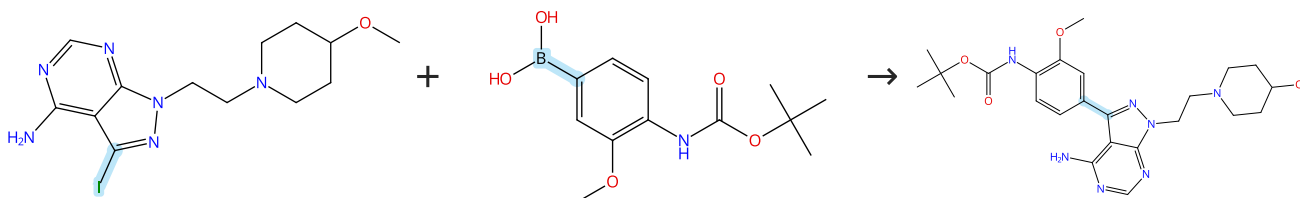
Chemistry - A European Journal (2022), 28(64), e202202207.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 85 °C

Experimental Protocols

Scheme 173 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (52)

Supplier (1)

31-179-CAS-11970336

Steps: 1 Yield: 97%

Rapid Discovery and Structure-Activity Relationships of Pyrazolopyrimidines That Potently Suppress Breast Cancer Cell Growth via SRC Kinase Inhibition with Exceptional Selectivity over ABL Kinase

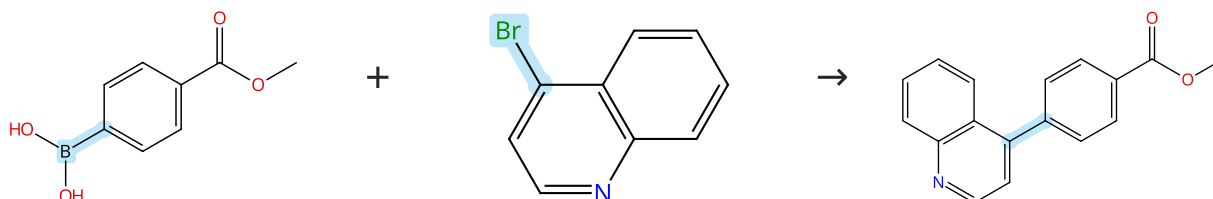
By: Fraser, Craig; et al

Journal of Medicinal Chemistry (2016), 59(10), 4697-4710.

1.1 **Reagents:** Potassium carbonate
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: 1,4-Dioxane, Water; 30 min, 120 °C

Scheme 174 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (108)

Suppliers (75)

31-614-CAS-33853805

Steps: 1 Yield: 97%

Copper-Catalyzed Oxidative C-H Annulation of Quinolines with Dichloroethane toward Benzoquinoliniziniums Using an In Situ Activation Strategy

By: Ji, Jinwen; et al

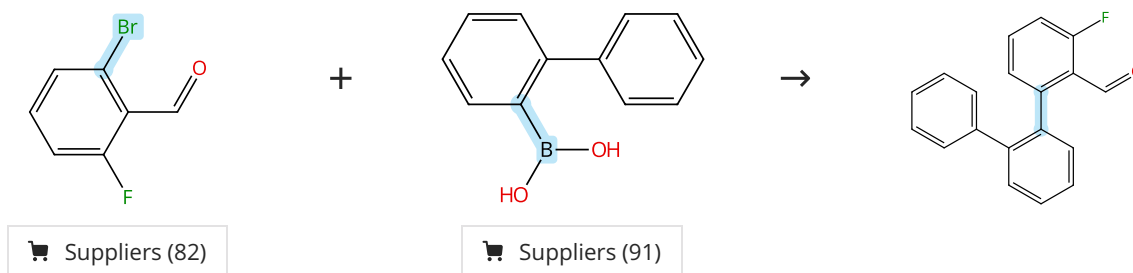
Organic Letters (2022), 24(34), 6256-6260.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: Toluene; 2 min, rt; 4 h, 100 °C

Experimental Protocols

Scheme 175 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-36484789

Steps: 1 Yield: 97%

Atroposelective brominations to access chiral biaryl scaffolds using high-valent Pd-catalysis

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Methanol, 1,2-Dimethoxyethane, Water; overnight, 80 °C; 80 °C → rt

By: Linde, Sif T.; et al

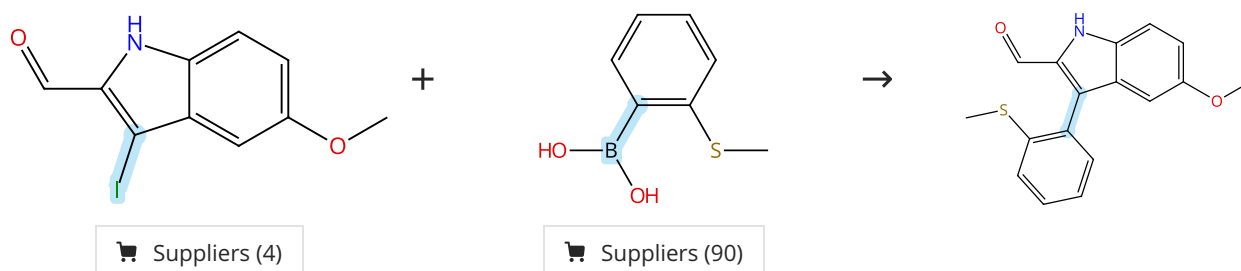
Chemical Science (2023), 14(13), 3676-3681.

- 1.2 **Reagents:** Water

Experimental Protocols

Scheme 176 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-37664346

Steps: 1 Yield: 97%

Dynamic Kinetic Resolution of Indole-Based Sulfonylated Heterobiaryls by Rhodium-Catalyzed Atroposelective Reductive Aldol Reaction

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; overnight, 70 °C; 70 °C → rt
- 1.2 **Reagents:** Sodium chloride
Solvents: Water

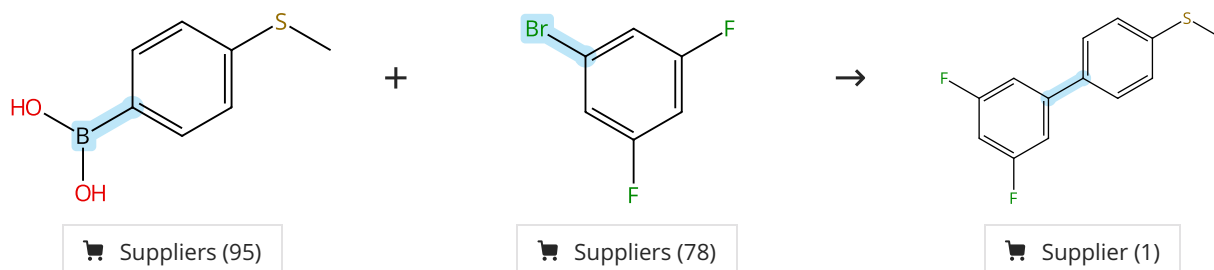
By: Rodriguez-Franco, Carlos; et al

ACS Catalysis (2023), 13(18), 12134-12141.

Experimental Protocols

Scheme 177 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-31102213

Steps: 1 Yield: 97%

Direct C-H carboxylation forming polyfunctionalized aromatic carboxylic acids by combined Bronsted bases

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 13 h, 90 °C

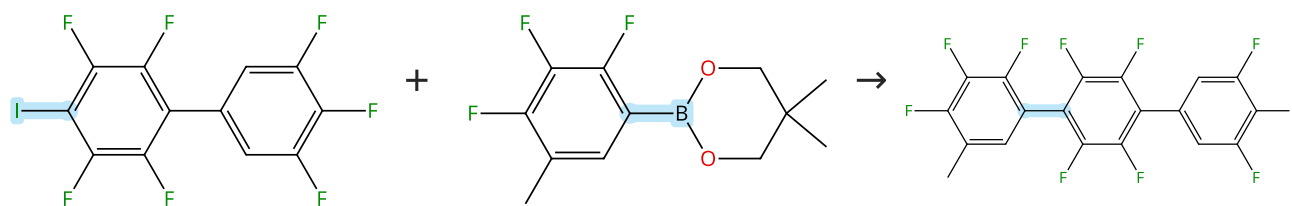
By: Shigeno, Masanori; et al

Organic Letters (2022), 24(3), 809-814.

Experimental Protocols

Scheme 178 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (3)

31-614-CAS-38216242

Steps: 1 Yield: 97%

A halogen bonding molecular tweezer

- 1.1 **Catalysts:** Tris(dibenzylideneacetone)dipalladium, 2-(Dicyclohexylphosphino)biphenyl
Solvents: Toluene, Tetrahydrofuran; 1 h, rt → 50 °C
- 1.2 **Reagents:** Potassium carbonate
Solvents: Water; 60 h, 50 °C → reflux; reflux → rt
- 1.3 **Reagents:** Ammonium chloride
Solvents: Water

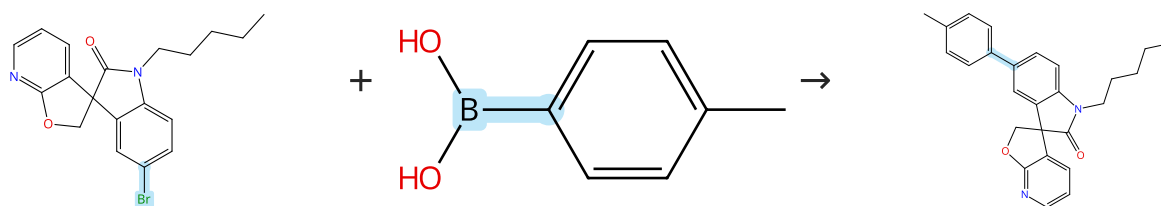
By: Vonnemann, Christoph J.; et al

Organic Chemistry Frontiers (2023), 10(24), 6146-6152.

Experimental Protocols

Scheme 179 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (114)

31-179-CAS-9148108

Steps: 1 Yield: 97%

Suzuki-Miyaura Reactions of Halospirooxindole Derivatives

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene; 10 min, 150 °C

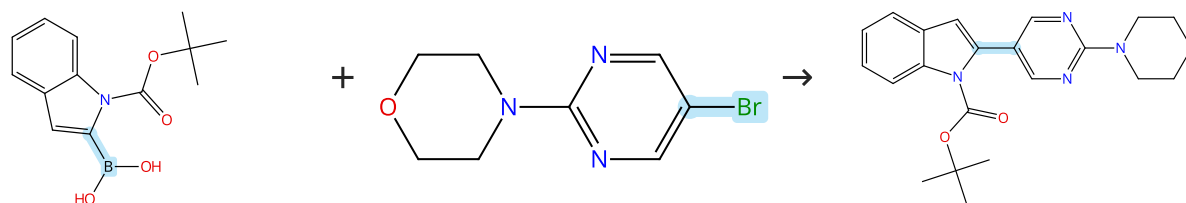
By: El Bouakher, Abderrahman; et al

European Journal of Organic Chemistry (2015), 2015(16), 3450-3461.

Experimental Protocols

Scheme 180 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (90)

Suppliers (75)

31-179-CAS-19379159

Steps: 1 Yield: 97%

EvanPhos: a ligand for ppm level Pd- catalyzed Suzuki-Miyaura couplings in either organic solvent or water

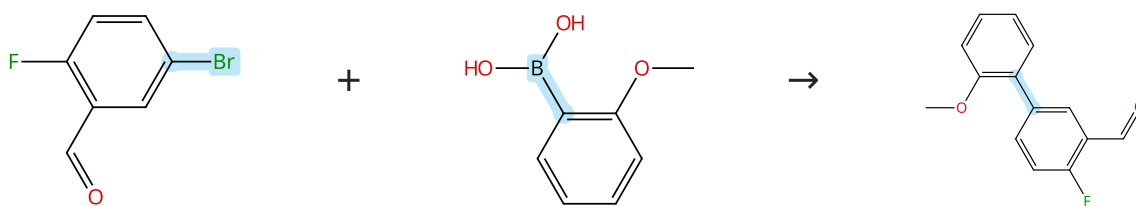
- 1.1 **Reagents:** Diisobutylaluminum hydride, Tripotassium phosphate, Poly(oxy-1,2-ethanediyl), α-[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy-
Catalysts: 2234278-60-3
Solvents: Toluene, Water; 30 min, rt; 3 h, 45 °C

By: Landstrom, Evan B.; et al

Green Chemistry (2018), 20(15), 3436-3443.

Scheme 181 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (93)

Suppliers (110)

Suppliers (7)

31-614-CAS-41686026

Steps: 1 Yield: 97%

The Substituent Effects of Suzuki Coupling in Aqueous Micelles

- 1.1 **Reagents:** Triethylamine, Polyethylene glycol lauryl ether
Catalysts: (*SP*-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino- κ P]ferrocene]dichloropalladium
Solvents: Water; 1 h, rt

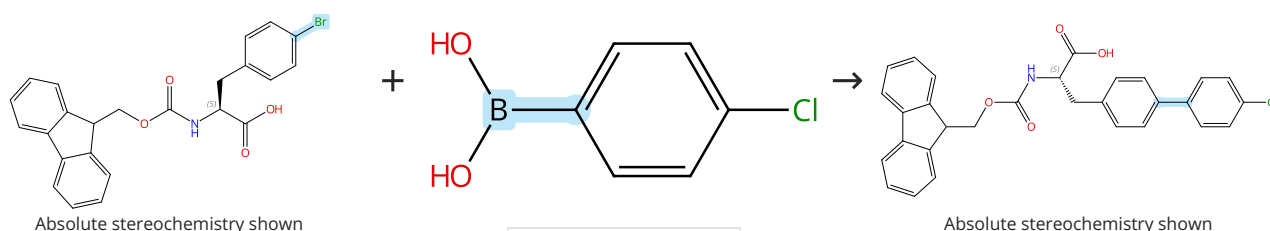
By: Liu, Haijing; et al

Journal of Organic Chemistry (2024), 89(17), 11930-11938.

Experimental Protocols

Scheme 182 (1 Reaction)

Steps: 1 Yield: 97%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (69)

Suppliers (110)

Suppliers (32)

31-179-CAS-16149169

Steps: 1 Yield: 97%

Synthesis of Fmoc-protected arylphenylalanines (Bip Derivatives) via nonaqueous Suzuki-Miyaura cross-coupling reactions

- 1.1 **Solvents:** Tetrahydrofuran; > 1 min, rt
 1.2 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, 1,1'-Bis(di-*tert*-butylphosphino)ferrocene; > 1 min, rt; 16 - 24 h, 50 °C; cooled
 1.3 **Reagents:** Citric acid
Solvents: Water; acidified, cooled

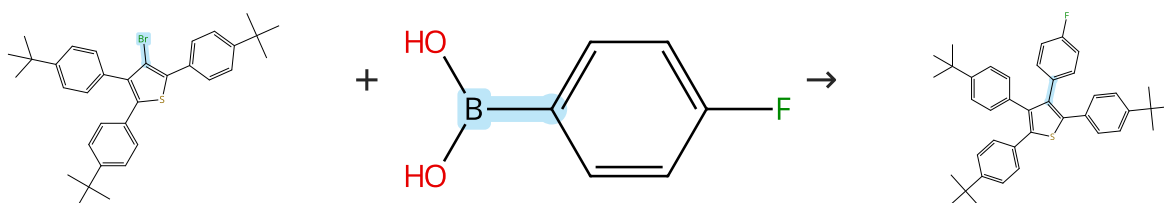
By: Qiao, Jennifer X.; et al

Journal of Organic Chemistry (2016), 81(19), 9499-9506.

Experimental Protocols

Scheme 183 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (104)

31-614-CAS-38944234

Steps: 1 Yield: 97%

Crafting diverse tetraphenylthiophene designs through comprehensive 'classical' Suzuki-Miyaura synthesis and electrochemical exploration

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Methanol, Toluene; 16 h, 80 °C; 80 °C → rt
 1.2 **Reagents:** Water; rt

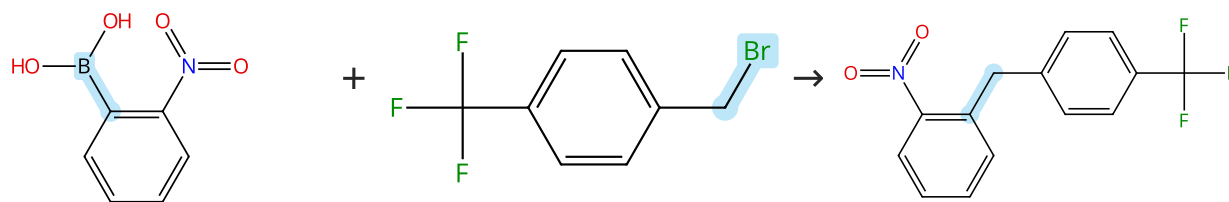
By: Maddala, Sudhakar; et al

Tetrahedron (2024), 151, 133807.

Experimental Protocols

Scheme 184 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (93)

Suppliers (94)

Supplier (1)

31-614-CAS-34078031

Steps: 1 Yield: 97%

A switchable redox annulation of 2-nitroarylethanols affording N-heterocycles: photoexcited nitro as a multifunctional handle

By: Wang, Bin; et al

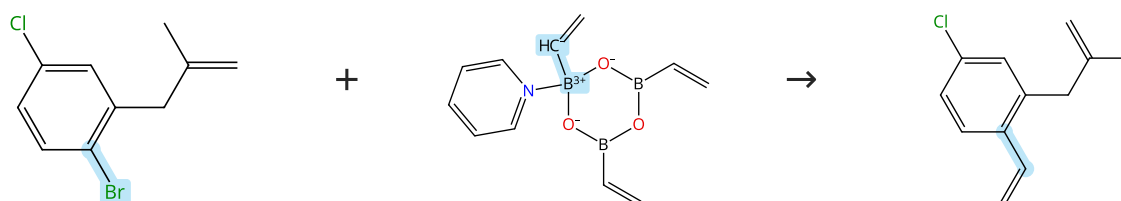
Chemical Science (2022), 13(37), 11074-11082.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 2 - 4 h, 60 - 90 °C

Experimental Protocols

Scheme 185 (1 Reaction)

Steps: 1 Yield: 97%



Supplier (1)

Suppliers (79)

31-614-CAS-32010091

Steps: 1 Yield: 97%

Ir-Catalyzed cyclization of α,ω -dienes with an N-methyl group via two C-H activation steps

By: Tanaka, Katsumasa; et al

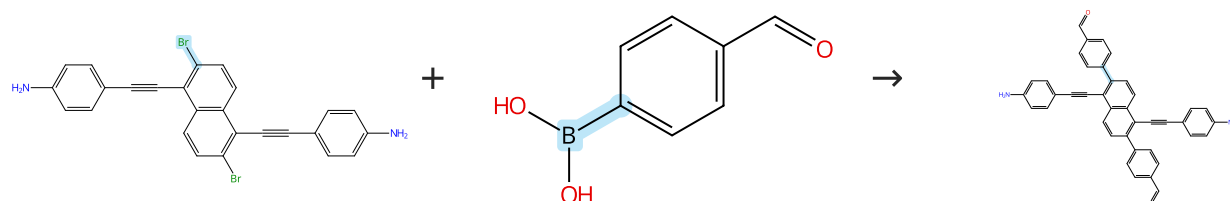
Chemical Communications (Cambridge, United Kingdom) (2022), 58(35), 5371-5374.

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 20 min, rt
 1.2 **Reagents:** Potassium carbonate
Solvents: Water; overnight, 100 °C

Experimental Protocols

Scheme 186 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (106)

31-614-CAS-35858586

Steps: 1 Yield: 97%

Targeted Synthesis of Isomeric Naphthalene-Based 2D Kagome Covalent Organic Frameworks

By: Li, Yusen; et al

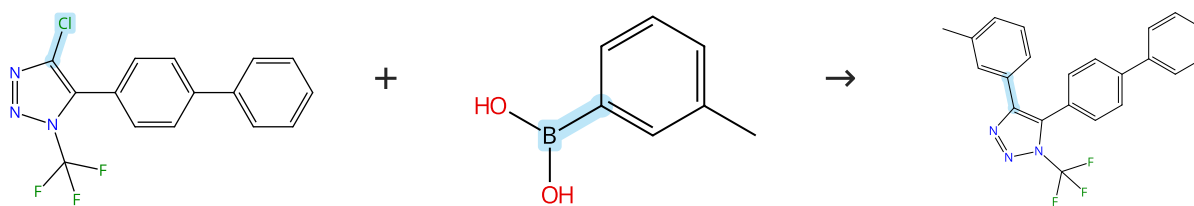
Angewandte Chemie, International Edition (2023), 62(10), e202216795.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 24 h, 100 °C

Experimental Protocols

Scheme 187 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (103)

31-614-CAS-40743664

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Potassium hydroxide
Catalysts: Tetrabutylammonium bromide, [1,3-Bis[2,6-bis(1-methylethyl)phenyl]tetrahydro-2(1*H*)-pyrimidinylidene]chloro [(1,2,3- η)-1-phenyl-2-propen-1-yl]palladium
Solvents: Water; 10 h, 140 °C

Experimental Protocols

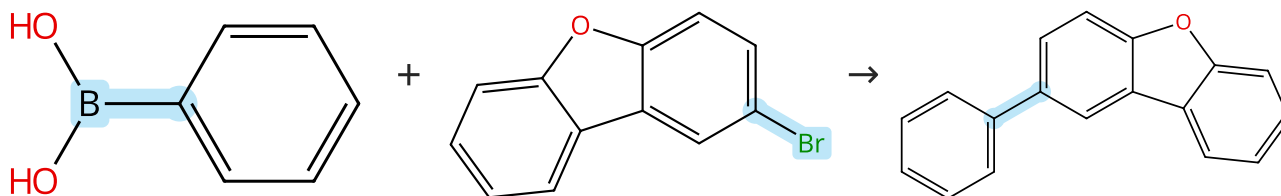
Controllable Regioselective [3+2] Cyclizations of N-CF₃ Imidoyl Chlorides and Ph₃PNNC: Divergent Synthesis of N-CF₃ Triazoles

By: Gao, Yan Fang; et al

Organic Letters (2024), 26(24), 5087-5091.

Scheme 188 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

Suppliers (77)

Suppliers (13)

31-614-CAS-36968184

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Benzene, diethenyl-, polymer with 1-(1,1-dimethylethyl)-4-ethenylbenzene, Pyridine, 4-ethenyl-, polymer with 1-(1,1-dimethylethyl)-4-ethenylbenzene
Solvents: Ethanol, Tetrahydrofuran, Water; 5 h, 70 °C

- 1.2 **Solvents:** Water

Experimental Protocols

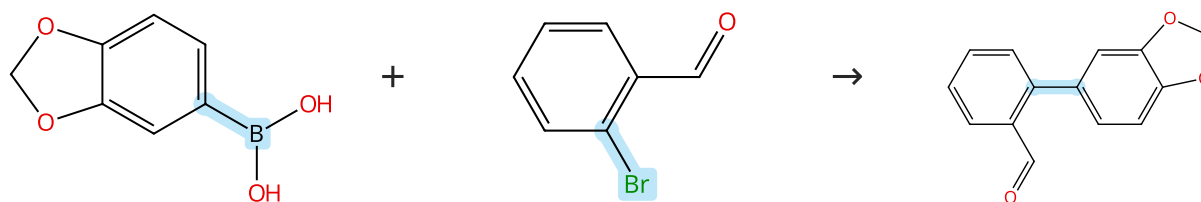
Continuous-Flow Suzuki-Miyaura Coupling in Water and Organic Solvents Promoted by Blends of Stabilized Convuluted Polymeric Palladium Catalysts and Polymeric Auxiliary Materials

By: Zhang, Zhenzhong; et al

Chemistry - A European Journal (2023), 29(34), e202300494.

Scheme 189 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (98)

Suppliers (89)

Suppliers (6)

31-179-CAS-23575684

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 25 h, rt → 100 °C

Experimental Protocols

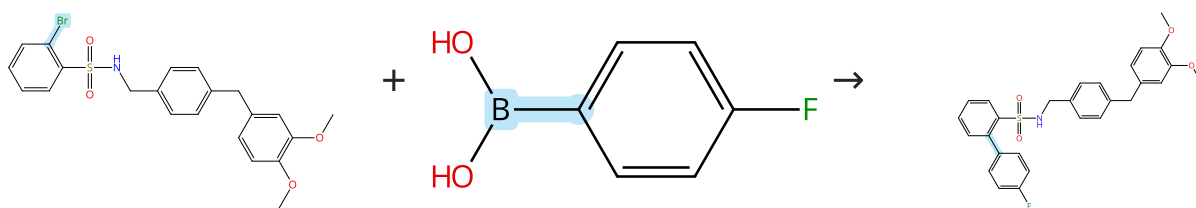
Imine as a linchpin approach for meta- C-H functionalization

By: Bag, Sukdev; et al

Nature Communications (2021), 12(1), 1393.

Scheme 190 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (104)

31-614-CAS-36210263

Steps: 1 Yield: 97%

Tuning liver pyruvate kinase activity up or down with a new class of allosteric modulators

By: Nain-Perez, Amalyn; et al

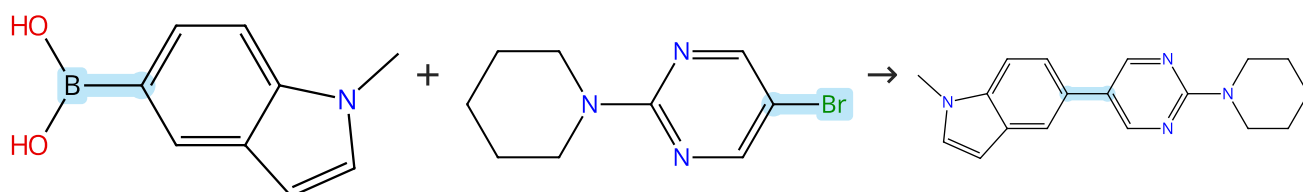
European Journal of Medicinal Chemistry (2023), 250, 115177.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 80 °C

Experimental Protocols

Scheme 191 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (73)

Suppliers (56)

31-179-CAS-20985222

Steps: 1 Yield: 97%

N,C-Disubstituted Biaryl palladacycles as Precatalysts for ppm Pd-Catalyzed Cross Couplings in Water under Mild Conditions

By: Thakore, Ruchita R.; et al

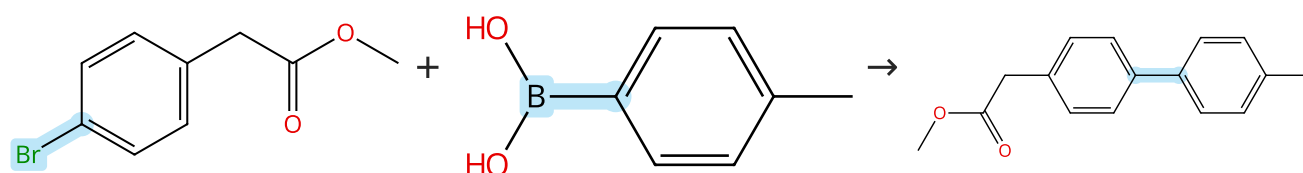
ACS Catalysis (2019), 9(12), 11647-11657.

1.1 **Reagents:** Tripotassium phosphate, Poly(oxy-1,2-ethanediyl), α -[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyl tridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]- ω -methoxy-
Catalysts: [Dicyclohexyl[2,6-dimethoxy-3-(2-methoxy-1-naphthalenyl)phenyl]phosphine- κP][4-(1-methylethyl)-2'-[(1-methylethyl)amino- κM][1,1'-biphenyl]-2-yl- κC](1,1,1-trifluoro methanesulfonato- κO)palladium
Solvents: Water; 18 h, 55 °C

Experimental Protocols

Scheme 192 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (72)

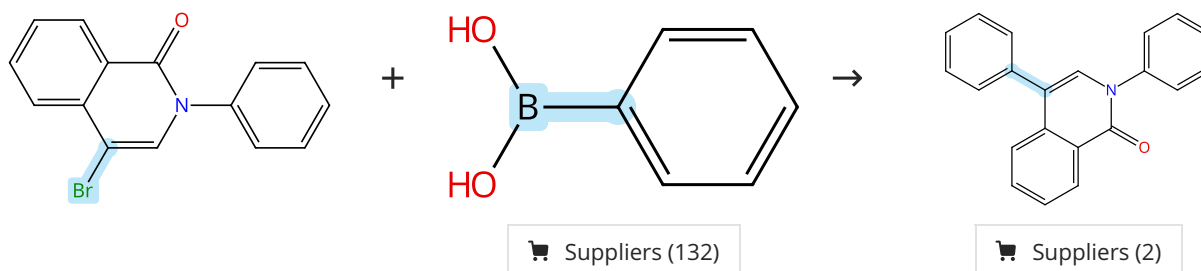
Suppliers (114)

Suppliers (18)

31-179-CAS-16597996 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate, Phenylboronic acid pinacol ester Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 1 h, rt → 70 °C; 70 °C → 0 °C 1.2 Reagents: Hydrogen peroxide Solvents: Water; 0 °C → rt; 1 h, rt 1.3 Reagents: Sodium pyrosulfite	Chemoselective Suzuki-Miyaura Cross-Coupling via Kinetic Transmetalation By: Fyfe, James W. B.; et al Angewandte Chemie, International Edition (2017), 56(5), 1249-1253.
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Scheme 193 (1 Reaction)

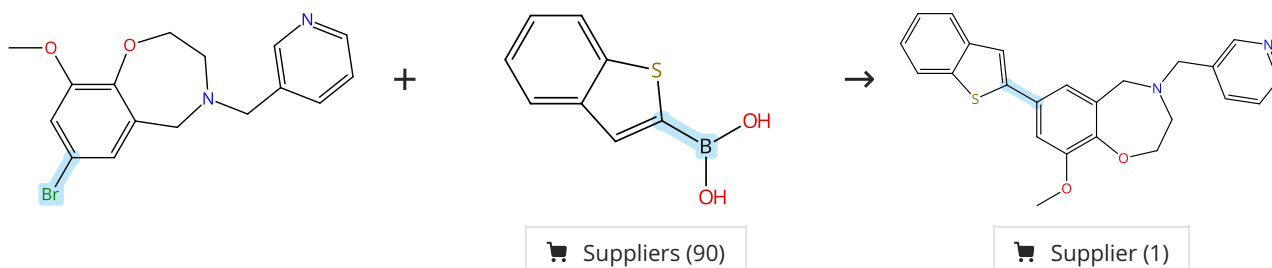
Steps: 1 Yield: 97%



31-179-CAS-21183132 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 20 h, 70 °C Experimental Protocols	Tandem Reaction Approaches to Isoquinolones from 2-Vinylbenzaldehydes and Anilines via Imine Formation-6π-Electrocyclization-Aerobic Oxidation Sequence By: Lee, Jiyeon; et al Organic Letters (2020), 22(2), 474-478.
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Scheme 194 (1 Reaction)

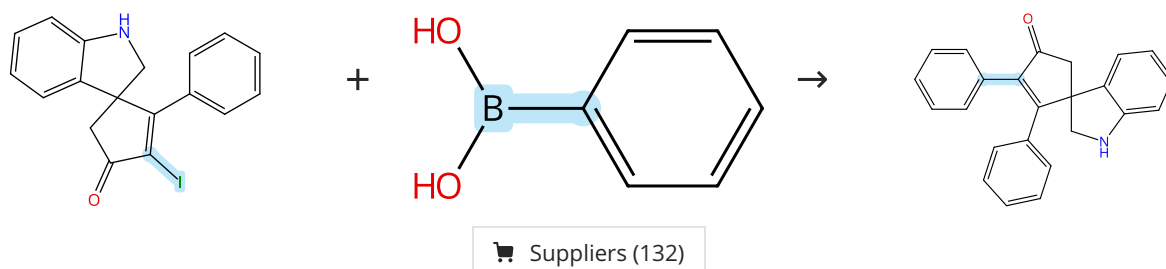
Steps: 1 Yield: 97%



31-179-CAS-15327044 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: Ethanol, 1,4-Dioxane, Water; 48 h, 80 °C Experimental Protocols	A Selective Prostaglandin E₂ Receptor Subtype 2 (EP2) Antagonist Increases the Macrophage-Mediated Clearance of Amyloid-Beta Plaques By: Fox, Brian M.; et al Journal of Medicinal Chemistry (2015), 58(13), 5256-5273.
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Scheme 195 (1 Reaction)

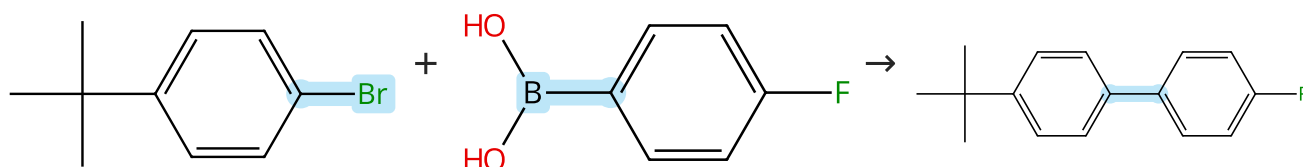
Steps: 1 Yield: 97%



31-179-CAS-18665408 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Triphenylphosphine, Palladium diacetate Solvents: 1,4-Dioxane, Water; 2 h, 60 °C	Synthesis of spiroindolenines by intramolecular ipso-iodocyclization of indol ynones By: Fedoseev, Pavel; et al Chemical Communications (Cambridge, United Kingdom) (2018), 54(29), 3625-3628.
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Scheme 196 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (79)

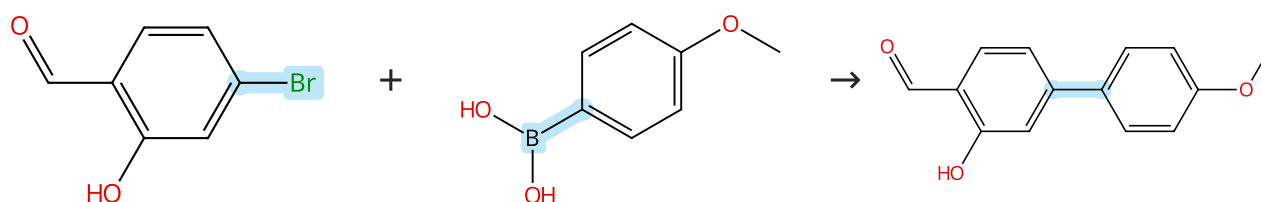
Suppliers (104)

Suppliers (11)

31-179-CAS-1016577 Steps: 1 Yield: 97% 1.1 Reagents: Diisopropylethylamine Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Isopropanol, Water; 17 h, 100 °C Experimental Protocols	Ni/Cu-Catalyzed Defluoroborylation of Fluoroarenes for Diverse C-F Bond Functionalizations By: Niwa, Takashi; et al Journal of the American Chemical Society (2015), 137(45), 14313-14318.
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Scheme 197 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (90)

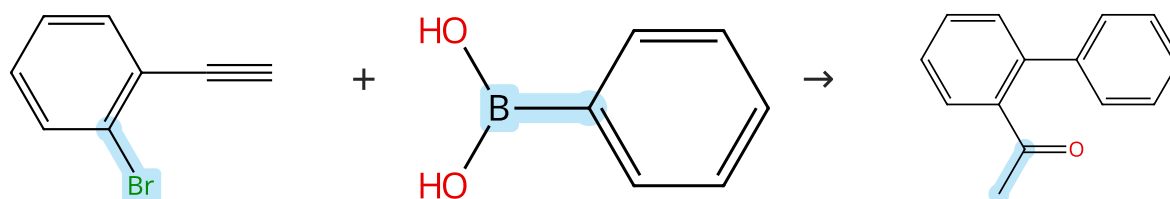
Suppliers (112)

Suppliers (26)

31-614-CAS-23946022 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; overnight, 60 °C Experimental Protocols	Dual-State Emissive π-Extended Salicylaldehyde Fluorophores: Synthesis, Photophysical Properties and First-Principle Calculations By: Stoerkler, Timothee; et al European Journal of Organic Chemistry (2021), 2021(26), 3726-3736.
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Scheme 198 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (73)

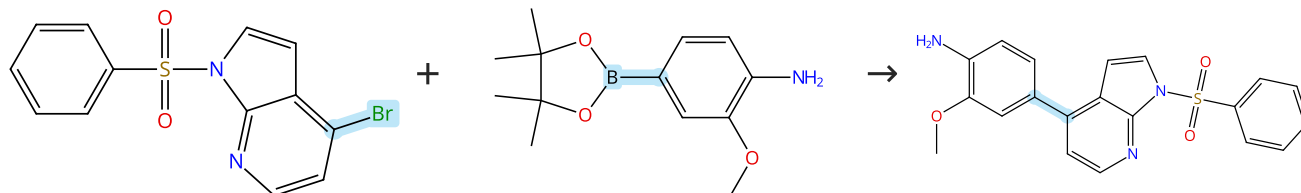
Suppliers (132)

Suppliers (62)

31-179-CAS-23182658 Steps: 1 Yield: 97% 1.1 Reagents: Water Catalysts: 2586061-76-7 (solid solution with M OF's containing unsubstituted di-acid and di-acid...) Solvents: Methanol; 24 h, 100 °C 1.2 Reagents: Triethylamine; 8 h, 100 °C Experimental Protocols	Dual-Metal N-Heterocyclic Carbene Complex (M = Au and Pd) - Functionalized UiO-67 MOF for Alkyne Hydration-Suzuki Coupling Tandem Reaction By: Dong, Ying; et al Journal of Organic Chemistry (2021), 86(2), 1818-1826.
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Scheme 199 (1 Reaction)

Steps: 1 Yield: 97%



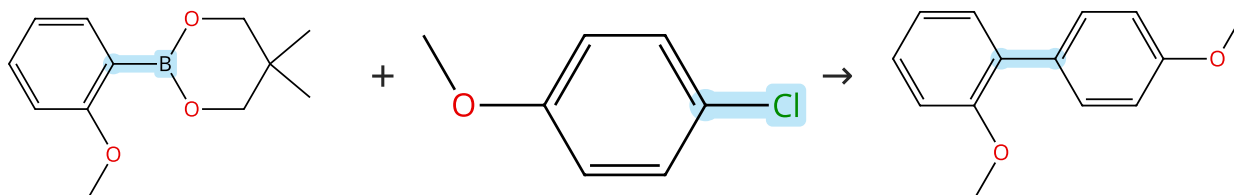
Suppliers (66)

Suppliers (75)

31-614-CAS-39404857 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 2 h, 110 °C Experimental Protocols	Discovery and Characterization of Selective, First-in-Class Inhibitors of Citron Kinase By: Maw, Joshua J.; et al Journal of Medicinal Chemistry (2024), 67(4), 2631-2666.
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Scheme 200 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (40)

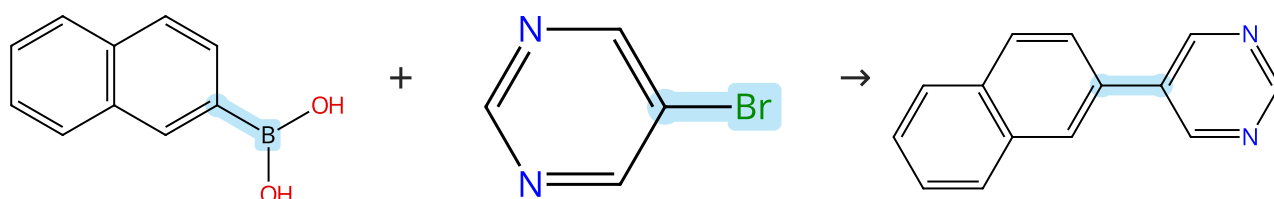
Suppliers (71)

Suppliers (13)

31-179-CAS-23481409 Steps: 1 Yield: 97% 1.1 Reagents: Potassium <i>tert</i>-butoxide Catalysts: Cobalt chloride (CoCl₂), 1<i>H</i>-Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-, chloride (1:1) Solvents: Tetrahydrofuran; 24 h, 60 °C 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Cobalt-Catalyzed Coupling of Aryl Chlorides with Aryl Boron Esters Activated by Alkoxides By: Tailor, Sanita B.; et al ACS Catalysis (2021), 11(7), 3856-3866.
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Scheme 201 (1 Reaction)

Steps: 1 Yield: 97%



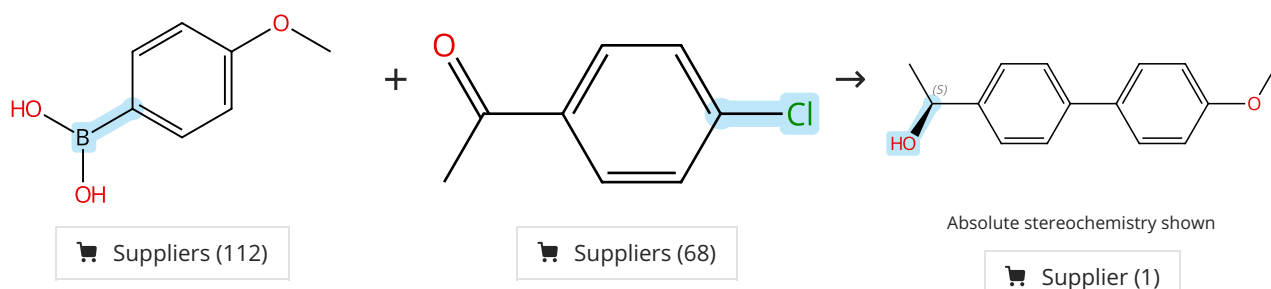
Suppliers (103)

Suppliers (100)

31-179-CAS-20985218 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate, Poly(oxy-1,2-ethanediyl), α-[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyl tridecyl)-2<i>H</i>-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy- Catalysts: [Dicyclohexyl[2,6-dimethoxy-3-(2-methoxy-1-naphthalenyl)phenyl]phosphine-κP][4-(1-methylethyl)-2'-[(1-methylethyl)amino-κM][1,1'-biphenyl]-2-yl-κC](1,1,1-trifluoro methanesulfonato-κO)palladium Solvents: Water; 17 h, 55 °C Experimental Protocols	N,C-Disubstituted Biaryl palladacycles as Precatalysts for ppm Pd-Catalyzed Cross Couplings in Water under Mild Conditions By: Thakore, Ruchita R.; et al ACS Catalysis (2019), 9(12), 11647-11657.
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Scheme 202 (1 Reaction)

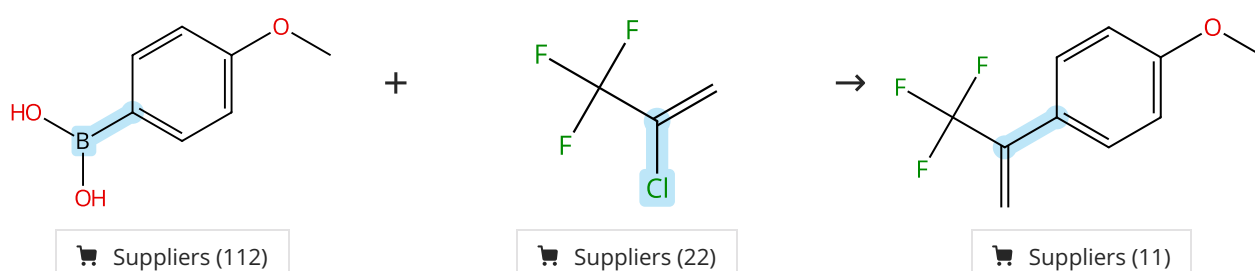
Steps: 1 Yield: 97%



31-179-CAS-19502800 Steps: 1 Yield: 97% 1.1 Reagents: Sodium formate, Cesium carbonate Catalysts: 2252264-54-1 (mesoporous hollow-shell-silica supported) Solvents: Isopropanol, Water; 0.5 - 1.5 h, 80 °C Experimental Protocols	An integrated immobilization strategy manipulates dual active centers to boost enantioselective tandem reactions By: Shu, Xiaomin; et al Chemical Communications (Cambridge, United Kingdom) (2018), 54(94), 13244-13247.
---	---

Scheme 203 (1 Reaction)

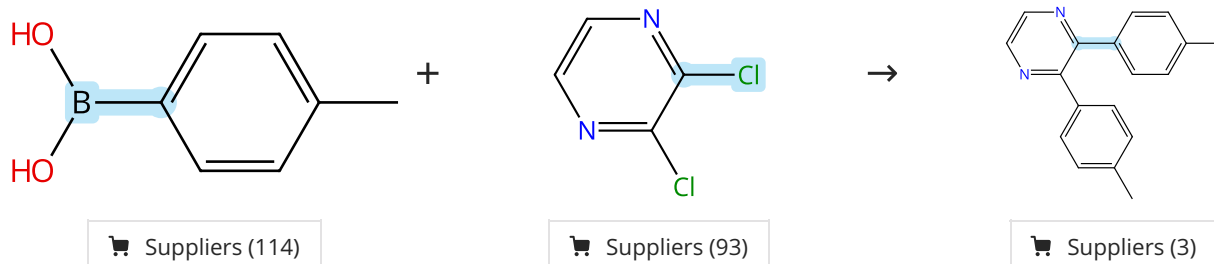
Steps: 1 Yield: 97%



31-179-CAS-16005843 Steps: 1 Yield: 97% 1.1 Reagents: Sodium bicarbonate Catalysts: Dichlorobis(triphenylphosphine)palladium, 2-(Dicyclohexylphosphino)-1-(2-methoxyphenyl)-1<i>H</i>-pyrrole Solvents: 1,4-Dioxane, Water; rt $\rightarrow$ -100 °C 1.2 5 h, 100 °C Experimental Protocols	Palladium-catalyzed Suzuki-Miyaura reaction of fluorinated vinyl chloride: a new approach for synthesis α and α,β-trifluoromethylstyrenes By: Li, Yang; et al Tetrahedron (2016), 72(37), 5684-5690.
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Scheme 204 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-15740767

Steps: 1 Yield: 97%

Synthesis of Chiral Piperazines via Hydrogenation of Pyrazines Activated by Alkyl Halides

- 1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 10 min, rt → 95 °C; 95 °C → rt
- 1.2 **Reagents:** Potassium fluoride; 95 °C

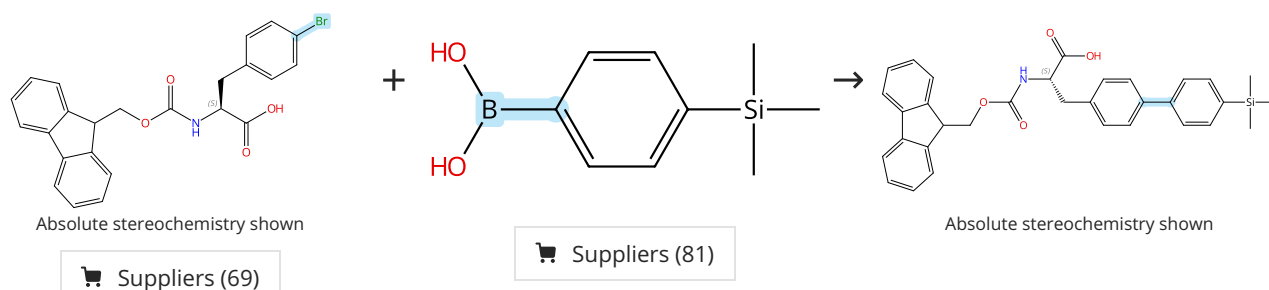
By: Huang, Wen-Xue; et al

Organic Letters (2016), 18(13), 3082-3085.

Experimental Protocols

Scheme 205 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-16149168

Steps: 1 Yield: 97%

Synthesis of Fmoc-protected arylphenylalanines (Bip Derivatives) via nonaqueous Suzuki-Miyaura cross-coupling reactions

- 1.1 **Solvents:** Tetrahydrofuran; > 1 min, rt
- 1.2 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, 1,1'-Bis(di-*tert*-butylphosphino)ferrocene; > 1 min, rt; 16 - 24 h, 50 °C; cooled
- 1.3 **Reagents:** Citric acid
Solvents: Water; acidified, cooled

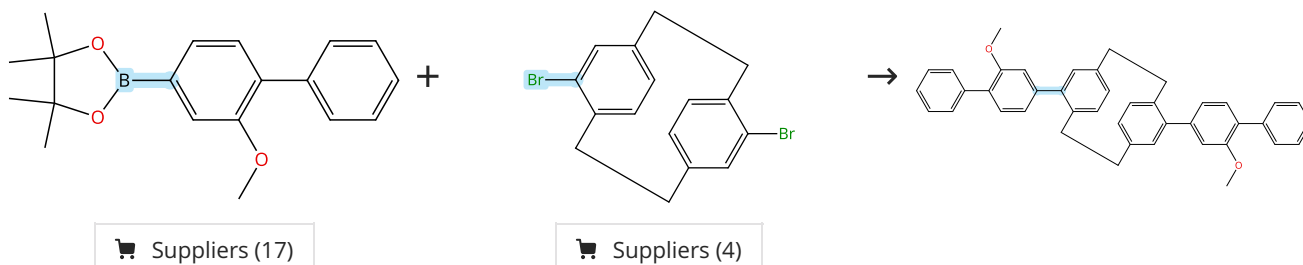
By: Qiao, Jennifer X.; et al

Journal of Organic Chemistry (2016), 81(19), 9499-9506.

Experimental Protocols

Scheme 206 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-20103241

Steps: 1 Yield: 97%

Design of Planar Chiral Phosphoric Acids with a [2.2] Paracyclopentadienyl Backbone as Organocatalysts for the Highly Enantioselective Aza-Friedel-Crafts Reaction

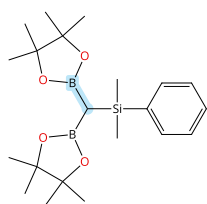
- 1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethyl sulfoxide, Water; 6 h, 90 °C

By: Xie, En; et al

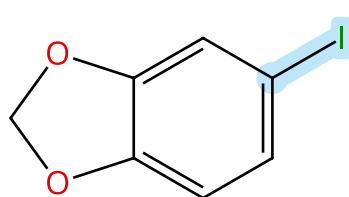
Organic Letters (2019), 21(10), 3682-3686.

Scheme 207 (1 Reaction)

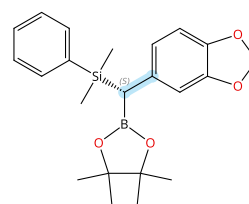
Steps: 1 Yield: 97%



+



→

Absolute stereochemistry shown,
Rotation (-)

Suppliers (73)

31-179-CAS-19670058

Steps: 1 Yield: 97%

- 1.1 **Catalysts:** Methanesulfonic acid, 1,1,1-trifluoro-, palladium(2+) salt (2:1), (2*R*)-1-[(1*R*)-1-[Bis[3,5-bis(trifluoromethyl)phenyl]phosphino]ethyl]-2-(dicyclohexylphosphino)ferrocene
Solvents: 1,4-Dioxane; 30 min, rt
- 1.2 **Reagents:** Sodium methoxide
Catalysts: Sodium iodide; 3 h, rt

Experimental Protocols

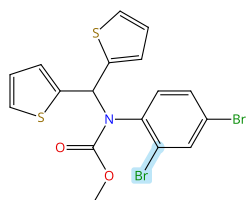
Access to Enantioenriched Benzylic 1,1-Silylboronate Esters by Palladium-Catalyzed Enantiotopic-Group Selective Suzuki-Miyaura Coupling of (Diborylmethyl)silanes with Aryl Iodides

By: Kim, Junghoon; et al

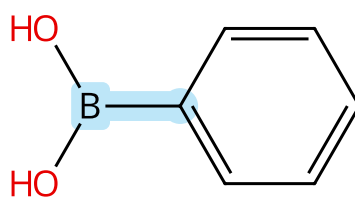
ACS Catalysis (2019), 9(1), 230-235.

Scheme 208 (1 Reaction)

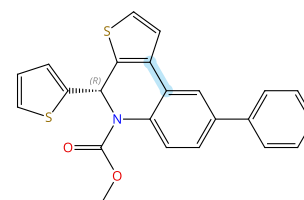
Steps: 1 Yield: 97%



+



→

Absolute stereochemistry shown,
Rotation (-)

Suppliers (132)

31-614-CAS-40026695

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 3024407-24-4
Solvents: Toluene; 48 h, 60 °C

Experimental Protocols

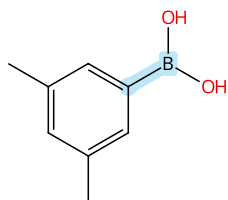
Access to chiral dihydrophenanthridines via a palladium(0)-catalyzed Suzuki coupling and C-H arylation cascade reaction using new chiral-bridged biphenyl bifunctional ligands

By: Chen, Bin; et al

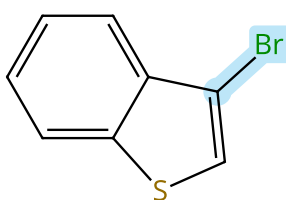
Chemical Science (2024), 15(18), 6884-6890.

Scheme 209 (1 Reaction)

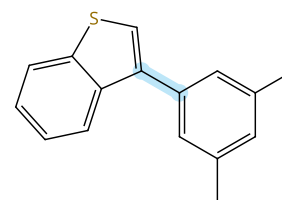
Steps: 1 Yield: 97%



+



→



Suppliers (95)

Suppliers (73)

Suppliers (9)

31-614-CAS-24823173

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 18 h, 110 °C
- 1.2 **Reagents:** Hydrochloric acid
Solvents: Ethyl acetate, Water; pH 2, rt

Experimental Protocols

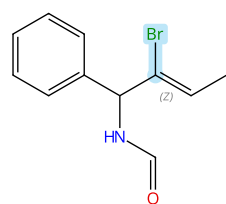
Chiral Benzothiophene Synthesis via Enantiospecific Coupling of Benzothiophene S-Oxides with Boronic Esters

By: Sang, Ruocheng; et al

Angewandte Chemie, International Edition (2021), 60(48), 25313-25317.

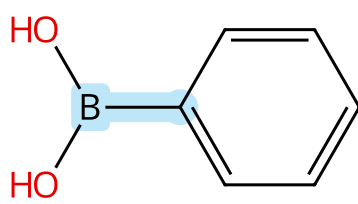
Scheme 210 (1 Reaction)

Steps: 1 Yield: 97%

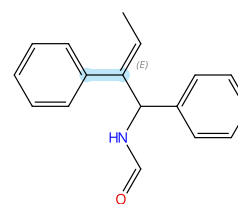


Double bond geometry shown

+



→



Double bond geometry shown

Suppliers (132)

31-614-CAS-36258069

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]
 palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 60 min, 100 °C

Experimental Protocols

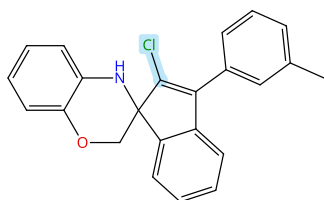
Synthesis of Chiral α -Amino Ketones via Transition Metal Catalyzed or Photoredox Cross-Coupling and Olefin Photo-Cleavage Reaction Sequence

By: Kobus-Bartoszewicz, Dominika; et al

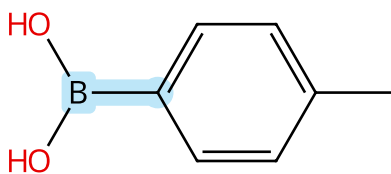
Advanced Synthesis & Catalysis (2023), 365(8), 1224-1237.

Scheme 211 (1 Reaction)

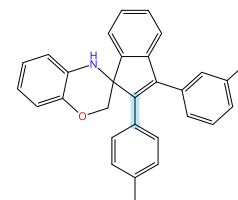
Steps: 1 Yield: 97%



+



→



Suppliers (114)

31-614-CAS-34634307

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine
Solvents: Tetrahydrofuran; 75 °C

Experimental Protocols

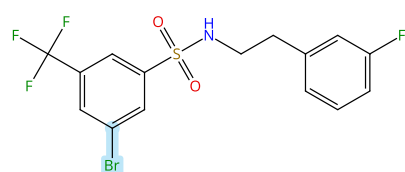
Rh-Catalyzed [3+2] Annulation of Cyclic Ketimines and Alkynyl Chloride: A Strategy for Accessing Unsymmetrically Substituted and Highly Functionalizable Indenes

By: Zhao, Xu; et al

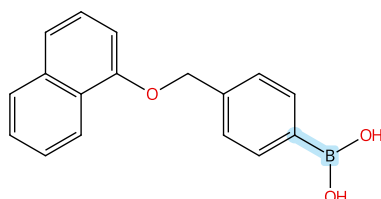
Organic Letters (2022), 24(50), 9169-9173.

Scheme 212 (1 Reaction)

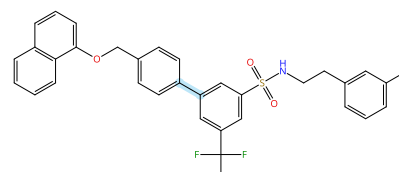
Steps: 1 Yield: 97%



+



→



Suppliers (42)

Supplier (1)

31-179-CAS-19076007

Steps: 1 Yield: 97%

- 1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 30 min, rt
 1.2 **Reagents:** Sodium carbonate
Solvents: Water; 16 h, 85 °C

Experimental Protocols

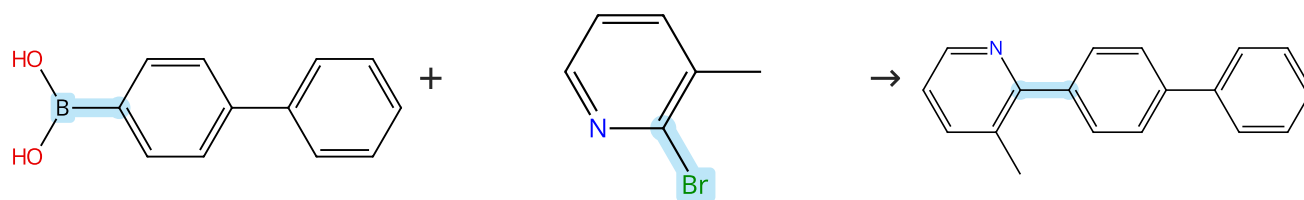
Discovery of new erbB4 inhibitors: Repositioning an orphan chemical library by inverse virtual screening

By: Giordano, Assunta; et al

European Journal of Medicinal Chemistry (2018), 152, 253-263.

Scheme 213 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (100)

Suppliers (96)

Suppliers (2)

31-614-CAS-44371650

Steps: 1 Yield: 97%

Rhenium-Catalyzed C(sp²)-H Silylalkenylation of Arenes: An Anti-Markovnikov Linchpin Strategy

By: Bhowmick, Suman; et al

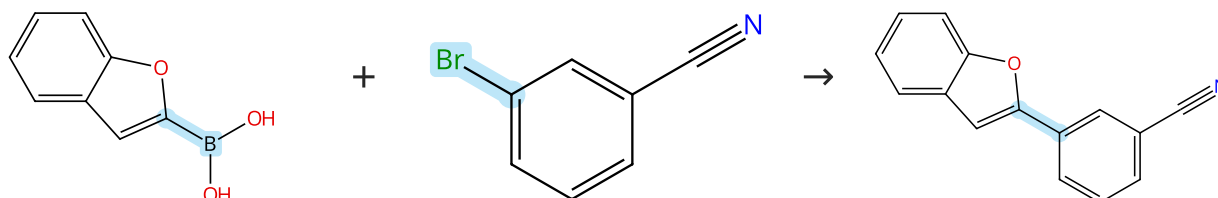
Organic Letters (2025), 27(10), 2289-2294.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 24 h, 110 °C

Experimental Protocols

Scheme 214 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (93)

Suppliers (80)

Suppliers (9)

31-179-CAS-20574056

Steps: 1 Yield: 97%

A new, substituted palladacycle for ppm level Pd-catalyzed Suzuki-Miyaura cross couplings in water

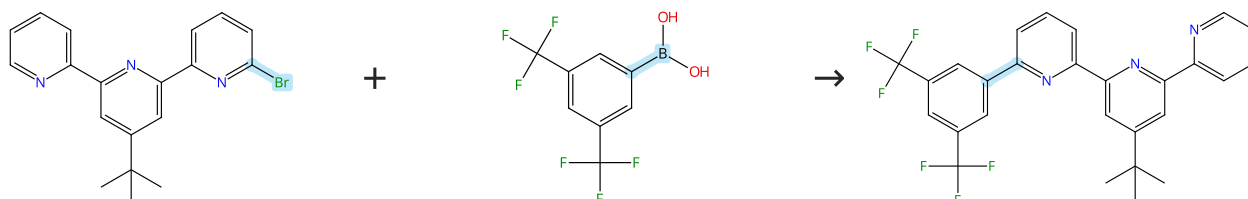
By: Takale, Balaram S.; et al

Chemical Science (2019), 10(38), 8825-8831.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: [2'-(Amino-κM)-4,4'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC][*rac*-(2*R*,3*S*)-4-(2,6-dimethoxyphenyl)-3-(1,1-dimethylethyl)-2,3-dihydro-2-[[2,4,6-tris(1-methylethyl)phenyl]methyl]-1,3-benzoxaphosphole-κP³](methanesulfonato-κO)palladium;
 8 h, 45 °C

Scheme 215 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (94)

31-179-CAS-23610798

Steps: 1 Yield: 97%

Ligand-Enabled, Iridium-Catalyzed ortho-Borylation of Fluoro arenes

By: Kuleshova, Olena; et al

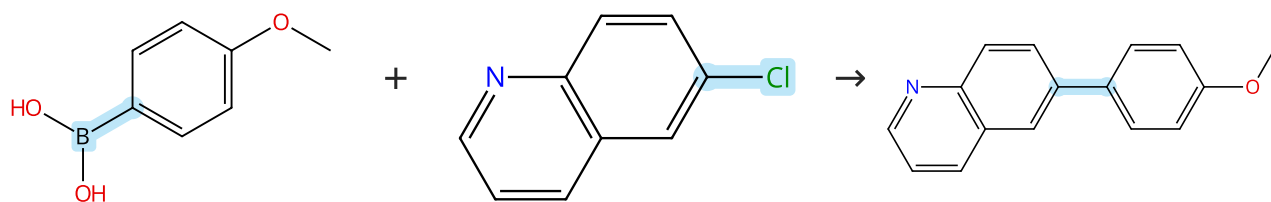
ACS Catalysis (2021), 11(10), 5968-5973.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; overnight, 100 °C

Experimental Protocols

Scheme 216 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (112)

Suppliers (84)

Suppliers (5)

31-614-CAS-41372140

Steps: 1 Yield: 97%

ProPhos: A Ligand for Promoting Nickel-Catalyzed Suzuki-Miyaura Coupling Inspired by Mechanistic Insights into Transmetalation

By: Yang, Jin; et al

Journal of the American Chemical Society (2024), 146(9), 6360-6368.

1.1 **Catalysts:** 3-(Diphenylphosphino)-1-propanol, Nickel dichloride

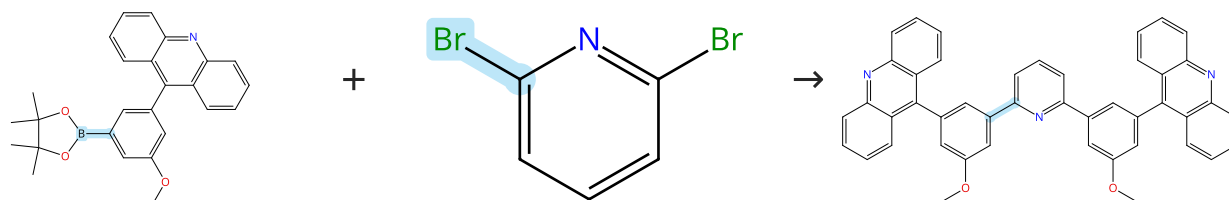
Solvents: Isopropanol; overnight, rt

1.2 **Reagents:** Tripotassium phosphate; 16 h, rt → 70 °C

Experimental Protocols

Scheme 217 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (87)

31-614-CAS-41680249

Steps: 1 Yield: 97%

Self-Assembled Bis-Acrininium Tweezer Equilibria Controlled by Multi-Responsive Properties

By: Hu, Johnny; et al

Chemistry - A European Journal (2024), 30(44), e202401866.

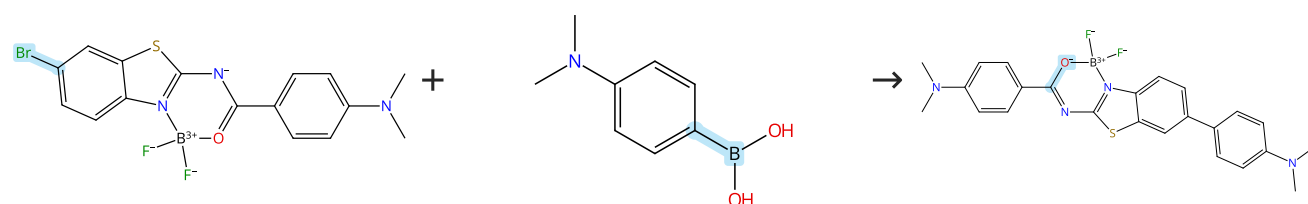
1.1 **Reagents:** Tripotassium phosphate

Catalysts: Tetrakis(triphenylphosphine)palladium**Solvents:** Dimethylformamide; 16 h, 80 °C

Experimental Protocols

Scheme 218 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (84)

31-179-CAS-20019971

Steps: 1 Yield: 97%

Application of the Suzuki-Miyaura Reaction for the Postfunctionalization of the Benzo[4,5]thiazolo[3,2-c][1,3,5,2]oxadiazaborinine Core: An Approach toward Fluorescent Dyes

By: Potopnyk, Mykhaylo A.; et al

Journal of Organic Chemistry (2019), 84(9), 5614-5626.

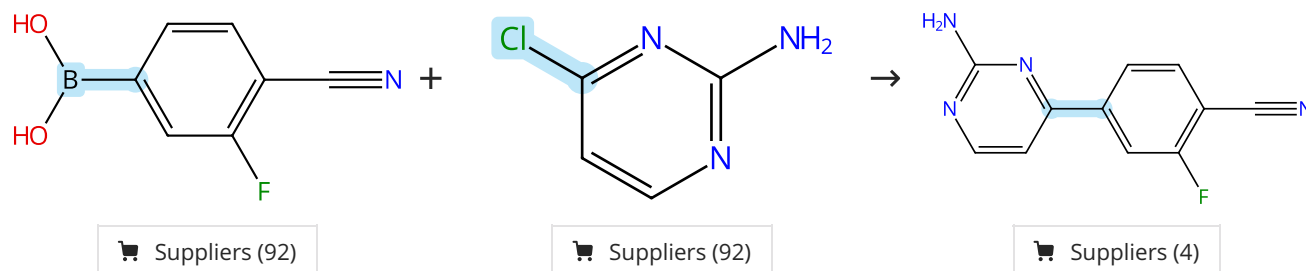
1.1 **Reagents:** Potassium carbonate

Catalysts: Dichlorobis(triphenylphosphine)palladium**Solvents:** 1,4-Dioxane, Water; 30 h, 80 °C

Experimental Protocols

Scheme 219 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-21561320

Steps: 1 Yield: 97%

Hit-to-Lead Optimization of Benzoxazepinoindazoles As Human African Trypanosomiasis Therapeutics

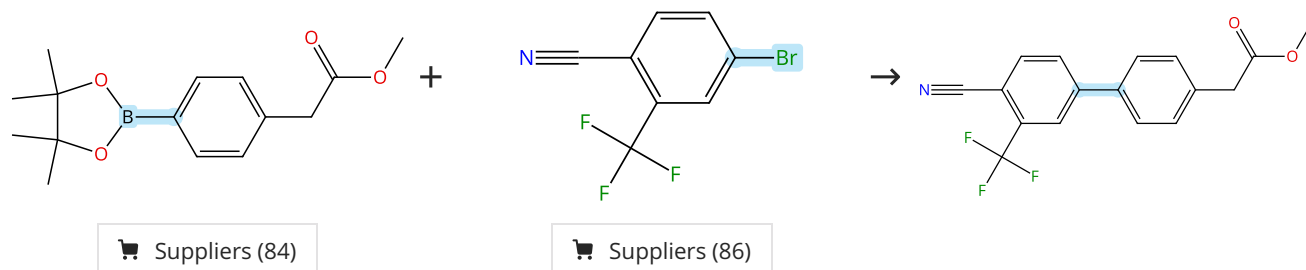
1.1 **Reagents:** Sodium bicarbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 3 h, 95 °C

By: Klug, Dana M.; et al

Journal of Medicinal Chemistry (2020), 63(5), 2527-2546.

Scheme 220 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-42307404

Steps: 1 Yield: 97%

Tuning RXR Modulators for PGC1α Recruitment

1.1 **Reagents:** Tripotassium phosphate
Catalysts: (*SP*-4-4)-[2'-(Amino-κN)[1,1'-biphenyl]-2-yl-κC]chloro [dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine]palladium
Solvents: 1,4-Dioxane, Water; 1.5 - 24 h, reflux

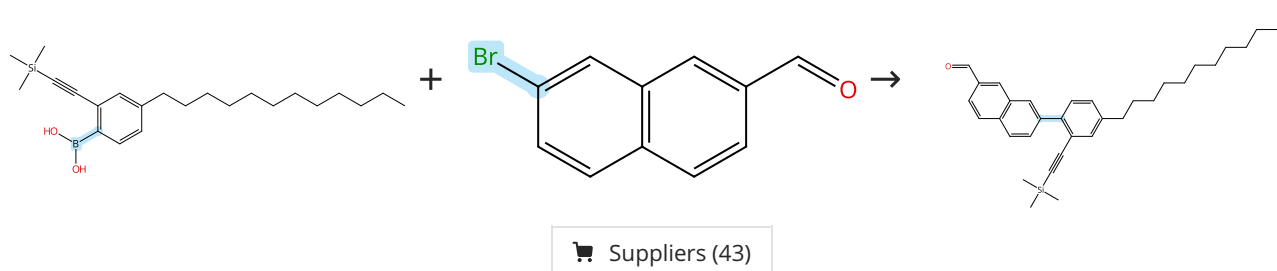
By: Nawa, Felix; et al

Journal of Medicinal Chemistry (2024), 67(18), 16338-16354.

Experimental Protocols

Scheme 221 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-17459137

Steps: 1 Yield: 97%

Synthesis of Dibenzo[hi,st]ovalene and Its Amplified Spontaneous Emission in a Polystyrene Matrix

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 6 h, 80 °C

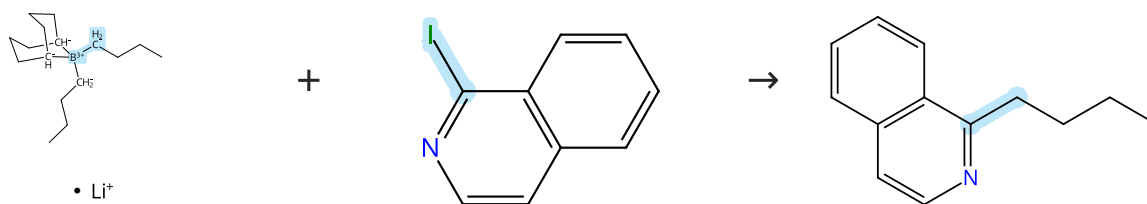
By: Paterno, Giuseppe M.; et al

Angewandte Chemie, International Edition (2017), 56(24), 6753-6757.

Experimental Protocols

Scheme 222 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (63)

Suppliers (3)

31-179-CAS-21201121

Steps: 1 Yield: 97%

1.1 **Reagents:** Borate(1-), 1,5-cyclooctanediyl dimethoxy-, lithium, (7-4)-

Catalysts: Cuprous iodide

Solvents: Dimethylformamide; 3 h, 100 °C

Experimental Protocols

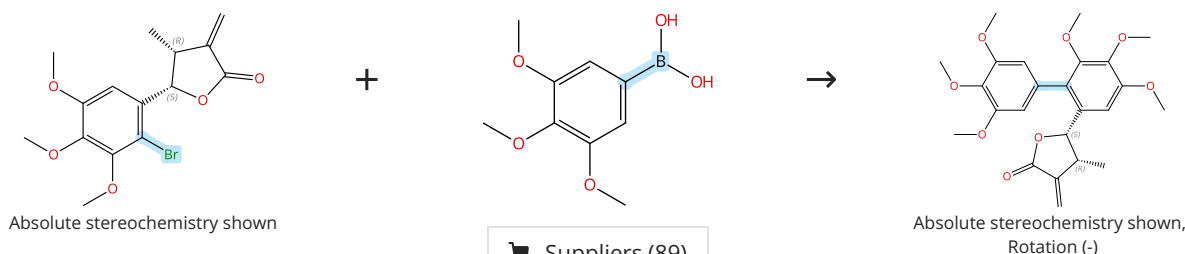
The copper-catalysed Suzuki-Miyaura coupling of alkylboron reagents: disproportionation of anionic (alkyl)(alkoxy)borates to anionic dialkylborates prior to transmetalation

By: Basnet, Prakash; et al

Chemical Communications (Cambridge, United Kingdom) (2016), 52(74), 11072-11075.

Scheme 223 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (89)

31-614-CAS-37099688

Steps: 1 Yield: 97%

1.1 **Catalysts:** Tris(dibenzylideneacetone)dipalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl

Solvents: Toluene; 48 h, 95 °C

Experimental Protocols

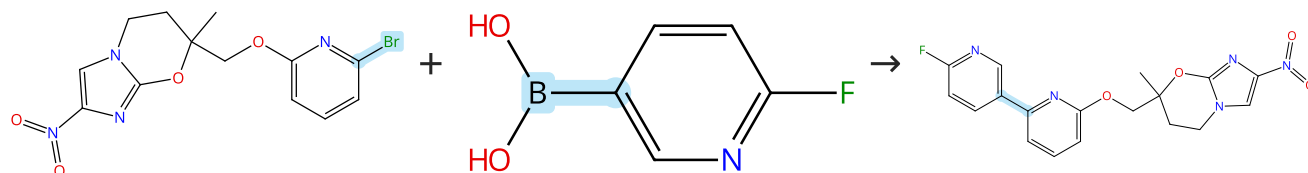
Asymmetric Synthesis of α -Methylene- γ -Butyrolactones via Tandem Allylboration/Lactonization: a Kinetic Resolution Process

By: Tan, Zheng; et al

Angewandte Chemie, International Edition (2023), 62(32), e202306146.

Scheme 224 (1 Reaction)

Steps: 1 Yield: 97%



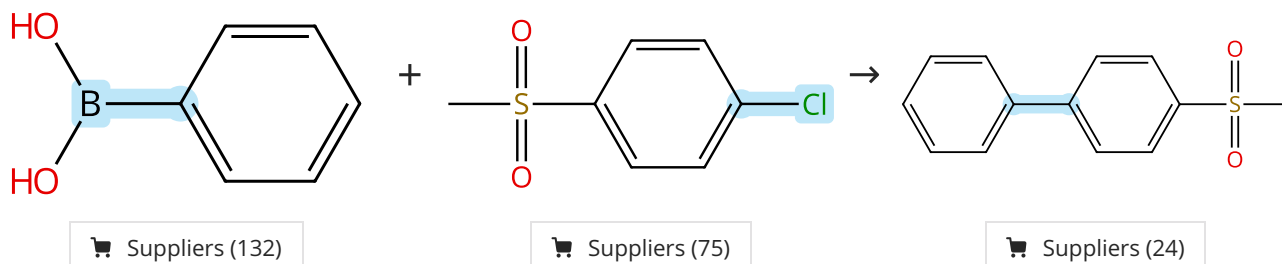
Suppliers (94)

Supplier (1)

31-179-CAS-17228841 Steps: 1 Yield: 97% 1.1 Reagents: Potassium bicarbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Ethanol, Dimethylformamide, Toluene, Water; 3 h, 82 °C 1.2 Reagents: Sodium bicarbonate Solvents: Water Experimental Protocols	7-Substituted 2-Nitro-5,6-dihydroimidazo[2,1-b][1,3]oxazines: Novel Antitubercular Agents Lead to a New Preclinical Candidate for Visceral Leishmaniasis By: Thompson, Andrew M.; et al Journal of Medicinal Chemistry (2017), 60(10), 4212-4233.
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Scheme 225 (1 Reaction)

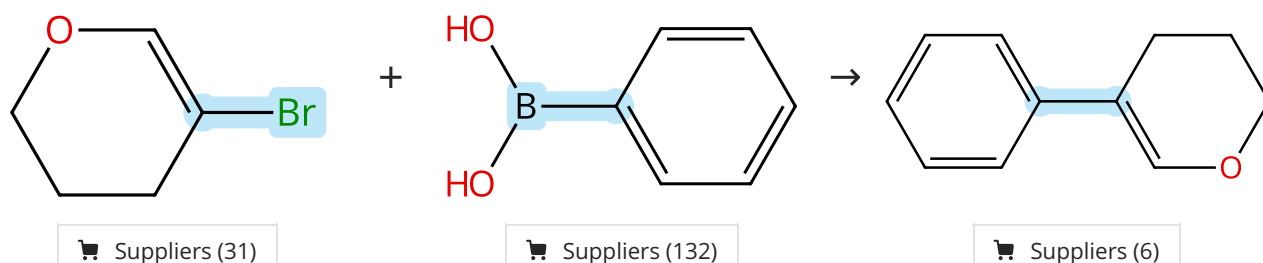
Steps: 1 Yield: 97%



31-179-CAS-3115466 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: stereoisomer of [1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2H-imidazol-2-ylidene]chloro[(1,2,3-η)-1-phenyl-2-propen-1-yl]palladium Solvents: Ethanol, Water; 4 h, 80 °C	The Suzuki-Miyaura Reaction Performed Using a Palladium-N-Heterocyclic Carbene Catalyst and a Weak Inorganic Base By: Izquierdo, Frederic; et al European Journal of Organic Chemistry (2015), 2015(9), 1920-1924.
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Scheme 226 (1 Reaction)

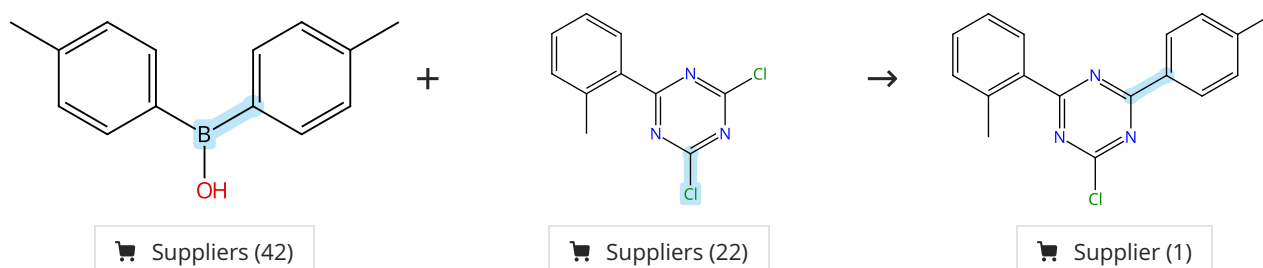
Steps: 1 Yield: 97%



31-179-CAS-23847298 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Bis(dibenzylideneacetone)palladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: Ethanol, Toluene, Water; 24 h, 80 °C Experimental Protocols	Broensted Acid-Catalyzed Enantioselective Iodocycloetherification Enabled by Triphenylphosphine Selenide Cocatalysis By: Guria, Sudip; et al Advanced Synthesis & Catalysis (2021), 363(15), 3852-3858.
---	---

Scheme 227 (1 Reaction)

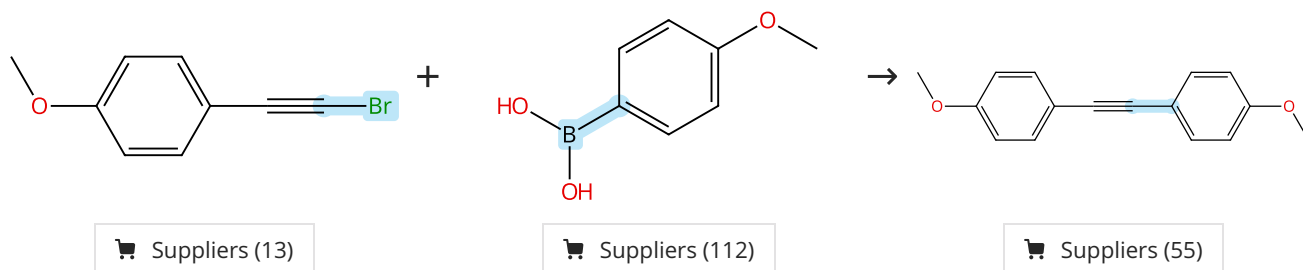
Steps: 1 Yield: 97%



31-179-CAS-21085753	Steps: 1 Yield: 97%	A Sequential Suzuki Coupling Approach to Unsymmetrical Aryl s-Triazines from Cyanuric Chloride By: Wang, Chen; et al Advanced Synthesis & Catalysis (2017), 359(14), 2514-2519.
1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Toluene, Water; 6 h, 60 °C		
Experimental Protocols		

Scheme 228 (1 Reaction)

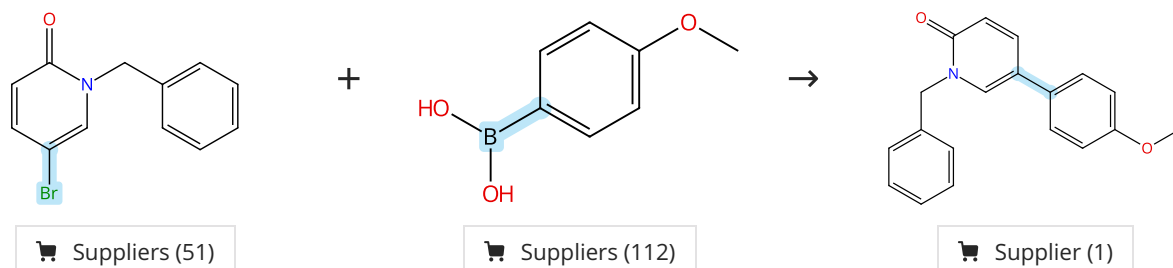
Steps: 1 Yield: 97%



31-179-CAS-17125072	Steps: 1 Yield: 97%	Synthesis of o-Allyloxy(ethynyl)benzene Derivatives by Cu-Catalyzed Suzuki-Miyaura-Type Reaction and Their Transformations into Heterocyclic Compounds By: Watanabe, Kohei; et al European Journal of Organic Chemistry (2017), 2017(16), 2359-2368.
1.1 Reagents: Tripotassium phosphate Catalysts: Cuprous iodide, (1 <i>E</i> ,2 <i>E</i>)-Ethanedial 1,2-bis(2-methyl-2-phenylhydrazone) Solvents: Isopropanol; 6 h, 40 °C		
Experimental Protocols		

Scheme 229 (1 Reaction)

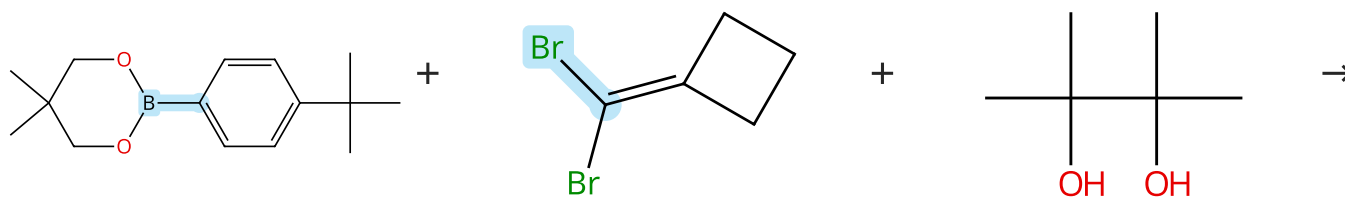
Steps: 1 Yield: 97%



31-614-CAS-40034585	Steps: 1 Yield: 97%	Copper-Catalyzed Regioselective Imdation of 2-Pyridones By: Murugesan, Tamilarasu; et al Organic Letters (2024), 26(15), 3048-3053.
1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; 16 h, 80 °C		
Experimental Protocols		

Scheme 230 (1 Reaction)

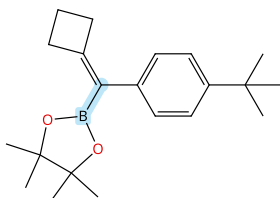
Steps: 1 Yield: 97%



Suppliers (17)

Suppliers (4)

Suppliers (85)



31-614-CAS-42386636

Steps: 1 Yield: 97%

Stereospecific alkenylidene homologation of organoboronates by S_NV reaction

By: Chen, Miao; et al

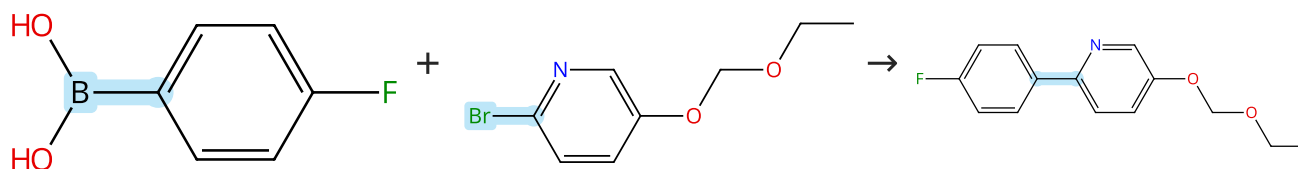
Nature (London, United Kingdom) (2024), 631(8020), 328-334.

- 1.1 **Reagents:** Butyllithium, 1,3-Dimethyl-3,4,5,6-tetrahydro-2(1*H*)-pyrimidinone, Lithium bromide
Solvents: Diethyl ether, Hexane; -110 °C; 30 min, -110 °C; -110 °C → rt; 30 min, rt
- 1.2 **Reagents:** Triethylamine
Solvents: Acetone; 12 h, rt → 65 °C

Experimental Protocols

Scheme 231 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (104)

Suppliers (5)

Supplier (1)

31-179-CAS-11549834

Steps: 1 Yield: 97%

Repositioning Antitubercular 6-Nitro-2,3-dihydroimidazo[2,1-b][1,3]oxazoles for Neglected Tropical Diseases: Structure-Activity Studies on a Preclinical Candidate for Visceral Leishmaniasis

By: Thompson, Andrew M.; et al

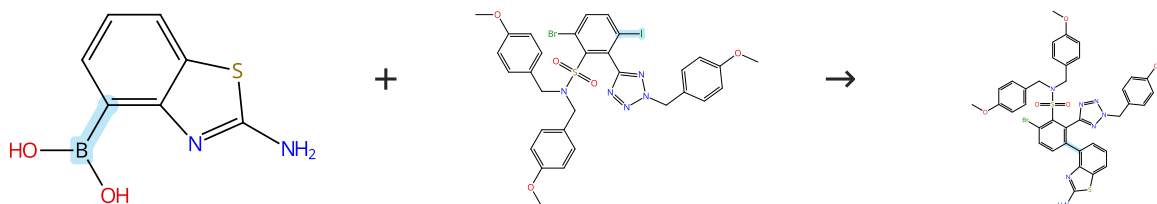
Journal of Medicinal Chemistry (2016), 59(6), 2530-2550.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Ethanol, Toluene, Water
- 1.2 4.5 h, 85 - 88 °C

Experimental Protocols

Scheme 232 (1 Reaction)

Steps: 1 Yield: 97%

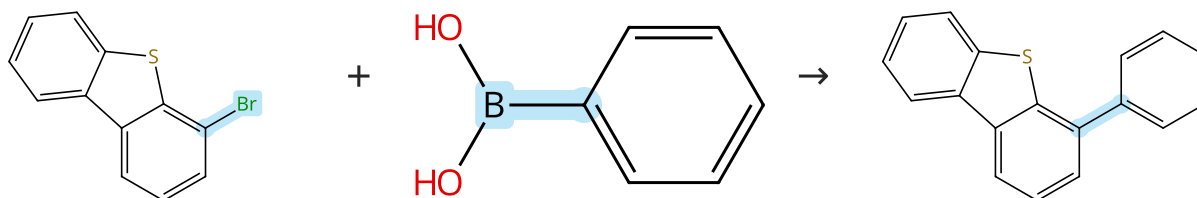


Suppliers (9)

31-614-CAS-39559001 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; overnight, 90 °C Experimental Protocols	Structure Guided Discovery of Novel Pan Metallo-β-Lactamase Inhibitors with Improved Gram-Negative Bacterial Cell Penetration By: Dong, Shuzhi; et al Journal of Medicinal Chemistry (2024), 67(5), 3400-3418.
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Scheme 233 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (77)

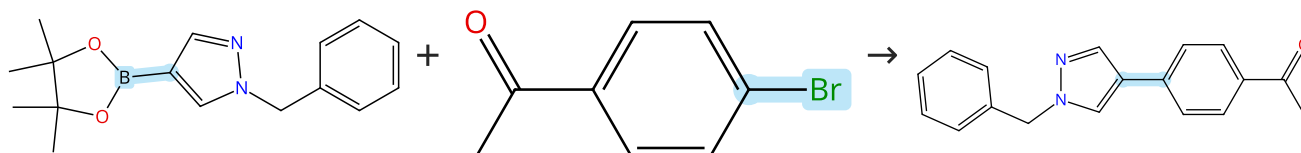
Suppliers (132)

Suppliers (44)

31-179-CAS-20413449 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 24 h, 80 °C	Transition-Metal-Free Cross-Coupling of Benzothiophenes and Styrenes in a Stereoselective Synthesis of Substituted (E, Z)-1,3-Dienes By: Siauciulis, Mindaugas; et al Angewandte Chemie, International Edition (2019), 58(26), 8779-8783.
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Scheme 234 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (94)

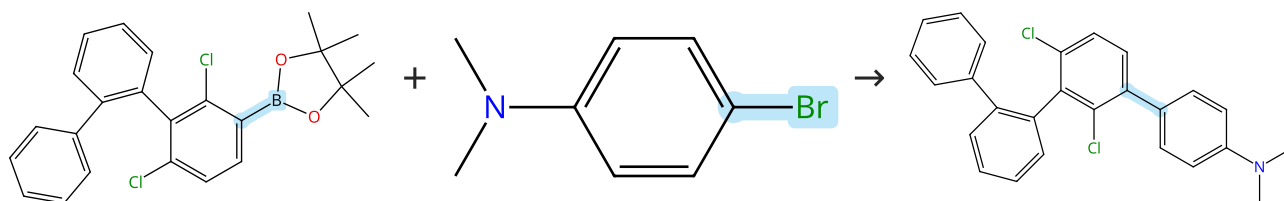
Suppliers (84)

Suppliers (4)

31-614-CAS-41372137 Steps: 1 Yield: 97% 1.1 Catalysts: Bis(1,5-cyclooctadiene)nickel, 3-(Diphenylphosphino)-1-propanol Solvents: 2-Methyltetrahydrofuran; 0.5 h, rt 1.2 Reagents: Tripotassium phosphate Solvents: Water; 16 h, rt → 70 °C Experimental Protocols	ProPhos: A Ligand for Promoting Nickel-Catalyzed Suzuki-Miyaura Coupling Inspired by Mechanistic Insights into Transmetalation By: Yang, Jin; et al Journal of the American Chemical Society (2024), 146(9), 6360-6368.
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Scheme 235 (1 Reaction)

Steps: 1 Yield: 97%

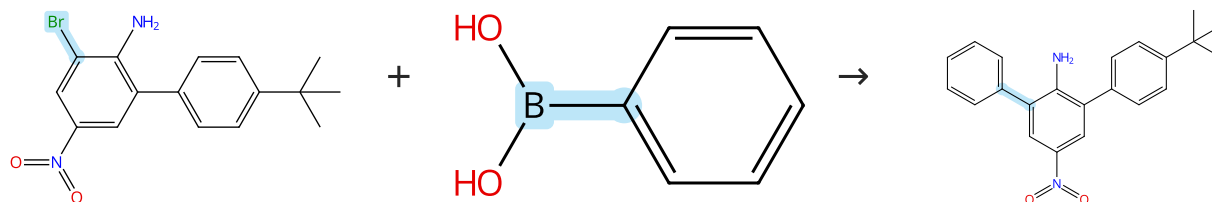


Suppliers (85)

31-614-CAS-40331279	Steps: 1 Yield: 97%	Aryl Chloride-Directed Enantioselective C(sp²)-H Borylation Enabled by Iridium Catalysis
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 5 h, 80 °C		By: Zhao, Hongliang; et al
Experimental Protocols		Journal of the American Chemical Society (2023), 145(46), 25214-25221.

Scheme 236 (1 Reaction)

Steps: 1 Yield: 97%

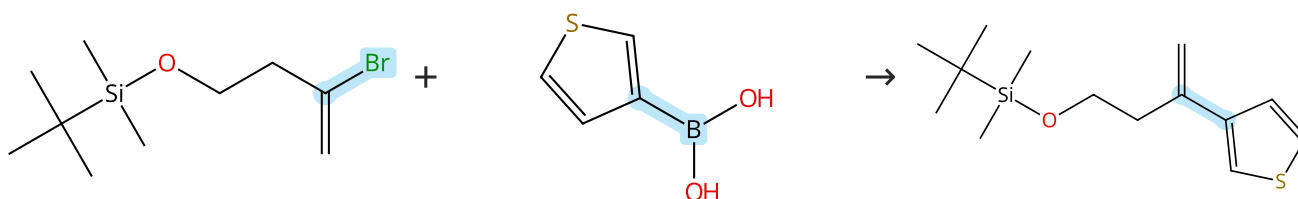


Suppliers (132)

31-179-CAS-5126603	Steps: 1 Yield: 97%	A Strategy towards the Multigram Synthesis of Uncommon Hexaarylbenzenes
1.1 Reagents: Sodium carbonate, Tetrabutylammonium bromide Catalysts: Palladium diacetate Solvents: Water; 30 min, 150 °C		By: Lungerich, Dominik; et al
Experimental Protocols		Angewandte Chemie, International Edition (2016), 55(18), 5602-5605.

Scheme 237 (1 Reaction)

Steps: 1 Yield: 97%



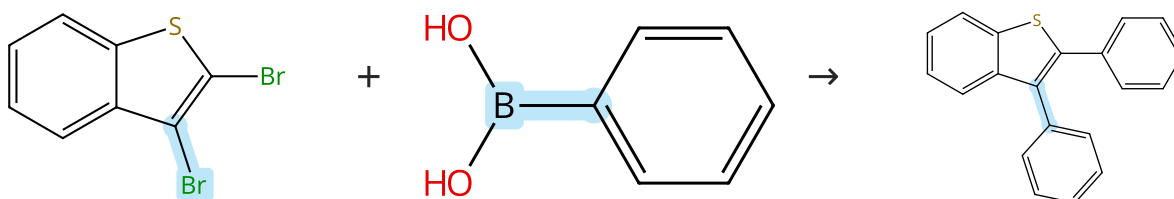
Suppliers (17)

Suppliers (101)

31-614-CAS-40919237	Steps: 1 Yield: 97%	Radical Homopolymerization of Linear α-Olefins Enabled by 1,4-Cyano Group Migration
1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, 1,4-Dioxane; 22 h, 90 °C		By: An, Bang; et al
Experimental Protocols		Angewandte Chemie, International Edition (2024), 63(25), e202402511.

Scheme 238 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (60)

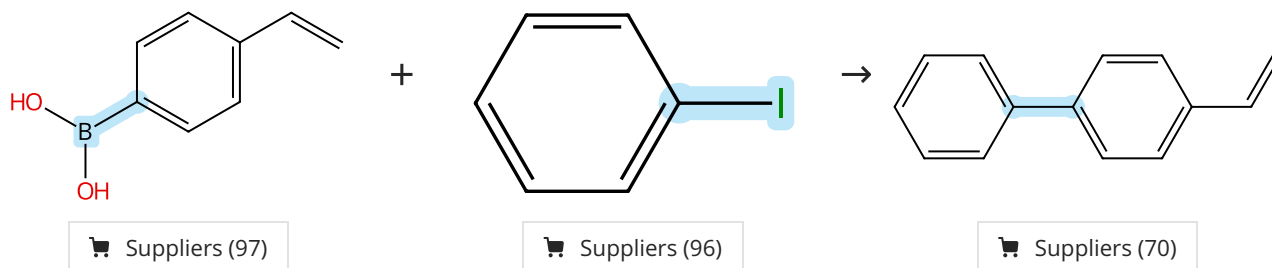
Suppliers (132)

Suppliers (3)

31-179-CAS-11419588	Steps: 1 Yield: 97%	Transition-Metal-Free Synthesis of Carbazoles and Indoles by an S_NAr-Based "Aromatic Metamorphosis" of Thiaarenes By: Bhanuchandra, M.; et al Angewandte Chemie, International Edition (2015), 54(35), 10234-10238.
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 48 h, 80 °C		
Experimental Protocols		

Scheme 239 (1 Reaction)

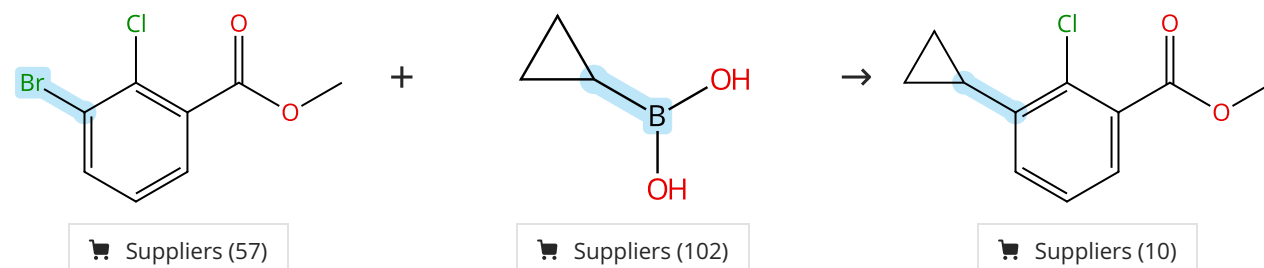
Steps: 1 Yield: 97%



31-614-CAS-34076996	Steps: 1 Yield: 97%	A palladium polyaniline complex: a simple and efficient catalyst for batch and flow Suzuki-Miyaura cross-couplings By: Zhang, Dongliang; et al Chemical Communications (Cambridge, United Kingdom) (2022), 58(77), 10845-10848.
1.1 Reagents: Potassium carbonate Catalysts: Palladium Solvents: Ethanol; 12 h, 100 °C		
Experimental Protocols		

Scheme 240 (1 Reaction)

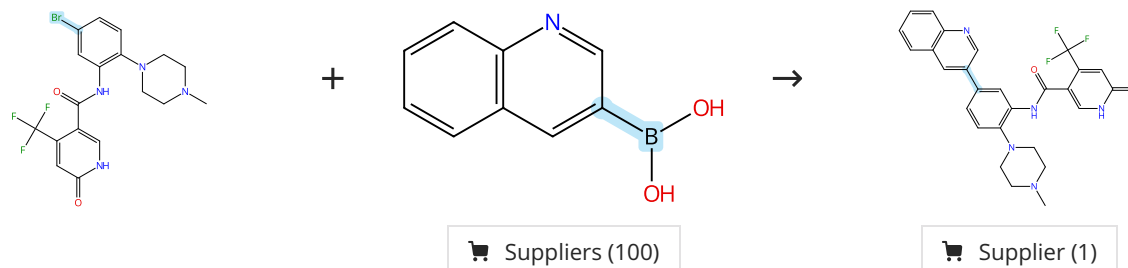
Steps: 1 Yield: 97%



31-179-CAS-20401783	Steps: 1 Yield: 97%	5-Aryl-1,3,4-oxadiazol-2-ylthioalkanoic Acids: A Highly Potent New Class of Inhibitors of Rho/Myocardin-Related Transcription Factor (MRTF)/Serum Response Factor (SRF)-Mediated Gene Transcription as Potential Antifibrotic Agents for Scleroderma By: Kahl, Dylan J.; et al Journal of Medicinal Chemistry (2019), 62(9), 4350-4369.
1.1 Reagents: Dipotassium phosphate Catalysts: Tricyclohexylphosphine Solvents: Toluene, Palladium diacetate, Water; 3 h, 100 °C		

Scheme 241 (1 Reaction)

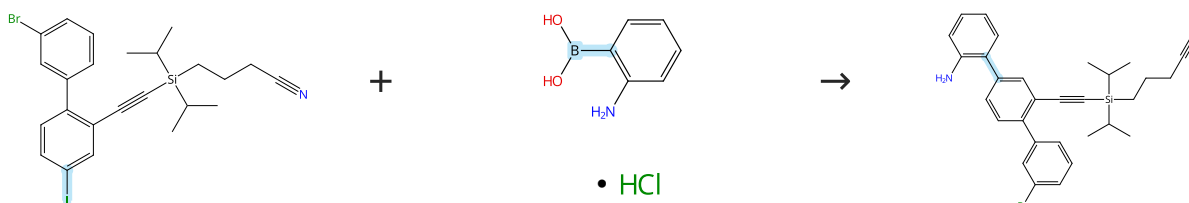
Steps: 1 Yield: 97%



31-179-CAS-15425359	Steps: 1 Yield: 97%	Structure-Based Optimization of a Small Molecule Antagonist of the Interaction Between WD Repeat-Containing Protein 5 (WDR5) and Mixed-Lineage Leukemia 1 (MLL1)
1.1 Reagents: Sodium carbonate Catalysts: X-Phos, (<i>SP</i> -4-4)-[2'-(Amino- κ M)[1,1'-biphenyl]-2-yl- κ C]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: 1,4-Dioxane, Water; 5 min; 60 min, 120 °C		By: Getlik, Matthaus; et al Journal of Medicinal Chemistry (2016), 59(6), 2478-2496.
Experimental Protocols		

Scheme 242 (1 Reaction)

Steps: 1 Yield: 97%

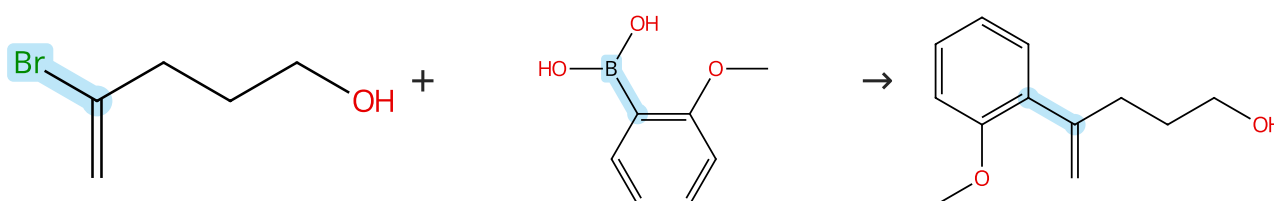


Suppliers (64)

31-179-CAS-23575507	Steps: 1 Yield: 97%	Bicyclic Phenyl-Ethynyl Architectures: Synthesis of a 1,4-Bis(phenylbuta-1,3-diyn-1-yl) Benzene Banister
1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 12 h, rt		By: Bannwart, Linda Maria; et al Chemistry - A European Journal (2021), 27(20), 6295-6307.
Experimental Protocols		

Scheme 243 (1 Reaction)

Steps: 1 Yield: 97%



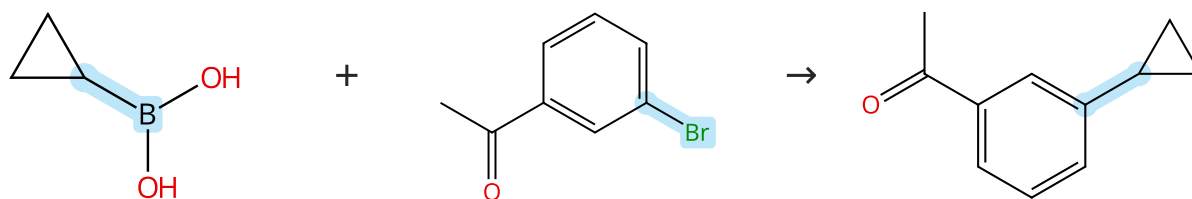
Suppliers (7)

Suppliers (110)

31-614-CAS-44358482	Steps: 1 Yield: 97%	Emergence of a distinct mechanism of C-N bond formation in photoenzymes
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; overnight, 75 °C		By: Raps, Felix C.; et al Nature (London, United Kingdom) (2025), 637(8045), 362-368.
Experimental Protocols		

Scheme 244 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

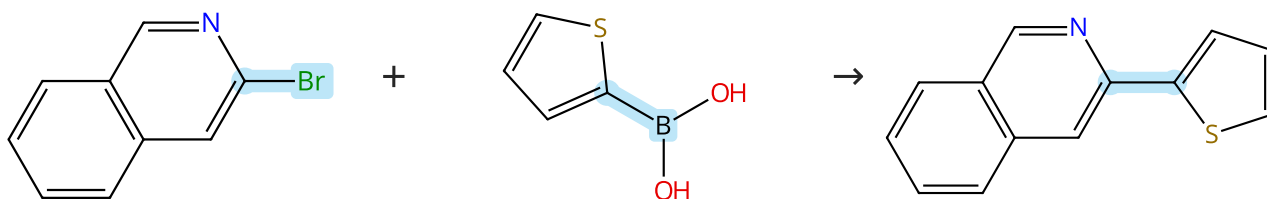
Suppliers (88)

Suppliers (60)

31-179-CAS-17501500	Steps: 1 Yield: 97%	Copper-catalyzed oxidative C(sp³)-H/C(sp²)-H cross-coupling en route to carbocyclic rings
1.1 Reagents: Tripotassium phosphate Catalysts: Tricyclohexylphosphine, Palladium diacetate Solvents: Toluene, Water; 15 h, 100 °C		By: Wang, Rui; et al Chemical Science (2017), 8(5), 3838-3842.

Scheme 245 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (72)

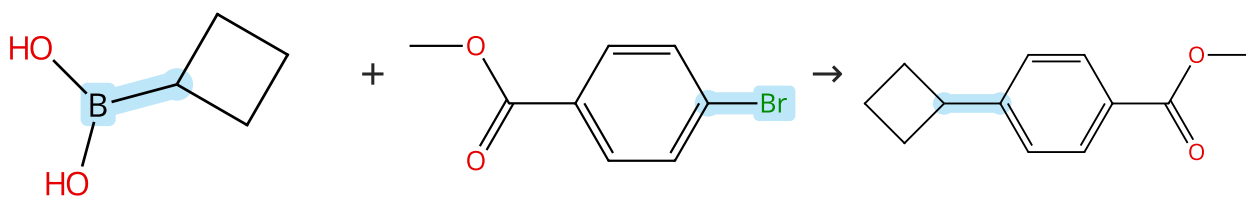
Suppliers (102)

Supplier (1)

31-179-CAS-23117393	Steps: 1 Yield: 97%	Regioselective Ru(II)/Pd(0) Dual Catalysis: One-Pot C-H Diaryl ation of Five-Membered Heterocyclic Derivatives
1.1 Reagents: Potassium carbonate, Triphenylphosphine Catalysts: Palladium diacetate Solvents: 1,4-Dioxane, Water; 24 h, 100 °C		By: Al Mamari, Hamad H.; et al Journal of Organic Chemistry (2021), 86(4), 3138-3151.
Experimental Protocols		

Scheme 246 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (67)

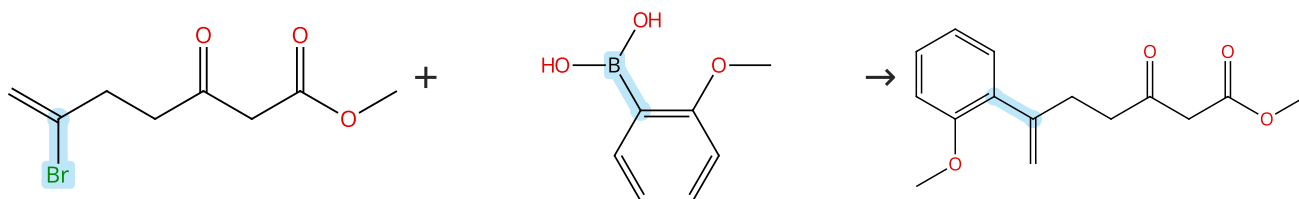
Suppliers (96)

Suppliers (35)

31-614-CAS-42144746	Steps: 1 Yield: 97%	Bromine radical release from a nickel-complex facilitates the activation of alkyl boronic acids: a boron selective Suzuki-Miyaura cross coupling
1.1 Reagents: Cesium carbonate Catalysts: Tetrabutylammonium bromide, 2,4,5,6-Tetra(9 <i>H</i> -carbazol-9-yl)isophthalonitrile, Nickel, [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-κ <i>N</i> ¹ ,κ <i>N</i> ¹]dibromo-, (<i>T</i> -4)- Solvents: 1,4-Dioxane; 24 h, 40 °C		By: Oliva, Monica; et al Chemical Science (2024), 15(42), 17490-17497.
Experimental Protocols		

Scheme 247 (1 Reaction)

Steps: 1 Yield: 97%



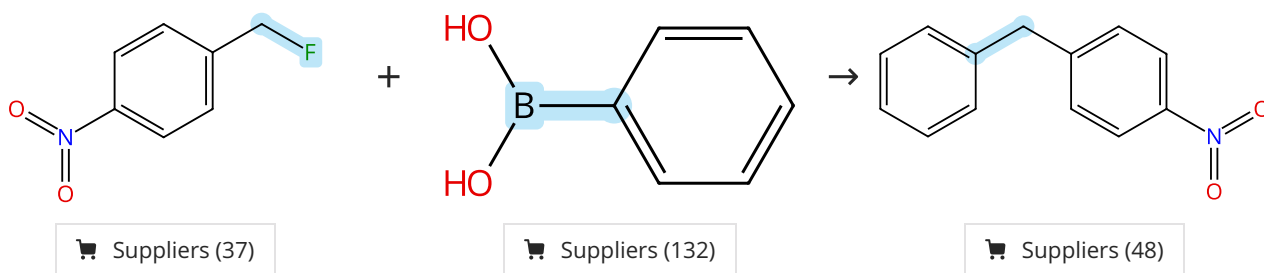
Supplier (1)

Suppliers (110)

31-179-CAS-21825235 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene; 3 h, 80 °C 1.2 Solvents: Water; pH 7 Experimental Protocols	Stepwise approach for sterically hindered C_{sp}3-C_{sp}3 bond formation by dehydrogenative O-alkylation and Lewis acid-catalyzed [1,3]-rearrangement towards the arylalkylcyclopentane skeleton of sesquiterpenes By: Fujitani, Ban; et al Chemical Communications (Cambridge, United Kingdom) (2020), 56(25), 3621-3624.
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Scheme 248 (1 Reaction)

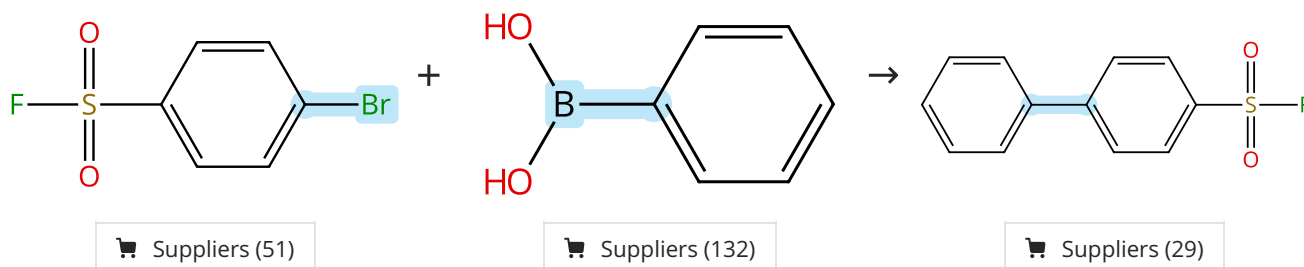
Steps: 1 Yield: 97%



31-614-CAS-24622826 Steps: 1 Yield: 97% 1.1 Reagents: Cesium carbonate, Lithium iodide Catalysts: Palladium diacetate, Tri-<i>o</i>-tolylphosphine Solvents: Toluene; 18 h, 100 °C Experimental Protocols	Palladium and Nickel Catalyzed Suzuki Cross-Coupling with Alkyl Fluorides By: Balaraman, Kaluvu; et al Organic Letters (2021), 23(22), 8994-8999.
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Scheme 249 (1 Reaction)

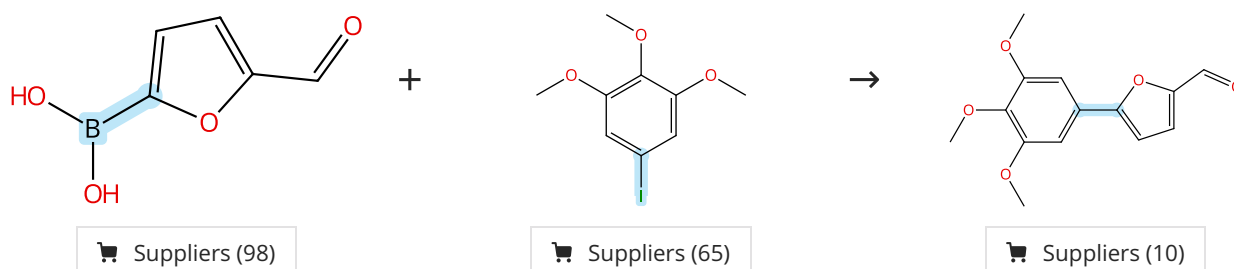
Steps: 1 Yield: 97%



31-179-CAS-12271513 Steps: 1 Yield: 97% 1.1 Reagents: Triethylamine Catalysts: Palladium diacetate Solvents: Water; 30 min, rt Experimental Protocols	On-Water Synthesis of Biaryl Sulfonyl Fluorides By: Chinthakindi, Praveen K.; et al Journal of Organic Chemistry (2016), 81(6), 2618-2623.
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Scheme 250 (1 Reaction)

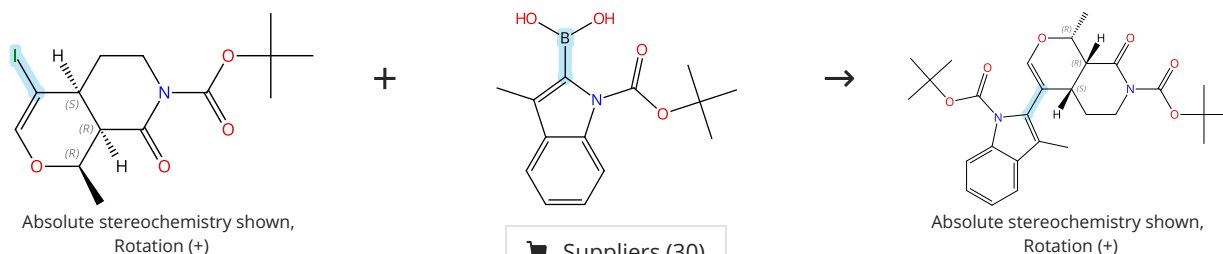
Steps: 1 Yield: 97%



31-179-CAS-16816930 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 1 - 3 h, rt → 65 °C Experimental Protocols	Novel inhibitors of Plasmodium falciparum based on 2,5-disubstituted furans By: Krake, Susann H.; et al European Journal of Medicinal Chemistry (2017), 126, 929-936.
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Scheme 251 (1 Reaction)

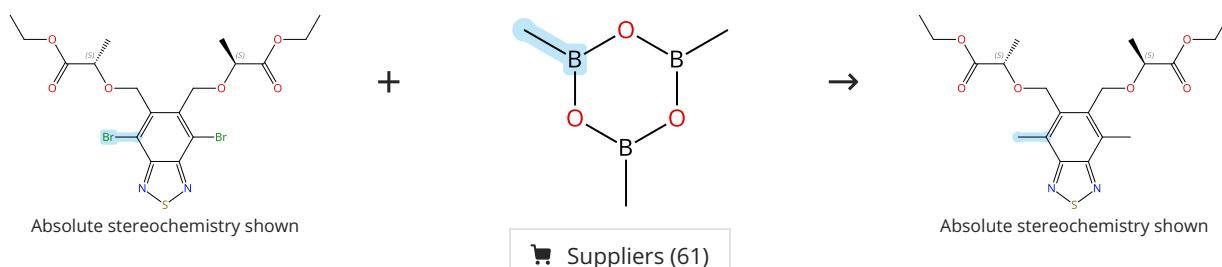
Steps: 1 Yield: 97%



31-179-CAS-19349924 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl(2',6'-dimethoxy[1,1'-biphenyl]-2-yl)phosphine-κP]palladium Solvents: Toluene, Water; 2 h, 60 °C Experimental Protocols	Enantioselective and Divergent Syntheses of Alstoscholarisines A, E and Their Enantiomers By: Hu, Lu; et al Organic Letters (2018), 20(19), 6202-6205.
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Scheme 252 (1 Reaction)

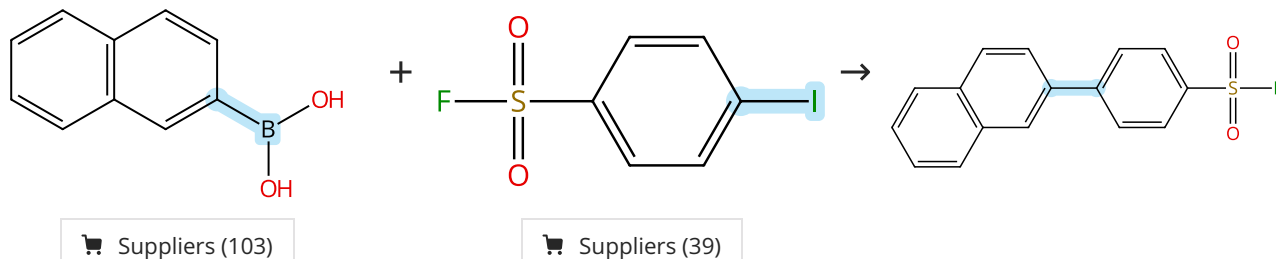
Steps: 1 Yield: 97%



31-179-CAS-6809503 Steps: 1 Yield: 97% 1.1 Reagents: Potassium fluoride Catalysts: Tri-<i>tert</i>-butylphosphine, Bis(dibenzylideneacetone)palladium Solvents: Tetrahydrofuran, Water; 3 h, reflux Experimental Protocols	Exerting Control over the Helical Chirality in the Main Chain of Sergeants-and-Soldiers-Type Poly(quinoxaline-2,3-diyl)s by Changing from Random to Block Copolymerization Protocols By: Nagata, Yuuya; et al Journal of the American Chemical Society (2015), 137(12), 4070-4073.
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Scheme 253 (1 Reaction)

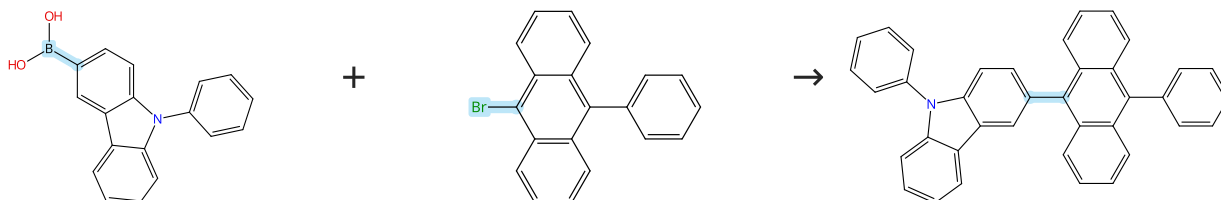
Steps: 1 Yield: 97%



31-179-CAS-9431811	Steps: 1 Yield: 97%	On-Water Synthesis of Biaryl Sulfonyl Fluorides
1.1 Reagents: Triethylamine Catalysts: Palladium diacetate Solvents: Water; 30 - 120 min, rt		By: Chinthakindi, Praveen K.; et al Journal of Organic Chemistry (2016), 81(6), 2618-2623.
Experimental Protocols		

Scheme 254 (1 Reaction)

Steps: 1 Yield: 97%



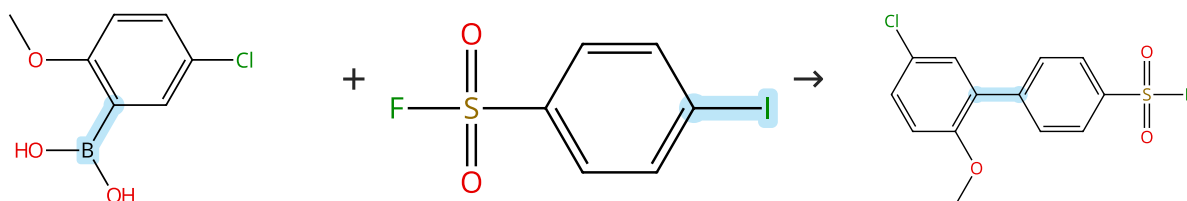
Suppliers (84)

Suppliers (84)

31-179-CAS-22039055	Steps: 1 Yield: 97%	Transition-Metal-Catalyzed Regioselective Functionalization of Monophosphino-o-Carboranes
1.1 Catalysts: Palladium diacetate, 2414549-20-3 Solvents: Toluene; 1 h, rt		By: Zhang, Zi-Yang; et al Chemistry - A European Journal (2020), 26(22), 5037-5050.
1.2 Reagents: Tripotassium phosphate Solvents: Ethanol, Toluene; 36 h, 120 °C		
Experimental Protocols		

Scheme 255 (1 Reaction)

Steps: 1 Yield: 97%



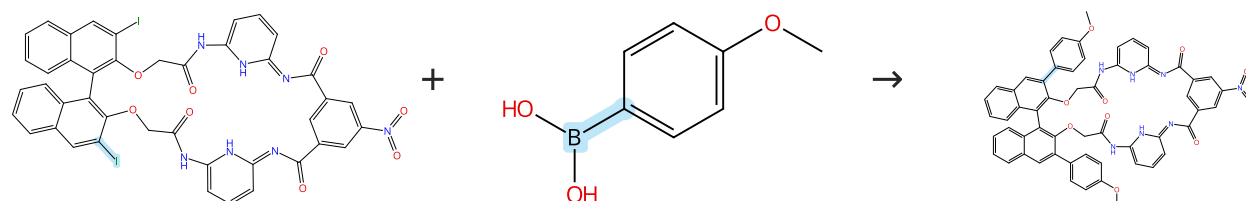
Suppliers (89)

Suppliers (39)

31-179-CAS-6972406	Steps: 1 Yield: 97%	On-Water Synthesis of Biaryl Sulfonyl Fluorides
1.1 Reagents: Triethylamine Catalysts: Palladium diacetate Solvents: Water; 30 - 120 min, rt		By: Chinthakindi, Praveen K.; et al Journal of Organic Chemistry (2016), 81(6), 2618-2623.
Experimental Protocols		

Scheme 256 (1 Reaction)

Steps: 1 Yield: 97%

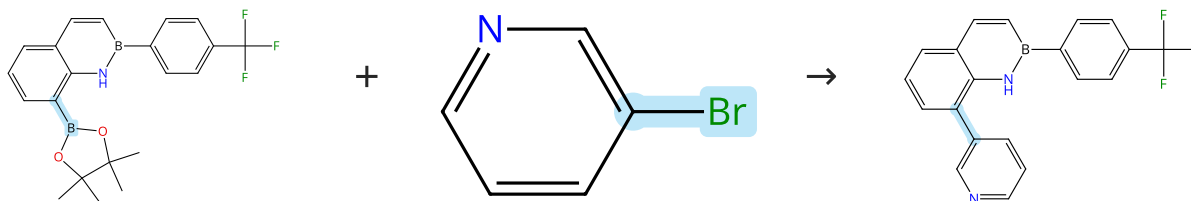


Suppliers (112)

31-179-CAS-20723777 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 6 h, 65 °C	Cross-Coupling Approach to an Array of Macrocyclic Receptors Functioning as Chiral Solvating Agents By: Ema, Tadashi; et al Journal of Organic Chemistry (2018), 83(18), 10762-10769.
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Scheme 257 (1 Reaction)

Steps: 1 Yield: 97%

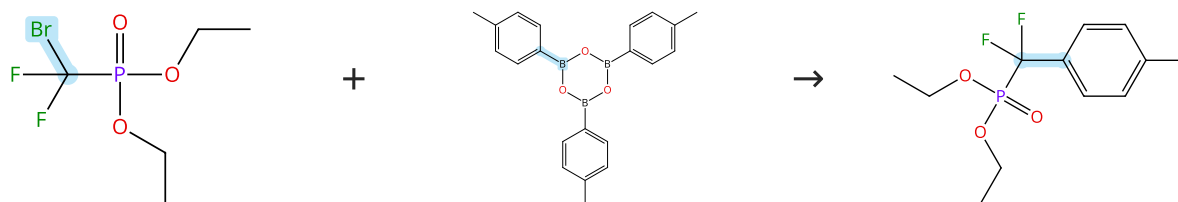


Suppliers (88)

31-179-CAS-17540644 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i>-4-4)-[2'-(Amino-κ<i>N</i>)[1,1'-biphenyl]-2-yl-κ<i>C</i>]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: <i>tert</i>-Butanol, Water; 18 h, 40 °C Experimental Protocols	Regioselective Diversification of 2,1-Borazaronaphthalenes: Unlocking Isosteric Space via C-H Activation By: Davies, Geraint H. M.; et al Journal of Organic Chemistry (2017), 82(15), 8072-8084.
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Scheme 258 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (78)

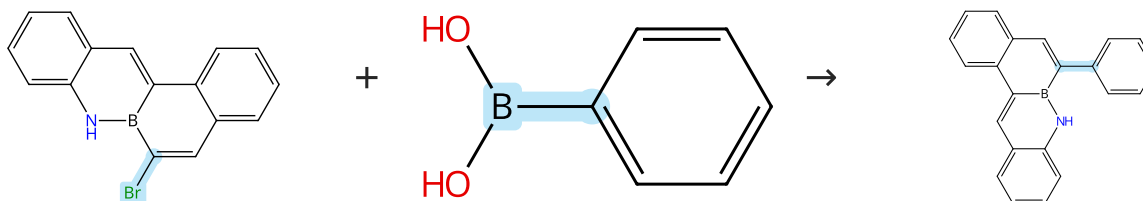
Suppliers (27)

Suppliers (2)

31-614-CAS-35372623 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i>-4-3)-Chloro-1-naphthalenylbis(triphenyl phosphine)nickel, 3,8-Diphenyl-1,10-phenanthroline Solvents: Ethyl acetate; 15 min, 80 °C; 80 °C → rt 1.2 5 h, 80 °C Experimental Protocols	Nickel-Catalyzed Cross-Coupling of Bromodifluoromethylphosphonates with Arylboron Reagents By: Kuriyama, Masami; et al Advanced Synthesis & Catalysis (2023), 365(1), 116-121.
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Scheme 259 (1 Reaction)

Steps: 1 Yield: 97%

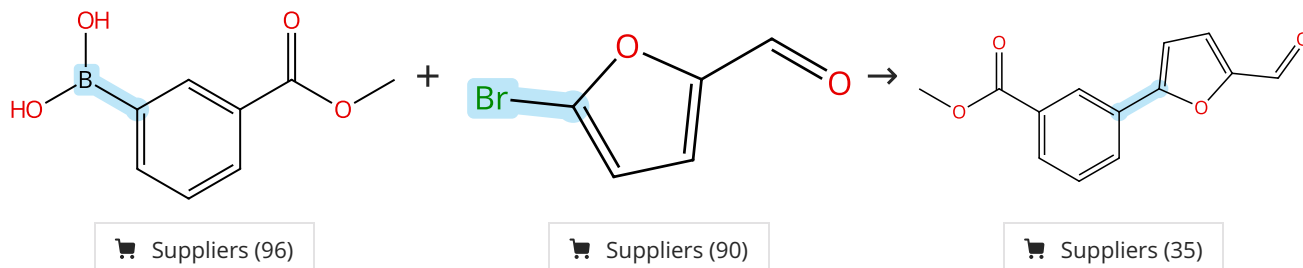


Suppliers (132)

31-614-CAS-25850709	Steps: 1 Yield: 97%	1,10a-Dihydro-1-aza-10a-boraphenanthrene and 6a,7-Dihydro-7-aza-6a-boratetraphene: Two New Fluorescent BN-PAHs
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene, Water; rt → 70 °C; 12 h, 70 °C		By: Valencia, Isabel; et al
Experimental Protocols		Journal of Organic Chemistry (2021), 86(23), 16259-16267.

Scheme 260 (1 Reaction)

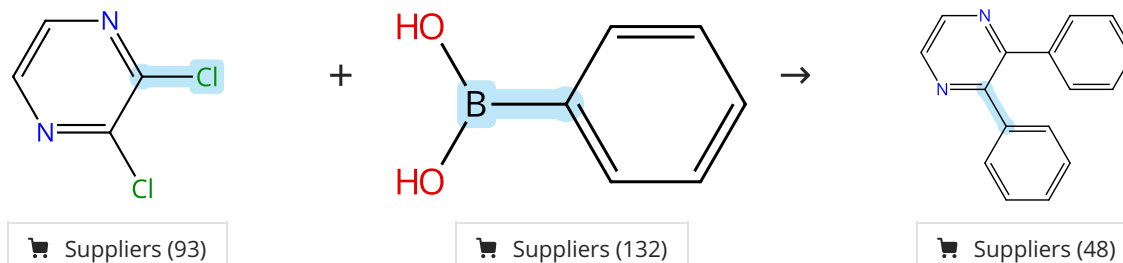
Steps: 1 Yield: 97%



31-179-CAS-5275808	Steps: 1 Yield: 97%	Artificial Ligands of Streptavidin (ALiS): Discovery, Characterization, and Application for Reversible Control of Intracellular Protein Transport
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,2-Dimethoxyethane, Water; 15 h, reflux		By: Terai, Takuya; et al
Experimental Protocols		Journal of the American Chemical Society (2015), 137(33), 10464-10467.

Scheme 261 (1 Reaction)

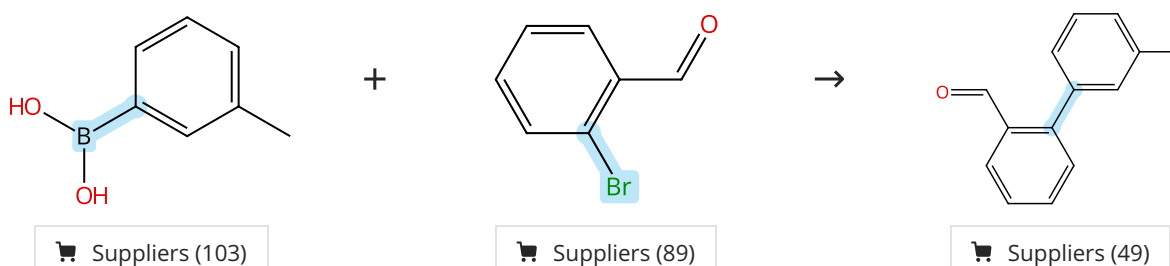
Steps: 1 Yield: 97%



31-179-CAS-15740755	Steps: 1 Yield: 97%	Synthesis of Chiral Piperazines via Hydrogenation of Pyrazines Activated by Alkyl Halides
1.1 Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; 10 min, rt → 95 °C; 95 °C → rt		By: Huang, Wen-Xue; et al
1.2 Reagents: Potassium fluoride; 95 °C		Organic Letters (2016), 18(13), 3082-3085.
Experimental Protocols		

Scheme 262 (1 Reaction)

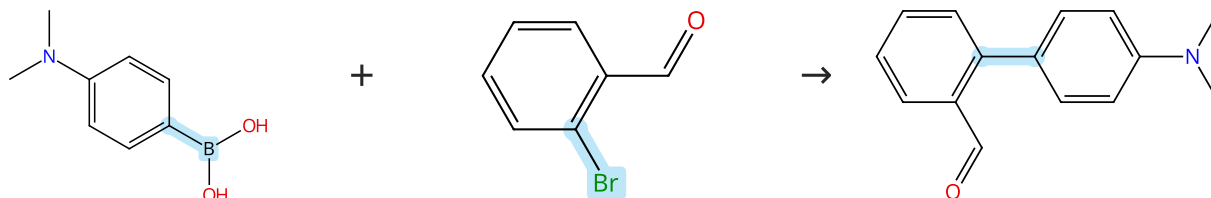
Steps: 1 Yield: 97%



31-179-CAS-23574845	Steps: 1 Yield: 97%	Imine as a linchpin approach for meta- C-H functionalization
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; 25 h, rt → 100 °C		By: Bag, Sukdev; et al Nature Communications (2021), 12(1), 1393.
Experimental Protocols		

Scheme 263 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (84)

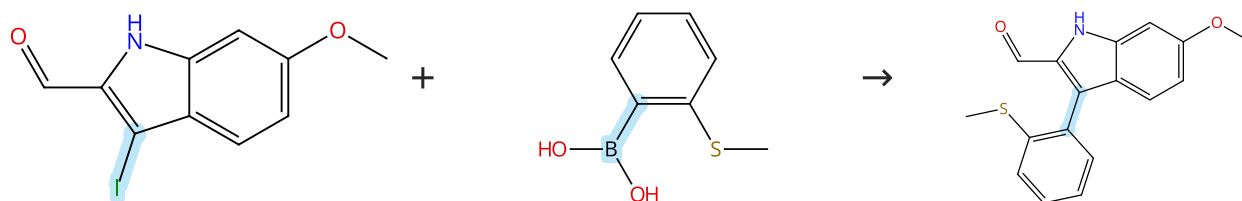
Suppliers (89)

Suppliers (10)

31-179-CAS-20129930	Steps: 1 Yield: 97%	Interplay between n→π* Interactions and Dynamic Covalent Bonds: Quantification and Modulation by Solvent Effects
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: 1,4-Dioxane; 12 h, 80 °C		By: Zheng, Hao; et al Journal of the American Chemical Society (2019), 141(22), 8825-8833.

Scheme 264 (1 Reaction)

Steps: 1 Yield: 97%



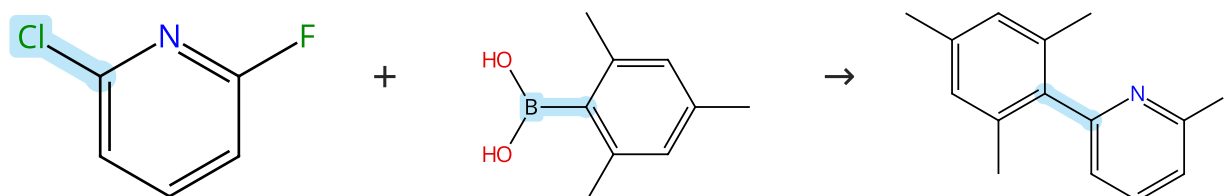
Supplier (1)

Suppliers (90)

31-614-CAS-37664336	Steps: 1 Yield: 97%	Dynamic Kinetic Resolution of Indole-Based Sulfenylated Heterobiaryls by Rhodium-Catalyzed Atroposelective Reductive Aldol Reaction
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene, Water; overnight, 70 °C; 70 °C → rt		By: Rodriguez-Franco, Carlos; et al ACS Catalysis (2023), 13(18), 12134-12141.
1.2 Reagents: Sodium chloride Solvents: Water		
Experimental Protocols		

Scheme 265 (1 Reaction)

Steps: 1 Yield: 97%



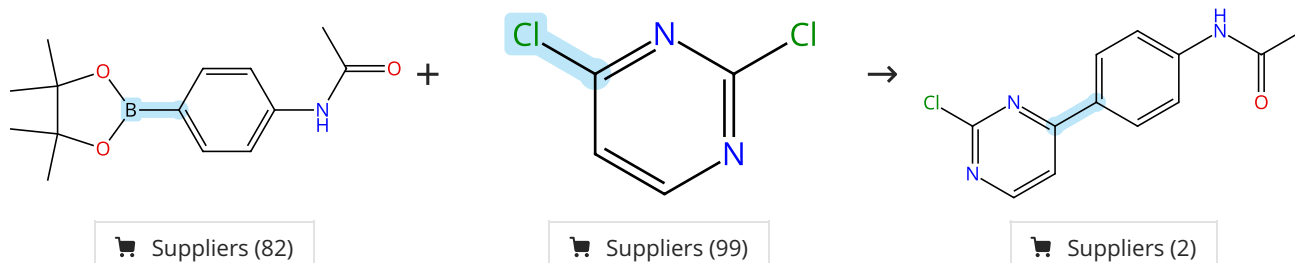
Suppliers (69)

Suppliers (91)

31-179-CAS-17819523 Steps: 1 Yield: 97% 1.1 Reagents: Potassium <i>tert</i>-butoxide Catalysts: (SP-4-1)-[7,9-Bis[2,6-bis(1-ethylpropyl)phenyl]-7,9-dihydro-8<i>H</i>-acenaphth[1,2-<i>d</i>]imidazol-8-ylidene]dichloro(3-chloropyridine-<i>κM</i>)palladium Solvents: <i>tert</i>-Butanol; 4 h, 80 °C Experimental Protocols	Bulky Yet Flexible Pd-PEPPSI-IPent^{An} for the Synthesis of Sterically Hindered Biaryls in Air By: Lu, Dong-Dong; et al Journal of Organic Chemistry (2017), 82(20), 10898-10911.
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Scheme 266 (1 Reaction)

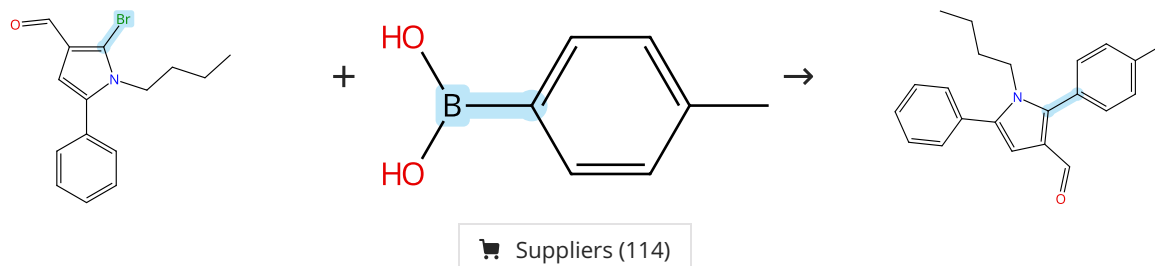
Steps: 1 Yield: 97%



31-179-CAS-19921458 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Acetonitrile, Water; overnight, reflux Experimental Protocols	Design and synthesis of potent dual inhibitors of JAK2 and HD AC based on fusing the pharmacophores of XL019 and vorinostat By: Chu-Farseeva, Yu-yi; et al European Journal of Medicinal Chemistry (2018), 158, 593-619.
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Scheme 267 (1 Reaction)

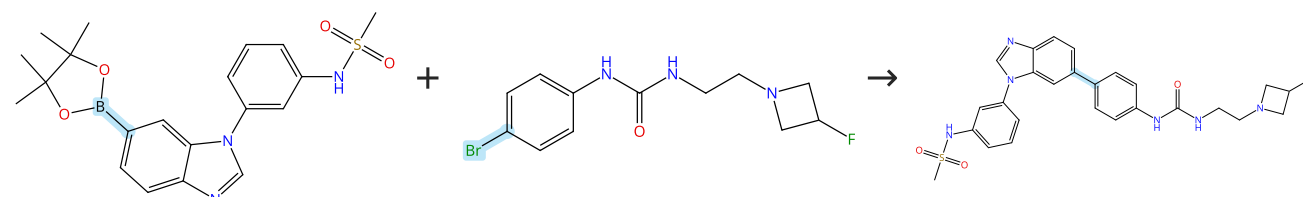
Steps: 1 Yield: 97%



31-614-CAS-31822967 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 16 h, 100 °C Experimental Protocols	Flexible Construction Approach to the Synthesis of 1,5-Substituted Pyrrole-3-carbaldehydes from 5-Bromo-1,2,3-triazine By: Wu, Chien-Chi; et al Organic Letters (2022), 24(15), 2889-2893.
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Scheme 268 (1 Reaction)

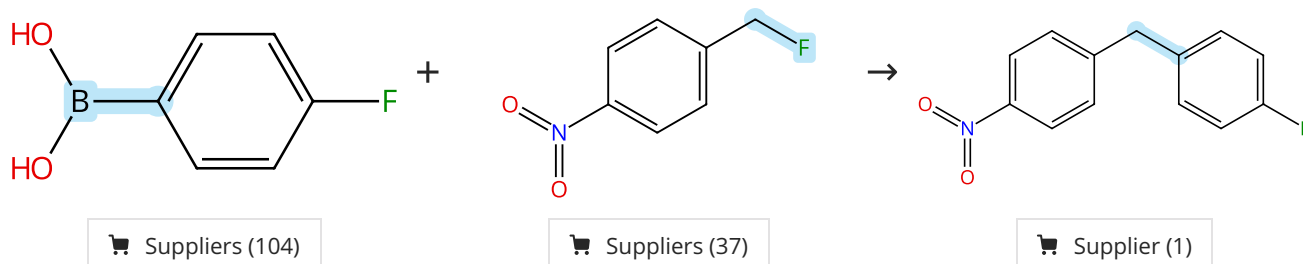
Steps: 1 Yield: 97%



31-614-CAS-36888940	Steps: 1 Yield: 97%	Design and Optimization of Novel Benzimidazole- and Imidazo[4,5-b]pyridine-Based ATM Kinase Inhibitors with Subnanomolar Activities By: Dimitrov, Teodor; et al Journal of Medicinal Chemistry (2023), 66(11), 7304-7330.
1.1 Reagents: Tripotassium phosphate Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[tris(1,1-dimethylethyl)phosphine]palladium Solvents: 1,4-Dioxane, Water; rt → 65 °C; overnight, 65 °C		
Experimental Protocols		

Scheme 269 (1 Reaction)

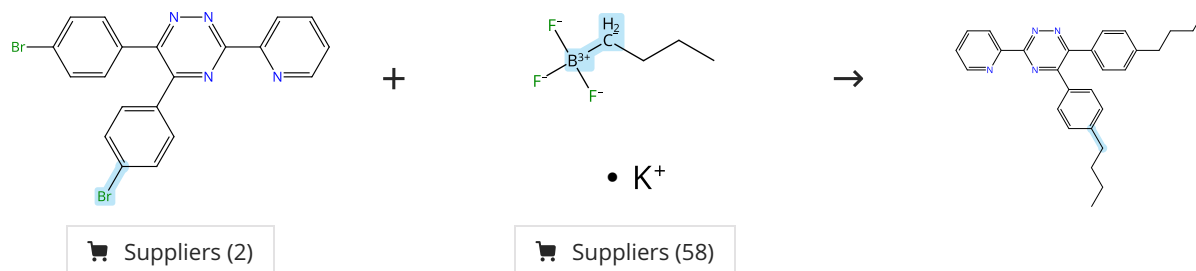
Steps: 1 Yield: 97%



31-614-CAS-24622838	Steps: 1 Yield: 97%	Palladium and Nickel Catalyzed Suzuki Cross-Coupling with Alkyl Fluorides By: Balaraman, Kaluvu; et al Organic Letters (2021), 23(22), 8994-8999.
1.1 Reagents: Cesium carbonate, Lithium iodide Catalysts: Palladium diacetate, Tri- <i>o</i> -tolylphosphine Solvents: Toluene; 18 h, 100 °C		
Experimental Protocols		

Scheme 270 (1 Reaction)

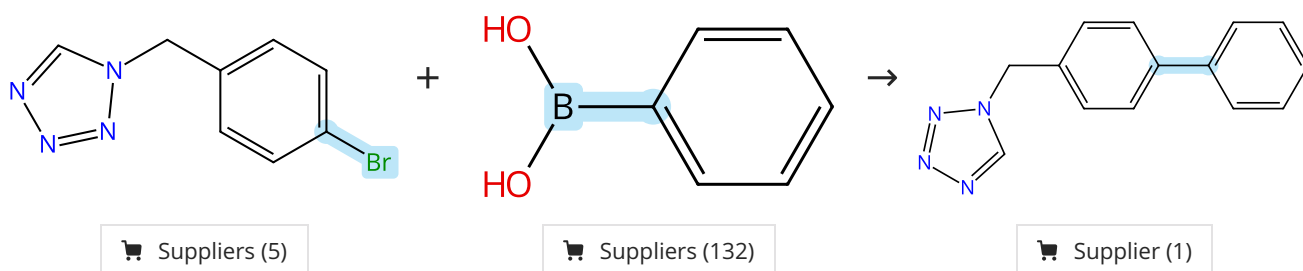
Steps: 1 Yield: 97%



31-179-CAS-18220416	Steps: 1 Yield: 97%	Modular Approaches to Diversified Soft Lewis Basic Complexants through Suzuki-Miyaura Cross-Coupling of Bromoheteroarenes with Organotrifluoroborates By: Chin, Ai Lin; et al Journal of Organic Chemistry (2016), 81(3), 1106-1115.
1.1 Reagents: Cesium carbonate Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine Solvents: Toluene, Water; rt → 115 °C; 16 h, 115 °C		
Experimental Protocols		

Scheme 271 (1 Reaction)

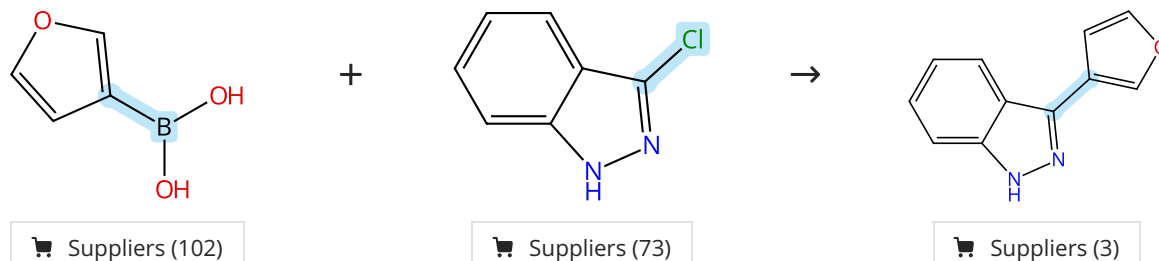
Steps: 1 Yield: 97%



31-179-CAS-17135979 Steps: 1 Yield: 97%	Aryl and Heteroaryl N1-Tetrazoles through Ligand-Free Suzuki-Reaction under Aerobic, Aqueous Conditions
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene; 18 h, 120 °C	By: Seifried, Marco; et al European Journal of Organic Chemistry (2017), 2017(17), 2416-2424.

Scheme 272 (1 Reaction)

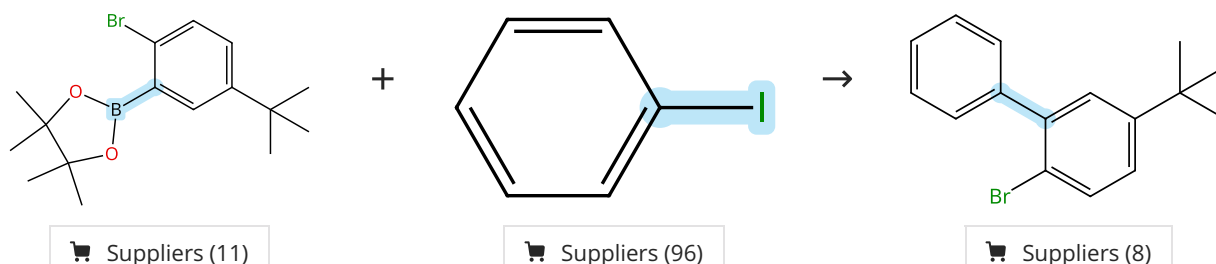
Steps: 1 Yield: 97%



31-179-CAS-12660794 Steps: 1 Yield: 97%	Design of a Versatile and Improved Precatalyst Scaffold for Palladium-Catalyzed Cross-Coupling: (η^3-1-^tBu-indenyl)₂(μ-Cl)₂ Pd₂
1.1 Reagents: Potassium carbonate Catalysts: Chloro[dicyclohexyl(2',6'-dimethoxy[1,1'-biphenyl]-2-yl)phosphine- κ P]((1,2,3- η)-1-(1,1-dimethylethyl)-1H-inden-1-yl)palladium Solvents: Methanol, 1,4-Dioxane; 15 h, 80 °C Experimental Protocols	By: Melvin, Patrick R.; et al ACS Catalysis (2015), 5(6), 3680-3688.

Scheme 273 (1 Reaction)

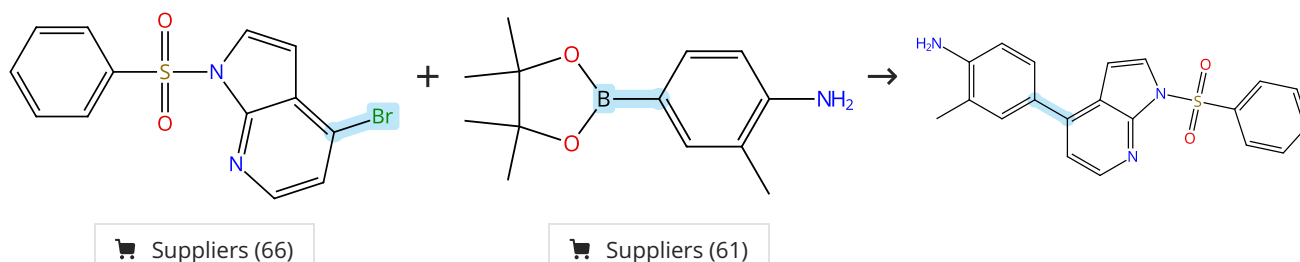
Steps: 1 Yield: 97%



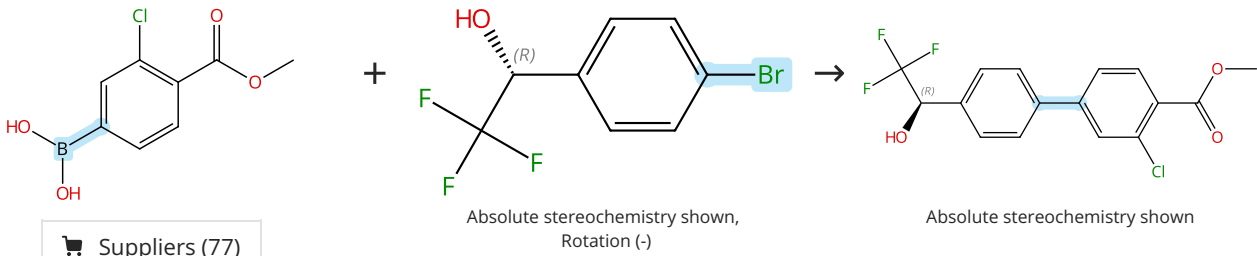
31-179-CAS-18964514 Steps: 1 Yield: 97%	A Water-Soluble Warped Nanographene: Synthesis and Applications for Photoinduced Cell Death
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Dimethylformamide; 24 h, 100 °C; 100 °C → rt Experimental Protocols	By: Lin, Hsing-An; et al Angewandte Chemie, International Edition (2018), 57(11), 2874-2878.

Scheme 274 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-39404854	Steps: 1 Yield: 97%	Discovery and Characterization of Selective, First-in-Class Inhibitors of Citron Kinase
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane, Water; 3 h, 110 °C		By: Maw, Joshua J.; et al Journal of Medicinal Chemistry (2024), 67(4), 2631-2666.
Experimental Protocols		

Scheme 275 (1 Reaction)	Steps: 1 Yield: 97%
	Absolute stereochemistry shown, Rotation (-) Absolute stereochemistry shown
 Suppliers (77)	 Suppliers (56)

31-614-CAS-36331519	Steps: 1 Yield: 97%	Automated, Capsule-Based Suzuki-Miyaura Cross Couplings
1.1 Reagents: Cesium carbonate Catalysts: Silica, Phosphine, diphenyl-, palladium(2+) salt (2:1) Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 3.5 h, 60 °C; 60 °C → rt		By: Coin, Guillaume; et al Organic Letters (2024), 26(14), 2708-2712.
1.2 Reagents: Silica		
1.3 Reagents: Methanaminium, <i>N,N,N</i> -trimethyl-, carbonate (2:1) (reagent supported); 2 h, rt		
Experimental Protocols		

Scheme 276 (1 Reaction)

Steps: 1 Yield: 97%

+

 Suppliers (112)

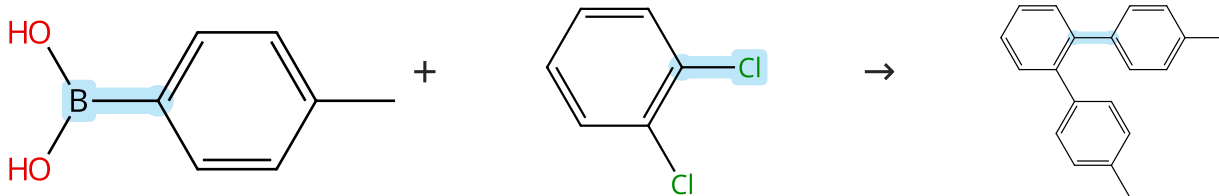
 Suppliers (69)

 Supplier (1)

31-179-CAS-3722041	Steps: 1 Yield: 97%	A general protocol for the efficient synthesis of polyarylated benzenes by multiple Suzuki-Miyaura reactions of polychlorinated benzenes
1.1 Reagents: Tripotassium phosphate Catalysts: Tris(dibenzylideneacetone)dipalladium, CataCXium A Solvents: Toluene; 24 h, 110 °C		By: Reimann, Sebastian; et al Tetrahedron (2016), 72(8), 1083-1094.

Scheme 277 (1 Reaction)

Steps: 1 Yield: 97%



Chemical reaction scheme showing the synthesis of 1,2-bis(4-methylphenyl)benzene. The reaction involves 4-methylphenylboronic acid (left) reacting with 1,2-dichlorobenzene (middle) to form 1,2-bis(4-methylphenyl)benzene (right). The product is shown with absolute stereochemistry.

 Suppliers (114)

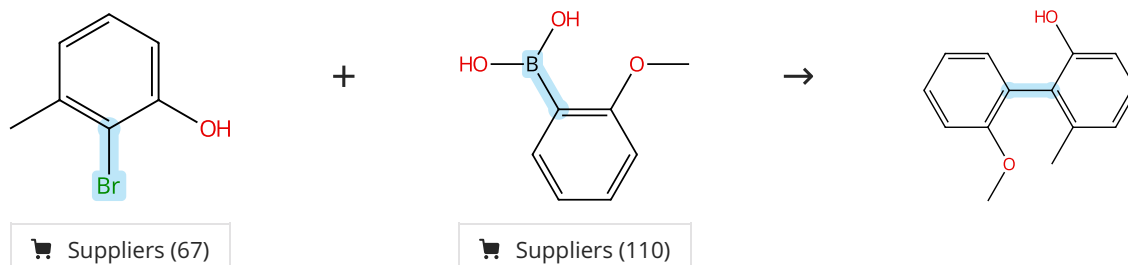
 Suppliers (132)

 Suppliers (10)

31-179-CAS-17819434 Steps: 1 Yield: 97% 1.1 Reagents: Potassium <i>tert</i>-butoxide Catalysts: (<i>SP</i>-4-1)-[7,9-Bis[2,6-bis(1-ethylpropyl)phenyl]-7,9-dihydro-8<i>H</i>-acenaphth[1,2-<i>d</i>]imidazol-8-ylidene]dichloro(3-chloropyridine-κM)palladium Solvents: <i>tert</i>-Butanol; 4 h, 80 °C Experimental Protocols	Bulky Yet Flexible Pd-PEPPSI-IPent^{An} for the Synthesis of Sterically Hindered Biaryls in Air By: Lu, Dong-Dong; et al Journal of Organic Chemistry (2017), 82(20), 10898-10911.
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Scheme 278 (1 Reaction)

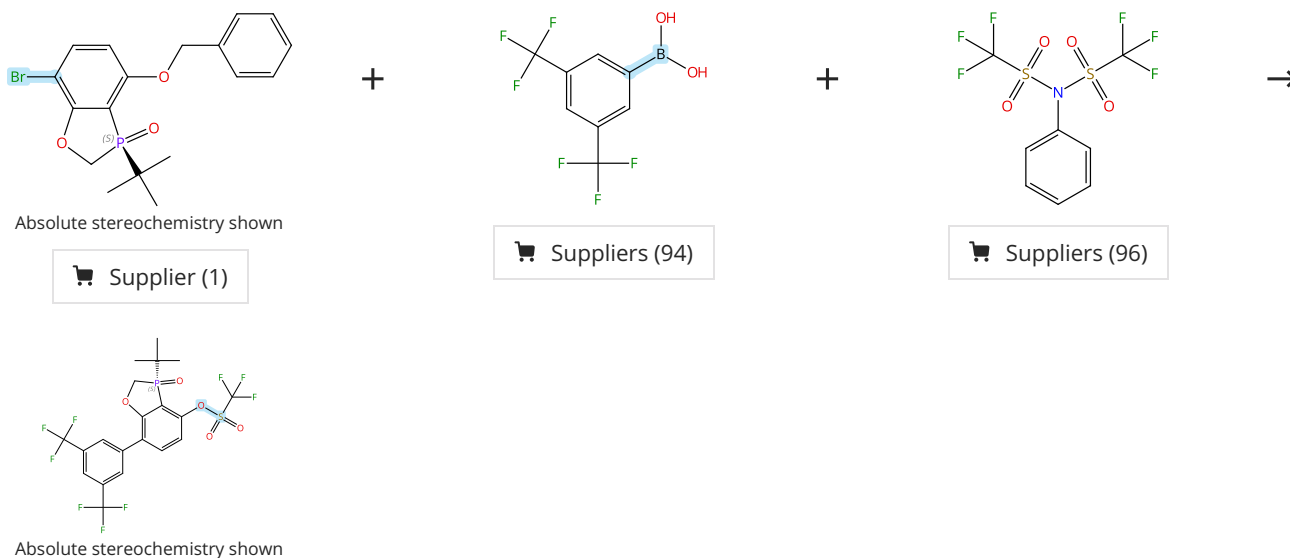
Steps: 1 Yield: 97%



31-614-CAS-35318567 Steps: 1 Yield: 97% 1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: Tetrahydrofuran, Water; 16 h, 70 °C Experimental Protocols	Dearomatization of Biaryls through Polarity Mismatched Radical Spirocyclization By: Azpilcueta-Nicolas, Carlos R.; et al Angewandte Chemie, International Edition (2023), 62(6), e202215422.
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Scheme 279 (1 Reaction)

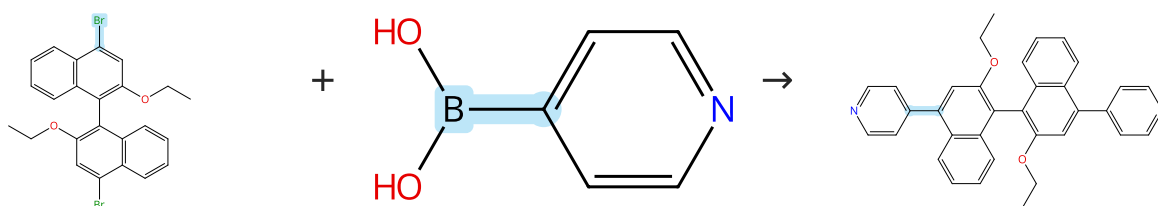
Steps: 1 Yield: 97%



31-179-CAS-18649036 Steps: 1 Yield: 97% 1.1 Reagents: Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: 1,4-Dioxane, Water; 2 h, rt → 80 °C 1.2 Reagents: Hydrogen Catalysts: Palladium Solvents: Methanol; 48 h, 300 psi, rt 1.3 Reagents: Triethylamine Solvents: Dichloromethane; 2 h, rt Experimental Protocols	BABIPhos Family of Biaryl Dihydrobenzooxaphosphole Ligands for Asymmetric Hydrogenation By: Li, Guisheng; et al Organic Letters (2018), 20(7), 1725-1729.
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Scheme 280 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

31-614-CAS-34984349

Steps: 1 Yield: 97%

A chiral binaphthyl-based coordination polymer as an enantioselective fluorescence sensor

By: Thoonen, Shannon; et al

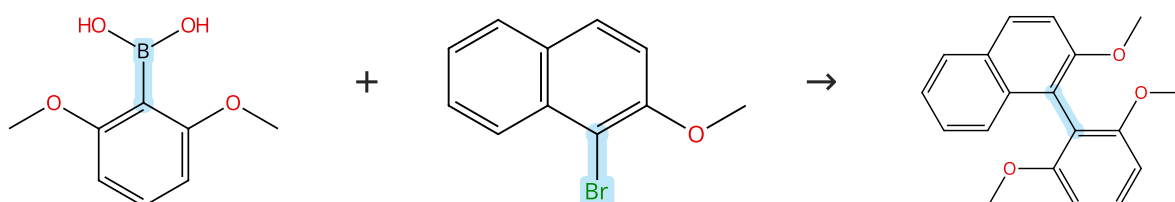
Chemical Communications (Cambridge, United Kingdom) (2022), 58(28), 4512-4515.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 16 h, rt → reflux

Experimental Protocols

Scheme 281 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (91)

Suppliers (76)

Suppliers (2)

31-614-CAS-46328575

Steps: 1 Yield: 97%

Highly Active Catalyst for Suzuki-Miyaura Coupling to Form Sterically Demanding Biaryls: Electronic Control of Pd through the Secondary Interaction Using a Buchwald-Type Ligand Bearing a Fluorinated Aryl Ring

By: Korenaga, Toshinobu; et al

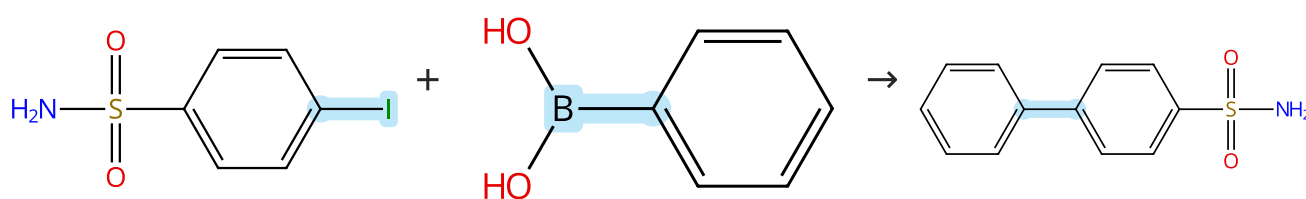
Organic Letters (2025), 27(25), 6588-6592.

1.1 **Reagents:** Potassium carbonate
Catalysts: 3096148-22-7
Solvents: Isopropanol, Water; 12 h, 60 °C

Experimental Protocols

Scheme 282 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (60)

Suppliers (132)

Suppliers (34)

31-179-CAS-7465123

Steps: 1 Yield: 97%

4-Arylbenzenesulfonamides as Human Carbonic Anhydrase Inhibitors (hCAIs): Synthesis by Pd Nanocatalyst-Mediated Suzuki-Miyaura Reaction, Enzyme Inhibition, and X-ray Crystallographic Studies

By: Cornelio, Benedetta; et al

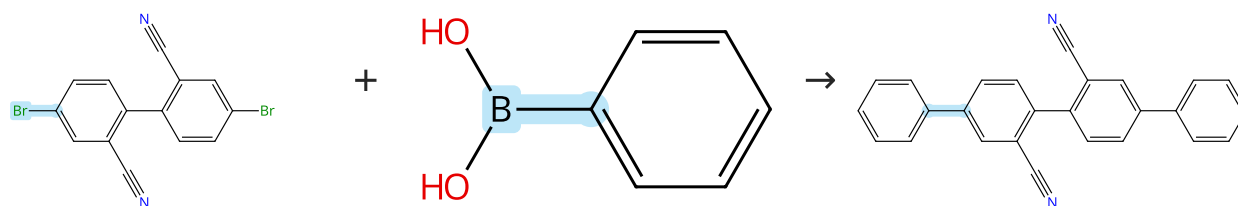
Journal of Medicinal Chemistry (2016), 59(2), 721-732.

1.1 **Reagents:** Sodium acetate
Catalysts: Palladium (dodecanethiol stabilized, adsorbed on multi walled carbon nanotubes)
Solvents: Methanol; 8 h, 70 °C; 70 °C → rt

Experimental Protocols

Scheme 283 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (2)

Suppliers (132)

31-179-CAS-21536041

Steps: 1 Yield: 97%

Intermolecular coupling and intramolecular cyclization of aryl nitriles on Au(111)

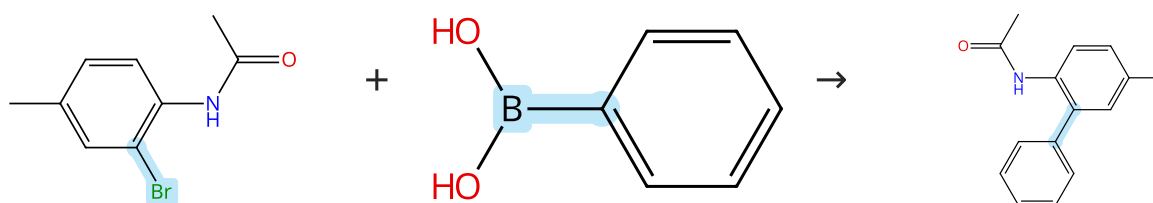
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; rt; 69 h, 80 °C

By: Klaasen, Henning; et al

Chemical Communications (Cambridge, United Kingdom) (2019), 55(77), 11611-11614.

Scheme 284 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (69)

Suppliers (132)

Suppliers (14)

31-614-CAS-37395346

Steps: 1 Yield: 97%

Pd(II)/Lewis Acid Catalyzed Intramolecular Oxidative C-H Amination to Construct Carbazoles with Dioxygen

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: Toluene; 90 °C

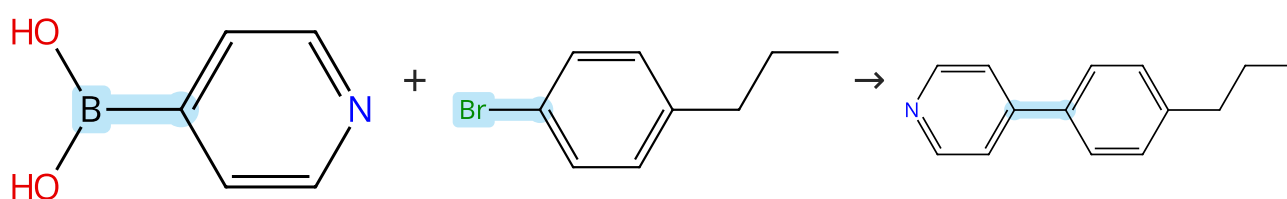
By: Jiang, Hongwu; et al

European Journal of Organic Chemistry (2023), 26(35), e202300598.

Experimental Protocols

Scheme 285 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

Suppliers (79)

Suppliers (10)

31-614-CAS-32528415

Steps: 1 Yield: 97%

Facile Synthesis of 2-Methylnicotinonitrile through Degenerate Ring Transformation of Pyridinium Salts

1.1 **Reagents:** Potassium carbonate
Catalysts: (*SP*4-2)-Dichloro[1,1'-(1,3-propanediyl)bis[1,1-diphenylphosphine-κ^P]]palladium
Solvents: 1,4-Dioxane, Water; overnight, reflux; reflux → rt

By: Duan, Xiaoxia; et al

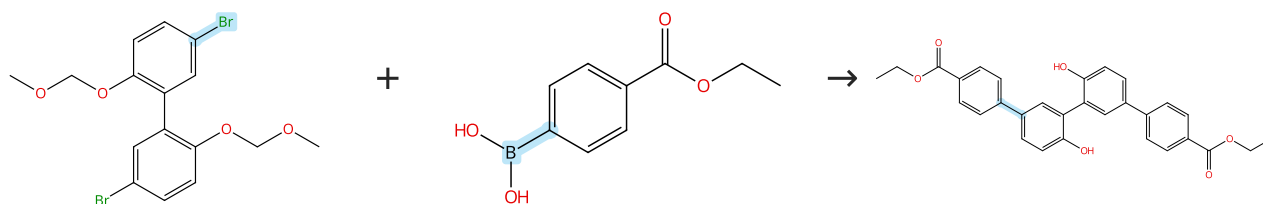
Journal of Organic Chemistry (2022), 87(12), 7975-7988.

1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 286 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (8)

Suppliers (95)

31-614-CAS-42499504

Steps: 1 Yield: 97%

Palladium-Catalyzed Double ipso-Defluoroetherification of Allylic gem-Difluorides with Phenols or Alcohols

By: Chen, Sheng-yu; et al

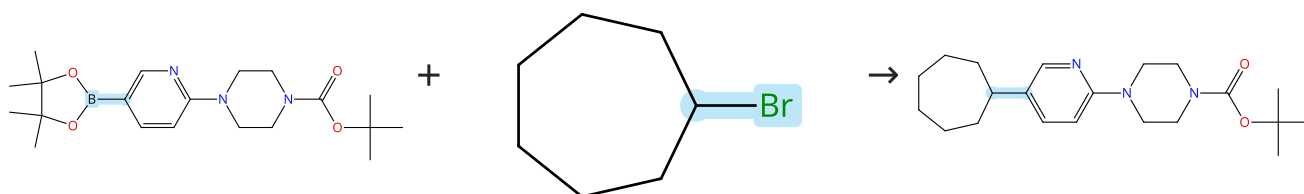
Organic Letters (2024), 26(45), 9799-9803.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane; 24 h, 80 °C
- 1.2 **Reagents:** *p*-Toluenesulfonic acid
Solvents: Methanol; 6 h, 60 °C

Experimental Protocols

Scheme 287 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (73)

Suppliers (81)

31-614-CAS-40989527

Steps: 1 Yield: 97%

A Rationally Designed Iron(II) Catalyst for C(sp³)-C(sp²) and C(sp³)-C(sp³) Suzuki-Miyaura Cross-Coupling

By: Chen, Donghuang; et al

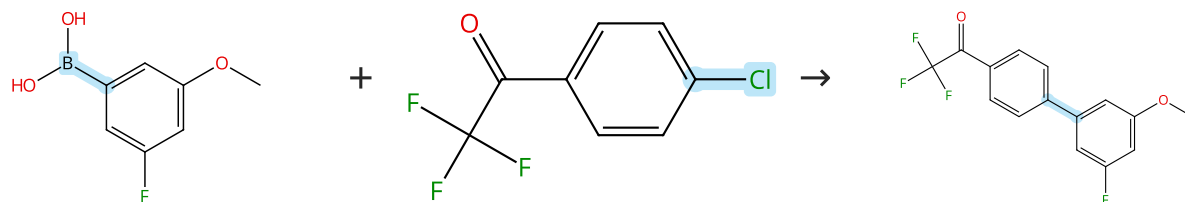
Angewandte Chemie, International Edition (2024), 63(32), e202408419.

- 1.1 **Reagents:** Ethanamine, *N*-methyl-, lithium salt
Catalysts: 3050668-44-2
Solvents: Benzene; 24 h, 25 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water; 25 °C

Experimental Protocols

Scheme 288 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (80)

Suppliers (72)

31-614-CAS-35010251

Steps: 1 Yield: 97%

Directing Group-Free Formal Suzuki-Miyaura Coupling of Simple Ketones Enabled by Activation of Unstrained C-C Bonds

By: Huang, Jiangkun; et al

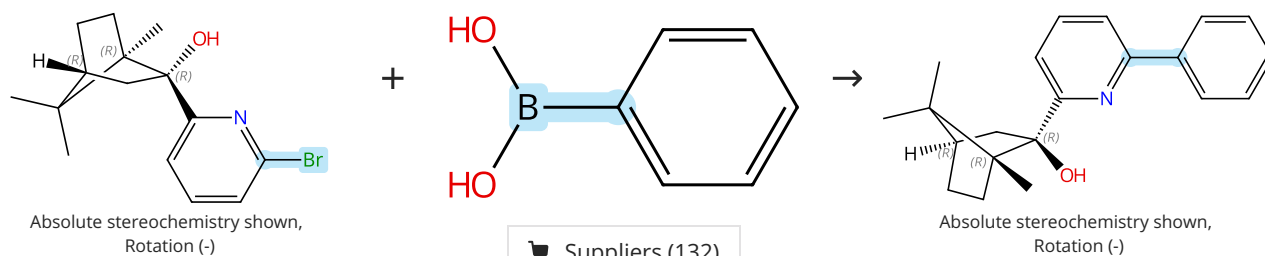
Angewandte Chemie, International Edition (2022), 61(49), e202211080.

- 1.1 **Reagents:** Potassium fluoride, Tri-*tert*-butylphosphonium tetrafluoroborate
Catalysts: Tris(dibenzylideneacetone)dipalladium
Solvents: Tetrahydrofuran; overnight, rt

Experimental Protocols

Scheme 289 (1 Reaction)

Steps: 1 Yield: 97%



Supplier (1)

Suppliers (132)

31-179-CAS-22989702

Steps: 1 Yield: 97%

Studies towards Pyridine-Based N,N,O-Gold(III) Complexes: Synthesis, Characterization and Application

By: Solvi, Thomas Nordboe; et al

European Journal of Organic Chemistry (2020), 2020(45), 7062-7068.

1.1 Reagents: Potassium carbonate

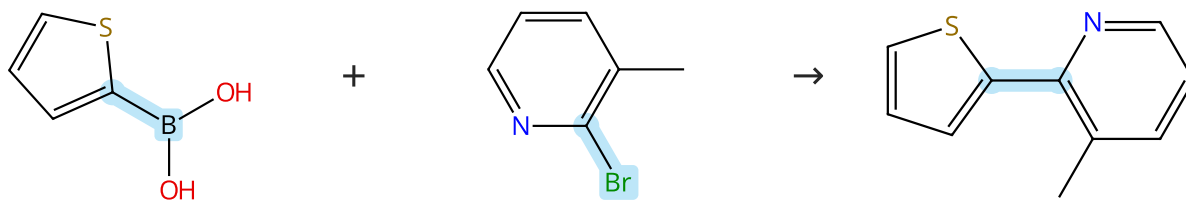
Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: 1,4-Dioxane, Water; 3 h, 70 °C

Experimental Protocols

Scheme 290 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

Suppliers (96)

Suppliers (4)

31-614-CAS-33767306

Steps: 1 Yield: 97%

Ru(II)-Catalyzed Hydroarylation of in situ Generated 3,3,3-Trifluoro-1-propyne by C-H Bond Activation: A Facile and Practical Access to β -Trifluoromethylstyrenes

By: Vuagnat, Martin; et al

Chemistry - A European Journal (2022), 28(55), e202201928.

1.1 Reagents: Sodium carbonate

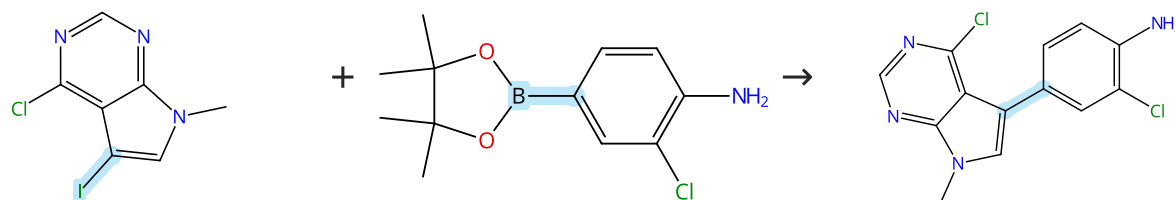
Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: Ethanol, Toluene, Water; 16 h, 100 °C

Experimental Protocols

Scheme 291 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (56)

Suppliers (68)

31-179-CAS-23716417

Steps: 1 Yield: 97%

Discovery of pyrrolo[2,3-d]pyrimidine derivatives as potent Axl inhibitors: Design, synthesis and biological evaluation

By: Xu, Dandan; et al

European Journal of Medicinal Chemistry (2021), 220, 113497.

1.1 Reagents: Sodium carbonate

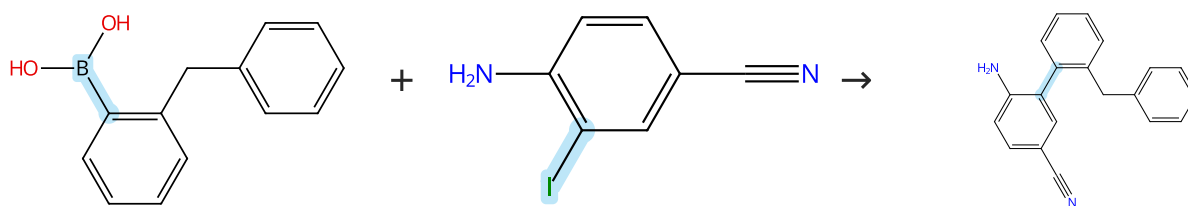
Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: 1,4-Dioxane, Water; 14 h, rt $\rightarrow$ 90 °C

Experimental Protocols

Scheme 292 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (54)

Suppliers (69)

31-179-CAS-17867791

Steps: 1 Yield: 97%

Coupling N-H Deprotonation, C-H Activation, and Oxidation: Metal-Free C(sp³)-H Aminations with Unprotected Anilines

By: Evoniuk, Christopher J.; et al

Journal of the American Chemical Society (2017), 139(45), 16210-16221.

1.1 Reagents: Potassium carbonate

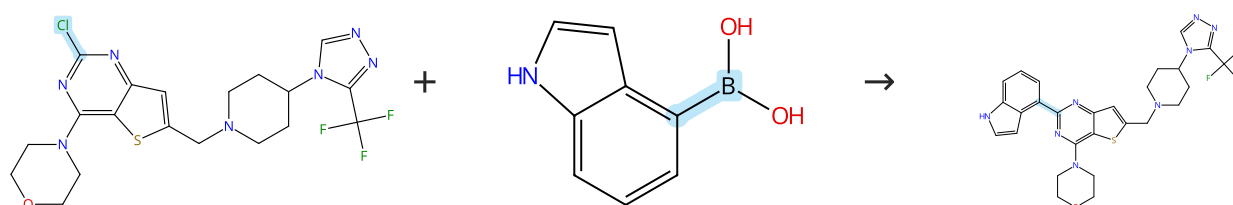
Solvents: Ethanol, Toluene; 0.5 h, rt

1.2 Catalysts: Tetrakis(triphenylphosphine)palladium; 8 h, reflux

1.3 Solvents: Ethyl acetate, Water; rt

Scheme 293 (1 Reaction)

Steps: 1 Yield: 97%



Supplier (1)

Suppliers (78)

Supplier (1)

31-179-CAS-16619291

Steps: 1 Yield: 97%

Design and Elaboration of a Tractable Tricyclic Scaffold To Synthesize Druglike Inhibitors of Dipeptidyl Peptidase-4 (DPP-4), Antagonists of the C-C Chemokine Receptor Type 5 (CCR5), and Highly Potent and Selective Phosphoinositol-3 Kinase δ (PI3K δ) Inhibitors

By: Schwehm, Carolin; et al

Journal of Medicinal Chemistry (2017), 60(4), 1534-1554.

1.1 Reagents: Sodium carbonate

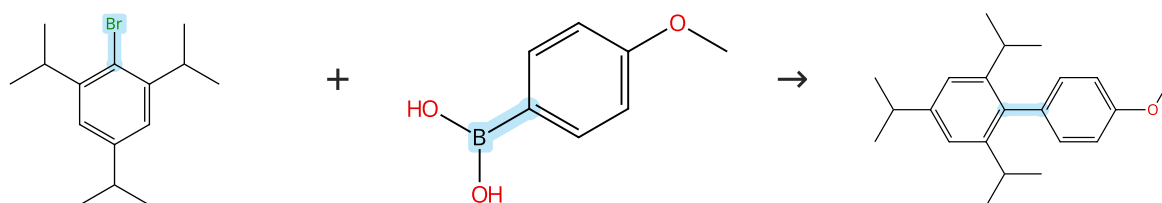
Catalysts: Dichlorobis(triphenylphosphine)palladium

Solvents: Ethanol, Water; 45 min, 125 °C

Experimental Protocols

Scheme 294 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (87)

Suppliers (112)

Suppliers (5)

31-179-CAS-16868996

Steps: 1 Yield: 97%

D_{3h}-Symmetrical Shape-Persistent Macrocycles Consisting of Pyridine-Acetylene-Phenol Conjugates as an Efficient Host Architecture for Saccharide Recognition

By: Abe, Hajime; et al

Chemistry - A European Journal (2016), 22(52), 18944-18952.

1.1 Reagents: Tripotassium phosphate

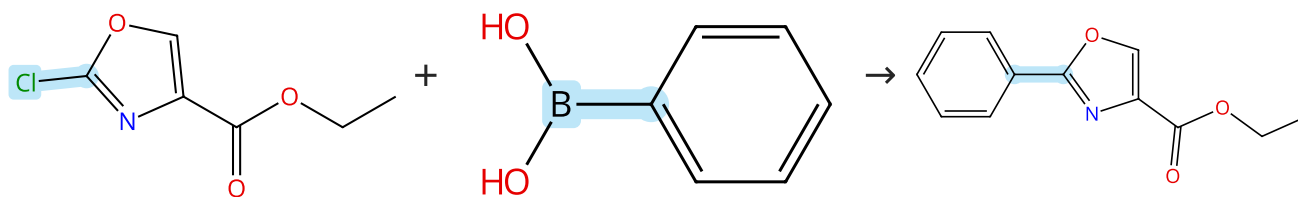
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl

Solvents: Toluene; 13 h, reflux

Experimental Protocols

Scheme 295 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (83)

Suppliers (132)

Suppliers (56)

31-179-CAS-19636491

Steps: 1 Yield: 97%

Asymmetric synthesis and biological evaluation of imidazole- and oxazole-containing synthetic lipoxin A₄ mimetics (sLXms)

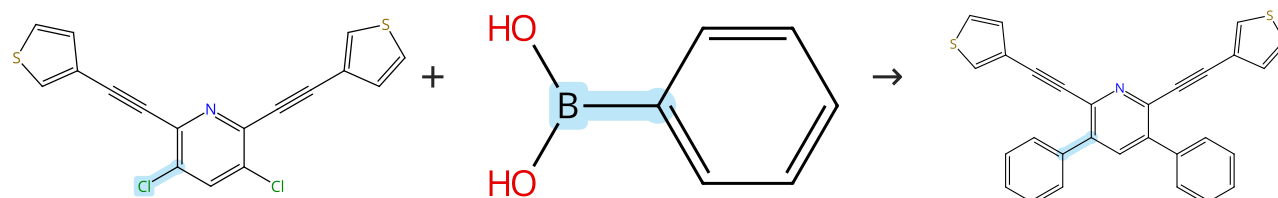
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene; 2.5 h, 90 °C

By: de Gaetano, Monica; et al

European Journal of Medicinal Chemistry (2019), 162, 80-108.

Scheme 296 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

31-179-CAS-22754043

Steps: 1 Yield: 97%

Synthesis and Properties of Dibenzo [a,j]acridines

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tris(dibenzylideneacetone)dipalladium, Butyltricyclo[3.3.1.1^{3,7}]dec-1-yltricyclo[3.3.1.1^{3,7}]dec-2-ylphosphine
Solvents: Dimethylformamide; 22 h, 130 °C

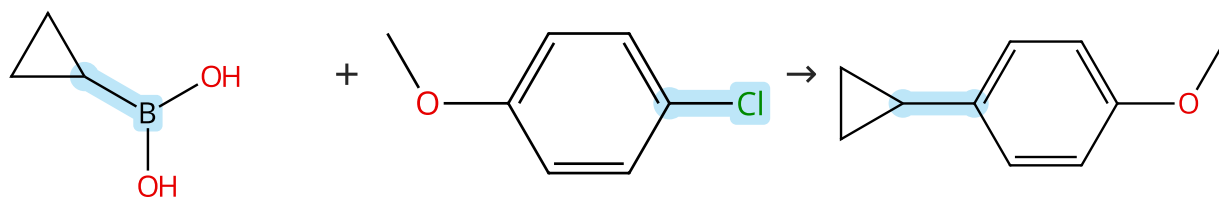
By: Ammon, Erich; et al

European Journal of Organic Chemistry (2020), 2020(36), 5867-5875.

Experimental Protocols

Scheme 297 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

Suppliers (71)

Suppliers (32)

31-614-CAS-24822657

Steps: 1 Yield: 97%

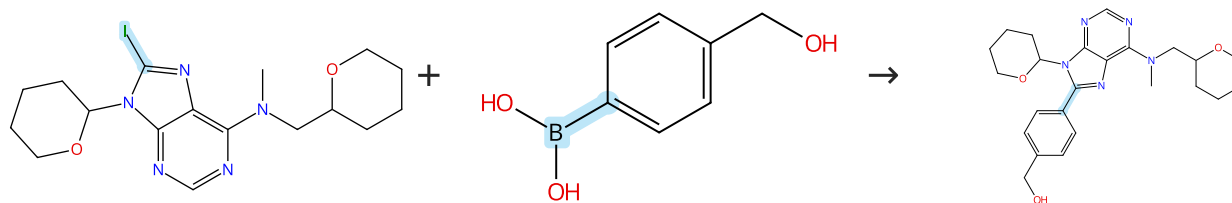
Halogen-Bridged Methyl-naphthyl Palladium Dimers as Versatile Catalyst Precursors in Coupling Reactions

1.1 **Reagents:** Potassium carbonate, Tetradecane
Catalysts: Tri-*tert*-butylphosphonium tetrafluoroborate, Palladium, di-μ-bromobis(η³-1-naphthalenylmethyl)di-
Solvents: Toluene, Water; 11 h, 80 °C

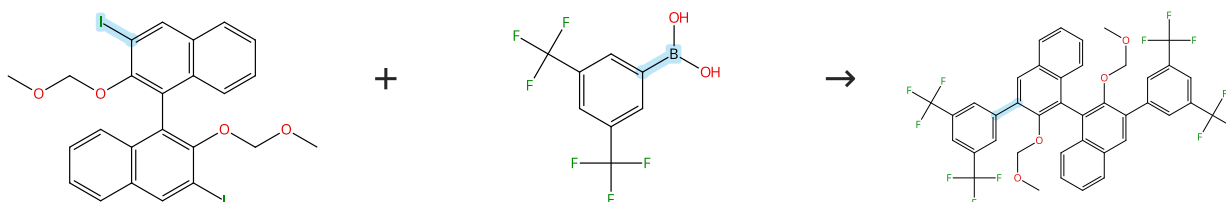
By: Sivendran, Nardana; et al

Angewandte Chemie, International Edition (2021), 60(47), 25151-25160.

Experimental Protocols

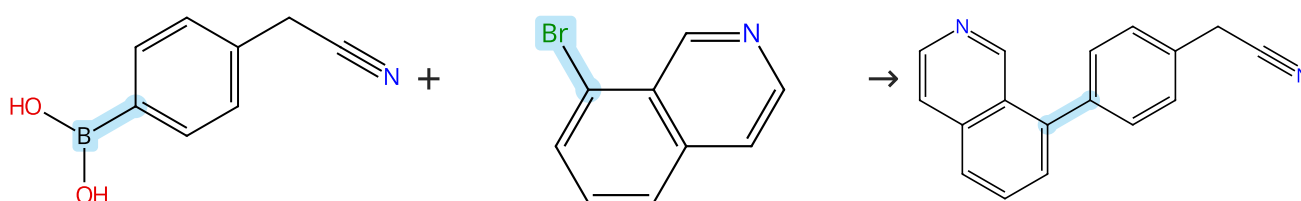
Steps: **1** Yield: **97%**

Experimental Protocols

Steps: **1** Yield: **97%**

 Suppliers (94)

ACS Catalysis (2021), 11(10), 6270-6275.

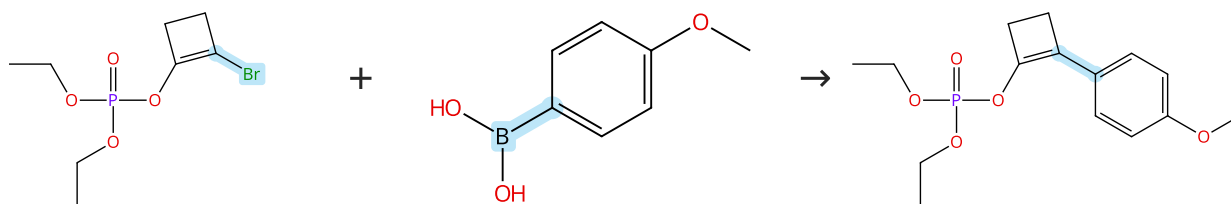
Steps: **1** Yield: **97%**

 Suppliers (2)

Experimental Protocols

Scheme 301 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (112)

31-179-CAS-19668296

Steps: 1 Yield: 97%

Novel Selective Approach to Terminally Substituted [n]Dendralenes

By: Polak, Peter; et al

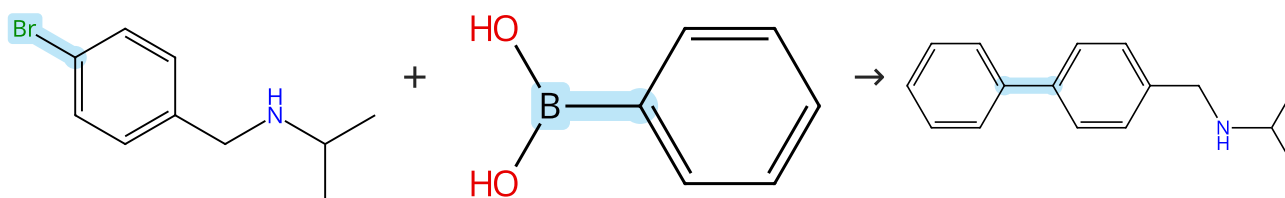
European Journal of Organic Chemistry (2019), 2019(5), 957-968.

1.1 **Catalysts:** Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine**Solvents:** Toluene; 1 min, rt1.2 **Reagents:** Tripotassium phosphate**Solvents:** Water; 18 h, 60 °C

Experimental Protocols

Scheme 302 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (49)

Suppliers (132)

Suppliers (12)

31-179-CAS-16136895

Steps: 1 Yield: 97%

Aqueous Suzuki Coupling Reactions of Basic Nitrogen-Containing Substrates in the Absence of Added Base and Ligand: Observation of High Yields under Acidic Conditions

By: Li, Zhao; et al

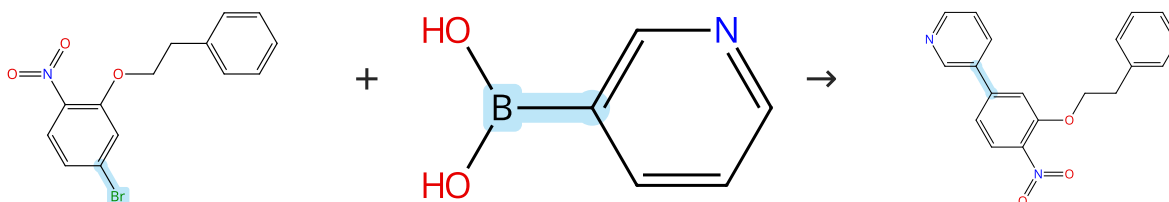
Journal of Organic Chemistry (2016), 81(18), 8520-8529.

1.1 **Catalysts:** Palladium diacetate**Solvents:** Water; 4 h, 8.7 atm → 3.6 atm, 100 °C; 100 °C → rt1.2 **Reagents:** Sodium hydroxide**Solvents:** Water; > pH 12, rt

Experimental Protocols

Scheme 303 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (109)

Suppliers (2)

31-179-CAS-277668

Steps: 1 Yield: 97%

Identification of 4-(4-nitro-2-phenethoxyphenyl)pyridine as a promising new lead for discovering inhibitors of both human and rat 11β-Hydroxylase

By: Hu, Qingzhong; et al

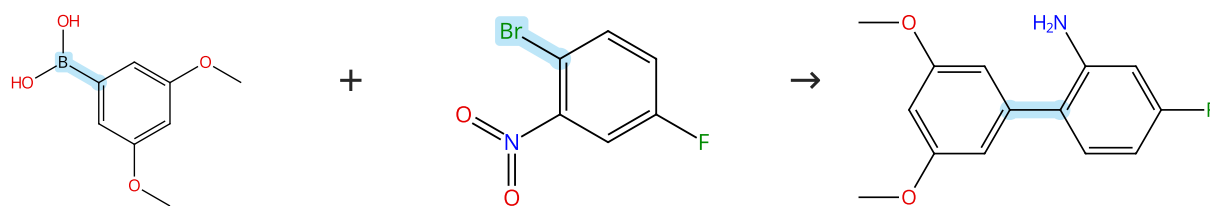
European Journal of Medicinal Chemistry (2015), 96, 139-150.

1.1 **Reagents:** Sodium carbonate**Catalysts:** [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium**Solvents:** 1,2-Dimethoxyethane, Water; overnight, reflux

Experimental Protocols

Scheme 304 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (100)

Suppliers (92)

Supplier (1)

31-179-CAS-17120622

Steps: 1 Yield: 97%

Organocatalytic Enantioselective Synthesis of 6-Aryl-5,6-dihydrophenanthridines by a Modified Pictet-Spengler Reaction of Biphenyl-2-amines and Aromatic Aldehydes

By: Wang, Chaoceng; et al

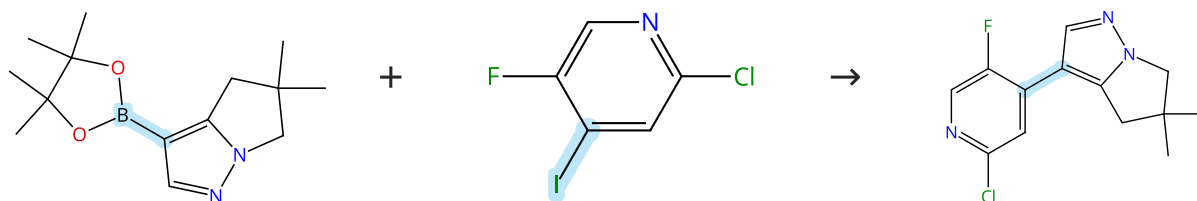
European Journal of Organic Chemistry (2017), 2017(14), 1865-1869.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol; 24 h, reflux
- 1.2 **Reagents:** Zinc, Ammonium chloride
Solvents: Methanol; 24 h, reflux

Experimental Protocols

Scheme 305 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (37)

Suppliers (74)

Supplier (1)

31-614-CAS-31188336

Steps: 1 Yield: 97%

Discovery of AZD4573, a Potent and Selective Inhibitor of C D K9 That Enables Short Duration of Target Engagement for the Treatment of Hematological Malignancies

By: Barlaam, Bernard; et al

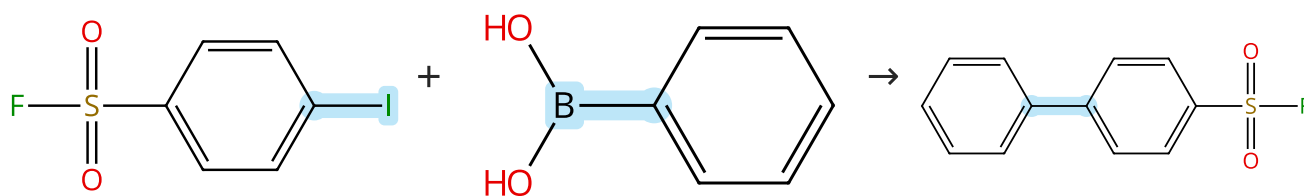
Journal of Medicinal Chemistry (2020), 63(24), 15564-15590.

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: (*SP*-4-4)-[2'-(Amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*]chloro [dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl] phosphine]palladium
Solvents: 1,4-Dioxane, Water; 21 °C; 3 h, 80 °C

Experimental Protocols

Scheme 306 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (39)

Suppliers (132)

Suppliers (29)

31-179-CAS-5149593

Steps: 1 Yield: 97%

On-Water Synthesis of Biaryl Sulfonyl Fluorides

By: Chinthakindi, Praveen K.; et al

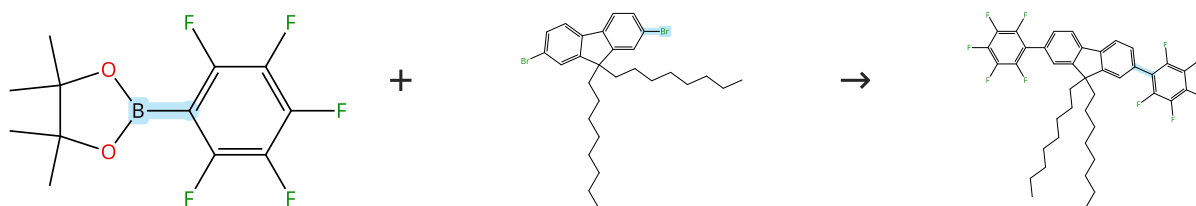
Journal of Organic Chemistry (2016), 81(6), 2618-2623.

- 1.1 **Reagents:** Triethylamine
Catalysts: Palladium diacetate
Solvents: Water; 30 min, rt

Experimental Protocols

Scheme 307 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (51)

Suppliers (69)

31-179-CAS-18702665

Steps: 1 Yield: 97%

An "On-Cycle" Precatalyst Enables Room-Temperature Polyfluoroarylation Using Sensitive Boronic Acids

By: Chen, Liye; et al

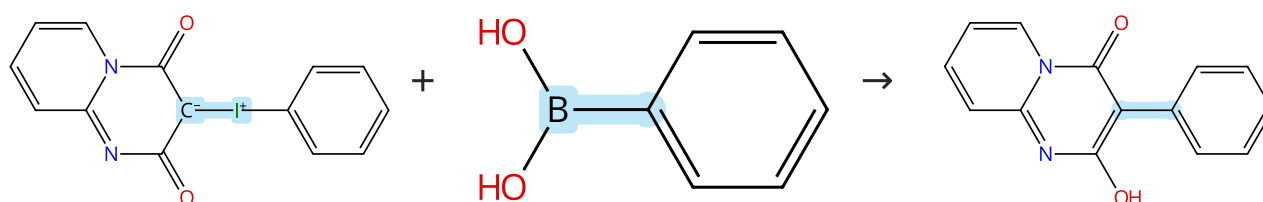
ACS Catalysis (2018), 8(4), 2989-2994.

1.1 **Reagents:** Triethylamine, Sodium sulfate
Catalysts: Bromo(4-fluorophenyl)[tris(tricyclo[3.3.1.1^{3,7}]dec-1-yl)phosphine]palladium
Solvents: Toluene; 2 h, rt

Experimental Protocols

Scheme 308 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (132)

Suppliers (29)

31-614-CAS-39973609

Steps: 1 Yield: 97%

Catalytic and Base-free Suzuki-type α -Arylation of Cyclic 1,3-Dicarbonyls via a Cyclic Iodonium Ylide Strategy

By: Liang, Mingxuan; et al

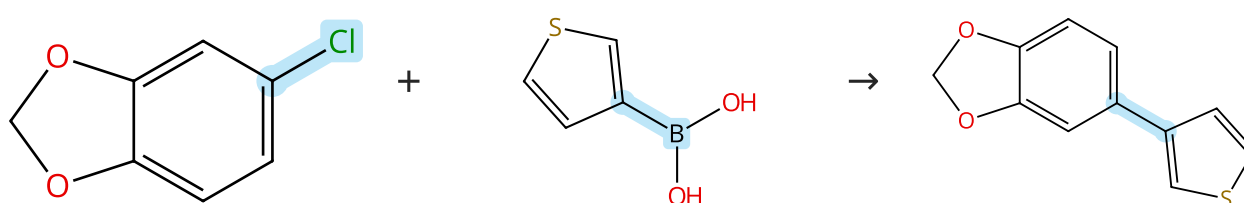
Angewandte Chemie, International Edition (2024), 63(17), e202400741.

1.1 **Catalysts:** Bis[dichloro(η^5 -(pentamethylcyclopentadienyl))rhodium]
Solvents: Water; 30 min, 50 °C

Experimental Protocols

Scheme 309 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (50)

Suppliers (101)

Supplier (1)

31-179-CAS-17819543

Steps: 1 Yield: 97%

Bulky Yet Flexible Pd-PEPPSI-IPent^{An} for the Synthesis of Sterically Hindered Biaryls in Air

By: Lu, Dong-Dong; et al

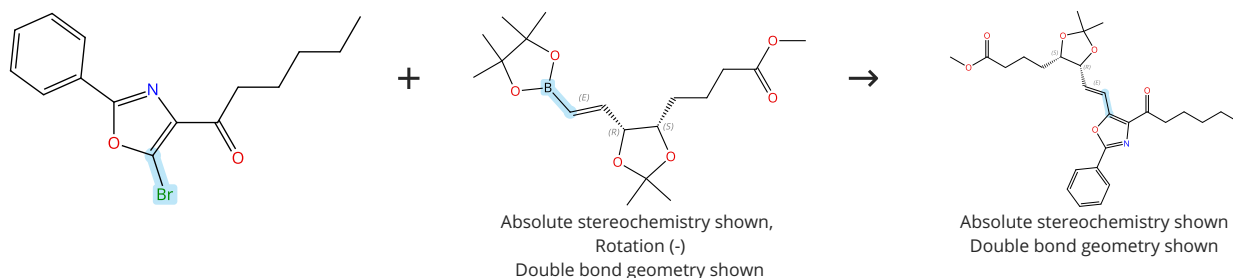
Journal of Organic Chemistry (2017), 82(20), 10898-10911.

1.1 **Reagents:** Potassium carbonate
Catalysts: (*SP*-4-1)-[7,9-Bis[2,6-bis(1-ethylpropyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]dichloro(3-chloropyridine-*k*M)palladium
Solvents: Ethanol; 4 h, 80 °C

Experimental Protocols

Scheme 310 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (8)

31-177-CAS-19636495

Steps: 1 Yield: 97%

Asymmetric synthesis and biological evaluation of imidazole- and oxazole-containing synthetic lipoxin A₄ mimetics (sLXms)

By: de Gaetano, Monica; et al

European Journal of Medicinal Chemistry (2019), 162, 80-108.

1.1 Reagents: Sodium carbonate

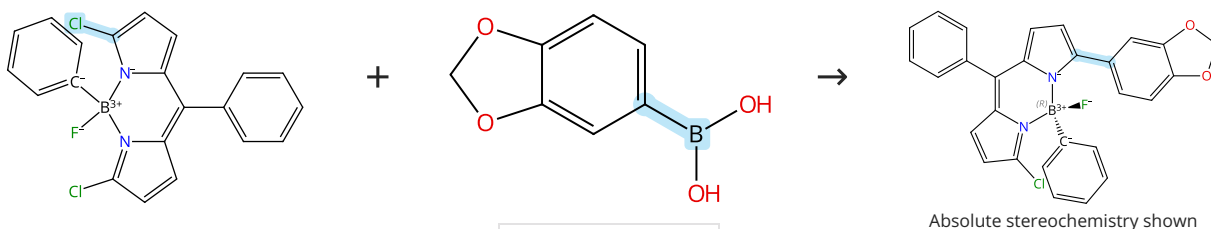
Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: Dimethylformamide, Water; 30 min, rt → 100 °C

Experimental Protocols

Scheme 311 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (98)

31-614-CAS-42013237

Steps: 1 Yield: 97%

Modular enantioselective assembly of multi-substituted boron-stereogenic BODIPYs

By: Ren, Li-Qing; et al

Nature Chemistry (2025), 17(1), 83-91.

1.1 Catalysts: Bis(dibenzylideneacetone)palladium, *N,N*-Bis[(1*R*)-1-phenylethyl]dinaphtho[2,1-*a*:1',2'-*f*][1,3,2]dioxaphosphepin-4-amine

Solvents: Diethyl ether; 5 min, rt

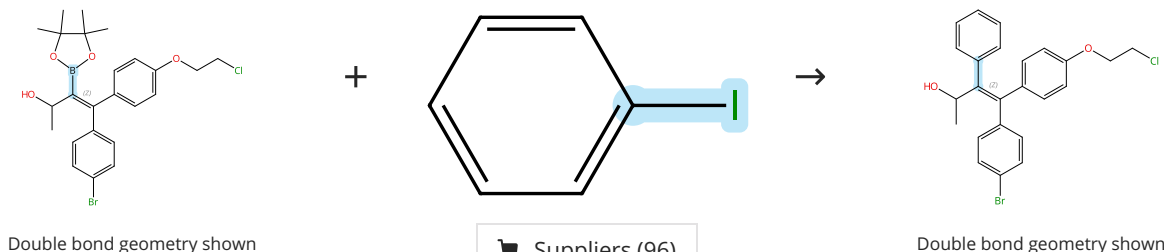
1.2 Reagents: Potassium carbonate

Solvents: Diethyl ether; 24 h, 80 °C

Experimental Protocols

Scheme 312 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (96)

31-177-CAS-20015618

Steps: 1 Yield: 97%

Regioselective trans-Carboboration of Propargyl Alcohols

By: Jin, Hongming; et al

Organic Letters (2019), 21(9), 3446-3450.

1.1 Reagents: Potassium hydroxide

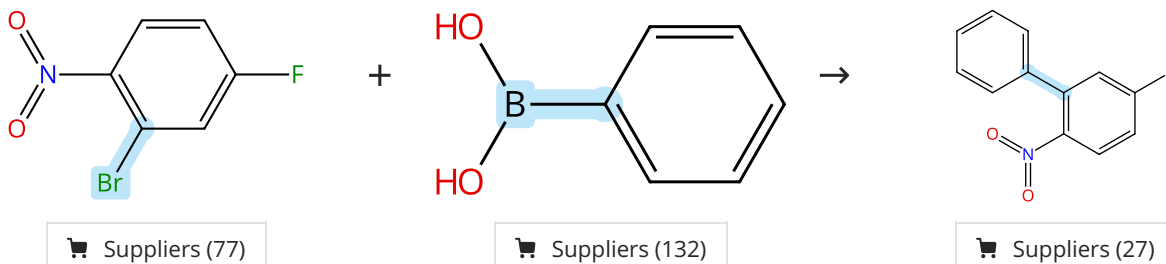
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium

Solvents: Tetrahydrofuran, Water; 6 h, rt

Experimental Protocols

Scheme 313 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-22911167

Steps: 1 Yield: 97%

¹⁹F- and ¹⁸F-arene deoxyfluorination via organic photoredox-catalysed polarity-reversed nucleophilic aromatic substitution

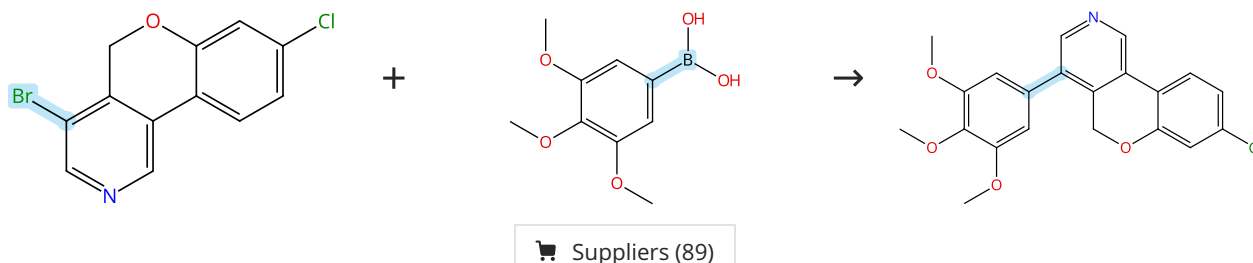
- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 6 h, reflux; cooled
- 1.2 **Reagents:** Ammonium chloride; cooled

By: Tay, Nicholas E. S.; et al
 Nature Catalysis (2020), 3(9), 734-742.

Experimental Protocols

Scheme 314 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-40803897

Steps: 1 Yield: 97%

Discovery of Conformationally Constrained ALK2 Inhibitors

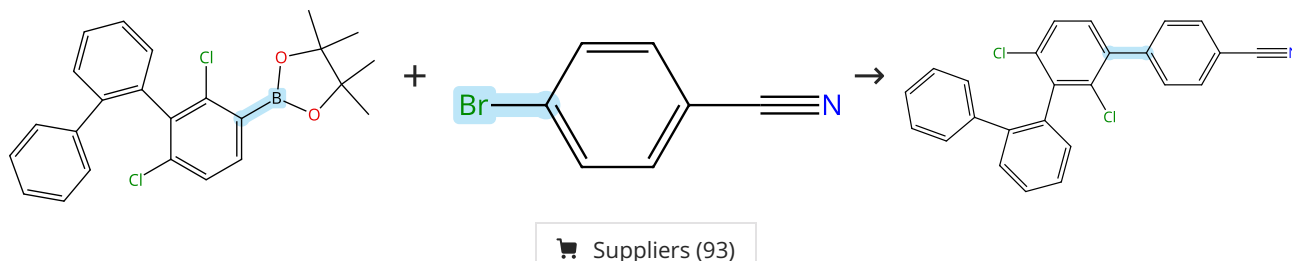
- 1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; rt → 100 °C; 3 h, 100 °C; 100 °C → rt
- 1.2 **Solvents:** Water; rt

By: Gonzalez-Alvarez, Hector; et al
 Journal of Medicinal Chemistry (2024), 67(6), 4707-4725.

Experimental Protocols

Scheme 315 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-40331273

Steps: 1 Yield: 97%

Aryl Chloride-Directed Enantioselective C(sp²)-H Borylation Enabled by Iridium Catalysis

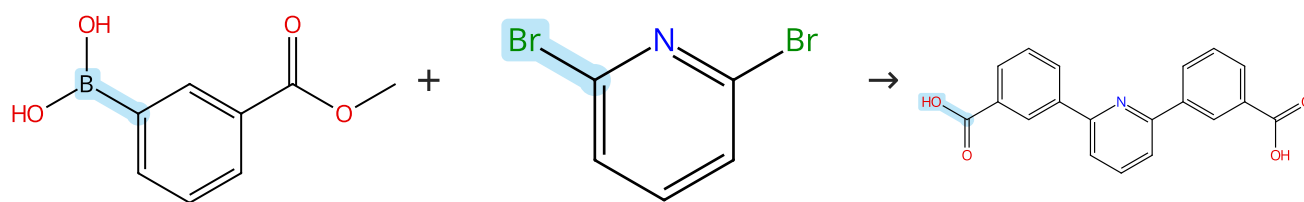
- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 5 h, 80 °C

By: Zhao, Hongliang; et al
 Journal of the American Chemical Society (2023), 145(46), 25214-25221.

Experimental Protocols

Scheme 316 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (96)

Suppliers (87)

Suppliers (38)

31-179-CAS-22581124

Steps: 1 Yield: 97%

Synthesis and characterization of low-nuclearity lantern-type porous coordination cages

By: Taggart, Garrett A.; et al

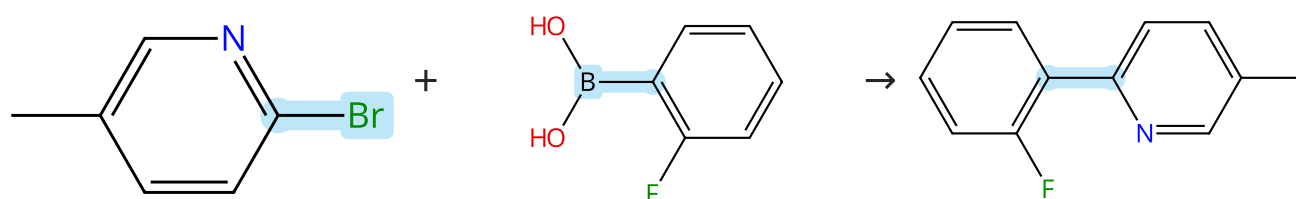
Chemical Communications (Cambridge, United Kingdom) (2020), 56(63), 8924-8927.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Tetrahydrofuran, Water; overnight, rt → 80 °C
- 1.2 **Reagents:** Sodium hydroxide
Solvents: Water; rt → 100 °C
- 1.3 **Reagents:** Hydrochloric acid; acidified, rt

Experimental Protocols

Scheme 317 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (104)

Suppliers (113)

Suppliers (12)

31-179-CAS-18765283

Steps: 1 Yield: 97%

Accelerated skin wound healing by selective 11β-Hydroxylase (CYP11B1) inhibitors

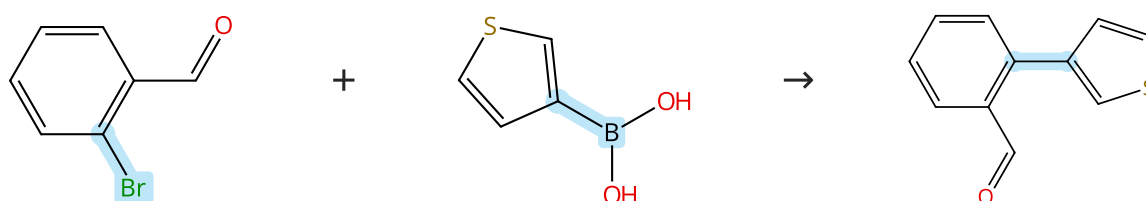
By: Emmerich, Juliette; et al

European Journal of Medicinal Chemistry (2018), 143, 591-597.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 20 h, 100 °C

Scheme 318 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (89)

Suppliers (101)

Suppliers (56)

31-179-CAS-15751015

Steps: 1 Yield: 97%

Exploring the Reactivity of N-Alkynylated Sulfoximines: Acid-Catalyzed Cyclizations

By: Pirwerdjan, Ramona; et al

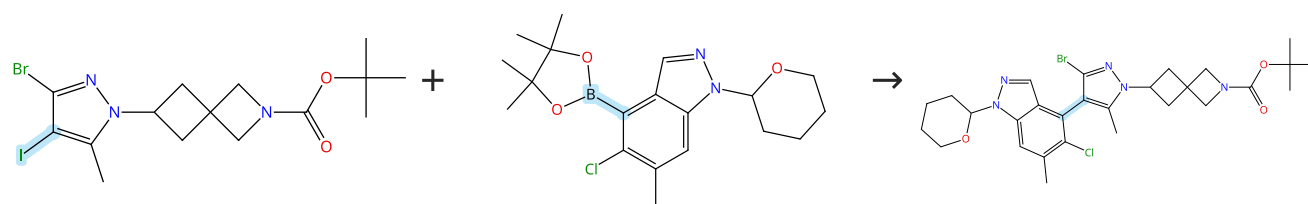
Organic Letters (2016), 18(14), 3307-3309.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 2 h, rt → 100 °C

Experimental Protocols

Scheme 319 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (20)

Suppliers (49)

Suppliers (32)

31-614-CAS-34716758

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine, [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][[2',6'-bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine-κP](methanesulfonato-κO)palladium
Solvents: 1,4-Dioxane, Water; 1 h, 80 °C

Experimental Protocols

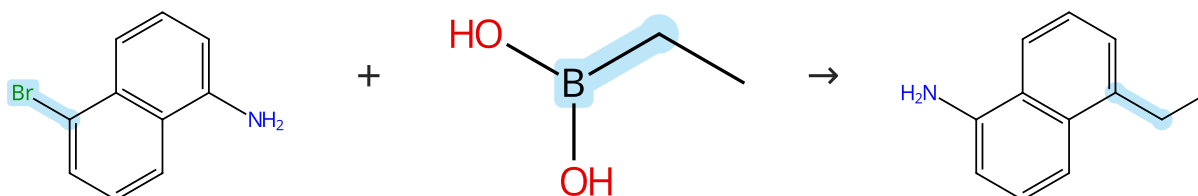
JDQ443, a structurally novel, pyrazole-based, covalent inhibitor of KRAS^{G12C} for the treatment of solid tumors

By: Lorthiois, Edwige; et al

Journal of Medicinal Chemistry (2022), 65(24), 16173-16203.

Scheme 320 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (64)

Suppliers (95)

Suppliers (18)

31-614-CAS-44554655

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene]palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 12 h, 80 °C

Experimental Protocols

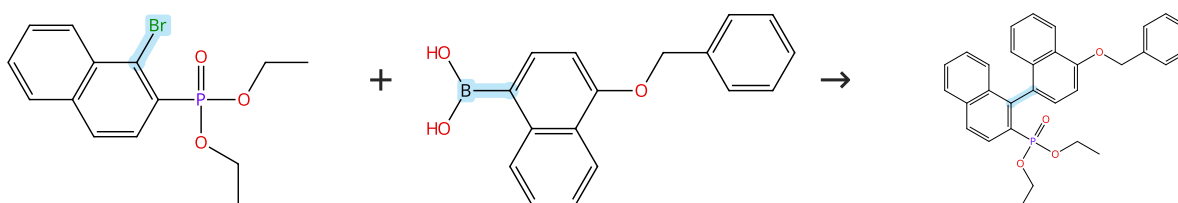
Enantioselective Acylation of Benzylic C(sp³)-H Bond Enabled by a Cooperative Photoredox and N-Heterocyclic Carbene Catalysis

By: Li, Cao-Bo; et al

Angewandte Chemie, International Edition (2025), 64(19), e202421151.

Scheme 321 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (2)

Suppliers (38)

31-179-CAS-17121457

Steps: 1 Yield: 97%

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Tris(dibenzylideneacetone)dipalladium, [(6S,7S,12aS)-6,7-Dihydro-6,7-dimethyldibenzo[e,g][1,4]dioxocin-1-yl]bis(3,5-dimethylphenyl)phosphine
Solvents: Tetrahydrofuran; 60 h, 40 °C

Experimental Protocols

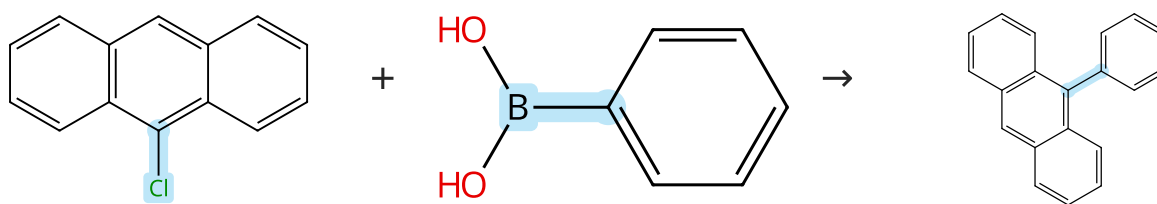
Synthesis of Chiral-Bridged Atropisomeric Monophosphine Ligands with Tunable Dihedral Angles and their Applications in Asymmetric Suzuki-Miyaura Coupling Reactions

By: Xia, Wang; et al

Advanced Synthesis & Catalysis (2017), 359(10), 1656-1662.

Scheme 322 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (64)

Suppliers (132)

Suppliers (76)

31-179-CAS-16474868

Steps: 1 Yield: 97%

An N-Heterocyclic Carbene-Nickel Half-Sandwich Complex as a Precatalyst for Suzuki-Miyaura Coupling of Aryl/Heteroaryl Halides with Aryl/Heteroarylboronic Acids

By: Ando, Shin; et al

Journal of Organic Chemistry (2017), 82(2), 1266-1272.

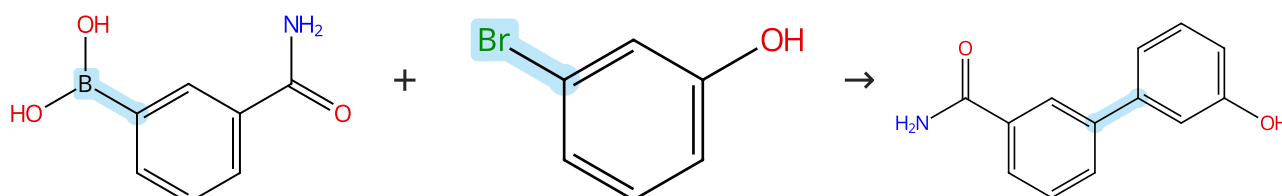
1.1 Reagents: Tripotassium phosphate
 Catalysts: Triphenylphosphine, Chloro(η^5 -2,4-cyclopentadien-1-yl)[*rel*-(3*aR*,9*aS*)-1,3,3*a*,4,9,9*a*-hexahydro-1,3-bis(1-methylethyl)-4,9[1',2']-benzeno-2*H*-naphth[2,3-*d'*]imidazol-2-ylidene]nickel
 Solvents: 1-Pentanol, Toluene; 25 °C; 24 h, 110 °C

1.2 Solvents: Water; 0 °C

Experimental Protocols

Scheme 323 (2 Reactions)

Steps: 1 Yield: 97%



Suppliers (86)

Suppliers (74)

Suppliers (55)

31-179-CAS-19253347

Steps: 1 Yield: 97%

Identification of Bivalent Ligands with Melatonin Receptor Agonist and Fatty Acid Amide Hydrolase (FAAH) Inhibitory Activity That Exhibit Ocular Hypotensive Effect in the Rabbit

By: Spadoni, Gilberto; et al

Journal of Medicinal Chemistry (2018), 61(17), 7902-7916.

1.1 Reagents: Sodium carbonate
 Catalysts: Tetrakis(triphenylphosphine)palladium
 Solvents: Acetonitrile, Water; 1.5 h, 90 °C

Experimental Protocols

31-179-CAS-17008827

Steps: 1 Yield: 97%

Design, Synthesis, Structure-Activity Relationship Studies, and Three-Dimensional Quantitative Structure-Activity Relationship (3D-QSAR) Modeling of a Series of O-Biphenyl Carbamates as Dual Modulators of Dopamine D3 Receptor and Fatty Acid Amide Hydrolase

By: De Simone, Alessio; et al

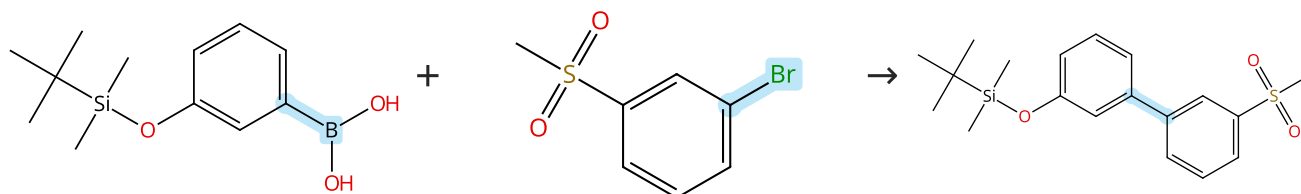
Journal of Medicinal Chemistry (2017), 60(6), 2287-2304.

1.1 Reagents: Potassium carbonate
 Catalysts: Palladium diacetate
 Solvents: 2-Methoxyethanol, Water; > 1 min, rt; 5 min, rt
 1.2 Reagents: Hydrochloric acid
 Solvents: Water; acidified, rt

Experimental Protocols

Scheme 324 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (83)

Suppliers (69)

31-614-CAS-36331540

Steps: 1 Yield: 97%

Automated, Capsule-Based Suzuki-Miyaura Cross Couplings

By: Coin, Guillaume; et al

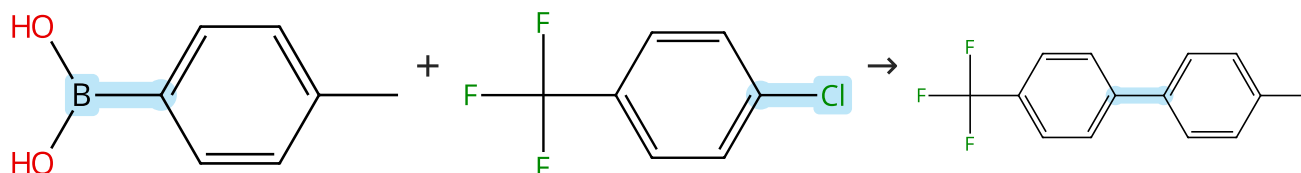
Organic Letters (2024), 26(14), 2708-2712.

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Silica, Phosphine, diphenyl-, palladium(2+) salt (2:1)
Solvents: Ethanol, 1,2-Dimethoxyethane, Water; 3.5 h, 60 °C; 60 °C → rt
- 1.2 **Reagents:** Silica
- 1.3 **Reagents:** Methanaminium, *N,N,N*-trimethyl-, carbonate (2:1) (reagent supported); 2 h, rt

Experimental Protocols

Scheme 325 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (114)

Suppliers (73)

Suppliers (30)

31-614-CAS-32738960

Steps: 1 Yield: 97%

An air-stable, well-defined palladium-BIAN-NHC chloro dimer: a fast-activating, highly efficient catalyst for cross-coupling

By: Zhang, Jin; et al

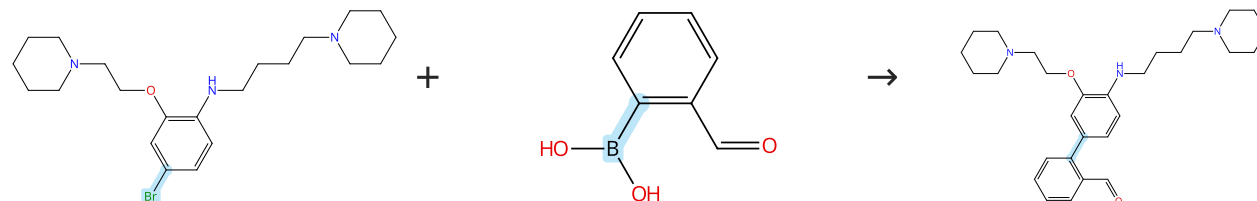
Chemical Communications (Cambridge, United Kingdom) (2022), 58(53), 7404-7407.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Bis[7,9-bis[2,6-bis(1-methylethyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]di-μ-chlorodic hlorodipalladium
Solvents: Ethanol; 12 h, 25 °C

Experimental Protocols

Scheme 326 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (102)

31-179-CAS-19762153

Steps: 1 Yield: 97%

A phenotypic approach to the discovery of compounds that promote non-amyloidogenic processing of the amyloid precursor protein: Toward a new profile of indirect β-secretase inhibitors

By: Gay, Marion; et al

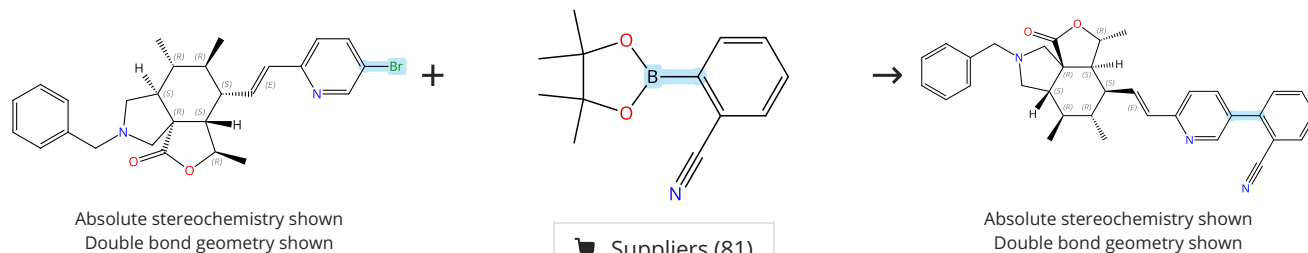
European Journal of Medicinal Chemistry (2018), 159, 104-125.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Ethanol, Toluene; 30 min, rt
- 1.2 **Catalysts:** Tri-*o*-tolylphosphine, Tris(dibenzylideneacetone) dipalladium; 18 h, reflux

Experimental Protocols

Scheme 327 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-40575614

Steps: 1 Yield: 97%

Lead Optimization to Advance Protease-Activated Receptor-1 Antagonists in Early Discovery

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 4 h, 100 °C

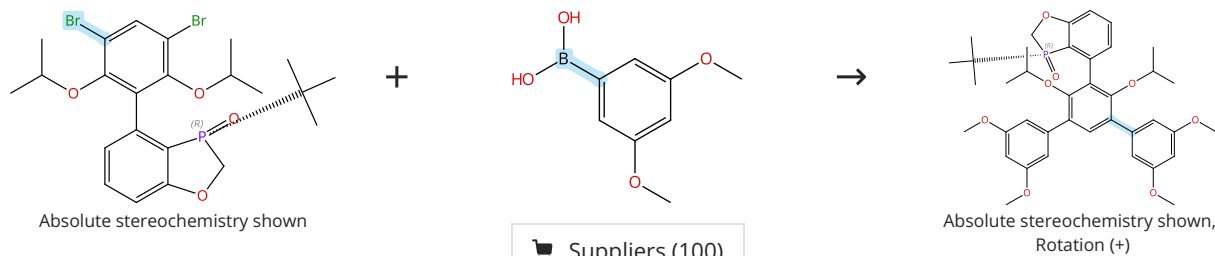
By: Mandal, Mihirbaran; et al

Journal of Medicinal Chemistry (2022), 65(7), 5575-5592.

Experimental Protocols

Scheme 328 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-19654872

Steps: 1 Yield: 97%

Computationally Assisted Mechanistic Investigation and Development of Pd-Catalyzed Asymmetric Suzuki-Miyaura and Negishi Cross-Coupling Reactions for Tetra-ortho-Substituted Biaryl Synthesis

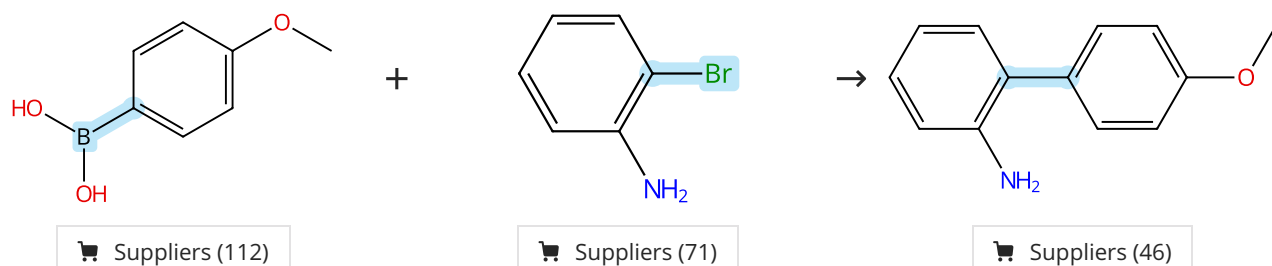
- 1.1 **Reagents:** Sodium carbonate
Catalysts: Palladium diacetate, 4-(2,6-Dimethoxyphenyl)-3-(1,1-dimethylethyl)-2,3-dihydro-1,3-benzoxaphosphole
Solvents: 2-Methyltetrahydrofuran, Water; 18 h, 70 °C

By: Patel, Nitinchandra D.; et al

ACS Catalysis (2018), 8(11), 10190-10209.

Scheme 329 (1 Reaction)

Steps: 1 Yield: 97%



31-179-CAS-19395320

Steps: 1 Yield: 97%

Visible Light-Induced Radical Cyclization of Tertiary Bromides with Isonitriles To Construct Trifluoromethylated Quaternary Carbon Center

- 1.1 **Reagents:** Potassium carbonate
Solvents: 1,2-Dimethoxyethane; 30 min, rt
 1.2 **Catalysts:** Dichlorobis(triphenylphosphine)palladium; 12 h, 80 °C; 80 °C → rt

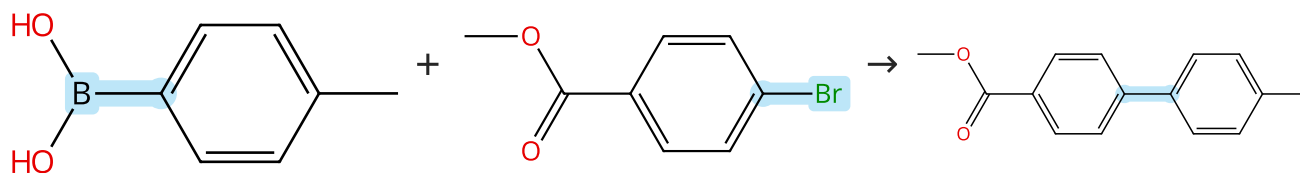
By: Wang, Wengui; et al

Journal of Organic Chemistry (2018), 83(23), 14588-14599.

Experimental Protocols

Scheme 330 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (114)

Suppliers (96)

Suppliers (51)

31-179-CAS-13937276

Steps: 1 Yield: 97%

A self-assembled bisoxazoline/Pd composite microsphere as an excellent catalyst for Suzuki-Miyaura coupling reactions

By: Wang, Junke; et al

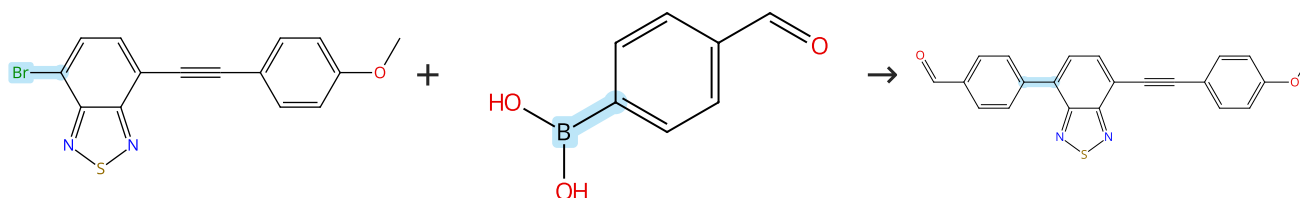
Green Chemistry (2016), 18(4), 967-973.

1.1 **Reagents:** Potassium carbonate, Oxygen
Catalysts: Polyethylene glycol, 1820646-53-4
Solvents: Water; 4 h, 70 °C

Experimental Protocols

Scheme 331 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (106)

31-179-CAS-16988956

Steps: 1 Yield: 97%

An investigation of the role the donor moiety plays in modulating the efficiency of 'donor-π-acceptor-π-acceptor' organic DSSCs

By: Park, Kwang-Won; et al

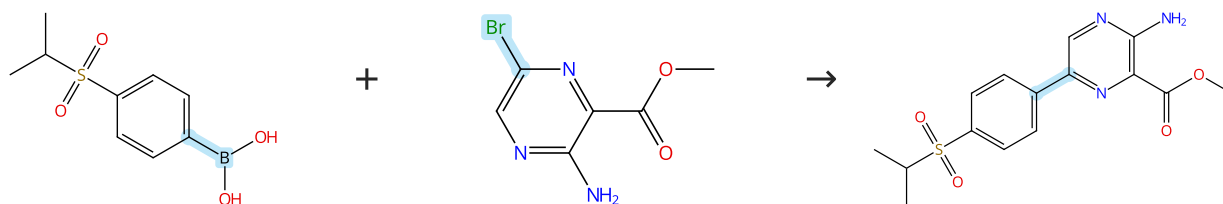
Tetrahedron (2017), 73(8), 1098-1104.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 18 h, 70 °C

Experimental Protocols

Scheme 332 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (66)

Suppliers (86)

Supplier (1)

31-179-CAS-21873412

Steps: 1 Yield: 97%

Rational Design of 5-(4-(Isopropylsulfonyl)phenyl)-3-(3-(4-((methylamino)methyl)phenyl)isoxazol-5-yl)pyrazin-2-amine (V X-970, M6620): Optimization of Intra- and Intermolecular Polar Interactions of a New Ataxia Telangiectasia Mutated and Rad3-Related (ATR) Kinase Inhibitor

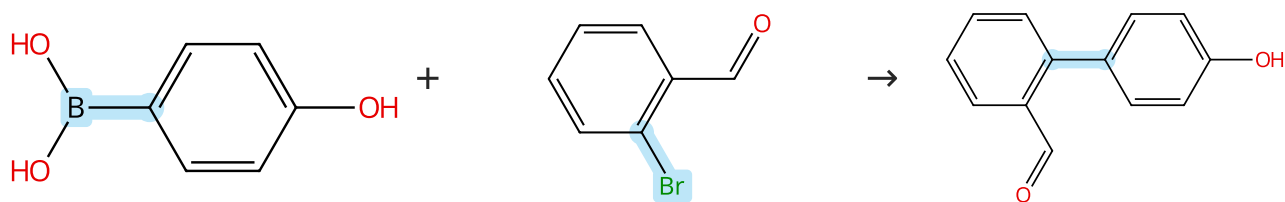
By: Knegtel, Ronald; et al

Journal of Medicinal Chemistry (2019), 62(11), 5547-5561.

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Bis(tri-*tert*-butylphosphine)palladium
Solvents: Acetonitrile, Water; rt; 2 h, 60 °C

Scheme 333 (4 Reactions)

Steps: 1 Yield: 97%



Suppliers (103)

Suppliers (89)

Suppliers (52)

31-614-CAS-43773537

Steps: 1 Yield: 97%

Stereocontrolled desymmetrization of 2,5-cyclohexadienones via organocatalytic domino sulfa-1,6-/1,4-addition or sulfa-1,6-/1,4-/sulfa-1,4-addition reactions

By: Sodhi, Vanisha; et al

Organic Chemistry Frontiers (2025), 12(4), 1144-1149.

- 1.1 Reagents: Sodium carbonate
Solvents: Dimethylformamide, Water; 5 min, rt
- 1.2 Catalysts: Palladium diacetate; rt

Experimental Protocols

31-614-CAS-43478835

Steps: 1 Yield: 97%

Asymmetric desymmetrization of 2,5-cyclohexadienones initiated by organocatalytic conjugate addition to 4-nitro-5-styrylisoxazoles

By: Sharma, Manisha; et al

Chemical Communications (Cambridge, United Kingdom) (2025), 61(7), 1455-1458.

- 1.1 Reagents: Sodium carbonate
Solvents: Dimethylformamide, Water; 5 min, rt
- 1.2 Catalysts: Palladium diacetate; rt

Experimental Protocols

31-614-CAS-38204957

Steps: 1 Yield: 97%

Asymmetric Synthesis of Hydrophenanthrenones Bearing Multiple Stereogenic Centers via Squaramide-Catalyzed Domino 1,4-/1,4-Addition Desymmetrization Sequence

By: Sharma, Manisha; et al

Organic Letters (2023), 25(43), 7911-7916.

- 1.1 Reagents: Sodium carbonate
Solvents: Dimethylformamide, Water; 5 min, rt
- 1.2 Catalysts: Palladium diacetate; rt

Experimental Protocols

31-614-CAS-31802952

Steps: 1 Yield: 97%

Asymmetric Synthesis of Cyclohexenone Fused Isochromans via Quinidine Catalyzed Domino Peroxyhemiacetalization/Oxa-Michael Addition Desymmetrization Sequence

By: Tamanna; et al

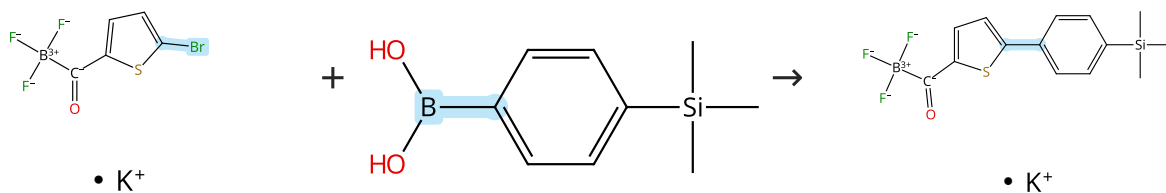
Journal of Organic Chemistry (2022), 87(9), 6397-6402.

- 1.1 Reagents: Sodium carbonate
Solvents: Dimethylformamide, Water; 5 min, rt
- 1.2 Catalysts: Palladium diacetate; rt

Experimental Protocols

Scheme 334 (1 Reaction)

Steps: 1 Yield: 97%

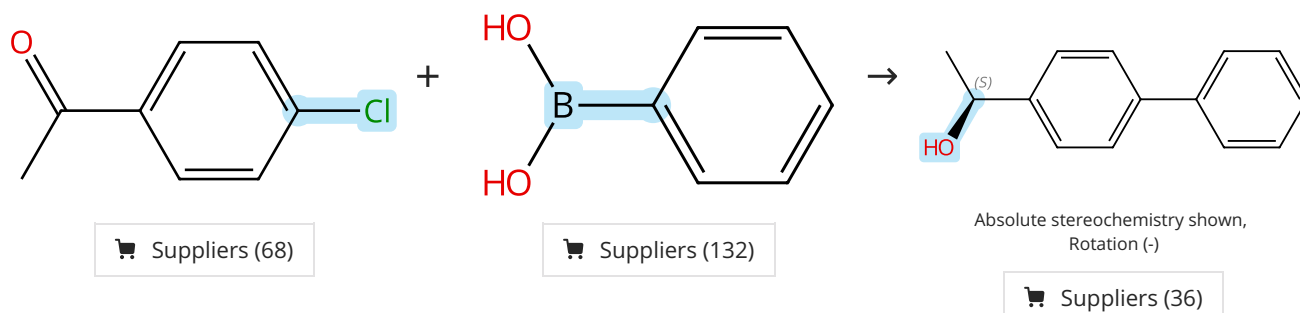
• K⁺• K⁺

Suppliers (81)

31-179-CAS-23285764 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: (SP-4-3)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato-κO)palladium Solvents: 1,2-Dimethoxyethane, Water; 18 - 24 h, 60 °C Experimental Protocols	Rapid Access to Diverse Potassium Acyltrifluoroborates (KATs) through Late-Stage Chemoselective Cross-Coupling Reactions By: Deng, Xingwang; et al Organic Letters (2021), 23(5), 1886-1890.
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Scheme 335 (1 Reaction)

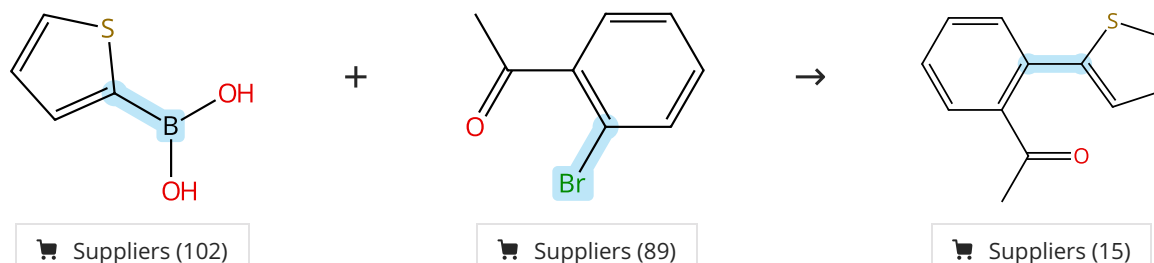
Steps: 1 Yield: 97%



31-179-CAS-19502789 Steps: 1 Yield: 97% 1.1 Reagents: Sodium formate, Cesium carbonate Catalysts: 2252264-54-1 (mesoporous hollow-shell-silica supported) Solvents: Isopropanol, Water; 0.5 - 1.5 h, 80 °C Experimental Protocols	An integrated immobilization strategy manipulates dual active centers to boost enantioselective tandem reactions By: Shu, Xiaomin; et al Chemical Communications (Cambridge, United Kingdom) (2018), 54(94), 13244-13247.
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Scheme 336 (1 Reaction)

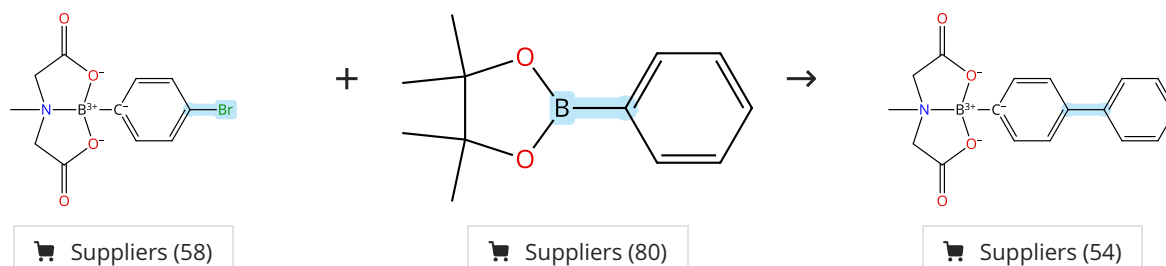
Steps: 1 Yield: 97%



31-614-CAS-35310279 Steps: 1 Yield: 97% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Ethanol, Toluene; 2 h, 85 °C 1.2 Reagents: Water Experimental Protocols	o-Nitrobenzyl Oxime Ethers Enable Photoinduced Cyclization Reaction to Provide Phenanthridines under Aqueous Conditions By: Okamura, Hidenori; et al Organic Letters (2023), 25(3), 466-470.
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Scheme 337 (1 Reaction)

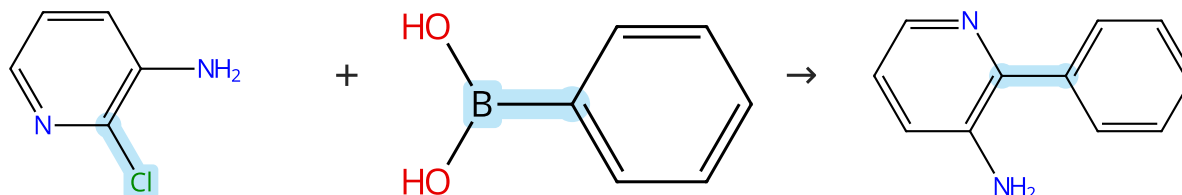
Steps: 1 Yield: 97%



31-179-CAS-294945	Steps: 1 Yield: 97%	Speciation control during Suzuki-Miyaura cross-coupling of haloaryl and haloalkenyl MIDA boronic esters
1.1 Reagents: Tripotassium phosphate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Tetrahydrofuran, Water; 24 h, rt		By: Fyfe, James W. B.; et al Chemistry - A European Journal (2015), 21(24), 8951-8964.
Experimental Protocols		

Scheme 338 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (101)

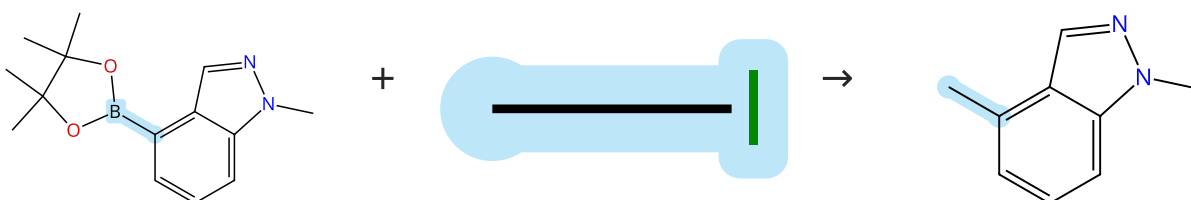
Suppliers (132)

Suppliers (68)

31-179-CAS-22943998	Steps: 1 Yield: 97%	Asymmetric Hydrogenation of 2-Aryl-3-phthalimidopyridinium Salts: Synthesis of Piperidine Derivatives with Two Contiguous Stereocenters
1.1 Reagents: Benzaldehyde, Sodium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Toluene, Water; 10 min, rt; 15 min, rt; overnight, rt → reflux; reflux → rt		By: Zheng, Long-Sheng; et al Organic Letters (2020), 22(22), 8882-8887.
Experimental Protocols		

Scheme 339 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (59)

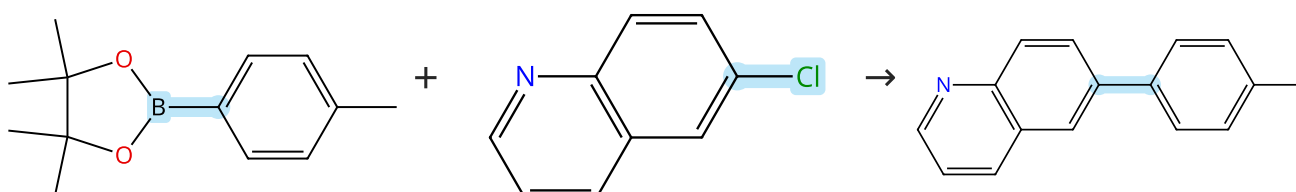
Suppliers (87)

Suppliers (12)

31-179-CAS-19857988	Steps: 1 Yield: 97%	Palladium-Catalyzed Methylation of Aryl, Heteroaryl, and Vinyl Boronate Esters
1.1 Reagents: Potassium <i>tert</i> -butoxide Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][bis(1,1-dimethylethyl)methylphosphine]chloropalladium Solvents: 2-Methyl-2-butanol; 5 min, rt		By: Haydl, Alexander M.; et al Organic Letters (2019), 21(5), 1337-1341.
1.2 18 h, 65 °C; 65 °C → rt		
1.3 Reagents: Sodium bicarbonate Solvents: Water; rt		
Experimental Protocols		

Scheme 340 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (86)

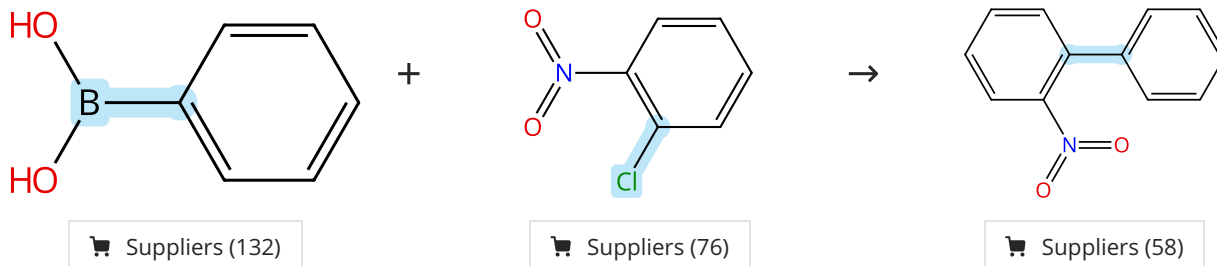
Suppliers (84)

Suppliers (8)

31-614-CAS-41372124	Steps: 1 Yield: 97%	ProPhos: A Ligand for Promoting Nickel-Catalyzed Suzuki-Miyaura Coupling Inspired by Mechanistic Insights into Transmetalation By: Yang, Jin; et al Journal of the American Chemical Society (2024), 146(9), 6360-6368.
1.1 Catalysts: Bis(1,5-cyclooctadiene)nickel, 3-(Diphenylphosphino)-1-propanol Solvents: 2-Methyltetrahydrofuran; 0.5 h, rt		
1.2 Reagents: Tripotassium phosphate Solvents: Water; 16 h, rt → 70 °C		
Experimental Protocols		

Scheme 341 (2 Reactions)

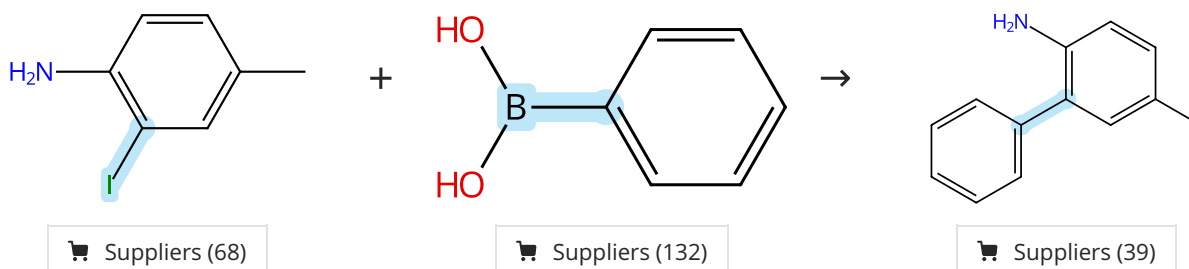
Steps: 1 Yield: 97%



31-614-CAS-44132953	Steps: 1 Yield: 97%	Using Common Alcohols to Transform N-H NHC-Palladium Complexes into Reactive Catalysts for the Suzuki-Miyaura Reaction with Aryl Chlorides By: Larson, Alexandra J. S.; et al Journal of Organic Chemistry (2025), 90(7), 2806-2810.
1.1 Reagents: Sodium hydroxide Catalysts: Palladium chloride, 1-[2,6-Bis(1-methylethyl)phenyl]-2-(diphenylphosphino)-1 <i>H</i> -imidazole Solvents: <i>tert</i> -Butanol; 30 min, 40 °C		
1.2 Reagents: Sodium hydride Solvents: 1,4-Dioxane; 16 h, 40 °C → 100 °C		
Experimental Protocols		
31-614-CAS-24285558	Steps: 1 Yield: 97%	Pd-Catalysed Suzuki-Miyaura cross-coupling of aryl chlorides at low catalyst loadings in water for the synthesis of industrially important fungicides By: Orecchia, Patrizio; et al Green Chemistry (2021), 23(20), 8169-8180.
1.1 Reagents: Sodium carbonate Catalysts: Palladium diacetate, <i>tert</i> -Butyldicyclohexylphosphine Solvents: Cyclohexane, Water; 3 h, 100 °C		
Experimental Protocols		

Scheme 342 (1 Reaction)

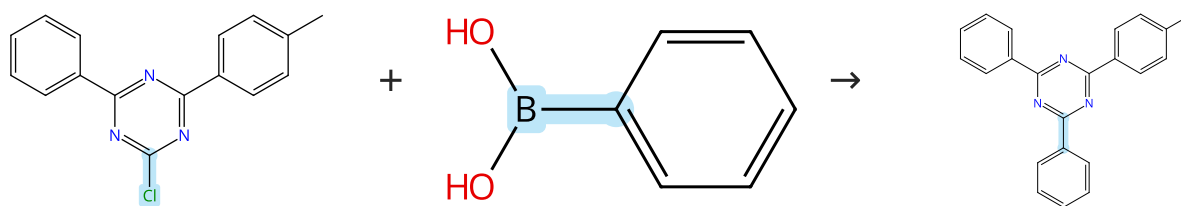
Steps: 1 Yield: 97%



31-179-CAS-4664524	Steps: 1 Yield: 97%	An Approach to Tetraphenylenes via Pd-Catalyzed C-H Functionalization By: Jiang, Hang; et al Organic Letters (2016), 18(9), 2032-2035.
1.1 Reagents: Potassium carbonate Solvents: Acetone, Water; rt → 65 °C		
1.2 Catalysts: Palladium diacetate Solvents: Acetone; overnight, 65 °C		

Scheme 343 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (3)

Suppliers (132)

Suppliers (4)

31-614-CAS-40745574

Steps: 1 Yield: 97%

Synthesis of unsymmetrical trisubstituted 1,3,5-triazines compounds

1.1 Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: 1,4-Dioxane; 15 min, 25 °C

1.2 Reagents: Sodium hydroxide

Solvents: Water; 45 h, 110 °C

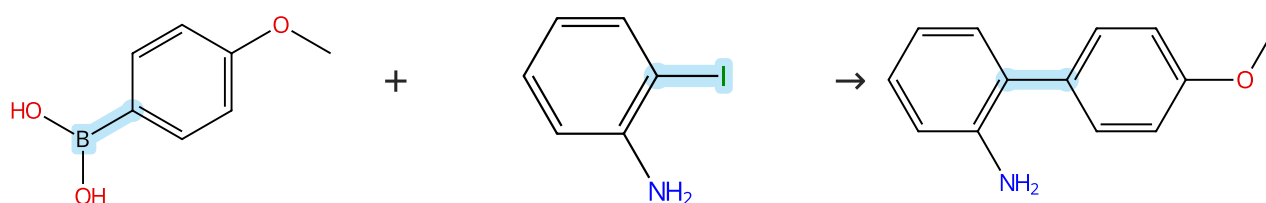
By: Zhao, Jing; et al

Tetrahedron (2024), 159, 134009.

Experimental Protocols

Scheme 344 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (112)

Suppliers (98)

Suppliers (46)

31-179-CAS-17952255

Steps: 1 Yield: 97%

Photochemical intramolecular amination for the synthesis of heterocycles

1.1 Reagents: Potassium carbonate

Catalysts: Dichlorobis(triphenylphosphine)palladium

Solvents: 1,2-Dimethoxyethane, Water; 18 h, 80 °C

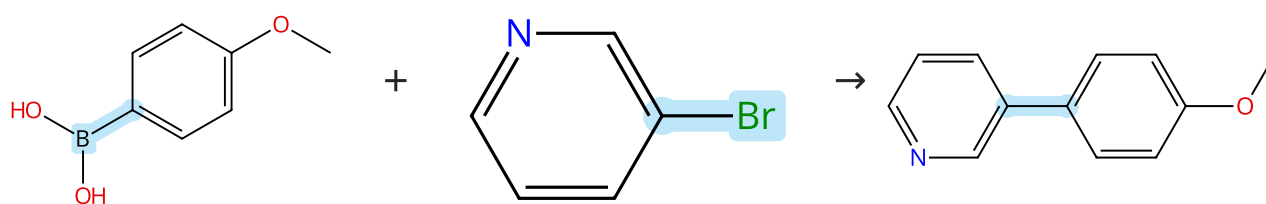
By: Parisien-Collette, Shawn; et al

Green Chemistry (2017), 19(20), 4798-4803.

Experimental Protocols

Scheme 345 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (112)

Suppliers (88)

Suppliers (54)

31-179-CAS-20010581

Steps: 1 Yield: 97%

Room-Temperature Aqueous Suzuki-Miyaura Cross-Coupling Reactions Catalyzed via a Recyclable Palladium@Haloysite Nanocomposite

1.1 Reagents: Cesium carbonate

Catalysts: Palladium (halloysite nanocomposite-supported)

Solvents: 1-Propanol, Water; 10 min, rt; 1 h, rt

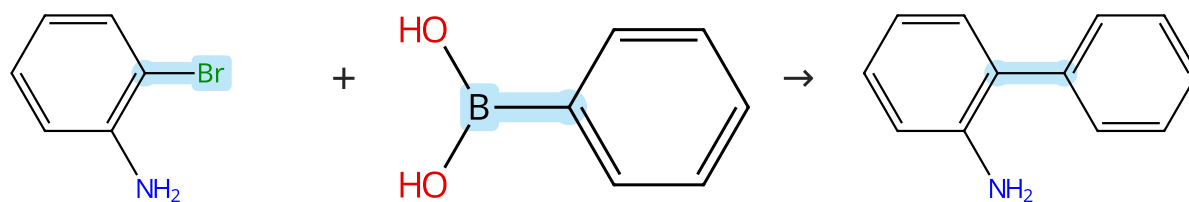
By: Hamdi, Jumanah; et al

Organic Letters (2019), 21(10), 3471-3475.

Experimental Protocols

Scheme 346 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (71)

Suppliers (132)

Suppliers (73)

31-179-CAS-20573973

Steps: 1 Yield: 97%

A new, substituted palladacycle for ppm level Pd- catalyzed Suzuki-Miyaura cross couplings in water

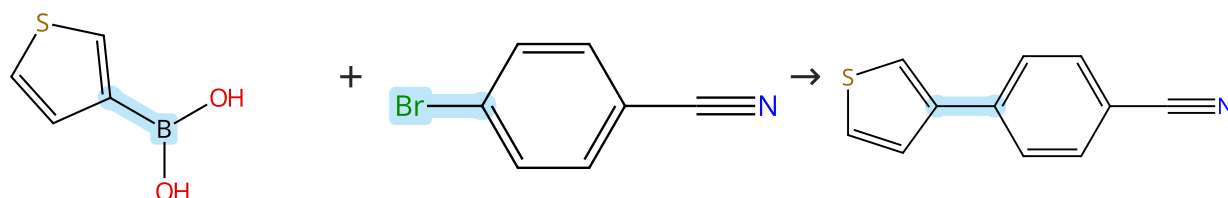
By: Takale, Balaram S.; et al

Chemical Science (2019), 10(38), 8825-8831.

- 1.1 **Reagents:** Tripotassium phosphate, Poly(oxy-1,2-ethanediyl), α -[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyl tridecyl)-2H-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]- ω -methoxy-
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: Tetrahydrofuran; 24 h, 45 °C

Scheme 347 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (101)

Suppliers (93)

Suppliers (45)

31-179-CAS-17820355

Steps: 1 Yield: 97%

Suzuki-Miyaura Micellar Cross-Coupling in Water, at Room Temperature, and under Aerobic Atmosphere

By: Mattiello, Sara; et al

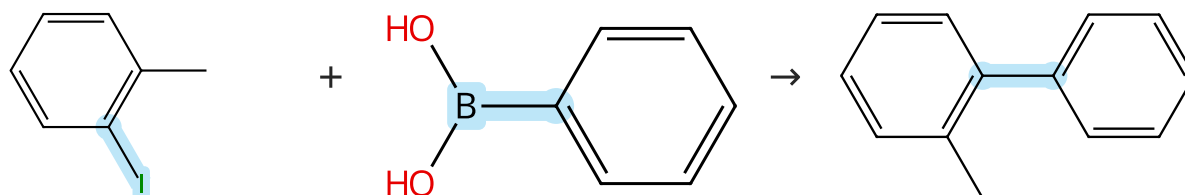
Organic Letters (2017), 19(3), 654-657.

- 1.1 **Reagents:** Triethylamine
Catalysts: (*SP*-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino- κ P]ferrocene]dichloropalladium
Solvents: Water; 1 h, rt

Experimental Protocols

Scheme 348 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (77)

Suppliers (132)

Suppliers (63)

31-614-CAS-42942514

Steps: 1 Yield: 97%

Heterogenization of palladium trimer and nanoparticles through polymerization boosted catalytic efficiencies in recyclable coupling and reduction reactions

By: Wang, Miaomiao; et al

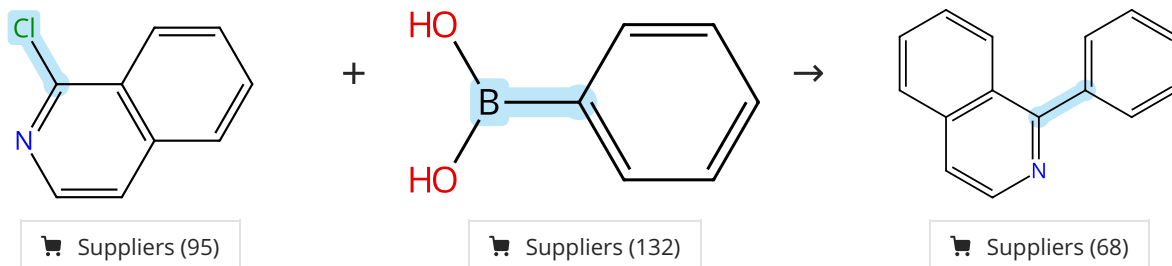
Chemistry - A European Journal (2024), 30(72), e202403447.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: 3066141-61-2
Solvents: Acetonitrile, Water; 80 °C

Experimental Protocols

Scheme 349 (1 Reaction)

Steps: 1 Yield: 97%



31-614-CAS-40811937

Steps: 1 Yield: 97%

Exogenous photocatalyst-free aryl radical generation from diaryliodonium salts and use in metal-catalyzed C-H arylation

By: Galicia, Jonathan; et al

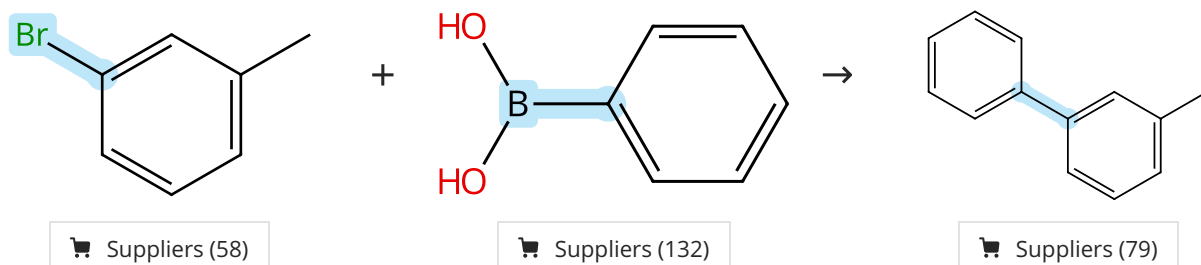
Chemical Communications (Cambridge, United Kingdom) (2024), 60(54), 6929-6932.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 5 h, 80 °C

Experimental Protocols

Scheme 350 (2 Reactions)

Steps: 1 Yield: 97%



31-614-CAS-42942524

Steps: 1 Yield: 97%

Heterogenization of palladium trimer and nanoparticles through polymerization boosted catalytic efficiencies in recyclable coupling and reduction reactions

By: Wang, Miaomiao; et al

Chemistry - A European Journal (2024), 30(72), e202403447.

1.1 **Reagents:** Potassium carbonate
Catalysts: 3066141-61-2
Solvents: Dimethylformamide, Water; 80 °C

Experimental Protocols

31-179-CAS-18949889

Steps: 1 Yield: 97%

Synthesis of Ultrafine and Highly Dispersed Metal Nanoparticles Confined in a Thioether-Containing Covalent Organic Framework and Their Catalytic Applications

By: Lu, Shuanglong; et al

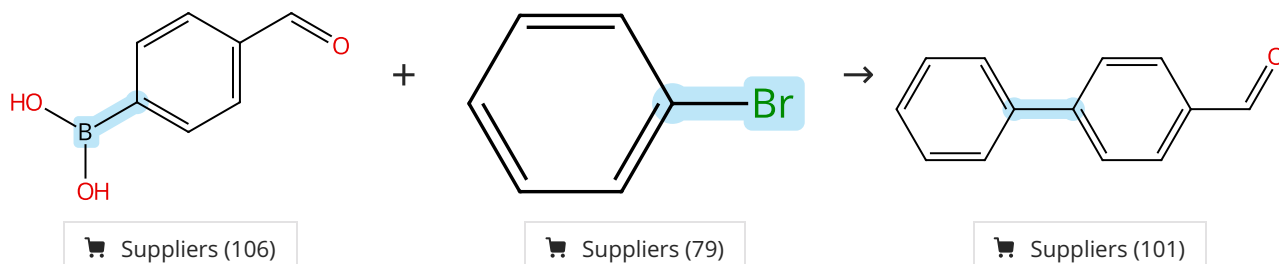
Journal of the American Chemical Society (2017), 139(47), 17082-17088.

1.1 **Reagents:** Potassium carbonate
Catalysts: Palladium, 2231404-38-7
Solvents: Dimethylformamide, Water; 3 h, 50 °C

Experimental Protocols

Scheme 351 (1 Reaction)

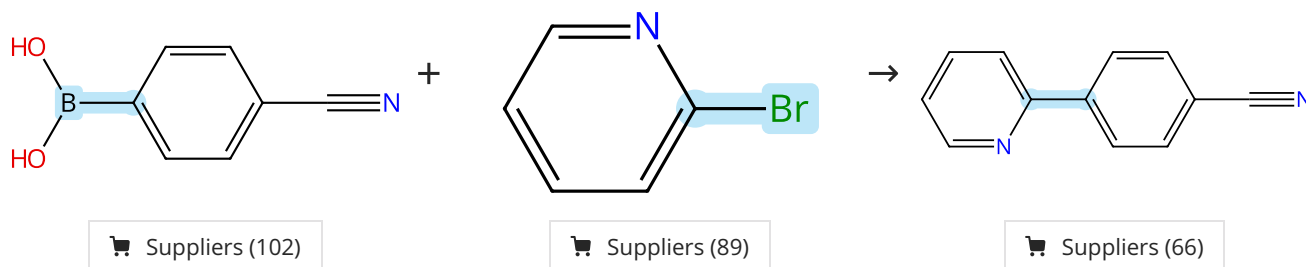
Steps: 1 Yield: 96%



31-179-CAS-21608059 Steps: 1 Yield: 96%	Pd@COF-QA: A phase transfer composite catalyst for aqueous Suzuki-Miyaura coupling reaction
1.1 Reagents: Triethylamine Catalysts: Palladium (supported on paraffin-chain quaternary ammonium salt decorated covalen...), 2412753-38-7 Solvents: Water; 6 h, 50 °C	By: Wang, Jian-Cheng; et al Green Chemistry (2020), 22(4), 1150-1155.

Scheme 352 (1 Reaction)

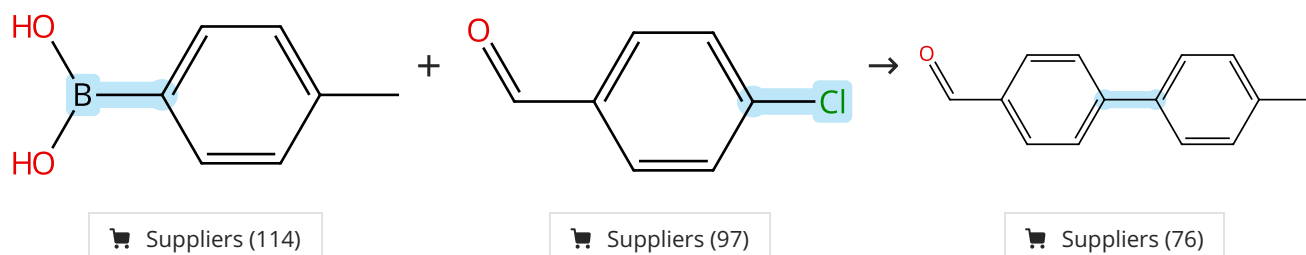
Steps: 1 Yield: 96%



31-614-CAS-44288346 Steps: 1 Yield: 96%	Nitrile-tolerant Iridium-catalysed Hydrogen Isotope Exchange
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate Solvents: Methanol, Acetonitrile; 24 h, rt → 65 °C Experimental Protocols	By: Knight, Nathan M. L.; et al Chemistry - A European Journal (2025), 31(20), e202500449.

Scheme 353 (1 Reaction)

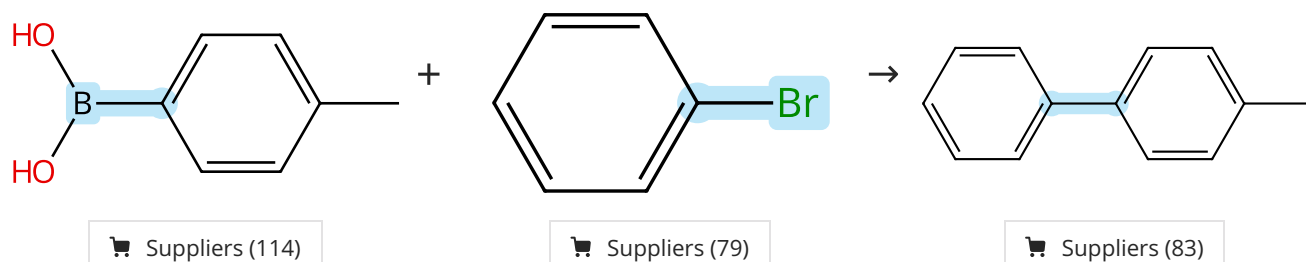
Steps: 1 Yield: 96%



31-614-CAS-32738965 Steps: 1 Yield: 96%	An air-stable, well-defined palladium-BIAN-NHC chloro dimer: a fast-activating, highly efficient catalyst for cross-coupling
1.1 Reagents: Potassium carbonate Catalysts: Bis[7,9-bis[2,6-bis(1-methylethyl)phenyl]-7,9-dihydro-8 <i>H</i> -acenaphth[1,2- <i>d</i>]imidazol-8-ylidene]di-μ-chlorodiclorodipalladium Solvents: Ethanol; 12 h, 25 °C Experimental Protocols	By: Zhang, Jin; et al Chemical Communications (Cambridge, United Kingdom) (2022), 58(53), 7404-7407.

Scheme 354 (1 Reaction)

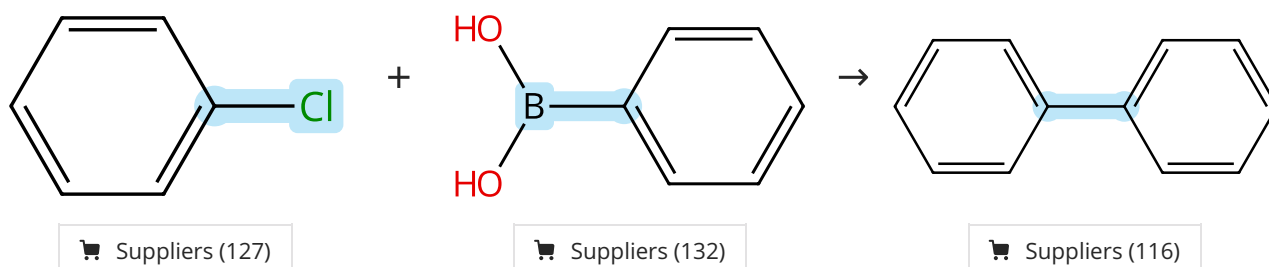
Steps: 1 Yield: 96%



31-179-CAS-16860442 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate, Oxygen Catalysts: Palladium(1+), μ-chlorobis[μ-(diphenylphosphino)] tris(triphenylphosphine)tri-, <i>triangulo</i>, (OC-6-11)-hexafluoro antimonate(1-) (1:1) (independent or solid solution with μ-Br analog) Solvents: Isopropanol, Water; 15 min, 21 - 24 °C Experimental Protocols	A Robust and Efficient Pd₃ Cluster Catalyst for the Suzuki Reaction and Its Odd Mechanism By: Fu, Fangyu; et al ACS Catalysis (2017), 7(3), 1860-1867.
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Scheme 355 (1 Reaction)

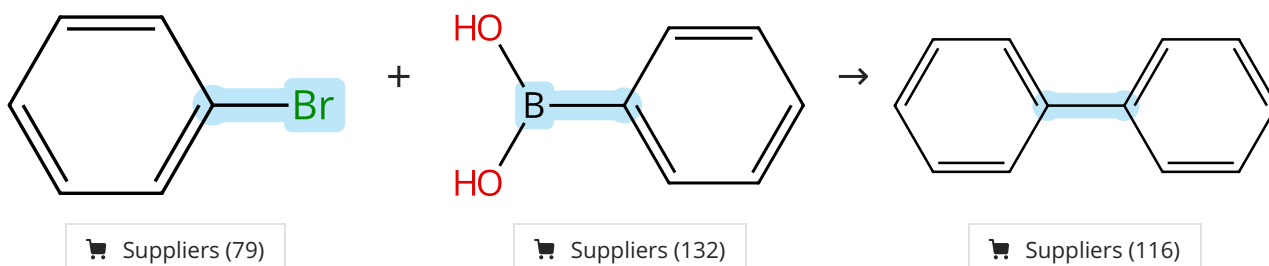
Steps: 1 Yield: 96%



31-179-CAS-20341270 Steps: 1 Yield: 96% 1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate Solvents: <i>p</i>-Xylene; 1.5 h, 128 °C; 128 °C → rt 1.2 Solvents: Water; 24 h, 130 °C	Ligand-free Suzuki-Miyaura coupling reaction of aryl chlorides using a continuous irradiation type microwave and a palladium nanoparticle catalyst: effect of a co-existing solid By: Yamada, Makito; et al Green Chemistry (2019), 21(16), 4541-4549.
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Scheme 356 (3 Reactions)

Steps: 1 Yield: 96%



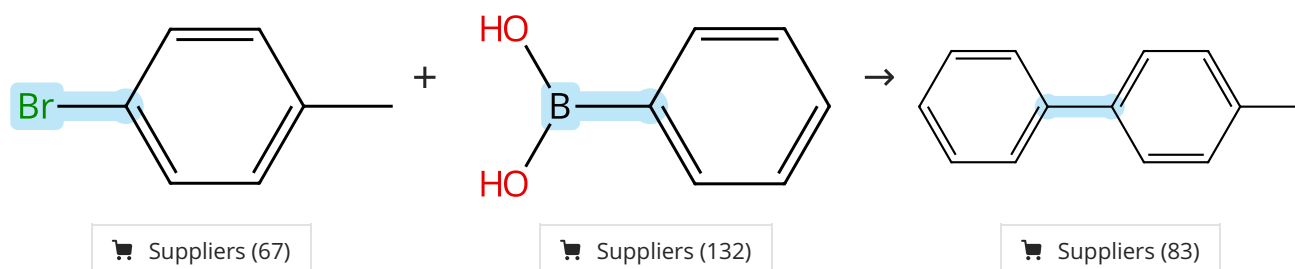
31-179-CAS-17995091 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: 2130870-97-0 Solvents: Isopropanol, Water; 10 min, rt 1.2 Reagents: Water Experimental Protocols	Synthesis of sterically hindered N-benzyladamantyl substituted benzimidazol-2-ylidene palladium complexes and investigation of their catalytic activity in aqueous medium By: Dehimat, Zineb Imene; et al Tetrahedron (2017), 73(40), 5940-5945.
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31-179-CAS-14174912 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate, Oxygen Catalysts: Palladium Solvents: Water; 4 h, 80 °C Experimental Protocols	Activation of Aryl Chlorides in Water under Phase-Transfer Agent-Free and Ligand-Free Suzuki Coupling by Heterogeneous Palladium Supported on Hybrid Mesoporous Carbon By: Duan, Linlin; et al ACS Catalysis (2015), 5(2), 575-586.
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31-179-CAS-1699656	Steps: 1 Yield: 96%	Accessing stable zirconium carboxy-aminophosphonate nanosheets as support for highly active Pd nanoparticles
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, Zirconate(5-), [μ -[phosphato(3-)- $\kappa O:\kappa O'$]] [μ -[<i>N</i> -[(phosphono- $\kappa O:\kappa O$)methyl]- <i>N</i> -(phosphonomethyl)glycinato(5-))] [<i>N</i> -[(phosphono- κO)methyl]- <i>N</i> -(phosphonomethyl)glycinato(5-)] di-, hydrogen, hydrate (1:5:1) Solvents: Ethanol, Water; 30 h, 70 °C		By: Costantino, Ferdinando; et al Chemical Communications (Cambridge, United Kingdom) (2015), 51(88), 15990-15993.
Experimental Protocols		

Scheme 357 (7 Reactions)

Steps: 1 Yield: 96%

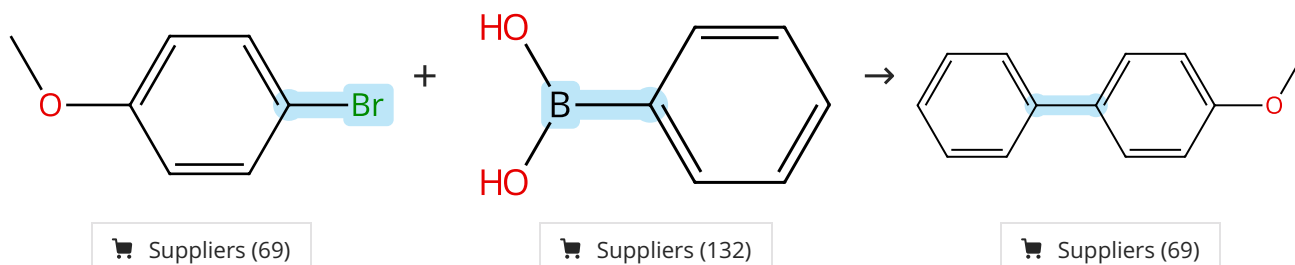


31-614-CAS-42942527	Steps: 1 Yield: 96%	Heterogenization of palladium trimer and nanoparticles through polymerization boosted catalytic efficiencies in recyclable coupling and reduction reactions
1.1 Reagents: Potassium carbonate Catalysts: 3066141-61-2 Solvents: Dimethylformamide, Water; 80 °C		By: Wang, Miaomiao; et al Chemistry - A European Journal (2024), 30(72), e202403447.
Experimental Protocols		
31-614-CAS-34078914	Steps: 1 Yield: 96%	Cyrene blends: a greener solvent system for organic syntheses
1.1 Reagents: Cesium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Dimethylformamide, (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; rt → 50 °C; 16 h, 50 °C		By: Sullivan, Caren; et al Green Chemistry (2022), 24(18), 7184-7193.
Experimental Protocols		
31-614-CAS-31041649	Steps: 1 Yield: 96%	Intermetallic ZrPd ₃ -Embedded Nanoporous ZrC as an Efficient and Stable Catalyst of the Suzuki Cross-Coupling Reaction
1.1 Reagents: Potassium carbonate Catalysts: Palladium diacetate, Zirconium oxychloride Solvents: Ethanol; 10 h, 60 °C		By: Lu, Yangfan; et al ACS Catalysis (2020), 10(24), 14366-14374.
Experimental Protocols		
31-179-CAS-22784840	Steps: 1 Yield: 96%	Anchoring Pd(OAc) ₂ on amide-bonded covalent organic frameworks: An efficient heterogeneous Pd@OC-MA catalyst for Suzuki-Miyaura coupling reactions in water
1.1 Reagents: Sodium bicarbonate Catalysts: Palladium diacetate Solvents: Water; 5 h, rt		By: Wu, Shang; et al Tetrahedron (2020), 76(48), 131664.
Experimental Protocols		
31-179-CAS-19332550	Steps: 1 Yield: 96%	Heteroatom Donor-Decorated Polymer-Immobilized Ionic Liquid Stabilized Palladium Nanoparticles: Efficient Catalysts for Room-Temperature Suzuki-Miyaura Cross-Coupling in Aqueous Media
1.1 Reagents: Potassium carbonate Catalysts: 2242655-81-6 Solvents: Ethanol, Water; 6 h, rt		By: Doherty, Simon; et al Advanced Synthesis & Catalysis (2018), 360(19), 3716-3731.
Experimental Protocols		

31-179-CAS-15895365 Steps: 1 Yield: 96% 1.1 Reagents: Potassium hydroxide Catalysts: 1,3-Benzenediol, 5-(hydroxymethyl)-, homopolymer (optionally carboxy-terminated) Solvents: Water; rt 1.2 Catalysts: Di-μ-chlorobis(η^3-2-propenyl)dipalladium; 15 min, rt 1.3 Reagents: Potassium carbonate Solvents: Water; 0 °C; 1 h, 100 °C	Suzuki-Miyaura reaction catalyzed by a dendritic phosphine-palladium complex By: Fujita, Ken-ichi; et al Tetrahedron (2016), 72(11), 1485-1492.
31-179-CAS-10727962 Steps: 1 Yield: 96% 1.1 Catalysts: Palladium diacetate, BINAP Solvents: Acetonitrile, Tetrahydrofuran; 3 h, 22 °C 1.2 Reagents: Tripotassium phosphate Solvents: Water; 40 min, 22 °C 1.3 Solvents: Methanol; 2.5 h, 22 °C Experimental Protocols	Mono-Oxidation of Bidentate Bis-phosphines in Catalyst Activation: Kinetic and Mechanistic Studies of a Pd/Xantphos-Catalyzed C-H Functionalization By: Ji, Yining; et al Journal of the American Chemical Society (2015), 137(41), 13272-13281.

Scheme 358 (2 Reactions)

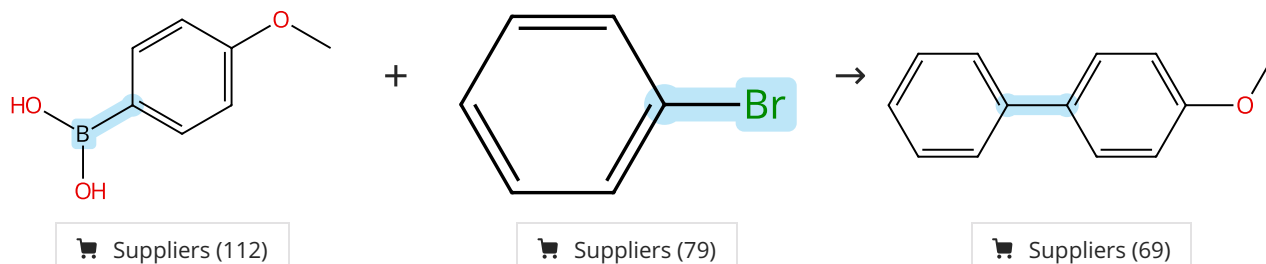
Steps: 1 Yield: 96%

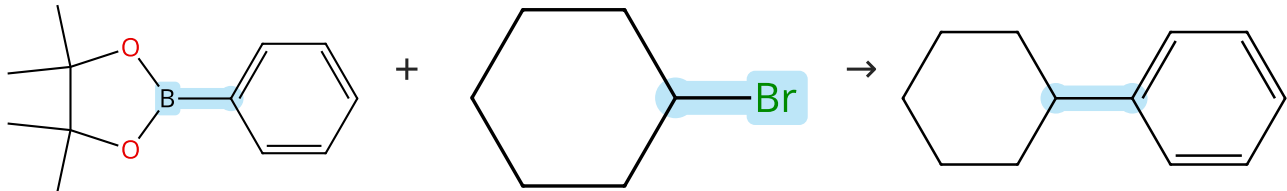
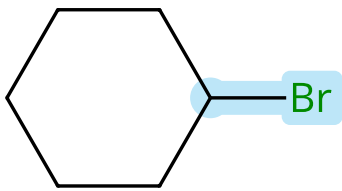
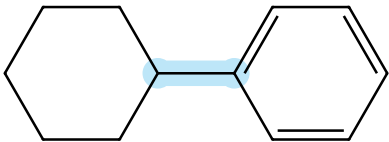


31-614-CAS-24553006 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: 4,8,12-Trioxa-2,10-diazatetradecane-5,17-diamine (palladium acetate complex) Solvents: <i>p</i>-Xylene; 2 h, 150 °C	Postsynthetic Metalation of a Robust Hydrogen-Bonded Organic Framework for Heterogeneous Catalysis By: Han, Bin; et al Journal of the American Chemical Society (2019), 141(22), 8737-8740.
31-179-CAS-17995088 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: 2130870-97-0 Solvents: Isopropanol, Water; 1 h, rt 1.2 Reagents: Water Experimental Protocols	Synthesis of sterically hindered N-benzyladamantyl substituted benzimidazol-2-ylidene palladium complexes and investigation of their catalytic activity in aqueous medium By: Dehimat, Zineb Imene; et al Tetrahedron (2017), 73(40), 5940-5945.

Scheme 359 (4 Reactions)

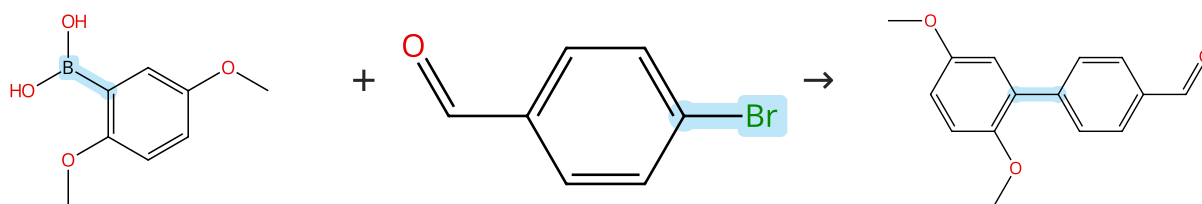
Steps: 1 Yield: 96%



31-614-CAS-46049895 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate, Tetrabutylammonium bromide Catalysts: 3091657-75-6 (reaction products iwth 6- aminoh exane-1-thiol hydrochloride and bis(hyd...) Solvents: Methanol, Water; 4 h, 90 °C	Palladium Complex Grafted on PEG-Based Amphiphilic Polymers as ppm Level Micellar Catalysts for Suzuki-Miyaura Coupling in Water By: Xiao, Song; et al Organic Letters (2025), 27(16), 4343-4348.
31-179-CAS-20991733 Steps: 1 Yield: 96% 1.1 Reagents: Tripotassium phosphate Catalysts: Palladium, [1,1'-(9,9-dimethyl-9<i>H</i>-xanthene-4,5-diyl) bis[1,1-diphenylphosphine-κ<i>P</i>]]bis[(trimethylsilyl)methyl]-, (<i>S</i> <i>P</i>-4-2)- Solvents: Tetrahydrofuran, Water; 6 h, rt → 80 °C	Air- and moisture-stable Xantphos-ligated palladium dialkyl complex as a precatalyst for cross-coupling reactions By: Takahashi, Rina; et al Chemical Communications (Cambridge, United Kingdom) (2020), 56(3), 407-410.
31-179-CAS-16584130 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: Palladium, Boron nitride (hydroxylated), 2-[[[3-(Trimethoxysilyl)propyl]imino]methyl]phenol (complexes with palladium, supported on hydroxylated boron nitride) Solvents: Water; 0.5 h, 70 °C Experimental Protocols	Ligand-Free C-C Coupling Reactions Promoted by Hexagonal Boron Nitride-Supported Palladium(II) Catalyst in Water By: Cheng, Xu; et al Advanced Synthesis & Catalysis (2017), 359(3), 454-466.
31-179-CAS-6262327 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: (<i>SP</i>-4-4)-Chloro[(<i>Z</i>)-[(2,6-dimethylphenyl)imino] [[(imino-κ<i>N</i>)phenylmethyl]phenylamino]methyl-κ<i>C</i>][2-(isocyano-κ<i>C</i>)-1,3-dimethylbenzene]palladium Solvents: Ethanol, Water; rt Experimental Protocols	Acyclic aminocarbene-like palladium complex-catalyzed Suzuki-Miyaura reaction at low catalyst loadings By: Fang, Yi; et al Tetrahedron (2015), 71(52), 9679-9683.
Scheme 360 (1 Reaction) Steps: 1 Yield: 96%  Suppliers (80)  Suppliers (75)  Suppliers (80)	
31-614-CAS-38930527 Steps: 1 Yield: 96% 1.1 Reagents: Sodium <i>tert</i>-butoxide Catalysts: 4,7-Diphenyl-1,10-phenanthroline, Bromo[1,1'-thiobis[methane]]copper Solvents: Benzene; 12 h, 80 °C Experimental Protocols	A General Copper Catalytic System for Suzuki-Miyaura Cross-Coupling of Unactivated Secondary and Primary Alkyl Halides with Arylborons By: Zhou, Yonglei; et al Journal of the American Chemical Society (2023), 145(51), 28146-28155.

Scheme 361 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (94)

Suppliers (92)

Suppliers (17)

31-614-CAS-41686025

Steps: 1 Yield: 96%

The Substituent Effects of Suzuki Coupling in Aqueous Micelles

By: Liu, Haijing; et al

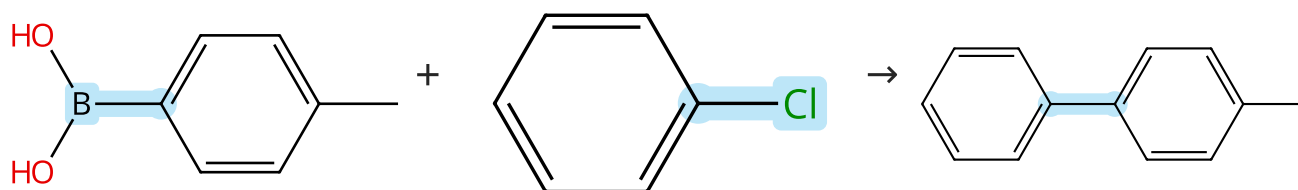
Journal of Organic Chemistry (2024), 89(17), 11930-11938.

- 1.1 **Reagents:** Triethylamine, Polyethylene glycol lauryl ether
Catalysts: (*SP*-4-2)-[1,1'-Bis[bis(1,1-dimethylethyl)phosphino- κ^2]ferrocene]dichloropalladium
Solvents: Water; 2 h, 40 °C

Experimental Protocols

Scheme 362 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (114)

Suppliers (127)

Suppliers (83)

31-179-CAS-11282143

Steps: 1 Yield: 96%

N-Heterocyclic carbene ligands bearing poly(ethylene glycol) chains: effect of the chain length on palladium-catalyzed coupling reactions employing aryl chlorides

By: Fujihara, Tetsuaki; et al

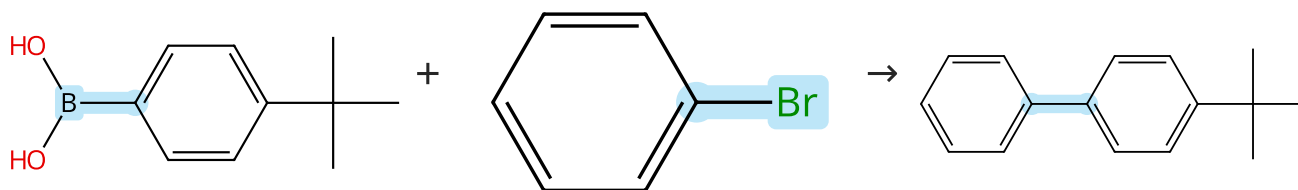
Chemical Communications (Cambridge, United Kingdom) (2015), 51(98), 17382-17385.

- 1.1 **Reagents:** Potassium bis(trimethylsilyl)amide
Catalysts: Di- μ -chlorobis(η^3 -2-propenyl)dipalladium, Poly(oxy-1,2-ethanediyl), α,α',α'' -[1*H*-imidazolium-1,3-diylbis(methylene-5,1,2,3-benzenetetrayl)]hexakis[ω -methoxy-, chloride (1:1)
Solvents: Tetrahydrofuran; 30 min, rt
- 1.2 **Reagents:** Tripotassium phosphate
Solvents: Tetrahydrofuran; 10 min, rt; 15 h, 45 °C

Experimental Protocols

Scheme 363 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (94)

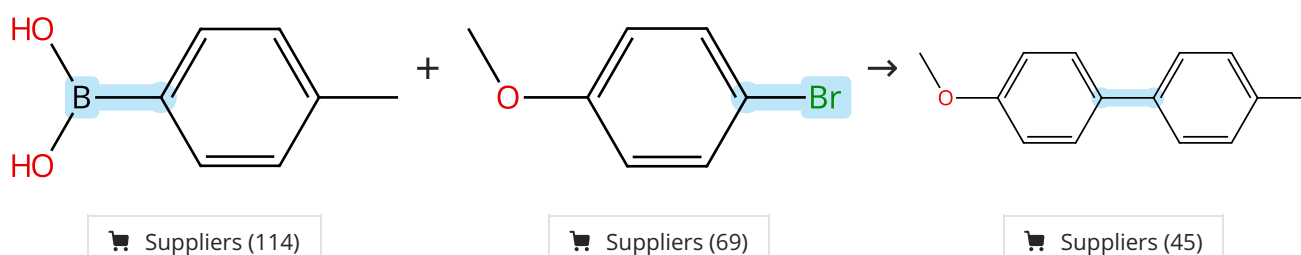
Suppliers (79)

Suppliers (57)

31-179-CAS-16860447 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate, Oxygen Catalysts: Palladium(1+), μ-chlorobis[μ-(diphenylphosphino)] tris(triphenylphosphine)tri-, <i>triangulo</i>, (OC-6-11)-hexafluoro antimonate(1-) (1:1) (independent or solid solution with μ-Br analog) Solvents: Isopropanol, Water; 15 min, 21 - 24 °C Experimental Protocols	A Robust and Efficient Pd₃ Cluster Catalyst for the Suzuki Reaction and Its Odd Mechanism By: Fu, Fangyu; et al ACS Catalysis (2017), 7(3), 1860-1867.
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Scheme 364 (1 Reaction)

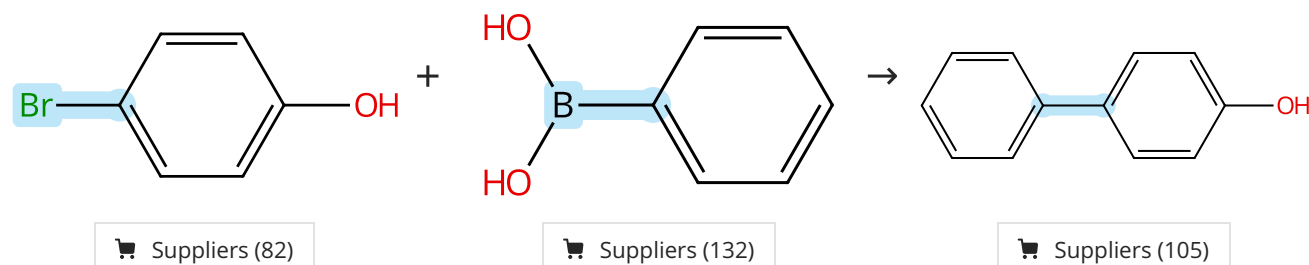
Steps: 1 Yield: 96%



31-614-CAS-23955906 Steps: 1 Yield: 96% 1.1 Catalysts: Dichlorobis(triphenylphosphine)palladium, 2696405-19-1 Solvents: Tetrahydrofuran; 5 min, rt 1.2 Reagents: Tripotassium phosphate Solvents: Tetrahydrofuran, Water; 10 h, 100 °C Experimental Protocols	Construction of Bulky Ligand Libraries by Ru^(III)-Catalyzed P^(III)-Assisted ortho-C-H Secondary Alkylation By: Li, Ming; et al Journal of Organic Chemistry (2021), 86(17), 11915-11925.
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Scheme 365 (1 Reaction)

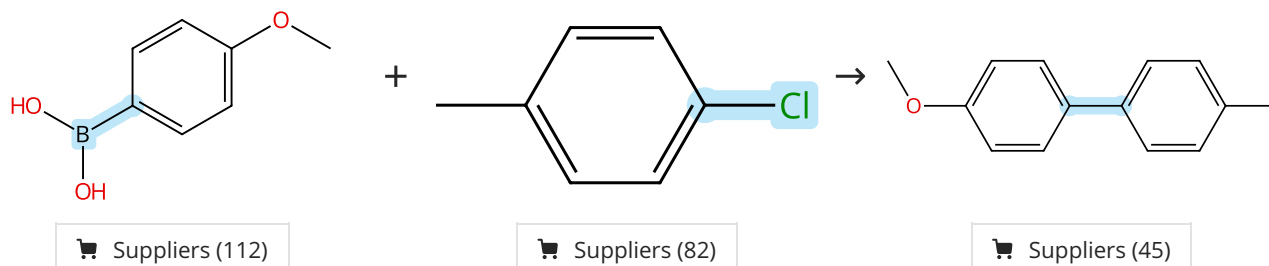
Steps: 1 Yield: 96%



31-614-CAS-24489498 Steps: 1 Yield: 96% 1.1 Reagents: Sodium hydroxide Catalysts: Graphene (post-treated with Pd(OAc)₂) Solvents: Water; 60 min, 100 °C Experimental Protocols	Palladium nanoparticles supported on graphene acid: a stable and eco-friendly bifunctional C-C homo- and cross-coupling catalyst By: Blanco, Matias; et al Green Chemistry (2019), 21(19), 5238-5247.
--	---

Scheme 366 (1 Reaction)

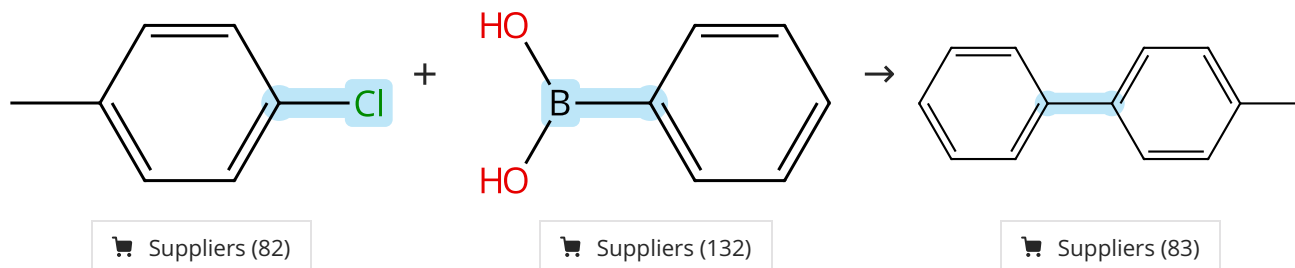
Steps: 1 Yield: 96%



31-179-CAS-21932608	Steps: 1 Yield: 96%	A Highly Efficient Monophosphine Ligand for Parts per Million Levels Pd-Catalyzed Suzuki-Miyaura Coupling of (Hetero)Aryl Chlorides
1.1 Catalysts: Palladium diacetate, 3-(Dicyclohexylphosphino)-2-(2,6-dimethoxyphenyl)-1-methyl-1 <i>H</i> -indole Solvents: 1,4-Dioxane; 10 min, rt		By: Choy, Pui Ying; et al
1.2 Reagents: Tripotassium phosphate; 24 h, 100 °C; 100 °C → rt		European Journal of Organic Chemistry (2020), 2020(19), 2846-2853.
1.3 Reagents: Water; rt		

Scheme 367 (2 Reactions)

Steps: 1 Yield: 96%

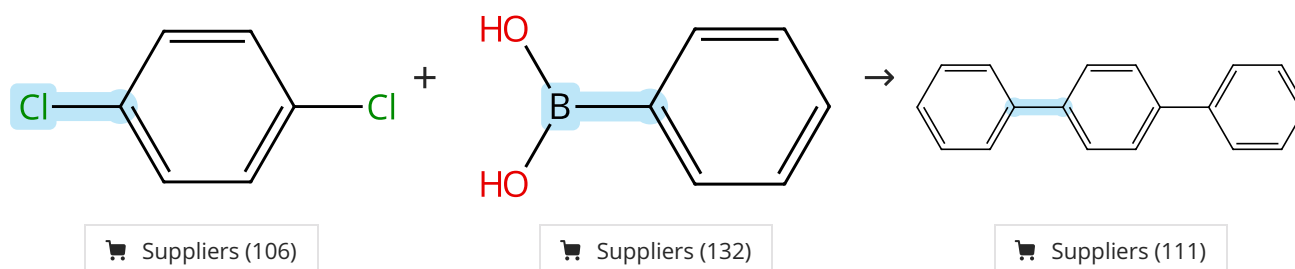


31-614-CAS-36762613	Steps: 1 Yield: 96%	Large acceleration of solvent-free Suzuki-Miyaura cross-coupling reactions at room temperature by the addition of small amounts of dioxane and water
1.1 Reagents: Potassium carbonate, Water Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine Solvents: 1,4-Dioxane; 45 min, 25 °C		By: Tanemura, Kiyoshi
Experimental Protocols		Tetrahedron (2023), 140, 133470.

31-179-CAS-22064190	Steps: 1 Yield: 96%	Activator-Promoted Aryl Halide-Dependent Chemoselective Buchwald-Hartwig and Suzuki-Miyaura Type Cross-Coupling Reactions
1.1 Reagents: 4-Methylaniline, Tripotassium phosphate Catalysts: Nickel iodide (NiI ₂) Solvents: 1,4-Dioxane; 24 h, 115 °C		By: Dhital, Raghu N.; et al
1.2 Reagents: Water		Organic Letters (2020), 22(12), 4797-4801.
Experimental Protocols		

Scheme 368 (1 Reaction)

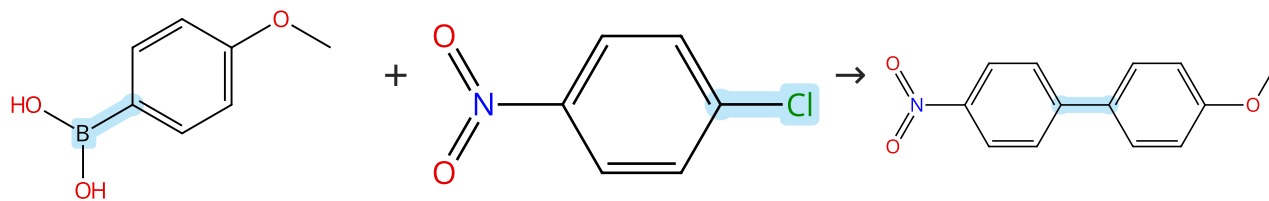
Steps: 1 Yield: 96%



31-179-CAS-19588912	Steps: 1 Yield: 96%	Sterically hindered N-aryl/benzyl substituted piperidoim idazolin-2-ylidene palladium complexes and their catalytic activities
1.1 Reagents: Potassium hydroxide Catalysts: 2247697-87-4 Solvents: Isopropanol; rt → 82 °C; 30 min, 82 °C		By: Gacal, Elif; et al
Experimental Protocols		Tetrahedron (2018), 74(47), 6829-6838.

Scheme 369 (2 Reactions)

Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (63)

Suppliers (55)

31-614-CAS-43871823

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Ethanolamine, Tetrabutylammonium bromide, Palladium chloride
Solvents: Water; 10 h, 90 °C

Experimental Protocols

ppm level Pd catalytic Suzuki- Miyaura coupling reaction in water and its application in the synthesis of Bixafen

By: Tan, Zhiyong; et al

Tetrahedron (2025), 173, 134464.

31-614-CAS-44132948

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Sodium hydroxide
Catalysts: Palladium chloride, 1-[2,6-Bis(1-methylethyl)phenyl]-2-(diphenylphosphino)-1*H*-imidazole
Solvents: *tert*-Butanol; 30 min, 40 °C
- 1.2 **Reagents:** Sodium hydride
Solvents: 1,4-Dioxane; 16 h, 40 °C → 100 °C

Experimental Protocols

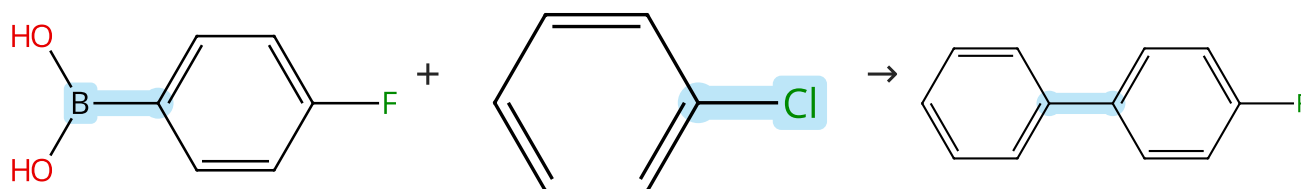
Using Common Alcohols to Transform N-H NHC-Palladium Complexes into Reactive Catalysts for the Suzuki-Miyaura Reaction with Aryl Chlorides

By: Larson, Alexandra J. S.; et al

Journal of Organic Chemistry (2025), 90(7), 2806-2810.

Scheme 370 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (104)

Suppliers (127)

Suppliers (78)

31-179-CAS-4943874

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Potassium bis(trimethylsilyl)amide
Catalysts: Di-μ-chlorobis(η³-2-propenyl)dipalladium, Poly(oxy-1,2-ethanediyl), α,α',α''-[1*H*-imidazolium-1,3-diylbis(methylene-5,1,2,3-benzenetetracyl)]hexakis[ω-methoxy-, chloride (1:1)
Solvents: Tetrahydrofuran; 30 min, rt
- 1.2 **Reagents:** Tripotassium phosphate
Solvents: Tetrahydrofuran; 10 min, rt; 15 h, 45 °C

Experimental Protocols

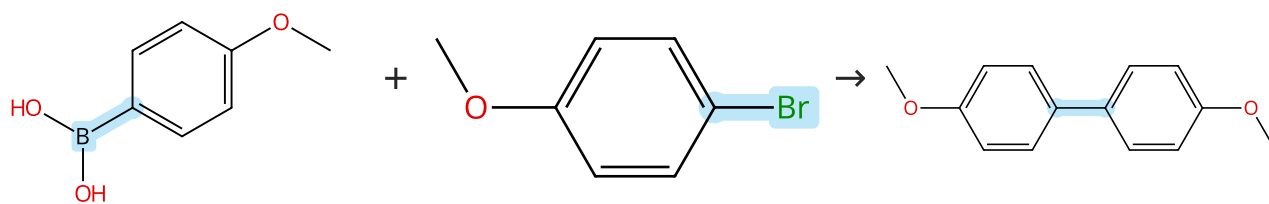
N-Heterocyclic carbene ligands bearing poly(ethylene glycol) chains: effect of the chain length on palladium-catalyzed coupling reactions employing aryl chlorides

By: Fujihara, Tetsuaki; et al

Chemical Communications (Cambridge, United Kingdom) (2015), 51(98), 17382-17385.

Scheme 371 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (69)

Suppliers (58)

31-614-CAS-37642581

Steps: 1 Yield: 96%

Mechanochemical Suzuki-Miyaura Cross-Coupling with Natural Deep Eutectic Solvent as Liquid-Assisted Grinding Additive: Merging Two Fields for a Greener Strategy

By: Thiery, Emilie; et al

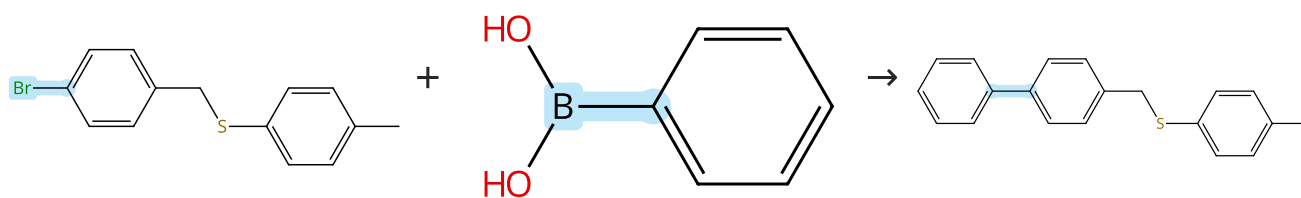
European Journal of Organic Chemistry (2023), 26(37), e202300727.

1.1 **Reagents:** Sodium carbonate
Catalysts: Palladium diacetate
Solvents: Ethanaminium, 2-hydroxy-*N,N,N*-trimethyl-, chloride (1:1), mixt. with 1,2,3-propanetriol; 10 min

Experimental Protocols

Scheme 372 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (12)

Suppliers (132)

31-614-CAS-38964278

Steps: 1 Yield: 96%

Organophosphorus-Catalyzed Direct Dehydroxylative Thioetherification of Alcohols with Hypervalent Organosulfur Compounds

By: Sun, Gang; et al

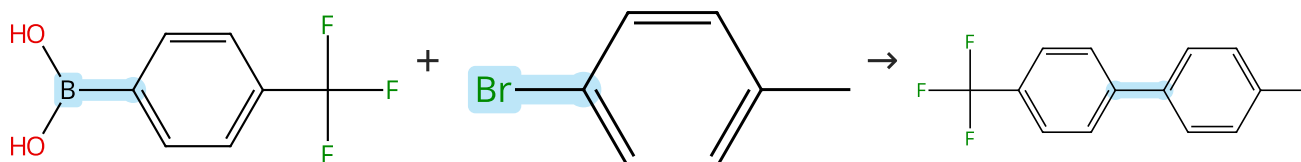
Journal of Organic Chemistry (2024), 89(2), 1083-1090.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Dimethylformamide; 6 h, 90 °C

Experimental Protocols

Scheme 373 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (113)

Suppliers (67)

Suppliers (30)

31-179-CAS-15788257

Steps: 1 Yield: 96%

Efficient Catalyst for Both Suzuki and Heck Cross-Coupling Reactions: Synthesis and Catalytic Behaviour of Geometry-Constrained Iminopyridylpalladium Chlorides

By: Tang, Yujie; et al

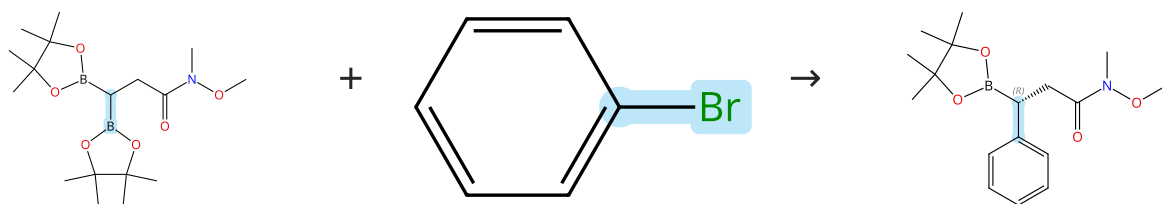
Advanced Synthesis & Catalysis (2016), 358(16), 2642-2651.

1.1 **Reagents:** Potassium carbonate
Catalysts: 1984856-10-1
Solvents: Toluene; 18 h, 110 °C

Experimental Protocols

Scheme 374 (1 Reaction)

Steps: 1 Yield: 96%



Absolute stereochemistry shown,
Rotation (-)

Suppliers (79)

31-179-CAS-3769972

Steps: 1 Yield: 96%

Optimization of Reaction and Substrate Activation in the Stereoselective Cross-Coupling of Chiral 3,3-Diboronate Amides

By: Lee, Jack Chang Hung; et al

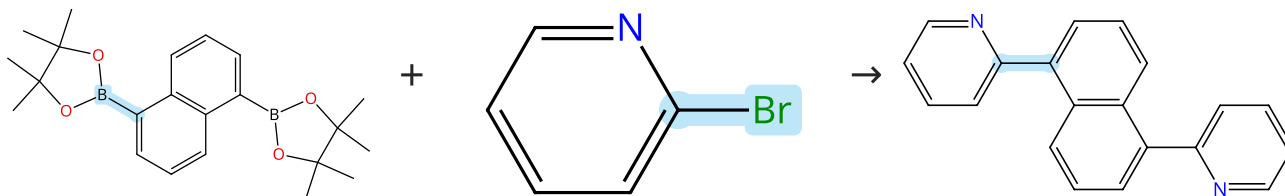
Journal of Organic Chemistry (2015), 80(14), 7134-7143.

1.1 **Reagents:** Phenol, Potassium carbonate
Catalysts: Bis(dibenzylideneacetone)palladium, X-Phos
Solvents: Toluene; 5 h, 80 °C; cooled

Experimental Protocols

Scheme 375 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (4)

Suppliers (89)

31-179-CAS-19398969

Steps: 1 Yield: 96%

Cationic Nitrogen-Doped Helical Nanographenes

By: Xu, Kun; et al

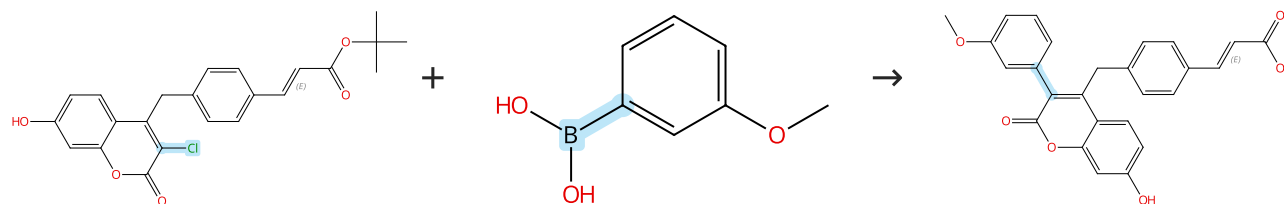
Angewandte Chemie, International Edition (2017), 56(50), 15876-15881.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Methanol, Toluene, Water; 15 h, 90 °C

Experimental Protocols

Scheme 376 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Suppliers (112)

Double bond geometry shown

Supplier (1)

31-179-CAS-14369767

Steps: 1 Yield: 96%

Investigation of (E)-3-[4-(2-Oxo-3-aryl-chromen-4-yl)oxyphenyl]acrylic Acids as Oral Selective Estrogen Receptor Down-Regulators

By: Degorce, Sebastien L.; et al

Journal of Medicinal Chemistry (2015), 58(8), 3522-3533.

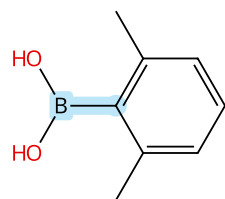
1.1 **Reagents:** Formic acid; 1 h, 25 °C
 1.2 **Catalysts:** Disodium tetrachloropalladate, 3-[Bis(1,1-dimethylethyl)phosphino]-1-propanesulfonic acid
Solvents: Water; 5 min, rt

1.3 **Reagents:** Potassium carbonate
Solvents: Water; overnight, 65 °C

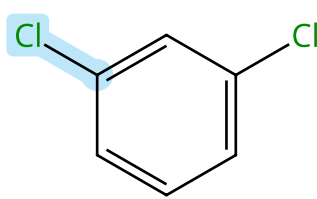
Experimental Protocols

Scheme 377 (1 Reaction)

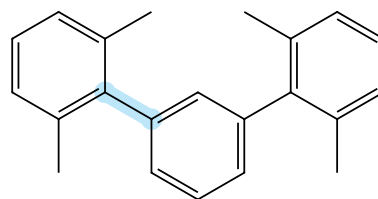
Steps: 1 Yield: 96%



+



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Suppliers (100)

Suppliers (85)

Suppliers (4)

31-179-CAS-17819431

Steps: 1 Yield: 96%

Bulky Yet Flexible Pd-PEPPSI-IPent^{An} for the Synthesis of Sterically Hindered Biaryls in Air

By: Lu, Dong-Dong; et al

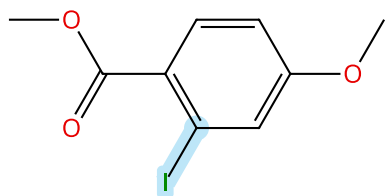
Journal of Organic Chemistry (2017), 82(20), 10898-10911.

1.1 **Reagents:** Potassium *tert*-butoxide
Catalysts: (SP-4-1)-[7,9-Bis[2,6-bis(1-ethylpropyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]dichloro(3-chloropyridine- κ M)palladium
Solvents: *tert*-Butanol; 4 h, 80 °C

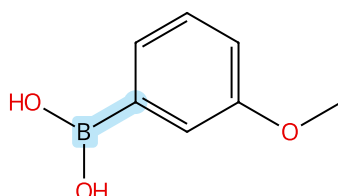
Experimental Protocols

Scheme 378 (1 Reaction)

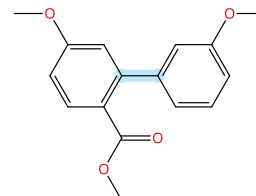
Steps: 1 Yield: 96%



+



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Suppliers (48)

Suppliers (112)

Suppliers (12)

31-614-CAS-44307275

Steps: 1 Yield: 96%

General Synthetic Methodologies for Building Blocks to Construct Molecular Motors

By: Doellerer, Daniel; et al

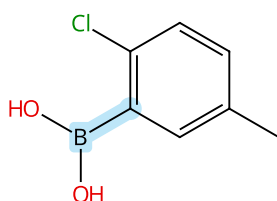
Journal of Organic Chemistry (2025), 90(10), 3519-3526.

1.1 **Reagents:** Sodium carbonate
Catalysts: Palladium diacetate
Solvents: Acetonitrile, Water; 18 h, 60 °C

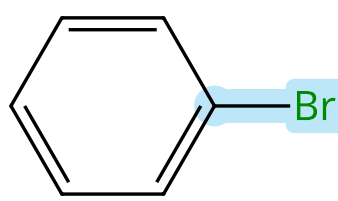
Experimental Protocols

Scheme 379 (1 Reaction)

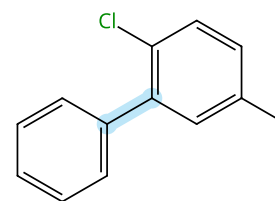
Steps: 1 Yield: 96%



+



→



Suppliers (78)

Suppliers (79)

Suppliers (15)

31-179-CAS-16848383

Steps: 1 Yield: 96%

Azaindoles as Zinc-Binding Small-Molecule Inhibitors of the JA MM Protease CSN5

By: Altmann, Eva; et al

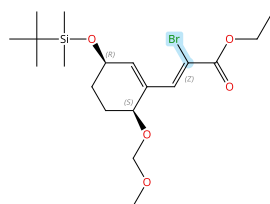
Angewandte Chemie, International Edition (2017), 56(5), 1294-1297.

1.1 **Reagents:** Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 16 h, 90 °C

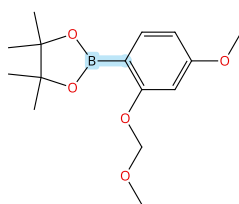
Experimental Protocols

Scheme 380 (1 Reaction)

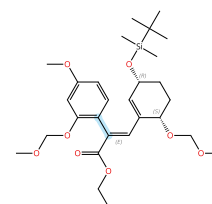
Steps: 1 Yield: 96%



+



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Suppliers (30)

31-614-CAS-34362346

Steps: 1 Yield: 96%

Asymmetric total synthesis of stachyodins A and B

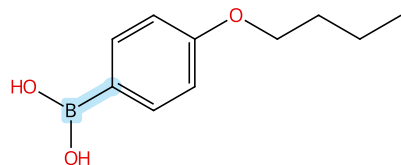
- 1.1 Reagents: Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 4 h, 90 °C
- 1.2 Reagents: Sodium chloride
Solvents: Water; rt

By: Kawamoto, Yuichiro; et al
Tetrahedron (2022), 106-107, 132595.

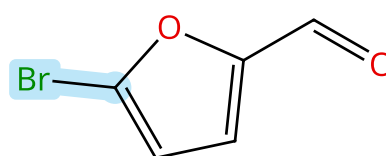
Experimental Protocols

Scheme 381 (1 Reaction)

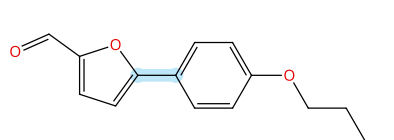
Steps: 1 Yield: 96%



+



→



Suppliers (87)

Suppliers (90)

Supplier (1)

31-179-CAS-16020278

Steps: 1 Yield: 96%

β -Furan-Fused bis(Difluoroboron)-1,2-bis((1H-pyrrol-2-yl)methylene)hydrazine Fluorescent Dyes in the Visible Deep-Red Region

- 1.1 Reagents: Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Toluene, Water; 24 h, rt → 80 °C

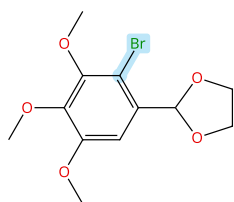
By: Zhou, Lin; et al

Journal of Organic Chemistry (2016), 81(17), 7439-7447.

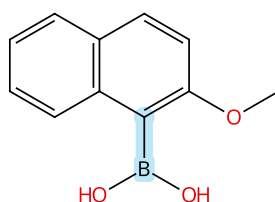
Experimental Protocols

Scheme 382 (1 Reaction)

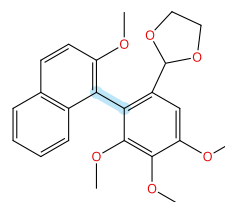
Steps: 1 Yield: 96%



+



→



Suppliers (9)

Suppliers (66)

31-614-CAS-33608028

Steps: 1 Yield: 96%

Atropisomeric Phosphine Ligands Bearing C-N Axial Chirality: Applications in Enantioselective Suzuki-Miyaura Cross-Coupling Towards the Assembly of Tetra-ortho-Substituted Biaryls

- 1.1 Reagents: Cesium carbonate
Catalysts: Bis(dibenzylideneacetone)palladium, 2816930-70-6
Solvents: Toluene; 48 h, 60 °C

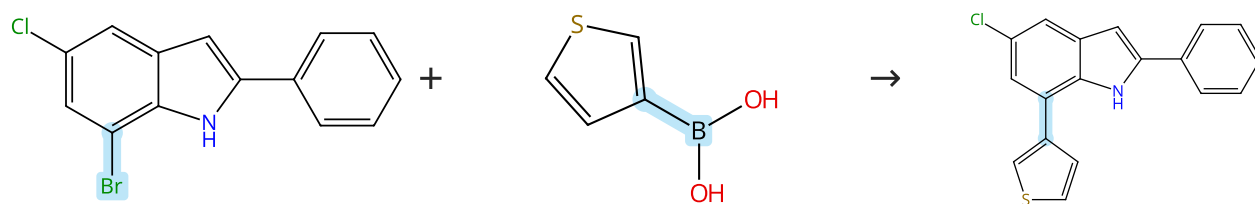
By: Gan, Kin Boon; et al

Journal of the American Chemical Society (2022), 144(32), 14864-14873.

Experimental Protocols

Scheme 383 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (8)

Suppliers (101)

31-179-CAS-9310058

Steps: 1 Yield: 96%

A facile palladium-catalyzed route to 2,5,7-trisubstituted indoles

- 1.1 **Reagents:** Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: 1,4-Dioxane; rt; 2 h, 100 °C

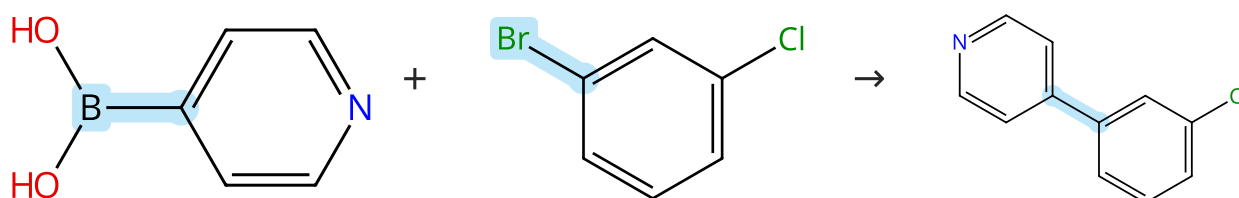
By: Cacchi, Sandro; et al

Tetrahedron (2015), 71(49), 9346-9356.

Experimental Protocols

Scheme 384 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (102)

Suppliers (81)

Suppliers (32)

31-614-CAS-32528428

Steps: 1 Yield: 96%

Facile Synthesis of 2-Methylnicotinonitrile through Degenerate Ring Transformation of Pyridinium Salts

- 1.1 **Reagents:** Potassium carbonate
Catalysts: (*SP*-4-2)-Dichloro[1,1'-(1,3-propanediyl)bis[1,1-diphenylphosphine-κP]]palladium
Solvents: 1,4-Dioxane, Water; overnight, reflux; reflux → rt

By: Duan, Xiaoxia; et al

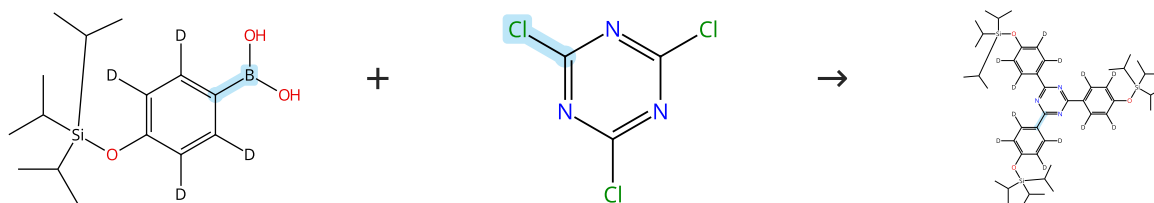
Journal of Organic Chemistry (2022), 87(12), 7975-7988.

- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 385 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (71)

31-614-CAS-40645163

Steps: 1 Yield: 96%

A Steric-Repulsion-Driven Clutch Stack of Triaryltriazines: Correlated Molecular Rotations and a Thermo-responsive Gearshift in the Crystalline Solid

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: Toluene, Water; 12 h, 100 °C

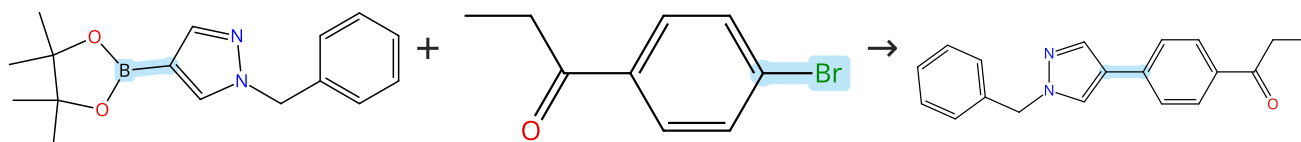
By: Jin, Mingoo; et al

Journal of the American Chemical Society (2023), 145(50), 27512-27520.

Experimental Protocols

Scheme 386 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (94)

Suppliers (64)

Suppliers (4)

31-614-CAS-34707383

Steps: 1 Yield: 96%

Enabling Suzuki-Miyaura coupling of Lewis-basic arylboronic esters with a nonprecious metal catalyst

By: Haibach, Michael C.; et al

Chemical Science (2022), 13(43), 12906-12912.

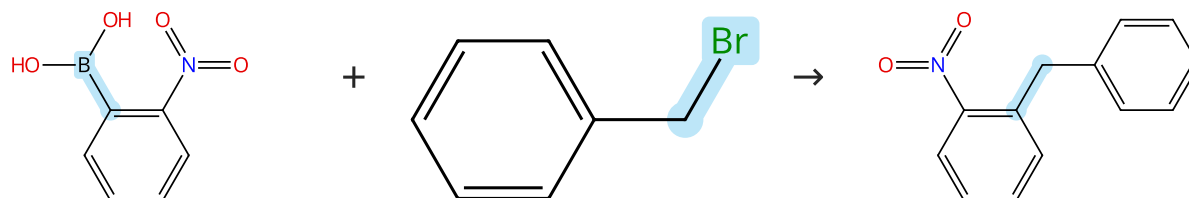
1.1 **Catalysts:** Butylmagnesium chloride, Methyldiphenylphosphine, Dichlorobis(methyldiphenylphosphine)nickel
Solvents: Tetrahydrofuran, 1,4-Dioxane; rt

1.2 **Reagents:** Tripotassium phosphate
Solvents: Water; 16 h, 90 °C

Experimental Protocols

Scheme 387 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (93)

Suppliers (66)

Suppliers (11)

31-614-CAS-42235793

Steps: 1 Yield: 96%

Modular Assembly of Acridines by Integrating Photo-Excitation of o-Alkyl Nitroarenes with Copper-Promoted Cascade Annulation

By: Huang, Haichao; et al

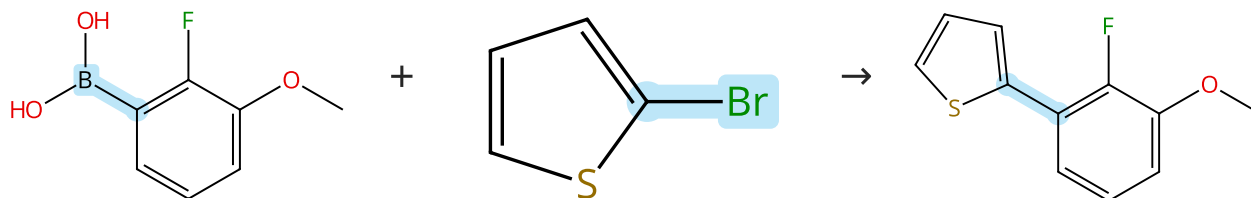
Angewandte Chemie, International Edition (2024), 63(42), e202409653.

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 12 h, reflux

Experimental Protocols

Scheme 388 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (97)

Suppliers (72)

Suppliers (10)

31-179-CAS-17146469

Steps: 1 Yield: 96%

First Dual Inhibitors of Steroid Sulfatase (STS) and 17β-Hydroxysteroid Dehydrogenase Type 1 (17β-HSD1): Designed Multiple Ligands as Novel Potential Therapeutics for Estrogen-Dependent Diseases

By: Salah, Mohamed; et al

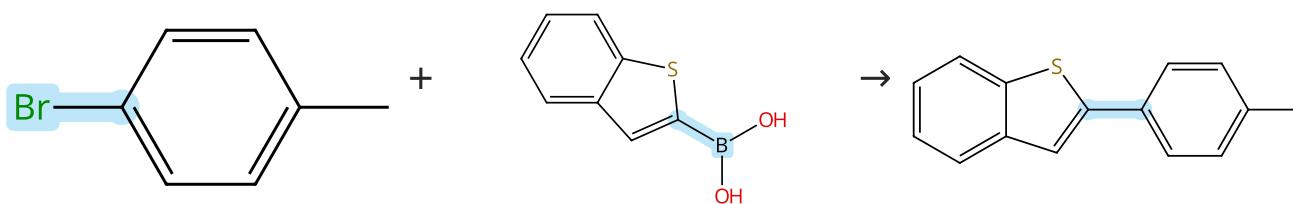
Journal of Medicinal Chemistry (2017), 60(9), 4086-4092.

1.1 **Reagents:** Cesium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 4 h, reflux

Experimental Protocols

Scheme 389 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (67)

Suppliers (90)

Suppliers (14)

31-179-CAS-17006086

Steps: 1 Yield: 96%

Iridium-Catalyzed Asymmetric Hydrogenation of Benzo[b]thiophene 1,1-Dioxides

By: Tosatti, Paolo; et al

Angewandte Chemie, International Edition (2017), 56(16), 4579-4582.

1.1 Catalysts: Tetrakis(triphenylphosphine)palladium

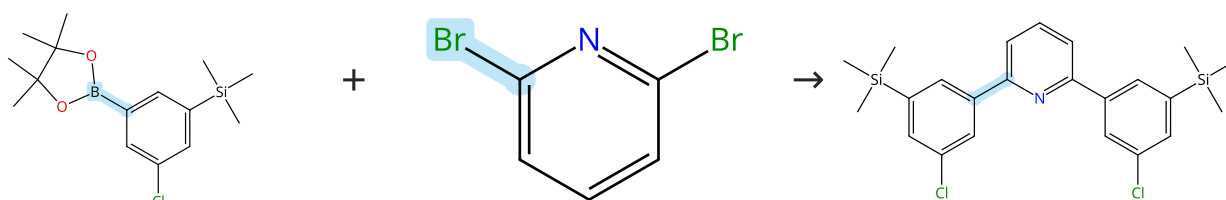
Solvents: 1,4-Dioxane; 20 min, rt

1.2 Reagents: Sodium bicarbonate

Solvents: Water; 1 h, reflux

Scheme 390 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (7)

Suppliers (87)

31-179-CAS-23467526

Steps: 1 Yield: 96%

A nitrogen-doped nanotube molecule with atom vacancy defects

By: Ikemoto, Koki; et al

Nature Communications (2020), 11(1), 1807.

1.1 Reagents: Potassium carbonate

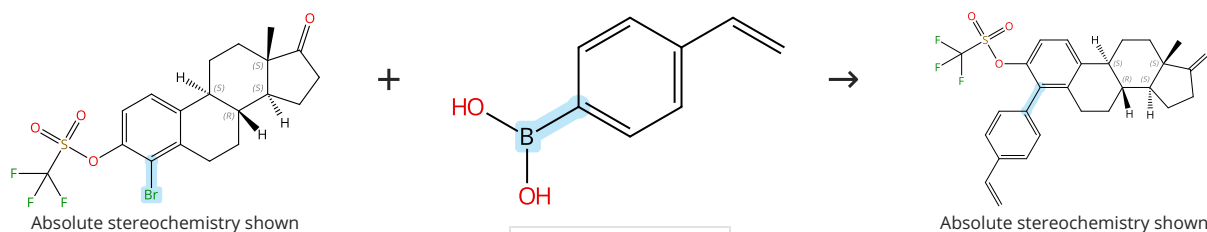
Catalysts: Tetrakis(triphenylphosphine)palladium

Solvents: Tetrahydrofuran, Water; 22 h, 70 °C

Experimental Protocols

Scheme 391 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (97)

31-179-CAS-18902659

Steps: 1 Yield: 96%

Chemoselective Suzuki-Miyaura reactions of 4-bromo-3-O-triflyl-estrone. Synthesis and atropisomerism of arylated estrones

By: Jopp, Stefan; et al

Tetrahedron (2018), 74(23), 2825-2836.

1.1 Reagents: Tripotassium phosphate

Catalysts: Palladium diacetate, CataCXium A

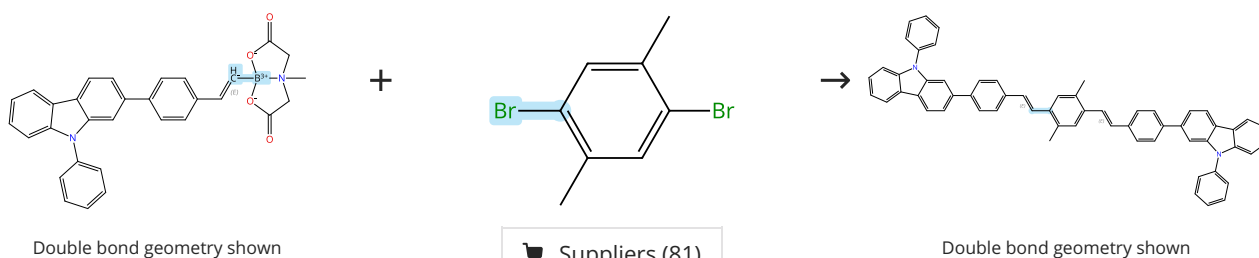
Solvents: Toluene; 20 h, 100 °C

1.2 Reagents: Water

Experimental Protocols

Scheme 392 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-40495615

Steps: 1 Yield: 96%

Delocalized, asynchronous, closed-loop discovery of organic laser emitters

- 1.1 **Reagents:** Sodium carbonate, Tripotassium phosphate
Catalysts: (*SP*-4-4)-[2'-(Amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium
Solvents: Tetrahydrofuran; 3 h, rt

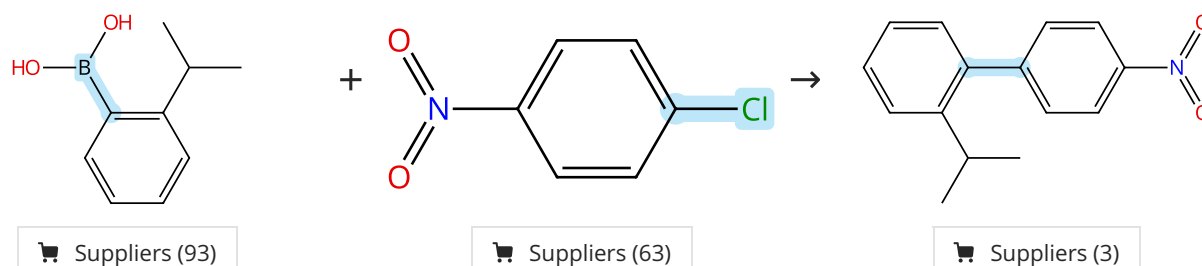
By: Strieth-Kalthoff, Felix; et al

Science (Washington, DC, United States) (2024), 384(6697), eadk9227.

Experimental Protocols

Scheme 393 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-44132960

Steps: 1 Yield: 96%

Using Common Alcohols to Transform N-H NHC-Palladium Complexes into Reactive Catalysts for the Suzuki-Miyaura Reaction with Aryl Chlorides

- 1.1 **Reagents:** Sodium hydroxide
Catalysts: Palladium chloride, 1-[2,6-Bis(1-methylethyl)phenyl]-2-(diphenylphosphino)-1*H*-imidazole
Solvents: *tert*-Butanol; 30 min, 40 °C
- 1.2 **Reagents:** Sodium hydride
Solvents: 1,4-Dioxane; 14 h, 40 °C → 100 °C

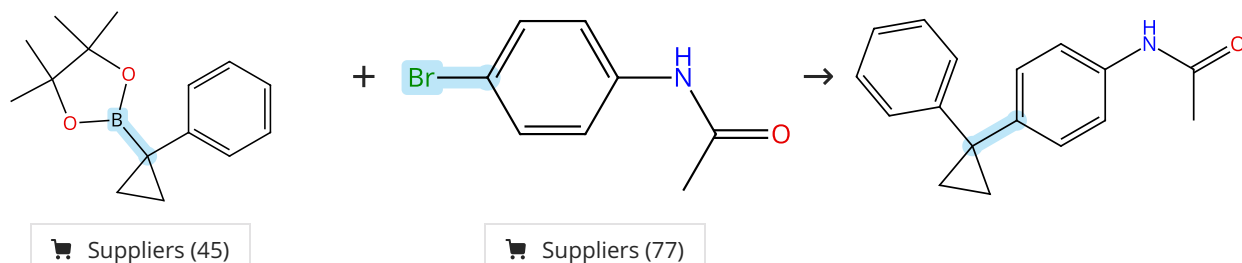
By: Larson, Alexandra J. S.; et al

Journal of Organic Chemistry (2025), 90(7), 2806-2810.

Experimental Protocols

Scheme 394 (1 Reaction)

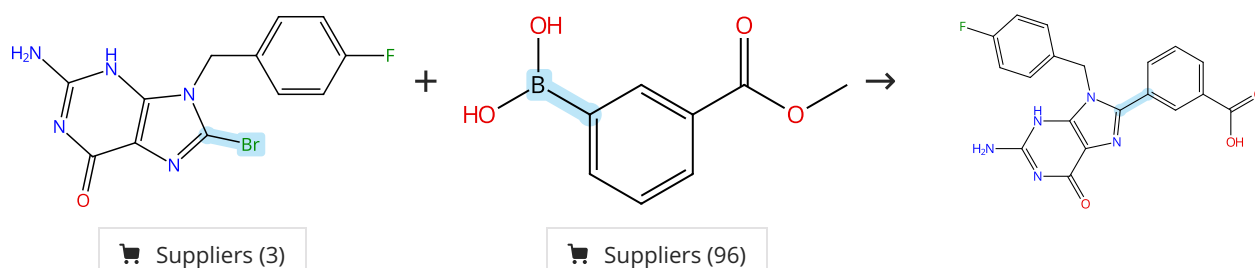
Steps: 1 Yield: 96%



31-614-CAS-32067915 Steps: 1 Yield: 96% 1.1 Reagents: Morpholine Catalysts: Iridium(1+), [4,4'-bis(1,1-dimethylethyl)-2,2'-bipyridine-$\kappa N^1, \kappa N^1$]bis[3,5-difluoro-2-[5-(trifluoromethyl)-2-pyridinyl-κN]phenyl-κC]-, (OC-6-33)-, hexafluorophosphate(1-)(1:1) Solvents: Dimethylformamide; rt 1.2 Reagents: Dichloro[1,2-di(methoxy-κO)ethane]nickel, 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine Solvents: Dimethylformamide; 30 s; 100 °C 1.3 2 h Experimental Protocols	ART - An Amino Radical Transfer Strategy for C (sp²)-C(sp³) Coupling Reactions, Enabled by Dual Photo/Nickel Catalysis By: Speckmeier, Elisabeth; et al Journal of the American Chemical Society (2022), 144(22), 9997-10005.
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Scheme 395 (1 Reaction)

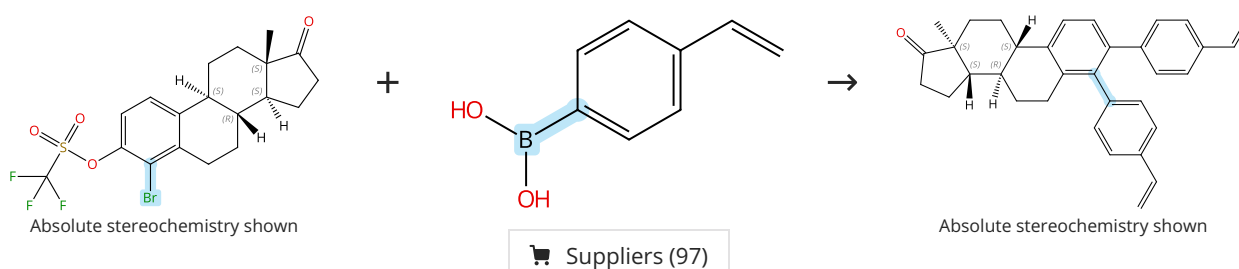
Steps: 1 Yield: 96%



31-614-CAS-37526827 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate, Benzenesulfonic acid, 3,3',3''-phosphinidynetris-, sodium salt (1:3) Catalysts: Palladium diacetate Solvents: Acetonitrile, Water; overnight, 100 °C 1.2 Reagents: Sodium hydroxide Solvents: Water; basified 1.3 Reagents: Hydrochloric acid Solvents: Water; pH 4 - 5 Experimental Protocols	Second-Generation AUTACs for Targeted Autophagic Degradation By: Takahashi, Daiki; et al Journal of Medicinal Chemistry (2023), 66(17), 12342-12372.
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Scheme 396 (1 Reaction)

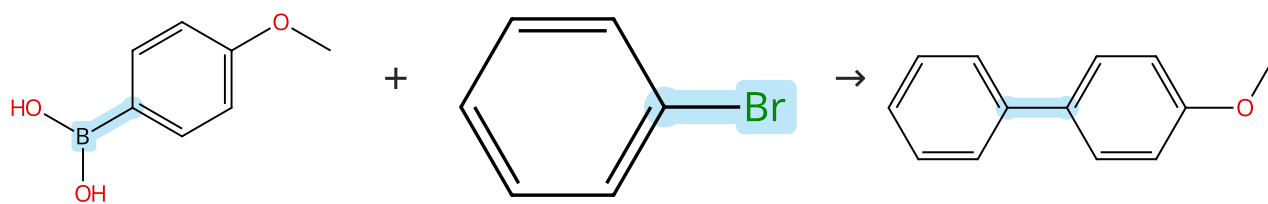
Steps: 1 Yield: 96%



31-179-CAS-18902666 Steps: 1 Yield: 96% 1.1 Reagents: Tripotassium phosphate Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl Solvents: 1,4-Dioxane; 20 h, 100 °C 1.2 Reagents: Water Experimental Protocols	Chemoselective Suzuki-Miyaura reactions of 4-bromo-3-O-triflyl-estrone. Synthesis and atropisomerism of arylated estrones By: Jopp, Stefan; et al Tetrahedron (2018), 74(23), 2825-2836.
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Scheme 397 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (79)

Suppliers (69)

31-614-CAS-36816772

Steps: 1 Yield: 96%

Palladium-bearing intermetallic electride as an efficient and stable catalyst for Suzuki cross-coupling reactions

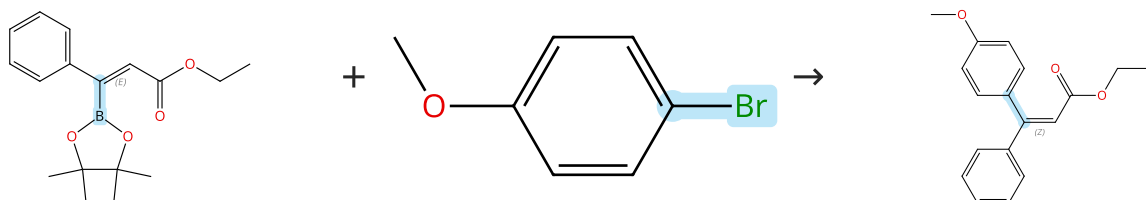
By: Ye, Tian-Nan; et al

Nature Communications (2019), 10(1), 5653.

1.1 **Reagents:** Potassium carbonate
Catalysts: Palladium, compd. with yttrium (2:3)
Solvents: Ethanol; 8 h, 60 °C

Scheme 398 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Suppliers (69)

Double bond geometry shown

Supplier (1)

31-177-CAS-18545127

Steps: 1 Yield: 96%

Phosphine-Catalyzed Anti-Hydroboration of Internal Alkynes

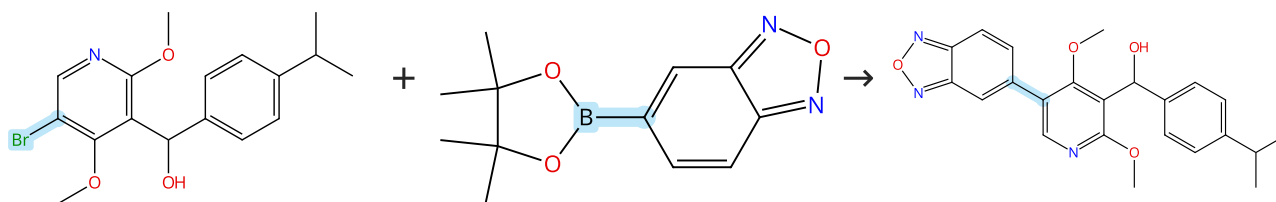
By: Nagao, Kazunori; et al

Organic Letters (2018), 20(7), 1861-1865.

1.1 **Reagents:** Water, Tripotassium phosphate
Catalysts: Palladium diacetate, 1,1'-Bis(di-*tert*-butylphosphino)ferrocene
Solvents: Toluene; rt; 5 h, 60 °C

Scheme 399 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (70)

31-179-CAS-19607219

Steps: 1 Yield: 96%

Discovery of 2,4-dimethoxypyridines as novel autophagy inhibitors

By: Robke, Lucas; et al

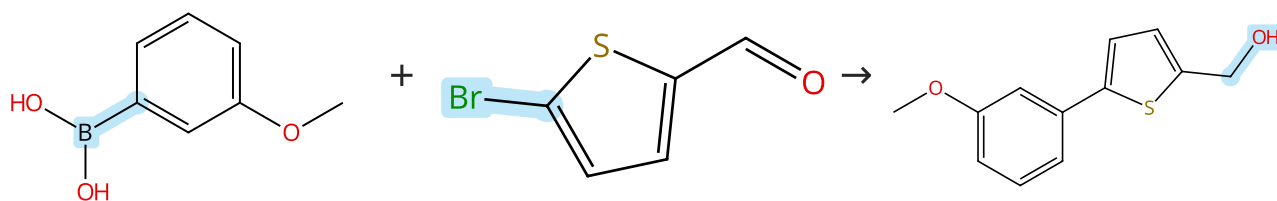
Tetrahedron (2018), 74(35), 4531-4537.

1.1 **Reagents:** Sodium carbonate
Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 16 h, 110 °C

Experimental Protocols

Scheme 400 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (90)

Suppliers (9)

31-179-CAS-20024226

Steps: 1 Yield: 96%

Metal-Free Aerobic Oxidative Selective C-C Bond Cleavage in Heteroaryl-Containing Primary and Secondary Alcohols

By: Xia, Anjie; et al

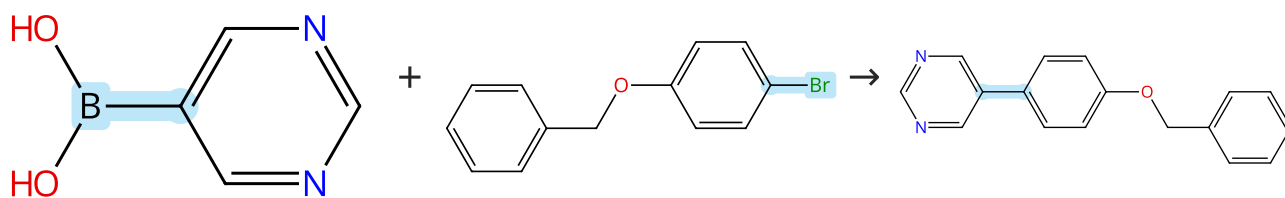
Organic Letters (2019), 21(9), 3028-3033.

- 1.1 Reagents: Potassium carbonate
Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: 1,4-Dioxane, Water; overnight, 100 °C
- 1.2 Reagents: Sodium borohydride
Solvents: Methanol; 1 h, rt

Experimental Protocols

Scheme 401 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (88)

31-179-CAS-22914700

Steps: 1 Yield: 96%

¹⁹F- and ¹⁸F-arene deoxyfluorination via organic photoredox-catalysed polarity-reversed nucleophilic aromatic substitution

By: Tay, Nicholas E. S.; et al

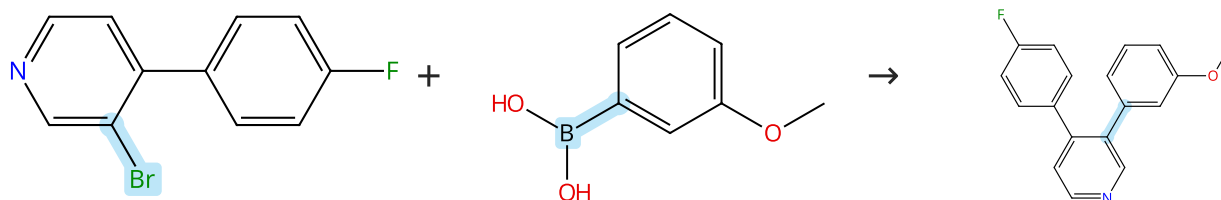
Nature Catalysis (2020), 3(9), 734-742.

- 1.1 Reagents: Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene, Water; 16 h, reflux; cooled
- 1.2 Reagents: Ammonium chloride; cooled

Experimental Protocols

Scheme 402 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (20)

Suppliers (112)

31-179-CAS-23044078

Steps: 1 Yield: 96%

Single point activation of pyridines enables reductive hydroxymethylation

By: Marinic, Bruno; et al

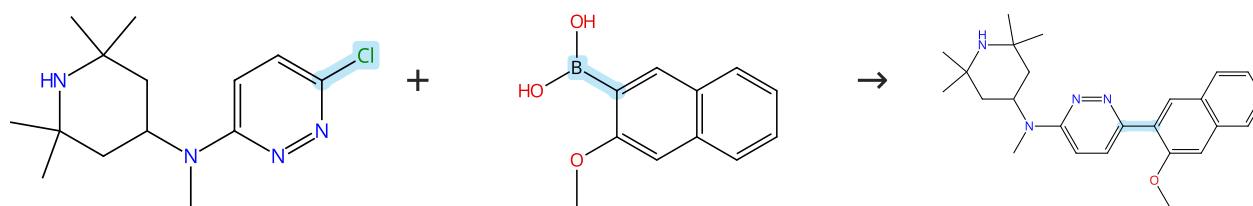
Chemical Science (2021), 12(2), 742-746.

- 1.1 Reagents: Potassium carbonate
Catalysts: Triphenylphosphine, Palladium diacetate
Solvents: 1,2-Dimethoxyethane, Water; 16 h, 85 °C

Experimental Protocols

Scheme 403 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (7)

Suppliers (63)

31-179-CAS-20385895

Steps: 1 Yield: 96%

Discovery of Small Molecule Splicing Modulators of Survival Motor Neuron-2 (SMN2) for the Treatment of Spinal Muscular Atrophy (SMA)

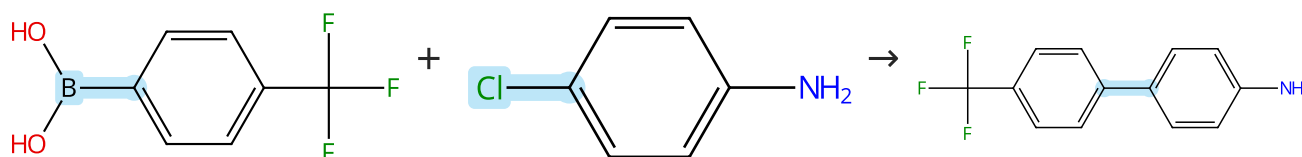
By: Cheung, Atwood K.; et al

Journal of Medicinal Chemistry (2018), 61(24), 11021-11036.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,2-Dimethoxyethane, Water; 30 - 45 min, 120 - 140 °C

Scheme 404 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (113)

Suppliers (98)

Suppliers (63)

31-179-CAS-16785485

Steps: 1 Yield: 96%

One-Pot Palladium-Catalyzed Cross-Coupling Treble of Borylation, the Suzuki Reaction and Amination

By: Jong, Howard; et al

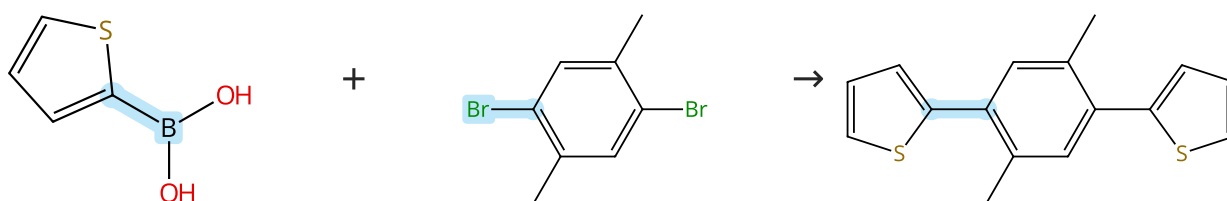
Advanced Synthesis & Catalysis (2017), 359(4), 616-622.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Dichlorobis[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1':3',1''-terphenyl]-2-yl]phosphine]palladium
Solvents: 1-Butanol; rt → 70 °C; 1.5 h, 70 °C

Experimental Protocols

Scheme 405 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (102)

Suppliers (81)

Suppliers (2)

31-614-CAS-34094093

Steps: 1 Yield: 96%

Achieving Record-Efficiency Organic Solar Cells upon Tuning the Conformation of Solid Additives

By: Li, Congqi; et al

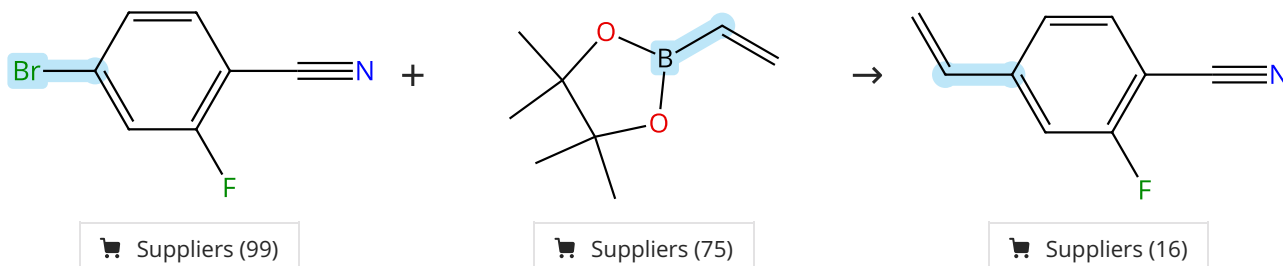
Journal of the American Chemical Society (2022), 144(32), 14731-14739.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 12 h, 60 °C

Experimental Protocols

Scheme 406 (1 Reaction)

Steps: 1 Yield: 96%



31-177-CAS-19387642

Steps: 1 Yield: 96%

Enantioselective, Catalytic Vicinal Difluorination of Alkenes

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium diacetate, 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: 1,4-Dioxane, Water; 14 h, 80 °C

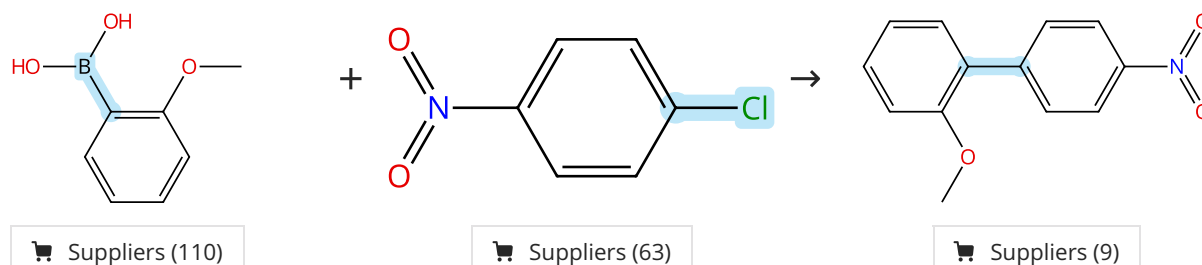
By: Scheidt, Felix; et al

Angewandte Chemie, International Edition (2018), 57(50), 16431-16435.

Experimental Protocols

Scheme 407 (1 Reaction)

Steps: 1 Yield: 96%



31-179-CAS-17240559

Steps: 1 Yield: 96%

Suzuki-Miyaura cross-coupling reaction of aryl chlorides with aryl boronic acids catalyzed by a palladium dichloride adduct of N-diphenylphosphanyl-2-aminopyridine

- 1.1 **Reagents:** Tripotassium phosphate
Catalysts: Palladium, dichloro(*P,P*-diphenyl-*N*-2-pyridinylphosphinous amide-*N,N,P*-, (*SP*-4-3)-
Solvents: Dimethylformamide; 15 h, 100 °C

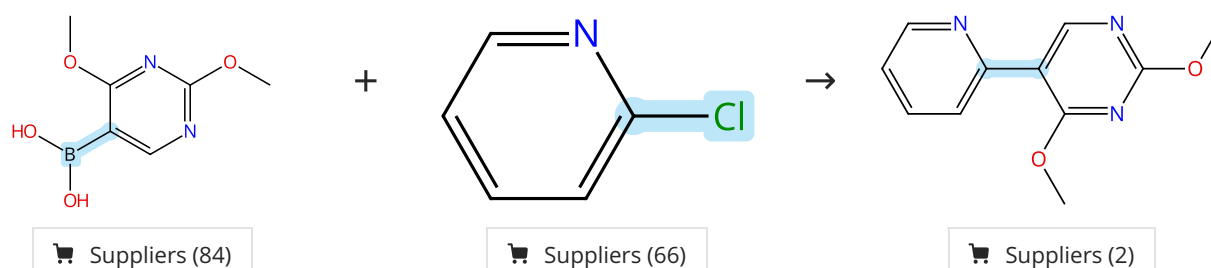
By: Xu, Lin-Yan; et al

Tetrahedron (2017), 73(22), 3125-3132.

Experimental Protocols

Scheme 408 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-41372139

Steps: 1 Yield: 96%

ProPhos: A Ligand for Promoting Nickel-Catalyzed Suzuki-Miyaura Coupling Inspired by Mechanistic Insights into Transmetalation

- 1.1 **Catalysts:** 3-(Diphenylphosphino)-1-propanol, Nickel dichloride
Solvents: Isopropanol; overnight, rt

By: Yang, Jin; et al

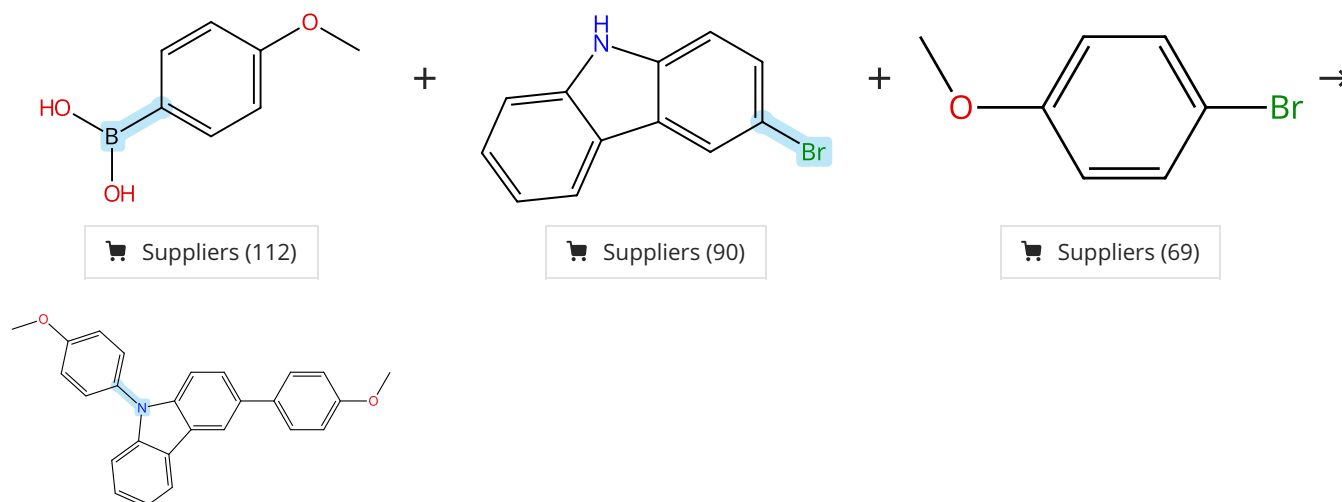
- 1.2 **Reagents:** Tripotassium phosphate; 16 h, rt → 70 °C

Journal of the American Chemical Society (2024), 146(9), 6360-6368.

Experimental Protocols

Scheme 409 (1 Reaction)

Steps: 1 Yield: 96%



31-172-CAS-22785431

Steps: 1 Yield: 96%

Concatenating Suzuki Arylation and Buchwald-Hartwig Amination by A Sequentially Pd-Catalyzed One-Pot Process-Consecutive Three-Component Synthesis of C,N-Diarylated Heterocycles

By: Mayer, Laura; et al

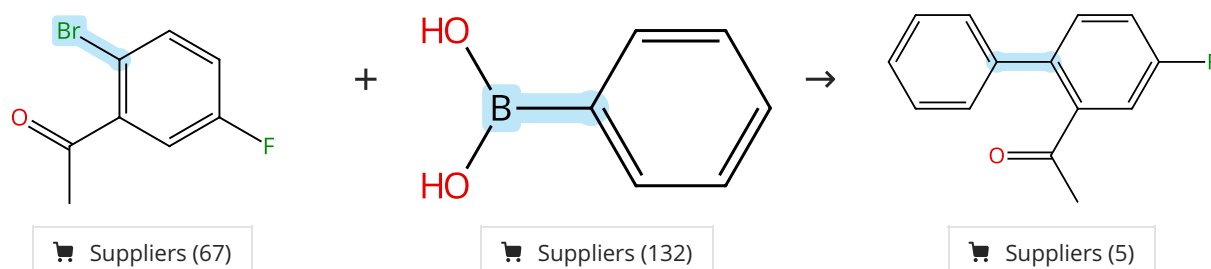
Chemistry - A European Journal (2020), 26(66), 15130-15134.

- 1.1 **Reagents:** Cesium fluoride
Catalysts: Bis(dibenzylideneacetone)palladium, Tri-*tert*-butylphosphonium tetrafluoroborate
Solvents: 1,4-Dioxane; 4 h, 120 °C; 120 °C → rt
- 1.2 **Reagents:** Potassium carbonate; 20 h, 120 °C

Experimental Protocols

Scheme 410 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-35310272

Steps: 1 Yield: 96%

o-Nitrobenzyl Oxime Ethers Enable Photoinduced Cyclization Reaction to Provide Phenanthridines under Aqueous Conditions

By: Okamura, Hidenori; et al

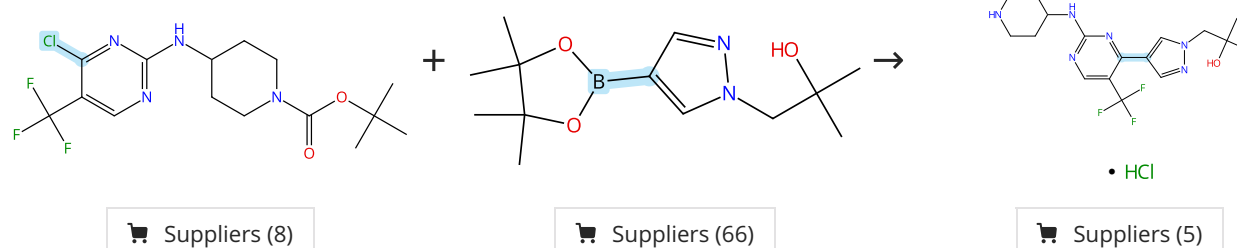
Organic Letters (2023), 25(3), 466-470.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Toluene; 26 h, reflux

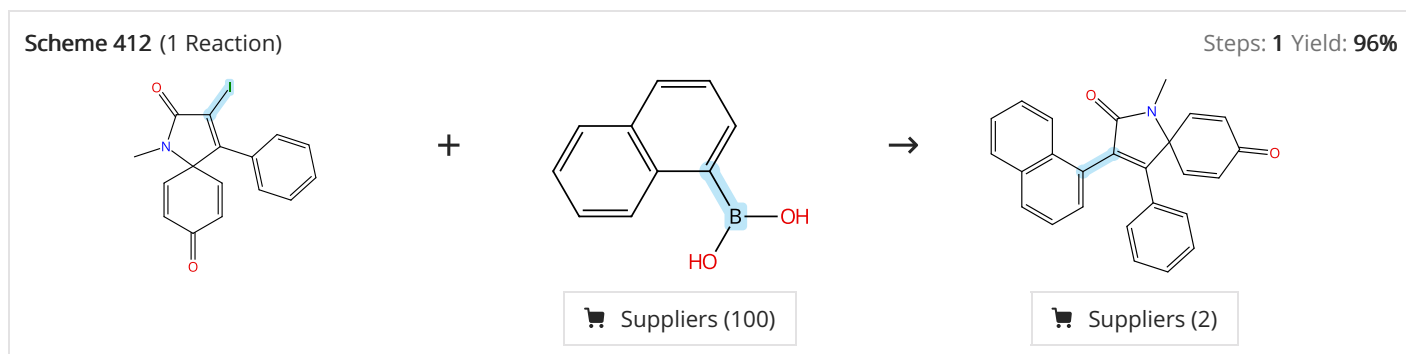
Experimental Protocols

Scheme 411 (1 Reaction)

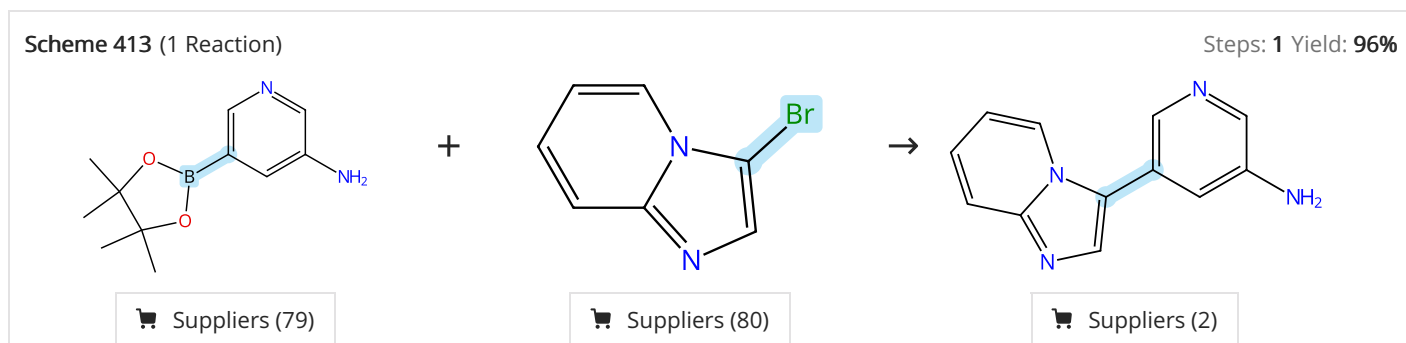
Steps: 1 Yield: 96%



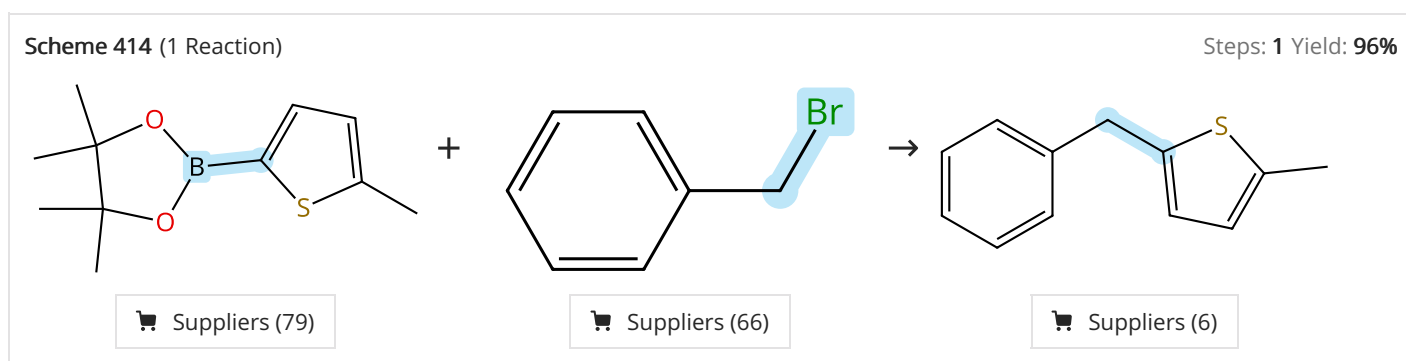
31-614-CAS-39486228 Steps: 1 Yield: 96% 1.1 Reagents: Sodium carbonate Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: 1,4-Dioxane, Water; 5 h, 100 °C; 100 °C → rt 1.2 Reagents: Sodium bicarbonate Solvents: Water; rt 1.3 Reagents: Hydrochloric acid Solvents: Methanol, Dichloromethane, 1,4-Dioxane; 2 h, rt Experimental Protocols	Discovery of (4-Pyrazolyl)-2-aminopyrimidines as Potent and Selective Inhibitors of Cyclin-Dependent Kinase 2 By: Hummel, Joshua R.; et al Journal of Medicinal Chemistry (2024), 67(4), 3112-3126.
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31-614-CAS-43022703 Steps: 1 Yield: 96% 1.1 Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Dimethylformamide; 30 min, rt 1.2 Reagents: Potassium carbonate Solvents: Water; 8 h, 85 °C Experimental Protocols	Halogen Radical-Enabled Dearomatization of N-Arylpropionamides via Photoinduced Sequential Halogenation/Spirocyclization/Oxidation Process By: Luo, Wenkun; et al Journal of Organic Chemistry (2024), 89(24), 18689-18697.
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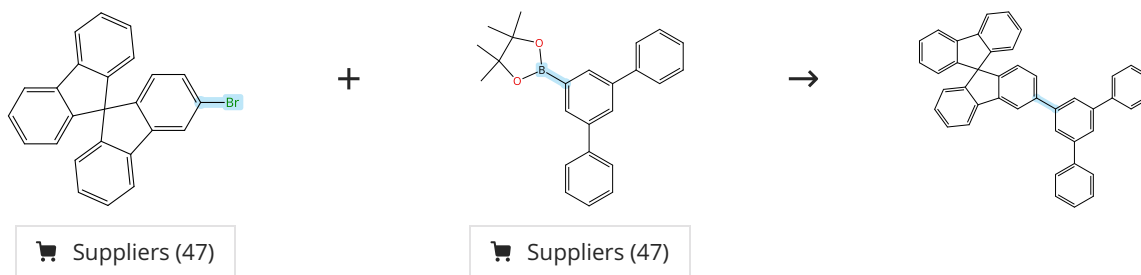
31-614-CAS-40478642 Steps: 1 Yield: 96% 1.1 Reagents: Sodium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: Acetonitrile, Water; 1 h, 110 °C Experimental Protocols	Identification of M4205-A Highly Selective Inhibitor of KIT Mutations for Treatment of Unresectable Metastatic or Recurrent Gastrointestinal Stromal Tumors By: Blum, Andreas; et al Journal of Medicinal Chemistry (2023), 66(4), 2386-2395.
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31-614-CAS-41473114 Steps: 1 Yield: 96% 1.1 Reagents: Potassium trimethylsilanolate Catalysts: (SP-4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium Solvents: Tetrahydrofuran; 5 min, rt 1.2 Reagents: Tetrabutylammonium bromide Catalysts: (SP-4-4)-[2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC]chloro[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine]palladium; 12 h, 50 °C Experimental Protocols	Halide Salts Alleviate TMSOK Inhibition in Suzuki-Miyaura Cross-Couplings By: Shi, Yao; et al ACS Catalysis (2024), 14(16), 12671-12680.
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Scheme 415 (1 Reaction)

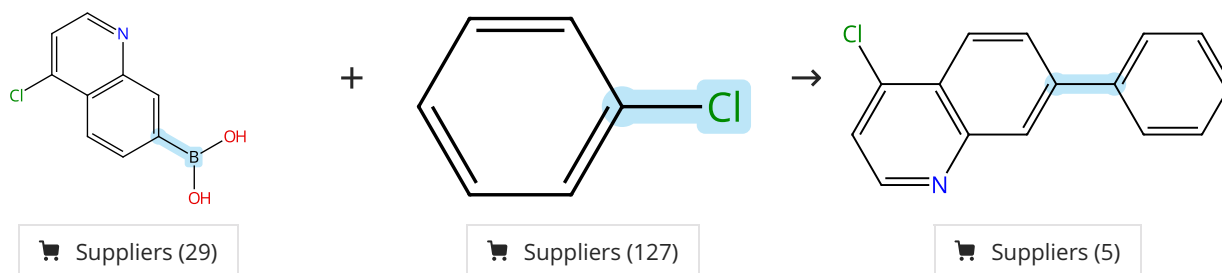
Steps: 1 Yield: 96%



31-179-CAS-23026584 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran, Water; 12 h, reflux	Evolution of pure hydrocarbon hosts: simpler structure, higher performance and universal application in RGB phosphorescent organic light-emitting diodes By: Wang, Qiang; et al Chemical Science (2020), 11(19), 4887-4894.
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Scheme 416 (1 Reaction)

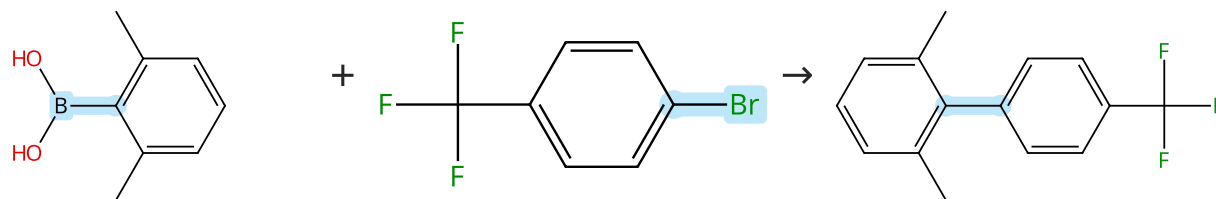
Steps: 1 Yield: 96%



31-179-CAS-17940825 Steps: 1 Yield: 96% 1.1 Reagents: Tripotassium phosphate Catalysts: 2162145-43-7 (silica coated iron oxide nanoparticles-supported) Solvents: Toluene; 24 h, 100 °C Experimental Protocols	A magnetic nanoparticle-supported N-heterocyclic carbene-palladacycle: an efficient and recyclable solid molecular catalyst for Suzuki-Miyaura cross-coupling of 9-chloroacridine By: Deng, Qinyue; et al Chemical Communications (Cambridge, United Kingdom) (2017), 53(97), 13063-13066.
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Scheme 417 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (100)

Suppliers (79)

Suppliers (10)

31-614-CAS-38504852

Steps: 1 Yield: 96%

It Is Not All about the Ligands: Exploring the Hidden Potentials of tBu₃P through Its Oxidative Addition Complex as the Precatalyst

By: Timsina, Yam N.; et al

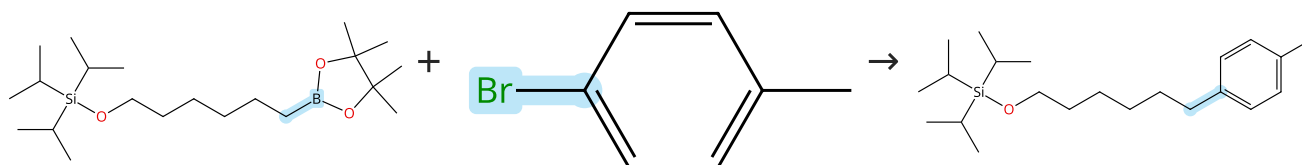
ACS Catalysis (2023), 13(12), 8106-8118.

1.1 **Reagents:** Triethylamine
Catalysts: Bromo[4-(trifluoromethyl)phenyl][tris(1,1-dimethylethyl)phosphine]palladium
Solvents: Toluene, Water; 24 h, rt

Experimental Protocols

Scheme 418 (1 Reaction)

Steps: 1 Yield: 96%



Supplier (1)

Suppliers (67)

31-179-CAS-21580817

Steps: 1 Yield: 96%

Rhodium-Catalyzed Remote C(sp³)-H Borylation of Silyl Enol Ethers

By: Li, Jie; et al

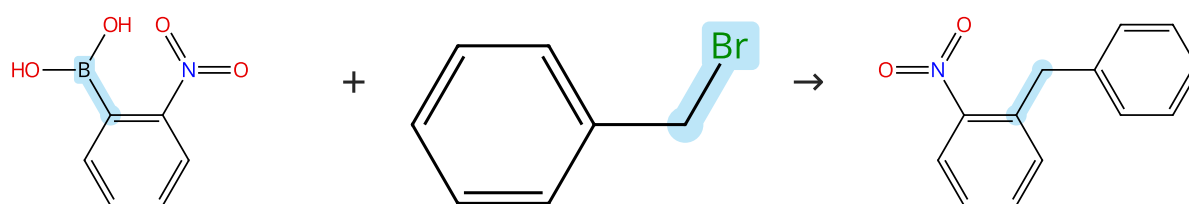
Angewandte Chemie, International Edition (2020), 59(6), 2360-2364.

1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: Tris(dibenzylideneacetone)dipalladium, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine
Solvents: Toluene, Water; 24 h, 80 °C

Experimental Protocols

Scheme 419 (2 Reactions)

Steps: 1 Yield: 96%



Suppliers (93)

Suppliers (66)

Suppliers (11)

31-614-CAS-34078021

Steps: 1 Yield: 96%

A switchable redox annulation of 2-nitroarylethanols affording N-heterocycles: photoexcited nitro as a multifunctional handle

By: Wang, Bin; et al

Chemical Science (2022), 13(37), 11074-11082.

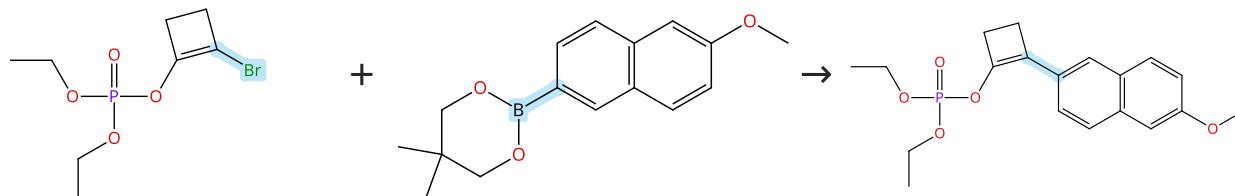
1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 12 h, reflux

Experimental Protocols

31-179-CAS-20469371	Steps: 1 Yield: 96%	Copper-Catalyzed One-Pot Synthesis of Dibenzo furans, Xanthenes, and Xanthenes from Cyclic Diphenyl Iodoniums
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran; 12 h, reflux		By: Zhu, Daqian; et al European Journal of Organic Chemistry (2019), 2019(28), 4566-4571.

Scheme 420 (1 Reaction)

Steps: 1 Yield: 96%

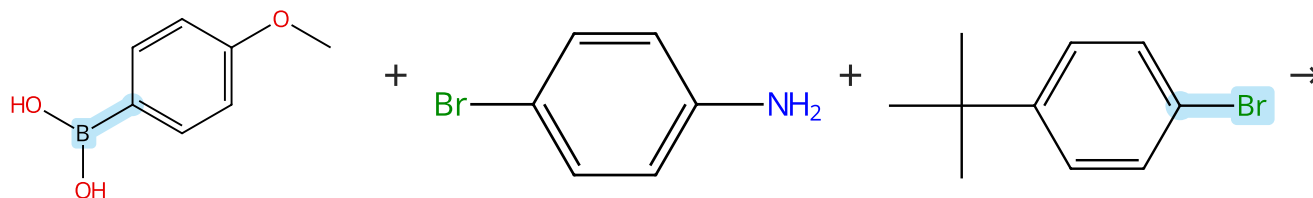


Suppliers (19)

31-179-CAS-19668300	Steps: 1 Yield: 96%	Novel Selective Approach to Terminally Substituted [n]Dendralenes
1.1 Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine Solvents: Toluene; 1 min, rt		By: Polak, Peter; et al European Journal of Organic Chemistry (2019), 2019(5), 957-968.
1.2 Reagents: Tripotassium phosphate Solvents: Water; 18 h, 60 °C		
Experimental Protocols		

Scheme 421 (1 Reaction)

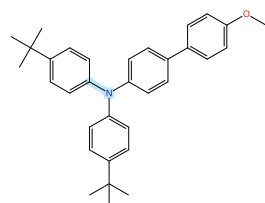
Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (83)

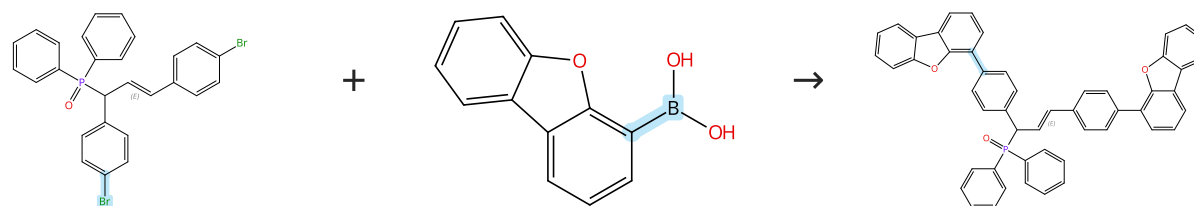
Suppliers (79)



31-614-CAS-39979370	Steps: 1 Yield: 96%	A Rational Design of Electrochemically and Photophysically Tunable Triarylamine Luminophores by Consecutive (Pseudo-)Four-Component Syntheses
1.1 Reagents: Cesium carbonate Catalysts: Bis(dibenzylideneacetone)palladium, Tri- <i>tert</i> -butylphosphonium tetrafluoroborate Solvents: 1,4-Dioxane; 1 h, 120 °C; 120 °C → rt		By: Kohlbecher, Regina; et al Chemistry - A European Journal (2024), 30(17), e202304119.
1.2 Reagents: Sodium <i>tert</i> -butoxide; 15 h, 120 °C		
Experimental Protocols		

Scheme 422 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Suppliers (93)

Double bond geometry shown

31-179-CAS-23462247

Steps: 1 Yield: 96%

Dehydrative allylation of P-H species under metal-free conditions

1.1 **Reagents:** Cesium fluoride
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Xylene; 12 h, 60 °C

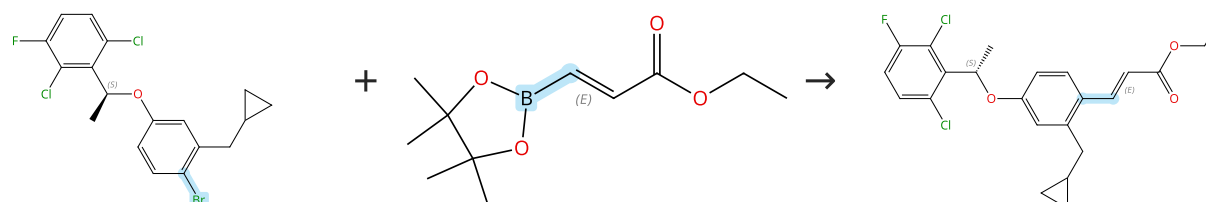
By: Yang, Xiaobo; et al

Green Chemistry (2021), 23(4), 1633-1637.

Experimental Protocols

Scheme 423 (1 Reaction)

Steps: 1 Yield: 96%



Absolute stereochemistry shown

Double bond geometry shown

Absolute stereochemistry shown
Double bond geometry shown

Suppliers (72)

31-614-CAS-35002936

Steps: 1 Yield: 96%

Discovery of Inhibitors of the Lipopolysaccharide Transporter MsbA: From a Screening Hit to Potent Wild-Type Gram-Negative Activity

1.1 **Reagents:** Cesium carbonate
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: 1,4-Dioxane, Water; 3 h, 90 °C

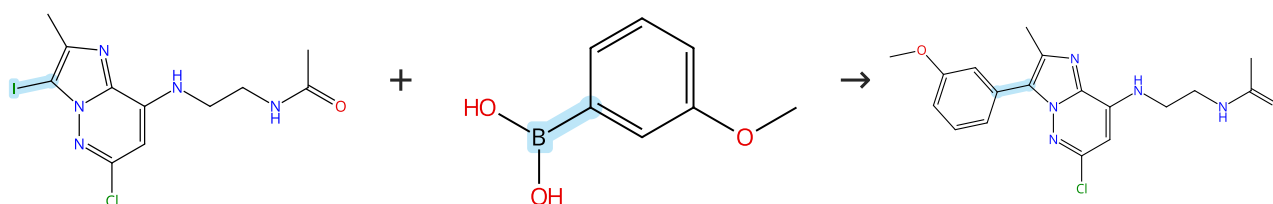
By: Verma, Vishal A.; et al

Journal of Medicinal Chemistry (2022), 65(5), 4085-4120.

Experimental Protocols

Scheme 424 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (2)

Suppliers (112)

Supplier (1)

31-179-CAS-13253473

Steps: 1 Yield: 96%

Highly Selective Phosphatidylinositol 4-Kinase IIIβ Inhibitors and Structural Insight into Their Mode of Action

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 2 h, reflux; overnight, 95 °C

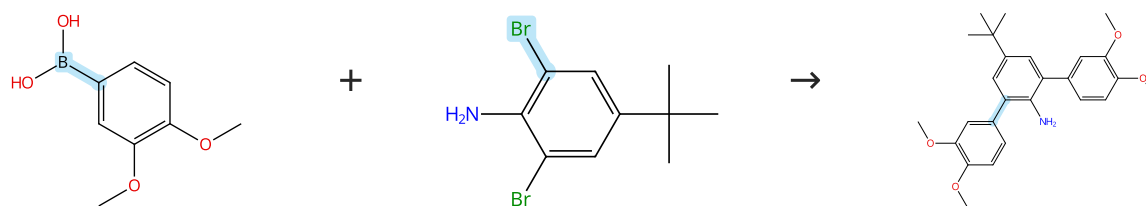
By: Mejdrova, Ivana; et al

Journal of Medicinal Chemistry (2015), 58(9), 3767-3793.

Experimental Protocols

Scheme 425 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (105)

Suppliers (53)

31-614-CAS-31388786

Steps: 1 Yield: 96%

A Near-Infrared Absorbing and Emissive Quadruple Helicene Enabled by the Scholl Reaction of Perylene

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Ethanol, Acetic acid, Toluene; 16 h, reflux

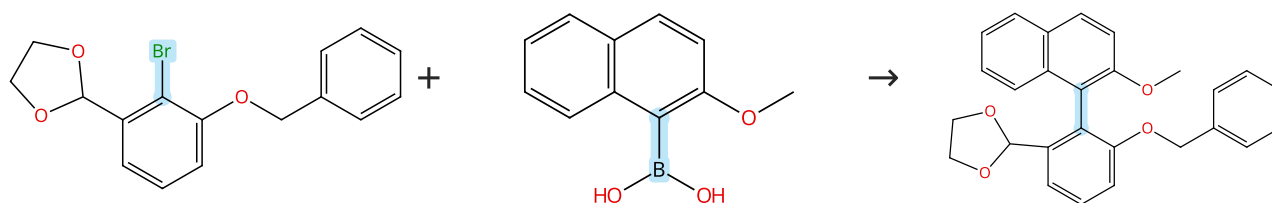
By: Pun, Sai Ho; et al

Angewandte Chemie, International Edition (2022), 61(8), e202113203.

Experimental Protocols

Scheme 426 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (3)

Suppliers (66)

31-614-CAS-33608024

Steps: 1 Yield: 96%

Atropisomeric Phosphine Ligands Bearing C-N Axial Chirality: Applications in Enantioselective Suzuki-Miyaura Cross-Coupling Towards the Assembly of Tetra-ortho-Substituted Biaryls

1.1 **Reagents:** Cesium carbonate
Catalysts: Bis(dibenzylideneacetone)palladium, 2816930-70-6
Solvents: Toluene; 48 h, 60 °C

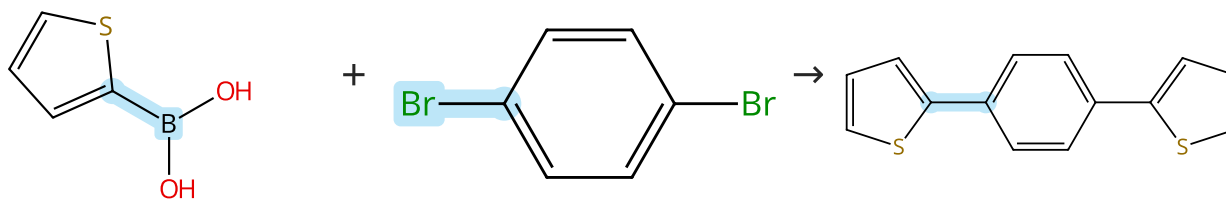
By: Gan, Kin Boon; et al

Journal of the American Chemical Society (2022), 144(32), 14864-14873.

Experimental Protocols

Scheme 427 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (102)

Suppliers (101)

Suppliers (43)

31-614-CAS-42957550

Steps: 1 Yield: 96%

Regulating Electron-Phonon Coupling by Solid Additive for Efficient Organic Solar Cells

1.1 **Reagents:** Potassium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran, Water; 12 h, 80 °C

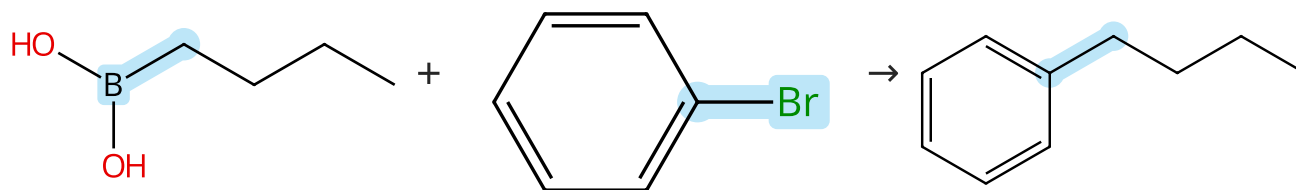
By: Ge, Zhongwei; et al

Angewandte Chemie, International Edition (2025), 64(1), e202413309.

Experimental Protocols

Scheme 428 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (101)

Suppliers (79)

Suppliers (68)

31-614-CAS-33544265

Steps: 1 Yield: 96%

Ru(II)-catalyzed P(III)-assisted C8-alkylation of naphthphosphines

By: Ma, Wen-Tao; et al

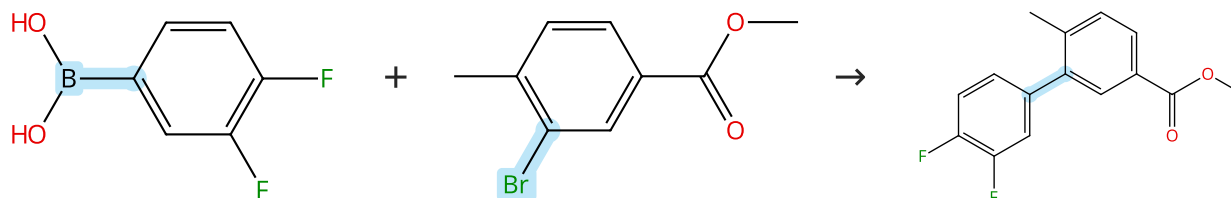
Chemical Communications (Cambridge, United Kingdom) (2022), 58(51), 7152-7155.

1.1 **Catalysts:** Palladium diacetate, Ethyl 8-(diphenylphosphino)-1-naphthalenepropanoate**Solvents:** Dichloromethane; 5 min, rt1.2 **Reagents:** Potassium carbonate**Solvents:** Toluene, Water; 16 h, 100 °C

Experimental Protocols

Scheme 429 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (111)

Suppliers (81)

Supplier (1)

31-179-CAS-13929743

Steps: 1 Yield: 96%

Rational Design of Selective Small-Molecule Inhibitors for β -Catenin/B-Cell Lymphoma 9 Protein-Protein Interactions

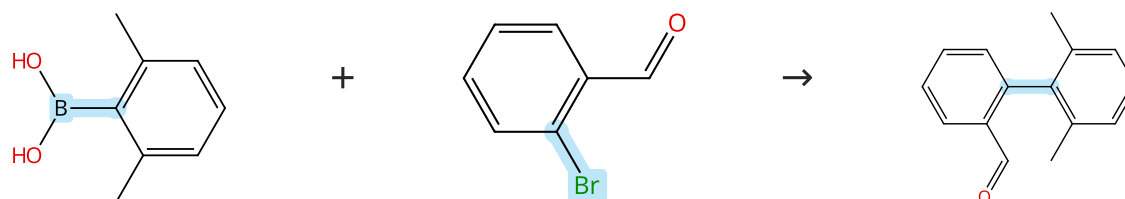
By: Hoggard, Logan R.; et al

Journal of the American Chemical Society (2015), 137(38), 12249-12260.

1.1 **Reagents:** Cesium carbonate**Catalysts:** Tetrakis(triphenylphosphine)palladium**Solvents:** Dimethylformamide; rt $\rightarrow$ 100 °C; 24 h, 100 °C

Scheme 430 (2 Reactions)

Steps: 1 Yield: 96%



Suppliers (100)

Suppliers (89)

Suppliers (11)

31-614-CAS-31847521

Steps: 1 Yield: 96%

Organocatalytic Asymmetric Dearomatizing Hetero-Diels-Alder Reaction of Nonactivated Arenes

By: Li, Kai; et al

Journal of the American Chemical Society (2022), 144(16), 7374-7381.

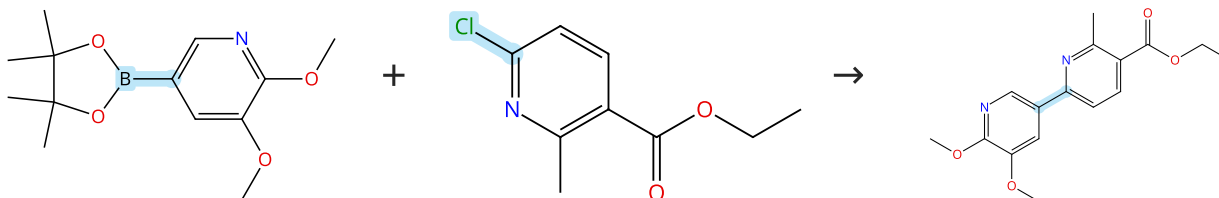
1.1 **Reagents:** Sodium carbonate**Catalysts:** Tetrakis(triphenylphosphine)palladium**Solvents:** Ethanol, Toluene, Water; 3 - 6 h, 90 °C1.2 **Reagents:** Ammonium chloride**Solvents:** Water; cooled

Experimental Protocols

31-179-CAS-23580638	Steps: 1 Yield: 96%	Imine as a linchpin approach for meta- C-H functionalization
1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; 25 h, rt → 100 °C		By: Bag, Sukdev; et al Nature Communications (2021), 12(1), 1393.
Experimental Protocols		

Scheme 431 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (69)

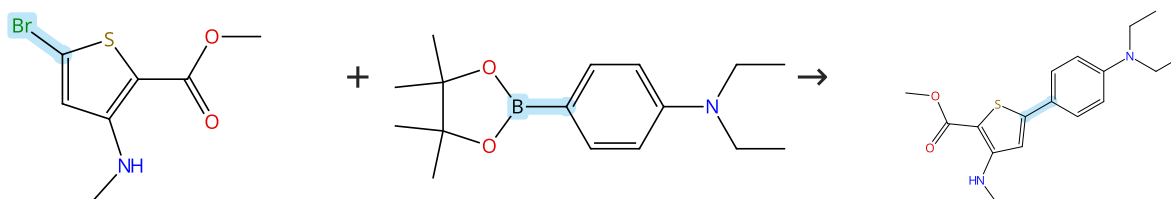
Suppliers (64)

Suppliers (2)

31-179-CAS-19035400	Steps: 1 Yield: 96%	Design and Synthesis of a Novel Series of Orally Bioavailable, CNS-Penetrant, Isoform Selective Phosphoinositide 3-Kinase γ (PI3K γ) Inhibitors with Potential for the Treatment of Multiple Sclerosis (MS)
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Acetonitrile, Water; 4 h, 90 °C		By: Come, Jon H.; et al Journal of Medicinal Chemistry (2018), 61(12), 5245-5256.

Scheme 432 (1 Reaction)

Steps: 1 Yield: 96%



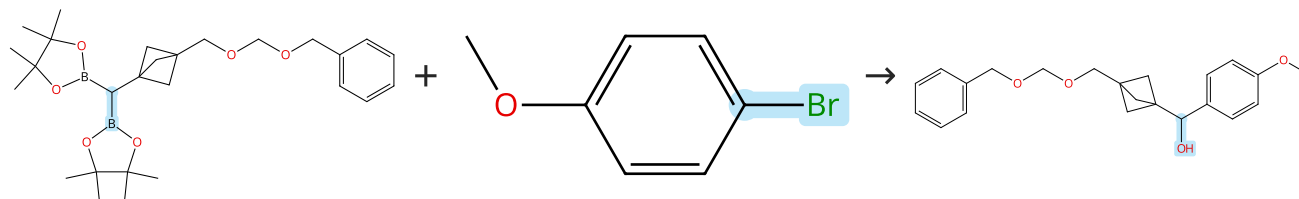
Suppliers (5)

Suppliers (63)

31-179-CAS-19478625	Steps: 1 Yield: 96%	Synthesis, Molecular Engineering, and Photophysical Properties of Fluorescent Thieno[3,2-b]pyridine-5(4H)-ones
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Toluene, Water; rt → 110 °C; 5 h, 110 °C		By: Sung, Dan-Bi; et al Journal of Organic Chemistry (2019), 84(1), 379-391.
1.2 Solvents: Water		
Experimental Protocols		

Scheme 433 (1 Reaction)

Steps: 1 Yield: 96%

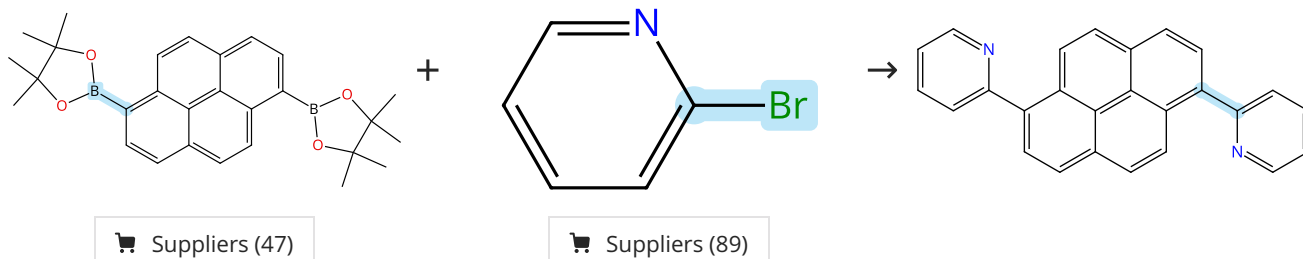


Suppliers (69)

31-614-CAS-35425056	Steps: 1 Yield: 96%	Synthesis of gem-Diboromethyl-Substituted Bicyclo[1.1.1]pentanes and Their Application in Palladium-Catalyzed Cross Couplings By: Ma, Xiaoshen; et al Journal of Organic Chemistry (2023), 88(3), 1665-1694.
1.1 Reagents: Potassium hydroxide Catalysts: Bis(tri- <i>tert</i> -butylphosphine)palladium Solvents: 1,4-Dioxane, Water; 30 min, 24 °C		
1.2 Reagents: Sodium hydroxide, Hydrogen peroxide Solvents: Tetrahydrofuran, Water; 0 °C; 30 min, 0 °C		
1.3 Reagents: Sodium thiosulfate Solvents: Water		
Experimental Protocols		

Scheme 434 (1 Reaction)

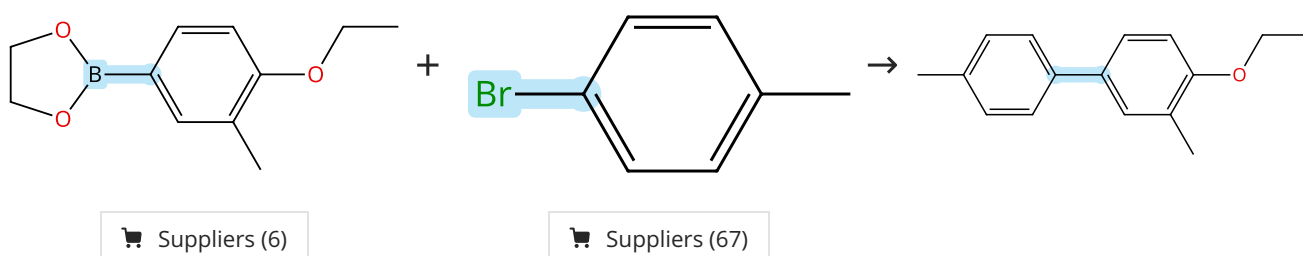
Steps: 1 Yield: 96%



31-179-CAS-19398971	Steps: 1 Yield: 96%	Cationic Nitrogen-Doped Helical Nanographenes By: Xu, Kun; et al Angewandte Chemie, International Edition (2017), 56(50), 15876-15881.
1.1 Reagents: Sodium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene, Water; 15 h, 90 °C		
Experimental Protocols		

Scheme 435 (1 Reaction)

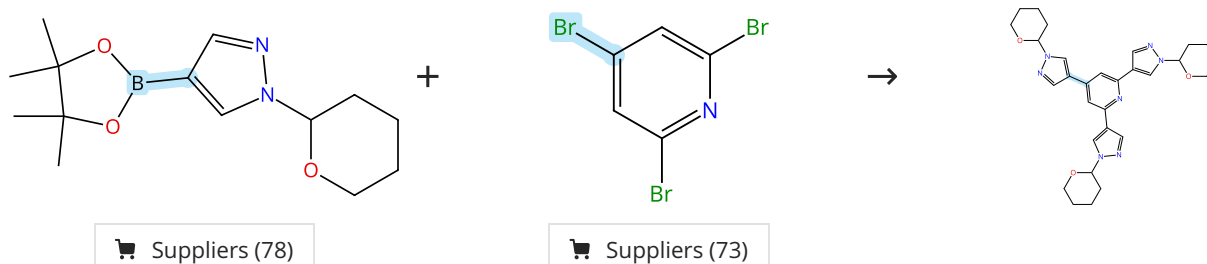
Steps: 1 Yield: 96%



31-179-CAS-6689880	Steps: 1 Yield: 96%	Superstable Palladium(0) Complex as an Air- and Thermo stable Catalyst for Suzuki Coupling Reactions By: Jakab, Alexandra; et al European Journal of Organic Chemistry (2015), 2015(1), 60-66.
1.1 Reagents: Potassium carbonate Catalysts: Tris[tris(3,5-bis(trifluoromethyl)phenyl)]phosphine-κP palladium		
Solvents: Methanol, Water; 1 h, 110 °C		
Experimental Protocols		

Scheme 436 (1 Reaction)

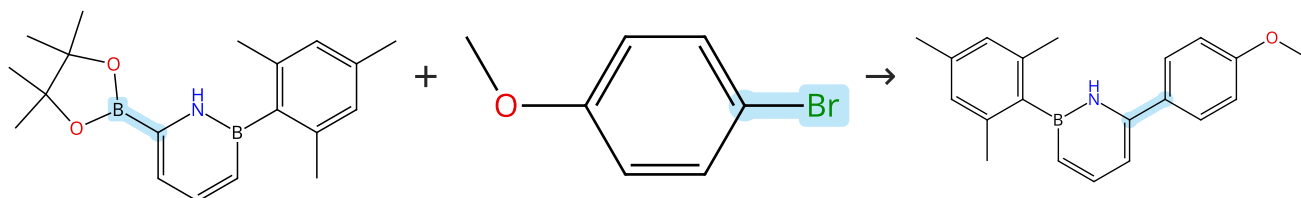
Steps: 1 Yield: 96%



31-614-CAS-37822304	A Microporous Hydrogen Bonded Organic Framework for Highly Selective Separation of Carbon Dioxide over Acetylene
1.1 Reagents: Potassium carbonate Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: 1,4-Dioxane; 24 h, reflux	By: Li, Yunbin; et al Angewandte Chemie, International Edition (2023), 62(39), e202311419.
Experimental Protocols	

Scheme 437 (1 Reaction)

Steps: 1 Yield: 96%

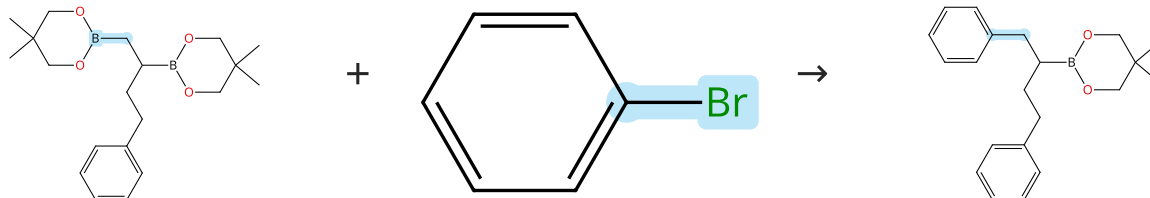


Suppliers (69)

31-177-CAS-5128543	Late-Stage Functionalization of 1,2-Dihydro-1,2-azaborines via Regioselective Iridium-Catalyzed C-H Borylation: The Development of a New N,N-Bidentate Ligand Scaffold
1.1 Reagents: Potassium hydroxide Catalysts: [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium Solvents: <i>tert</i> -Butyl methyl ether, Water; 12 h, 80 °C; cooled	By: Baggett, Andrew W.; et al Journal of the American Chemical Society (2015), 137(16), 5536-5541.
Experimental Protocols	

Scheme 438 (1 Reaction)

Steps: 1 Yield: 96%

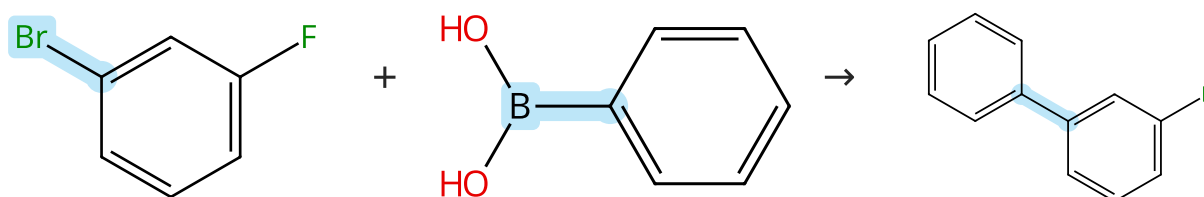


Suppliers (79)

31-614-CAS-37408177	Selective Formal Carbene Insertion into Carbon-Boron Bonds of Diboronates by N-Trisylhydrazones
1.1 Reagents: Potassium hydroxide Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine Solvents: Tetrahydrofuran, Water; 12 h, 70 °C	By: Bao, Zhicheng; et al Angewandte Chemie, International Edition (2023), 62(9), e202216356.
Experimental Protocols	

Scheme 439 (2 Reactions)

Steps: 1 Yield: 96%



Suppliers (77)

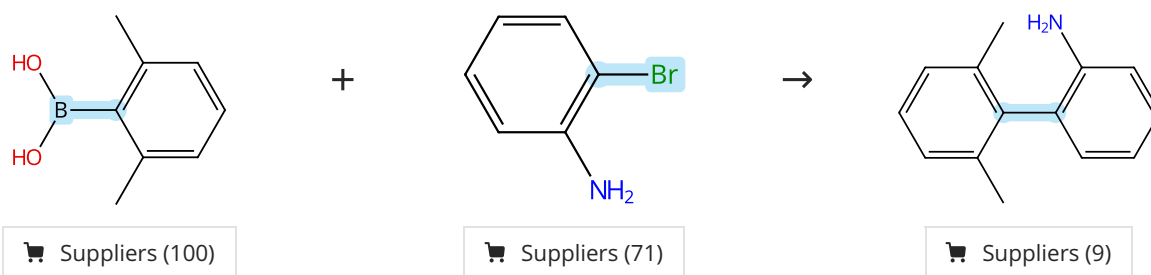
Suppliers (132)

Suppliers (50)

31-614-CAS-42289037 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: Tetrakis(triphenylphosphine)palladium Solvents: Methanol, Toluene; overnight, reflux; reflux → rt 1.2 Solvents: Water; 0 °C Experimental Protocols	Catalytic Concerted S_NAr Reactions of Fluoroarenes by an Organic Superbase By: Shigeno, Masanori; et al Journal of the American Chemical Society (2024), 146(47), 32452-32462.
31-179-CAS-17457199 Steps: 1 Yield: 96% 1.1 Reagents: Diisopropylethylamine Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct Solvents: Isopropanol, Water; 21 h, 100 °C Experimental Protocols	Copper-Catalyzed ipso-Borylation of Fluoroarenes By: Niwa, Takashi; et al ACS Catalysis (2017), 7(7), 4535-4541.

Scheme 440 (1 Reaction)

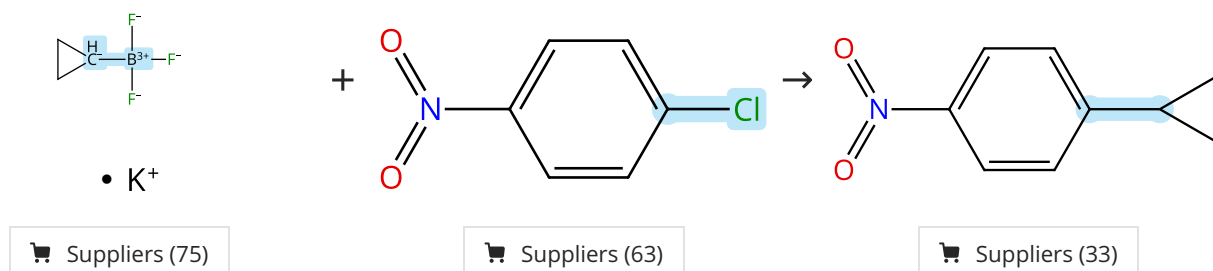
Steps: 1 Yield: 96%



31-179-CAS-23575818 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: Dichlorobis(triphenylphosphine)palladium Solvents: Dimethylformamide, Water; 24 h, 80 °C Experimental Protocols	Imine as a linchpin approach for meta- C-H functionalization By: Bag, Sukdev; et al Nature Communications (2021), 12(1), 1393.
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Scheme 441 (1 Reaction)

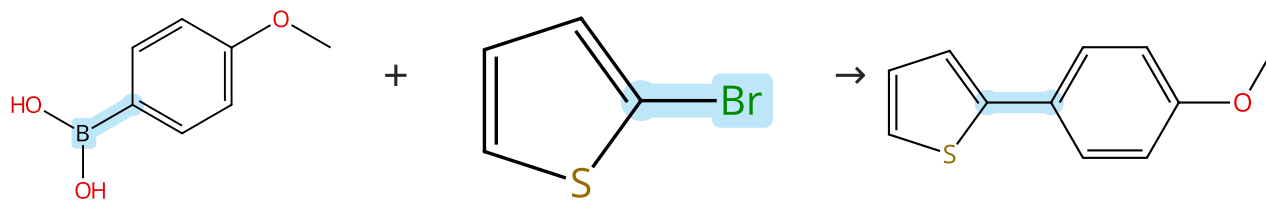
Steps: 1 Yield: 96%



31-179-CAS-18045813 Steps: 1 Yield: 96% 1.1 Reagents: Potassium carbonate Catalysts: Chloro[(1,2,3-η)-1-(1,1-dimethylethyl)-1<i>H</i>-inden-1-yl][tris(1,1-dimethylethyl)phosphine]palladium Solvents: Toluene, Water; 8 h, 80 °C Experimental Protocols	Design of a Versatile and Improved Precatalyst Scaffold for Palladium-Catalyzed Cross-Coupling: (η³-1-^tBu-indenyl)₂(μ-Cl)₂ Pd₂ By: Melvin, Patrick R.; et al ACS Catalysis (2015), 5(6), 3680-3688.
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Scheme 442 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (72)

Suppliers (41)

31-179-CAS-16954365

Steps: 1 Yield: 96%

Photo-oxidative Cross-Dehydrogenative Coupling-Type Reaction of Thiophenes with α -Position of Carbonyls Using a Catalytic Amount of Molecular Iodine

By: Sudo, Yusuke; et al

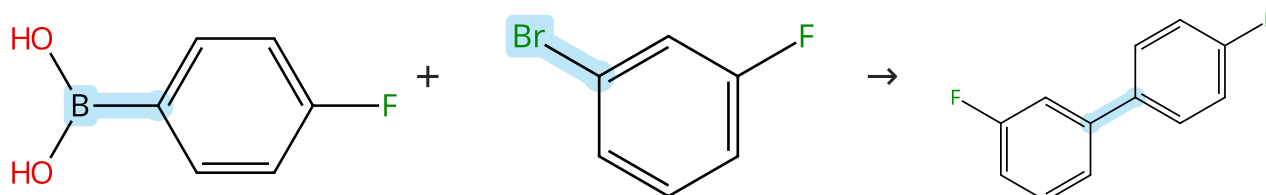
Organic Letters (2017), 19(7), 1610-1613.

1.1 **Reagents:** Sodium carbonate
Catalysts: Tetrakis(triphenylphosphine)palladium
Solvents: 1,2-Dimethoxyethane, Water; 20 h, reflux

Experimental Protocols

Scheme 443 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (104)

Suppliers (77)

Suppliers (9)

31-179-CAS-17457206

Steps: 1 Yield: 96%

Copper-Catalyzed ipso-Borylation of Fluoroarenes

By: Niwa, Takashi; et al

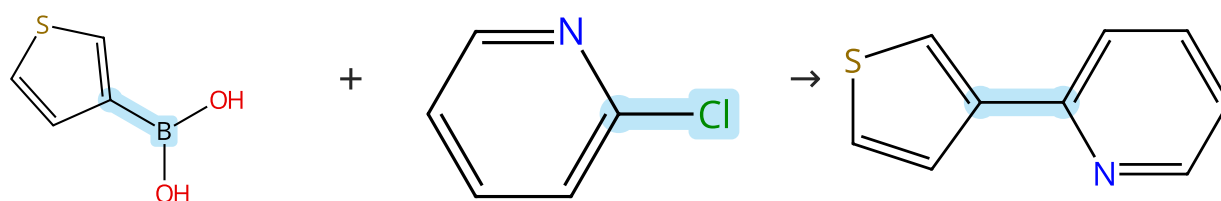
ACS Catalysis (2017), 7(7), 4535-4541.

1.1 **Reagents:** Diisopropylethylamine
Catalysts: Dichloro[1,1'-bis(diphenylphosphino)ferrocene] palladium(II) dichloromethane adduct
Solvents: Isopropanol, Water; 30 min, 120 °C

Experimental Protocols

Scheme 444 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (101)

Suppliers (66)

Suppliers (65)

31-179-CAS-16785478

Steps: 1 Yield: 96%

One-Pot Palladium-Catalyzed Cross-Coupling Treble of Borylation, the Suzuki Reaction and Amination

By: Jong, Howard; et al

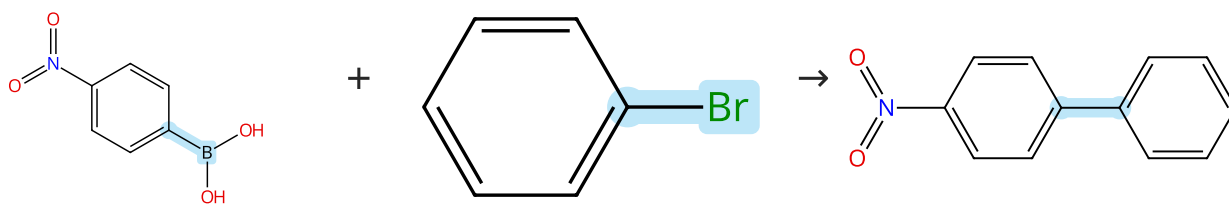
Advanced Synthesis & Catalysis (2017), 359(4), 616-622.

1.1 **Reagents:** Potassium hydroxide
Catalysts: Dichlorobis[dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1':3',1''-terphenyl]-2-yl]phosphine]palladium
Solvents: 1-Butanol; rt $\rightarrow$ 100 °C; 1 h, 100 °C

Experimental Protocols

Scheme 445 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (101)

Suppliers (79)

Suppliers (49)

31-614-CAS-46049894

Steps: 1 Yield: 96%

Palladium Complex Grafted on PEG-Based Amphiphilic Polymers as ppm Level Micellar Catalysts for Suzuki-Miyaura Coupling in Water

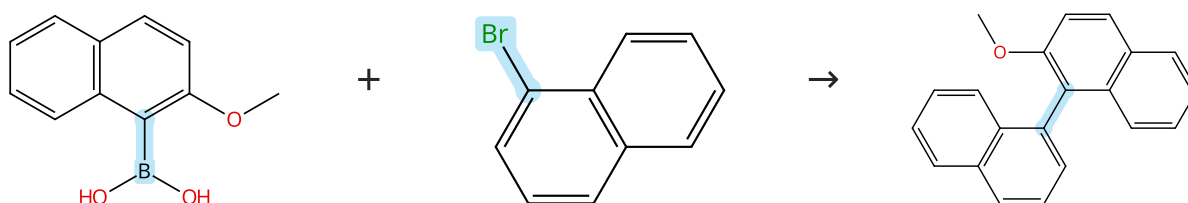
By: Xiao, Song; et al

Organic Letters (2025), 27(16), 4343-4348.

- 1.1 **Reagents:** Potassium carbonate, Tetrabutylammonium bromide
Catalysts: 3091657-75-6 (reaction products iwth 6- aminoh exane-1-thiol hydrochloride and bis(hyd...)
Solvents: Methanol, Water; 4 h, 90 °C

Scheme 446 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (66)

Suppliers (88)

Suppliers (7)

31-179-CAS-16307224

Steps: 1 Yield: 96%

Readily available catalysts for demanding Suzuki-Miyaura couplings under mild conditions

By: Demchuk, Oleg M.; et al

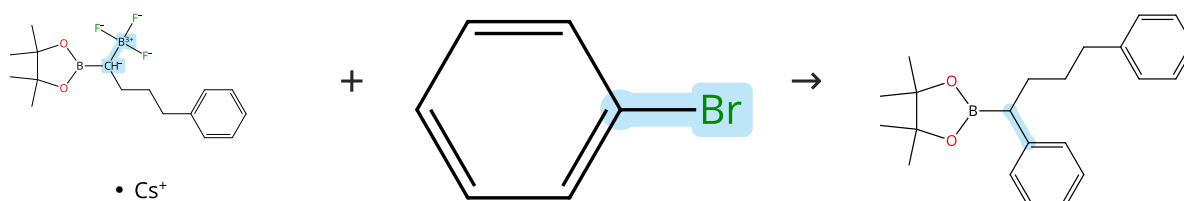
Tetrahedron (2016), 72(42), 6668-6677.

- 1.1 **Reagents:** Sodium carbonate
Catalysts: Polyethylene glycol monooleyl ether, 2015952-75-5
Solvents: Tetrahydrofuran, Water; 16 h, 75 °C

Experimental Protocols

Scheme 447 (1 Reaction)

Steps: 1 Yield: 96%

• Cs⁺

Suppliers (79)

31-179-CAS-22082569

Steps: 1 Yield: 96%

Stereoselective Desymmetrization of gem-Diborylalkanes by "Trifluorination"

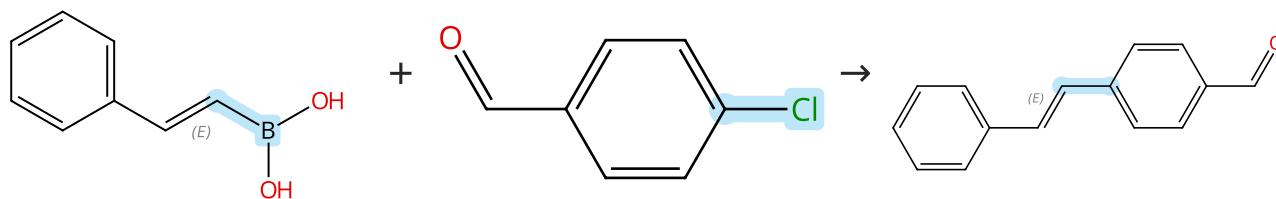
By: Kumar, Nivesh; et al

Chemistry - A European Journal (2019), 25(34), 8008-8012.

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Palladium diacetate, X-Phos
Solvents: Toluene, Water; 12 h, 80 °C

Scheme 448 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Suppliers (97)

Double bond geometry shown

Suppliers (60)

Suppliers (82)

31-177-CAS-19177778

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Potassium carbonate, Oxygen
Catalysts: 1*H*-Imidazolium, 1,3-bis[2,6-bis(1-methylethyl)phenyl]-, dichloro[(1,2,3- η)-1-phenyl-2-propen-1-yl]palladate (1-) (1:1)
Solvents: Ethanol; 1 h, 60 °C
- 1.2 4 h, 40 °C

Experimental Protocols

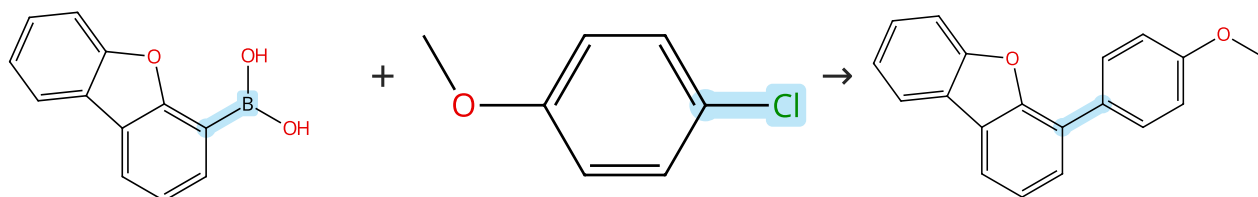
Towards environmentally friendlier Suzuki-Miyaura reactions with precursors of Pd-NHC (NHC = N-heterocyclic carbene) complexes

By: Zinser, Caroline M.; et al

Green Chemistry (2018), 20(14), 3246-3252.

Scheme 449 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (93)

Suppliers (71)

Supplier (1)

31-614-CAS-34388097

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Potassium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, 2849469-49-2
Solvents: 1,4-Dioxane, Water; 24 h, 60 °C

Experimental Protocols

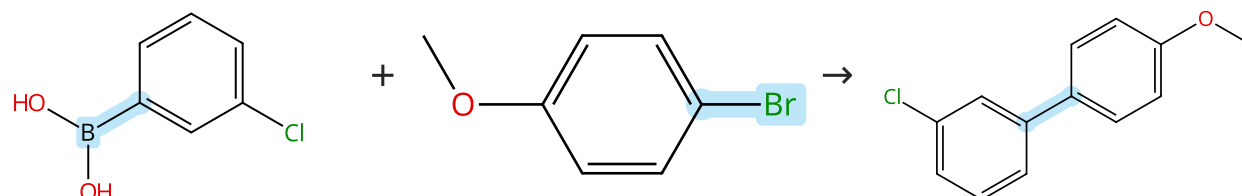
Ylide-Functionalized Diisopropyl Phosphine (prYPhos): A Ligand for Selective Suzuki-Miyaura Couplings of Aryl Chlorides

By: Wei, Xiao-Jing; et al

Advanced Synthesis & Catalysis (2022), 364(19), 3336-3341.

Scheme 450 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (112)

Suppliers (69)

Suppliers (11)

31-179-CAS-15474664

Steps: 1 Yield: 96%

- 1.1 **Reagents:** Sodium hydroxide
Catalysts: 1809605-48-8
Solvents: Water; 3 h, 100 °C
- 1.2 **Reagents:** Water

Experimental Protocols

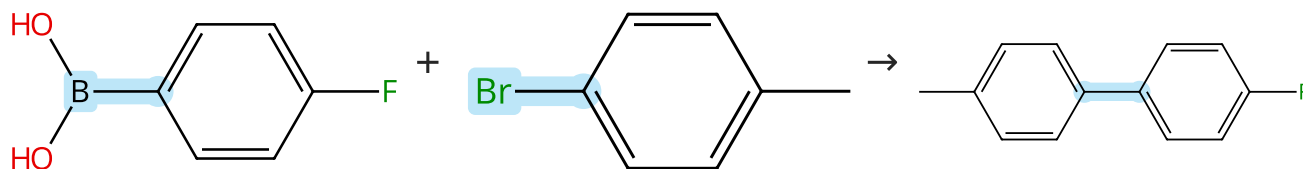
Water-soluble salen-Pd complex as an efficient catalyst for Suzuki-Miyaura reaction of sterically hindered substrates in pure water

By: Liu, Ya-shuai; et al

Tetrahedron (2015), 71(42), 7985-7989.

Scheme 451 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (104)

Suppliers (67)

Suppliers (49)

31-179-CAS-20010575

Steps: 1 Yield: 96%

Room-Temperature Aqueous Suzuki-Miyaura Cross-Coupling Reactions Catalyzed via a Recyclable Palladium@Halloysite Nanocomposite

By: Hamdi, Jumanah; et al

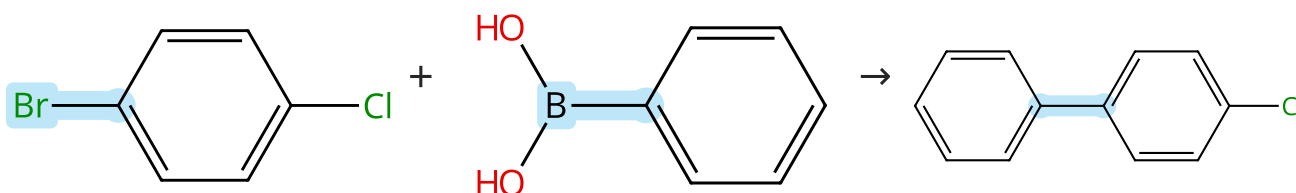
Organic Letters (2019), 21(10), 3471-3475.

1.1 **Reagents:** Cesium carbonate
Catalysts: Palladium (halloysite nanocomposite-supported)
Solvents: 1-Propanol, Water; 10 min, rt; 1 h, rt

Experimental Protocols

Scheme 452 (2 Reactions)

Steps: 1 Yield: 96%



Suppliers (82)

Suppliers (132)

Suppliers (59)

31-614-CAS-42942536

Steps: 1 Yield: 96%

Heterogenization of palladium trimer and nanoparticles through polymerization boosted catalytic efficiencies in recyclable coupling and reduction reactions

By: Wang, Miaomiao; et al

Chemistry - A European Journal (2024), 30(72), e202403447.

1.1 **Reagents:** Potassium carbonate
Catalysts: 3066141-61-2
Solvents: Dimethylformamide, Water; 80 °C

Experimental Protocols

31-614-CAS-36762625

Steps: 1 Yield: 96%

Large acceleration of solvent-free Suzuki-Miyaura cross-coupling reactions at room temperature by the addition of small amounts of dioxane and water

By: Tanemura, Kiyoshi

Tetrahedron (2023), 140, 133470.

1.1 **Reagents:** Potassium carbonate, Water
Catalysts: Palladium diacetate, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine
Solvents: 1,4-Dioxane; 1 h, 25 °C

Experimental Protocols